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◆ About the Journal

This is an annual journal publishing the best essays written by the top-performing students of the FitzEd Summer and Winter Schools, **which are the official educational programmes of Fitzwilliam College in the University of Cambridge for international students aged 14–18 year olds.** The journal covers 23 academic fields, including but not limited to: Mathematics, Physics, Artificial Intelligence, Microbiology, Elements of Mathematical Economics, Engineering, Psychology and Neuroscience, Business Economics, Chemistry, Cybersecurity and Medicine.

◆ About the Programme

The FitzEd Summer School is the official Summer School Programme of Fitzwilliam College in the University of Cambridge. **It is a unique educational programme in Cambridge where all courses are exclusively taught by academics from the University of Cambridge or one of its constituent colleges. The same academics who lecture and supervise undergraduate students at Cambridge teach undergraduate-level content at the FitzEd Summer School.** This programme is designed to provide students with a flavour of undergraduate study at Cambridge, and an opportunity to explore topics beyond what is covered within the school curriculum.

The core of Fitzwilliam's academic activities is a desire to retain 'the best of the old', while enthusiastically embracing 'the best of the new'. Fitzwilliam has always been characterised by discussion, debate and creativity of ideas and full participation should form a positive, rewarding and sustainable part of an academic course.

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01

Physics

The Evolution of Quantum Mechanics and its History of Development

XINRUI CAO

Introduction

One of the biggest scientific revolutions in history is quantum mechanics, which questions the deterministic outlook of classical physics. It is among the strangest theories in science, and no one has been able to properly explain it up to this point. With an emphasis on significant experiments and theoretical advancements that resulted in the formation of quantum theory, this essay will examine how physics changes from classical Newtonian mechanics to quantum mechanics – or rather, adapts to it. Along with discussing some recent theoretical advancements, the essay will also go over the conceptual challenges brought about by this new framework and how they were resolved.

Arise of a New Theory

Classical physics, with its deterministic framework, had long been successful in explaining macroscopic phenomena. However, as scientists began investigating the behaviour of microscopic particles, classical physics encountered significant limitations. “The beauty and clearness of the dynamical theory, which asserts heat and light to be models of motion, is at present obscured by two clouds.” (Kelvin) This is the well-known analogy that Lord Kelvin made in the 19th century, the first cloud referred to the failure to detect ether and upcoming mysteries of light propagation, while the second cloud referred to the ultraviolet catastrophe, a problem that arose in the study of black-body radiation. Unsolved problems include the blackbody radiation and the photoelectric effect. On account for the failure of classical mechanics to explain those phenomena (Feynman), the former cloud derived Einstein’s theory of special relativity, and the latter derived the quantum theory.

The Ultraviolet Catastrophe

The ultraviolet catastrophe refers to the incapability of classical physics to explain the patterns between the temperature of a blackbody and the wavelength of the emissions. Specifically, classical physics predicted that as the temperature of a black body increased, its emitted radiation would increase exponentially at higher frequencies, leading to infinite energy emission at ultraviolet wavelengths, even though the experimental observations has shown otherwise. This gave the name of the “ultraviolet catastrophe” . In 1901, Max Plank proposed his hypothesis, where energy at atomic scale is not emitted continuously, but in discrete amounts “quanta” . For radiations, each quantum of energy should follow the equation $E = h\nu$, which he introduced the Plank constant $h = 6.63 \times 10^{-34} \text{ Js}$ (Plank). This radical idea resolved the ultraviolet catastrophe by providing a formula that accurately described the radiation distribution. Despite he himself not being particularly proud of his theory, the idea of quantised energy profoundly influenced the subsequent developments of quantum mechanics.

Bohr’s Model of the Hydrogen Atom

Niels Bohr first proposed the idea of quantized energy levels in 1913 when he created a model of the hydrogen atom. Bohr believed that an atom’s electrons could only occupy certain orbits around the nucleus, each of which corresponded to a distinct energy level. This implies that the energy of electrons can only be in discrete amounts. The distinct lines in the hydrogen atomic spectrum can be explained by the discrete energy that an electron released or received in the form of light when it moved between different levels (Bohr). In order to bridge the gap between atomic structure and quantum theory, Bohr’s model was crucial in proving that quantum conceptions were necessary for comprehending atomic structure.

The Photoelectric Effect

The photoelectric effect is the phenomenon which electrons are emitted from a material when it is exposed in electromagnetic radiation with sufficient energy. The observations shown in the experiments demonstrated that the electrons are only emitted if the incident light has passed a certain threshold frequency, no matter how intense the light is. This is counterintuitive in classical mechanics, where originally it predicted the kinetic energy of electrons should depend on the intensity of light, as light is a wave, and its energy should distribute continuously over the wavefront. In 1905, Albert Einstein extended Planck's idea of quantized energy to light itself in his explanation of the photoelectric effect. He proposed that light could be thought of as being made up of discrete packets of energy called photons. The energy of a photon is also given by $E = h\nu$, so when light of a sufficient frequency hit the electron, the photons have enough energy, and they are able to knock off electrons. This explanation accounted for experimental observations that classical wave theory could not, including the fact that the intensity only increases the numbers of photons, so the increasing intensity increases the number of electrons ejected only if they pass the threshold frequency. (Einstein). Einstein's work provided further evidence for the quantum nature of light and contributed as evidence to wave-particle duality of light.

Wave-Particle Duality and Double Slit Interference

One of the most profound revelations of quantum mechanics was the wave-particle duality of matter. In 1924 French physicist Louis de Broglie suggested all particles have an associated wave, called a matter wave or a de Broglie wave. He extended the concept of wave-particle duality beyond light (photons) to all matter. Additionally, experiments such as the Davisson- Germer experiment (1927), which demonstrated electron diffraction, also showed that particles like electrons have both particle-like and wave-like behaviour. In quantum mechanics, this is explained to be the existence of probability waves, and how they collapse back to particles when they are observed.

Schrödinger's Wave Functions

Erwin Schrödinger created his wave equation in 1926 as a mathematical representation of quantum mechanics that explained how a particle's quantum state changed over time. He substituted a probabilistic description of particles for the deterministic trajectories of classical mechanics, meaning that we can never be absolutely certain of a particle's position. The probability distribution of a particle's potential location is given by the wave function, which is the solution to the wave equation. (Schrödinger) The wave function ψ fully describes a particle's position, momentum, and other characteristics at any given moment. This probabilistic interpretation completed the quantum theories with mathematical specifics and marked a substantial break from traditional deterministic theories.

Uncertainty Principle

In 1927, Werner Heisenberg formulated the uncertainty principle, which became one of the cornerstone concepts of quantum mechanics. He states that you can only have limited precision of measurements for position and momentum – it means the more precisely the position of a particle is measured, the less precisely its momentum can be determined, and vice versa. Satisfying the mathematical relation $\Delta x \cdot \Delta p \geq 2\hbar$. This uncertainty is present due to the fundamentals of quantum mechanics, that measurements (observations) directly interfere with the results of experiments.

Interpretations

The theory of quantum mechanics has been challenged throughout its history and in multiple ways. Till nowadays these debates are still ongoing and lack a definite answer. Multiple interpretations arise, which one of the most significant interpretations is the Copenhagen interpretation, developed primarily by Niels Bohr and Werner Heisenberg in 1920s. This interpretation believes particles exist as a superposition of all position states and only collapse into one specific outcome when measured. Bohr also suggests the principle of complementarity, that no single experiment can reveal the full nature

of the system. Examples like the Schrödinger's cat (the thought experiment showing the absurdity), and Einstein's famous quote "God does not play dice" criticises these opinions. On the other hand, the many-worlds interpretation suggests the wave function collapse will have all the possible outcomes happen in parallel, each having a different "branch" of universe. Another alternative is the hidden variables theory, challenging the Copenhagen interpretation of quantum mechanics to be incomplete, that is, they suggest there are still "hidden variables" that humans yet not discovered yet which determines the definitive status of a particle.

The EPR Paradox

In 1935, Albert Einstein, Boris Podolsky, and Nathan Rosen (EPR) published a paper that questioned the completeness of quantum mechanics. The EPR paradox, as it came to be known, arose from the concept of quantum entanglement and its apparent violation of the principle of locality – the idea that an object is only influenced by its immediate surroundings, and information cannot travel faster than the speed of light. It involves a thought experiment in which the measurement of one particle's properties (such as spin or position) will instantaneously determine the corresponding properties of the other one. It means the act of measurement affects the other particle. This is called quantum entanglement. Using Einstein's words, this "spooky action at a distance" may suggest that information can travel faster than light. He argued that quantum mechanics could not be a complete theory because it allowed such non-local interactions (Einstein, Podolsky, & Rosen, 1935) and would need "hidden variables" to preserve local realism (so that objects have definite properties independent of measurements). The EPR paradox has a crucial role in the future development of quantum mechanics, and it raised more deeper questions.

Current Research

Till today, the quantum theory is still waiting to be refined and developed, and there are still unsolved mysteries in this theory discovered 100 years ago. Researchers are also exploring quantum non-locality and whether it could lead to new insights into the fundamental nature of space and time. This area is closely related to Bell's theorem and experiments testing the Bell inequalities. The current experiment results violated the Bell inequalities, which supported the Copenhagen interpretation and the non-local realism, also rejecting the possibility of having a local hidden variable. (Bell's Theorem (Stanford Encyclopaedia of Philosophy)) Therefore, quantum entanglement is experimentally verified and is now an essential feature of quantum theory. Currently, researchers are also contributing to studying how to unify and bridge the gap between general relativity and quantum mechanics, aiming to explain how gravity operates at the quantum level, such as near black holes or in the early universe. Examples of popular theories include the string theory and the quantum gravity theory, but there is still no validated theory yet.

Conclusion

The transition from classical physics to quantum mechanics revolutionized our understanding of the physical world. Classical physics, with its deterministic laws, was ill-equipped to explain phenomena at the atomic and subatomic scales. The introduction of quantum theory, through key experiments and the development of new mathematical frameworks, provided a much-needed paradigm shift. However, quantum mechanics also brought with it a host of conceptual challenges, such as wave-particle duality, the uncertainty principle, and the measurement problem, which continue to be the subject of intense study and debate. It is just as one quote of Richard Feynman, "If you think you understand quantum mechanics, you don't understand quantum mechanics."

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The Photoelectric Effect: An Insight into Light-Matter Interaction

TIAN QIU

Introduction

When Heinrich Hertz first observed the photoelectric effect in 1887, the observation remained one of the most famous experiments that bridge classical and quantum physics. Although its original discovery baffled scientists, Albert Einstein's 1905 explanation in terms of light quanta not only eliminated the paradoxes in classical wave theory but also paved the road to modern quantum mechanics. Also notice that this paper pertaining to the photoelectric effect discusses both: (1) the historical context both with respect to the experimental setups as well as the challenge to classical physics with the overarching implication with the study of wave-particle duality in quantal systems and (2) the surrounding allegations with respect to technological innovation.

Experimental Observations and Classical Inadequacy

The photoelectric effect happens by shining light onto a metal surface, causing electrons to be emitted from that metal as long as the frequency of light surpasses a threshold specific for that material. Key observations include: Threshold Frequency: No electrons are emitted below a critical frequency ν_0 , regardless of light intensity. Instantaneous Emission: We view electrons get ejected as soon as the light is shone on them; there is no timestamp or delay. Kinetic Energy Dependence: The maximum kinetic energy of emitted electrons depends on the light's frequency, not its intensity. Classical electromagnetic theory, which treats light as a continuous wave, spectacularly failed to explain these phenomena., in James Clerk Maxwell's framework, energy transfer from light to electrons should depend on the wave's intensity (amplitude) rather than its frequency. Furthermore, it had also been unfavorable to explained that electrons had gained some energy when emitted after several previously attempts through forming a plasma before escaping from the metal—which was exactly the opposite of what was detected, which was instant emission. These inconsistencies pointed to a fundamental flaw in classical physics: light could not be fully described as a wave.

Einstein and the Quantum Hypothesis: Light as Particles

Einstein had a radical answer in 1905: Light is made of discrete packets of energy, called quanta (later called photons). The energy of each photon is related to its frequency:

$$E = h \nu$$

where h is Planck's constant. Einstein assumed a photon ejects its entire energy to a single electron. For this photon to lead an electron to escape from the metal, the photon's energy must be higher than the metal's work function (ϕ) that is the minimum energy needed to overcome atomic binding forces. Hence, the maximum KE of the electron is:

$$K_{\max} = h \nu - \phi$$

This model explained all experimental anomalies very elegantly:

- There is a threshold frequency $h \nu < \phi$, Beyond which electrons are emitted.
- Kinetic Energy Linearity: $K_{\max}$ does not depend on the intensity of light but only on ν .
- Instantaneity: Energy transfer Emission in the same place
- The process is instantaneous: Energy is transferred in discrete collisions between single-photon-electron pairs.

Einstein's theory faced a hostile reception, having contradicted for the first time the long-accepted wave paradigm. Nevertheless, Robert Millikan's careful measurements (1914–1916) confirmed the linear relationship between $K_{\max}$ and ν , effective confirmation of the quantum hypothesis that won Einstein the 1921 Nobel Prize.



Wider Implications for Quantum Theory

The photoelectric effect sparked a shift in paradigm in physics, with repercussions of vast scope:

Wave-Particle Duality:

The photons of Einstein added the idea that light behaves in both wave-like (e.g., interference) and particle-like (e.g., quantized energy) ways. Later, this duality was also applied to matter by Louis de Broglie in 1924, who theorized that particles were also wave-like, as may be the case with electrons.

Formalization of quantum mechanics:

The photoelectric effect highlighted the limitations in his current theoretical foundation. Schrödinger's wave equation (1926) and Heisenberg's matrix mechanics (1925) developed as complementary formulations of quantum theory, both grounded in quantized energy exchange.

Technological Innovations:

- Photovoltaic Cells: The conversion of solar energy utilizes the photoelectric effect; photons free electrons in semiconductors, which generates electric current.
- Photodetectors and Imaging: Modern CCD cameras and photomultiplier tubes use photon-electron interactions for the detection of light.
- Quantum Optics: Lasers and single-photon sources exploit the quantization of light-matter interactions for computing and cryptography applications.

Influence on Philosophy and Methodology

The photoelectric effect is an illustration of how experimental discrepancies motivate scientific upheavals. Einstein's ability to abandon even deep ingrained classical assumptions — well-tested over vast ranges of experience — illustrates a fundamental principle in propelling the sciences forward: that of the importance of conceptual audacity. It also illustrates the interplay between theory and experiment: Millikan's empirical confirmation was needed in order for the quantum hypothesis to be accepted, showing even revolutionary ideas need to be tested with rigorous experiments.

Conclusion

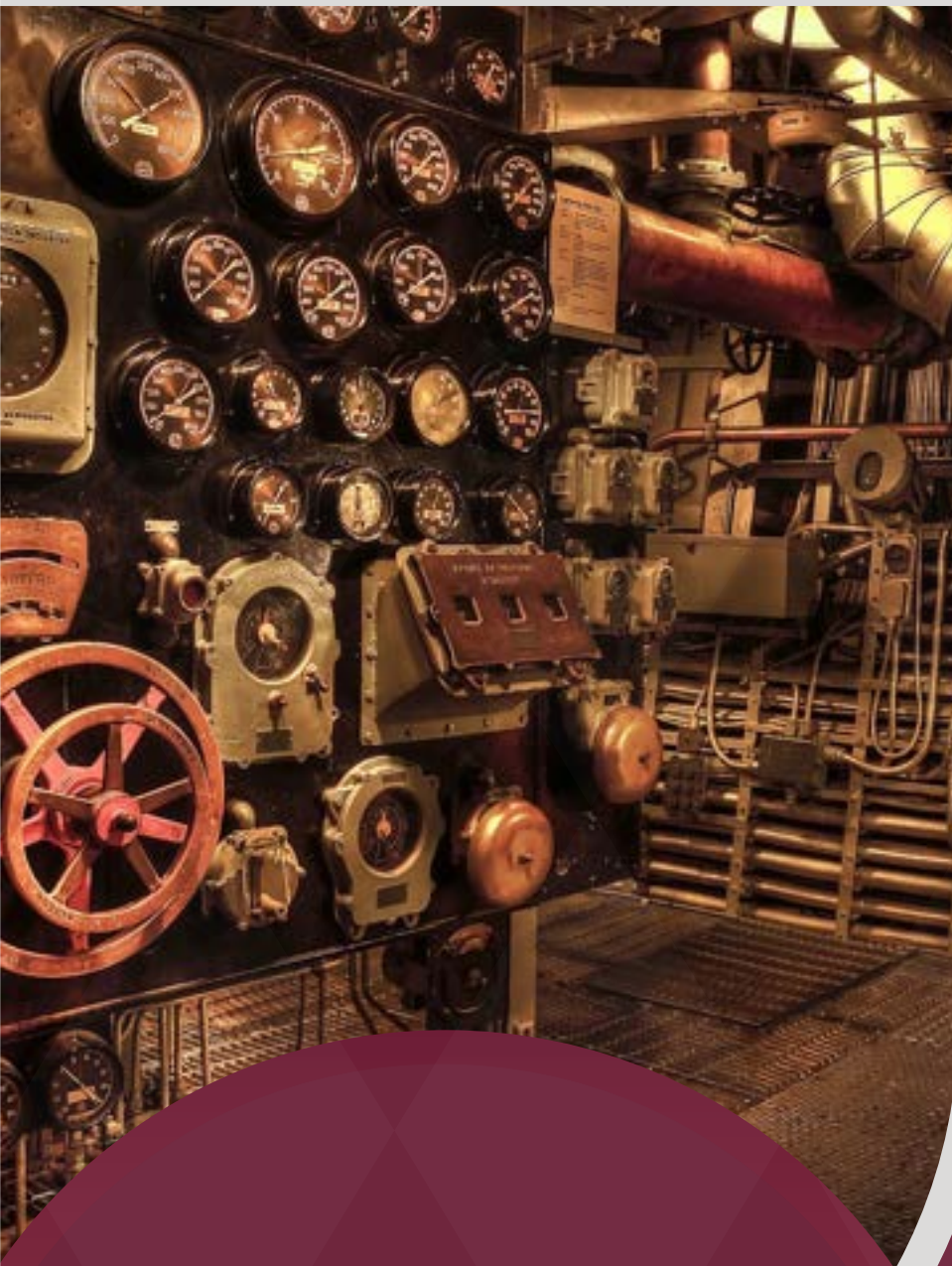
The photoelectric effect is not just a historical footnote; it became a fundamental concept in quantum theory, proof to power of rethinking physical principles. Connecting classical and quantum physics, it transformed our understanding of light and matter and helped catalyze technologies that shape the modern era. As we continue to probe quantum phenomena, from entanglement to quantum computing, the lessons of the photoelectric effect are ever relevant: each advancement often lies in questioning the givens that seem most unshakeable.

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20



Middle School – Engineering

Are Vacuum Tube Audio Amplifiers Better than Solid-state Audio Amplifiers?

SUNNI YANG

1. Abstract

There are many conflicting views on whether vacuum tube (VTAs) or solid-state audio amplifiers (SSAs) are the better choice for the various users of such technology, from audiophiles to musicians. Through analysis of the published literature, this paper seeks to clarify which approach to amplification is better for each end-user's needs. The fundamental technology of each amplification type is discussed, with advantages and disadvantages highlighted, affording insight into the choices available and their impact on artistic and mechanical priorities. Vacuum tube amplifiers, with their warm sound and harmonic richness, are ideal for musicians who are seeking expressive, dynamic tones. Solid-state amplifiers offer durability, accuracy, and efficiency, making them better suited for studio monitoring and other applications that need accuracy and consistency of sound reproduction.

2. Introduction

Since the invention of vacuum tube audio amplifiers and solid-state audio amplifiers, in the early 20th century, vacuum tube amplifiers (VTAs) were praised for their “warm” tonal qualities caused by harmonic distortion and dynamic compression. The unique sound they made was used by artists to define iconic guitar tones, cementing the foundation of different music genres (Braatz, 2020). However, VTAs face their own challenges: their fragile structure, thermal inefficiency, and high maintenance cost. On the other hand, the immediate, stable, and accurate performance of solid-state amplifiers (SSAs) enabled by semiconductor technology prioritises efficiency with low distortion. Through analysis of the published literature, this paper seeks to clarify which approach to amplification is better for each user's specification. The fundamental technology of each amplification type is discussed, with advantages and disadvantages highlighted for each application. Though neither technology is “better” than the other, this paper affords those interested in amplification technology an insight into the choices available and their implications in end-use, considering both artistic and mechanical priorities.

3. Vacuum Tube Audio Amplifiers

Vacuum tubes amplify signals by controlling electron flow in a vacuum, also called thermionic emission. Electrons are emitted from a heated cathode and are accelerated toward an anode under high voltage (CS Odessa, 2023). The grid between the cathode and anode controls the flow of electrons, which amplifies the signal (Jones, 2003).

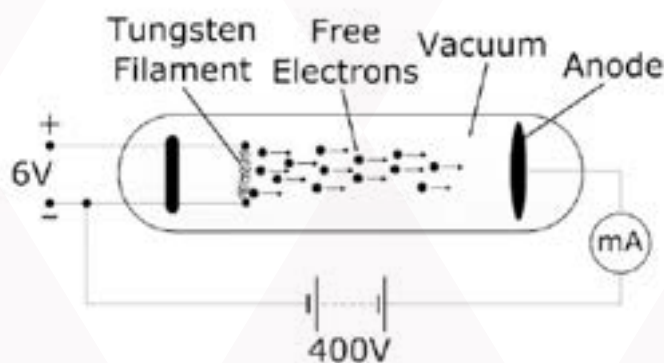


Figure 1.1 Thermionic Emission (data source: Approachables, 2023)

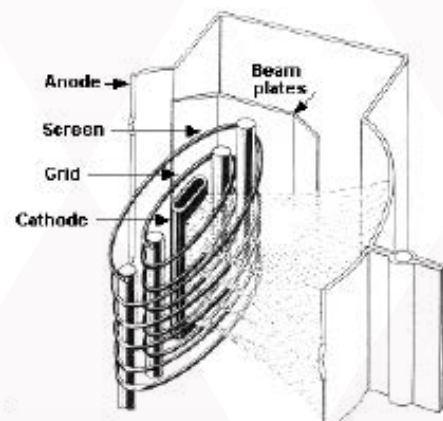


Figure 1.2 Vacuum Tube structure (data source: Burnett, 2024b)

The movement of electrons under high voltage requires tubes to operate at extreme temperatures, which necessitates strong thermal management. Heat-resistant and ventilated materials are essential, and high voltage capacitors that have discharged to a safe voltage (normally no more than 24 volts) should be used to prevent overheating (Conlan99, 2023). Amplification refers to a small amount of power at the input enlarged to the necessary level of output (loudspeaker). Critical design elements include high-voltage (300 – 500V) power supplies, biasing circuits, and output transformers (Horowitz & Hill, 2021). Beyond the thermal issues, vacuum tubes are fragile. Shocks or vibrations may easily cause damage to internal structures and performance degradation. To encounter this problem, shock-absorbing chassis are designed to protect them during transportation (Sweetwater, 2020).

When vacuum tube audio amplifiers were invented, they allowed the players to control touch dynamics and harmonic overdrive, which then became the standard way of sound amplification for electric guitars (Brief, 2025). This process generates harmonic distortion, which adds overtones to the original sound frequency. While distortion may seem like a disadvantage, the brass-like clipping effect, along with the controlled sustain effect formed by signal compression, represents the unique style of different music genres (e.g., blues, rock ...). Lately it has become a desirable characteristic for both professional and amateur guitarists (Barbour, 1998).

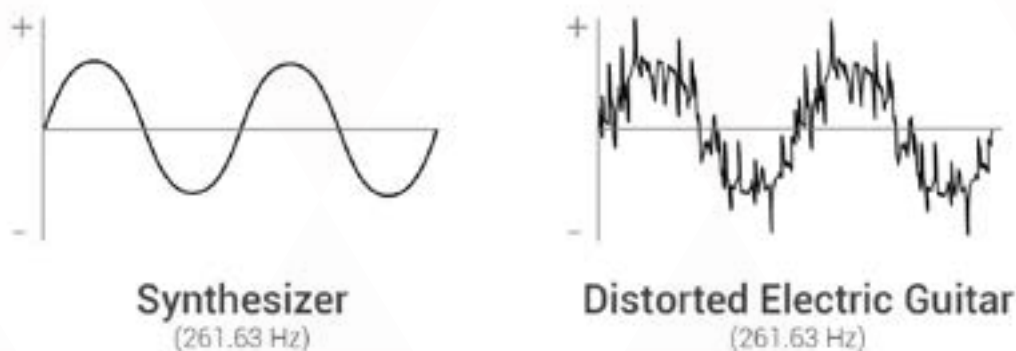


Figure 2 Harmonic Distortion (data source: Phongchit, 2017)

4. Solid-state Audio Amplifiers

SSAs amplify signals by using semiconductor devices and transistors. SSA consists of 3 sections: input stage, gain stage, and output stage (Arar, 2021). A bipolar junction transistor (BJT) is generally used as an amplifier. The bipolar transistor is manufactured with three semiconductor regions consisting of N-type and P-type semiconductors (especially pure silicon) (Takahashi, 2005). The base, collector, and emitter are doped differently. There are 2 types of bipolar transistors, pnp type and npn type. Both require correctly biased two pn junctions with an external DC voltage to operate properly

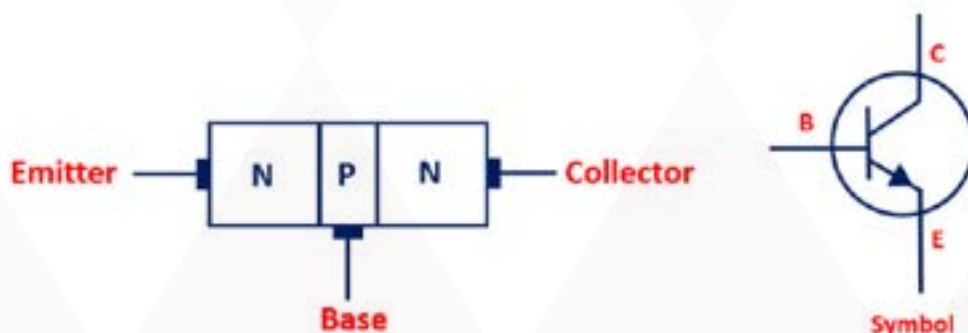


Figure 3 Bipolar Junction Transistor (data source: admin, 2020)

Compared to VTAs, SSA uses thermally conductive aluminum oxide dielectric layers, minimizing heat dissipation. The aluminium heatsink efficiently dissipates heat generated by transistors. A common SSA design called class-D amplifiers can achieve over 90% of efficiency, which is a huge improvement compared to < 30% efficiency of a VTA (Tong, 2022). SSAs ensure the accurate sound reproduction, especially for applications like studio monitoring. Besides, surface-mount packaging technology enables compact and lightweight designs for portable devices, ensuring reliability and adaptability of the SSAs (Muhammad Sufyan, 2025). Solid-state amplifiers are favoured by musicians due to their reliability, affordability, and consistent performance. SSA transistors generally have a lower failure rate than tubes, are also less expensive to manufacture, making them budget-friendly, and have low maintenance demand. Some people maintain that VTAs sound louder than SSAs; that is because solid-state hard-clipping, controlling the overall volume not to reach the limit. Which may sound nasty compared to the desirable distortion sound made by tubes (Low, 2015).

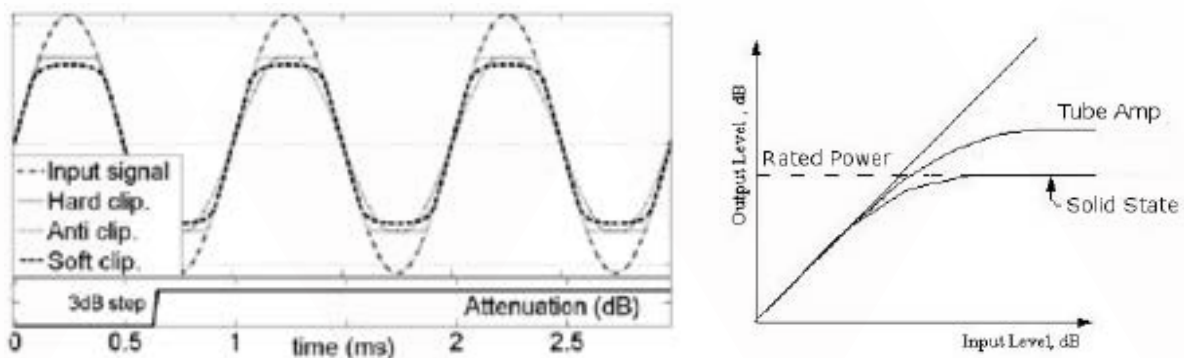


Figure 4 Hard clipping vs soft clipping (datasource: Low, 2015)

5. Comparative Analysis

Through the analysis of the published literature and past research, below is a comparative analysis of the two types of amplifiers with additional information and parameters. To make a comprehensive decision, both artistic and mechanical priorities should be considered.

Parameter	VTAs	SSAs
Efficiency	< 30%	>90%
Distortion	Harmonic Distortion	<0.1% THD (Total Harmonic Distortion)
Output Impedance	High (High-voltage power supply & transformer required)	Low (direct coupling)
Lifespan	Tube: 2000-5000 hours	Transistor: >100k hours
Dynamic Response	Interactive with sensitive feedback	Linear response, lack of "touch" feedback
Maintenance	Repair tube	Reliable, no need for further maintenance

Table 1. Comparative data of tube versus solid state amplifiers (data sources: Barbour 1998, Braatz 2020, Horowitz & Hill 2021, Reich 2005, Collection Of Data & Army 1952, Jones 2003)

6. Conclusion

In conclusion, neither vacuum tube nor solid-state amplifiers are “better” than the others. They each have benefits in different contexts. Vacuum tube amplifiers, with their warm sound and harmonic richness, are ideal for musicians who are seeking expressive, dynamic tones, expressing extended techniques easily in genres like rock and blues, providing artistic coloration of sound. On the other hand, solid-state amplifiers offer durability, accuracy, and efficiency, making them better suited for studio monitoring and other applications that need accuracy and consistency of sound reproduction. To sum up, the choice depends on the user’s priorities: artistic expression, or technical performance, highlighting the importance of matching amplifier technology to application.

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How Many Strokes? The Design of Internal Combustion Engines

CHULE SUN

Abstract

Internal combustion engines have designs with different numbers of strokes to fulfill critical needs with inevitable trade-offs. This report compares two, four, five, and six-stroke engines, analyzing thermal efficiency and inspecting the diminishing return of the number of strokes. The results show that the four-stroke engine is the most common one that best compromises production cost, structural complexity, emission, noise, and reliability. Nonetheless, the six-stroke engine with higher efficiency but similar configuration will soon take this role.

1. Introduction

The performance and feasibility of different stroke engines are affected by multiple factors, including friction, cost, and emissions. Establishing the appropriate number of strokes within an engine is an important consideration in its design and application. Through thermodynamic analysis and data of models, this report investigates the efficiency of engines and explores the diminishing point of the number of strokes.

1.1 Internal Combustion Engine

The internal combustion engine is a heat engine that releases chemical energy from fuel and converts it into mechanical energy, aiming to maximize thermal efficiency. From the earliest single-cylinder engine to advanced engines of the twenty-first century, the imperative challenge has been to reduce heat waste under thermodynamic laws, which determined that most of the energy from burnt fuel is lost during the process of conversion. A typical modern fourstroke car engine only turns 20 – 40% energy of burning gasoline into rotational motion [1]. However, the rest of the heat is being removed from the car as hot gas and wasted in the cooling system, resulting in energy inefficiency. The pursuit of high efficiency is strongly promoted by government regulations on fuel and carbon emissions, customers' demand, and the expectations of vehicle performance. These questions guide the engineers to design a variety of internal combustion engines with different strokes to experiment with heat transfer and improve energy usage.

1.2 Stroke

In the case of an internal combustion engine, a stroke is defined as a phase of the cycle, where the camshaft completes a half revolution and the piston travels from top dead center to bottom dead center, or in reverse, as per Figure 1.

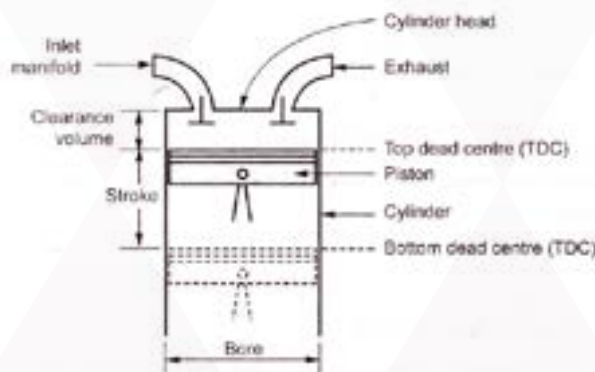


Figure 1: The structure of a combustion chamber[2]

This linear translation of the piston compresses and decompresses the air-fuel mixture, allowing combustion and air exchange to occur, that ensures the engine remains in a continuous and repeating cycle. Along with the increasing demand

on energy conversion efficiency, different stroke engines are appearing to redefine the design concept, in which a n -stroke engine has a number of n strokes in a cycle.

2. Thermodynamics of Reciprocating Engines

For a combustion engine, the primary purpose is to use as much heat as possible to generate rotational energy. In order to examine the percentage of heat produced that has been effectively utilized, this ratio can be calculated as per Equation 1 [3, p. 194].

$$\eta = \frac{W_{out}}{Q_{in}} = \frac{|Q_{in}| - |Q_{out}|}{|Q_{in}|} = 1 - \frac{|Q_{out}|}{|Q_{in}|} \quad (1)$$

Where η is thermal efficiency, Q_{in} is the heat input from the hot reservoir, Q_{out} is the remaining heat in the cold reservoir, and W_{out} is the mechanical work done. Ideally, the temperature in the cold reservoir should be 0 kelvin; however, it is impossible to reach absolute zero. This means η can never reach 100%, in addition, there are energy losses in case of a manufactured vehicle, causing the heat efficiency to be lower. First, the friction between the piston and the wall of the cylinder takes away energy, increasing with the number of strokes, which causes frictional force to be applied over a greater distance. Second, the surrounding of burnt fuel, the engine itself, conducts heat that makes the engine hot during operation. Third, the waste gases are not completely eliminated from the cylinder, which means they negatively influence the balance of the air-fuel mixture. Under these challenges, engines are invented for specific functions and conditions.

3. Different Stroke Comparison

This section compares different stroke engines using the four-stroke engine as a baseline.

3.1 4 Stroke Engine

A four-stroke engine is the most common heat engine used in automobiles and other land transport due to its reliable power, less noise than 2 stroke engine, and simpler structure than more stroke engines. This kind of engine has four stages as per Figure 2:

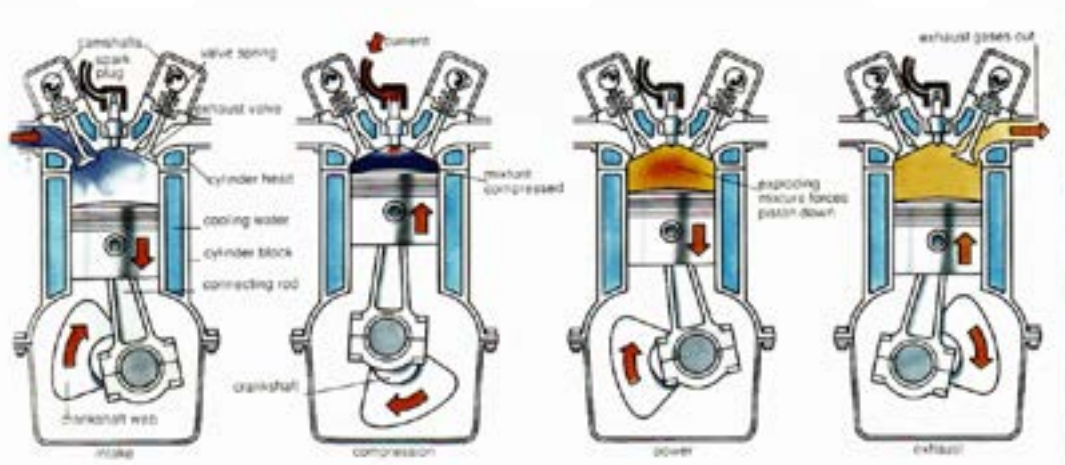


Figure 2: Combustion cycle of a 4-stroke engine

Intake The piston moves down and the intake valve opens, pulling air into the cylinder by creating a negative pressure.

Compression Both of the two valves are closed, and the piston begins to move upward, which increases the pressure, while the fuel injector sprays fuel into the air.

Power The spark plug ignites the fuel-air mixture, which creates an explosion and pushes the piston down.

Exhaust The exhaust valve opens and the piston travels up to remove waste gas.

As an illustration, Toyota Dynamic Force is a representative four-stroke engine in the industry, whose maximum thermal efficiency under gasoline reaches 40% and 41% with hybrid [4]. This is about the cap efficiency the industry can achieve with available technology in a four-stroke engine.

3.2 2 Stroke Engine

Another widespread engine is the two-stroke engine that is often used in motorcycles because its structure is simpler, which allows it to be half the size of a four-stroke engine [5]. The way it inputs and outputs gas depends on only one valve and relies heavily on pressure instead of two valves, which reduces necessary structures as per Figure 3.

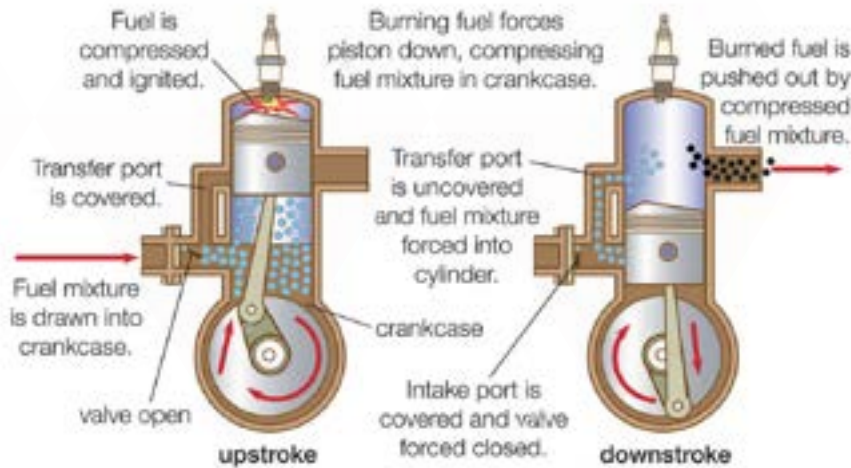


Figure 3: Combustion cycle of a 2-stroke engine

A two-stroke engine completes 2 cycles in one of the four-stroke engine, producing more power and achieving higher indicated thermal efficiency than a four-stroke engine under the same conditions [5].

3.3 5 Stroke Engine

Five-stroke engines are rare due to their complexity; unlike other engines, it is composed of two high-pressure chambers and one low-pressure chamber as per Figure 4.

The three cylinders are working in different phases during operation as shown in Table 1 [7].

In the range of 2600 to 3900 rpm estimated, the five-stroke engine shows greater torque and power, and lower fuel consumption in regard to a fourstroke engine [7]. This means its thermal efficiency is higher than the conventional four-stroke engine at a certain rotation speed.

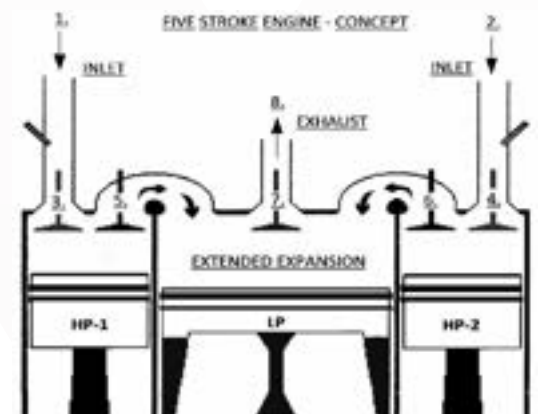


Figure 4: An illustration of five-stroke engine [6]

High-pressure cylinder	Low-pressure cylinder	High-pressure cylinder
Intake	Exhaust	Power
Compression	Additional Power	Exhaust into low-pressure cylinder
Power	Exhaust	Intake
Exhaust into low-pressure cylinder	Additional Power	Compression
Intake	Exhaust	Power

Table 1: The cycle of a five-stroke engine

3.4 6 Stroke Engine

The structure and cycle of a six-stroke engine are highly identical to a four-stroke engine that adds two phases to the four-stroke cycle, doing work twice in 3 revolutions compared to once in 2 revolutions of a four-stroke engine:

Recompression The exhaust from combustion is partially released, and water is injected while the piston travels up.

Steam Expansion Water vapors into steam and expand that pushes the piston down, then the rest of the waste is exhausted.

Through this cycle, additional energy is transferred to the piston by steam expansion, assisting the new Porsche six-stroke engine to achieve a thermal efficiency of 50% which is a significant increment from the four-stroke engine [8]. On the other hand, the pollution of NO_x is greatly reduced by 80% while HC is raised by 85% [9].

4. Application

Every different stroke engine has its own purpose and advantages that are deployed in vehicles with particular requirements. Four-stroke engines are usually installed in automobiles and trucks because they are reliable, by having a well-designed cylinder, and easily pass emission regulations. Two-stroke engines are often found in motorcycles in behalf of simple architecture and cheap manufacturing price, however, they produce more harmful gases since the combustion cycle is shorter. Five-stroke engines are not being adopted due to their complexity, contributing to more errors, and remarkably high cost. Sixstroke engines are yet to be implemented in any vehicle, but they have the potential for better thermal efficiency and lower emissions since the fuel undergoes more complete combustion. For more strokes, the mechanical structure will become excessively complicated and costly for manufacture while not improving a reasonable extent of thermal efficiency due to friction.

5. Conclusion

The performance of an internal combustion engine varies with the number of strokes in terms of thermal efficiency and toxic gas emission through utilized fuel consumption and enforced energy conversion. Nevertheless, performance correlates with the cost of production to determine the practicability of an engine design, which induces the question of diminishing return in respect to the number of strokes. This facilitates the success of four-stroke engines and the failure of five-stroke engines, yet this trend is soon to be shuffled by the six-stroke cycle, now waiting for scalable manufacturing protocols.

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03

Middle School -Mathematics

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Multinomial Distribution

ZUOYANG LIU

When rolling a 36-faced die 144 times, what is the probability that each face appears exactly 4 times?

To solve this kind of question, using multinomial distribution is a good and easy way.

1. Introduction

The multinomial distribution is a discrete probability distribution that generalizes the binomial distribution to experiments with more than two possible outcomes. Formally, it describes the probability of observing a specific combination of counts across multiple categories in a fixed number of independent trials, where each trial results in exactly one of the possible categories.

A classic example involves rolling multiple fair dice, where each die represents an independent trial with multiple faces (categories).

Multinomial distribution states that: If there are n fair dice (trials) and each of them has m faces (categories), and $\mathbf{X}_k$ is the number of times the k th face occurs after throwing the n dice. Let $\mathbf{A} = \{X_1 = i_1, X_2 = i_2, \dots, X_k = i_k, \dots, X_m = i_m\}$, then:

$$|\mathbf{A}| = \frac{n!}{i_1! i_2! \dots i_m!} \quad (\text{Can also be written as } \binom{n}{i_1, i_2, \dots, i_m})$$

$$P(\mathbf{A}) = \frac{n!}{i_1! i_2! \dots i_m!} \frac{1}{m^n}$$

where $0 \leq i_k \leq n$ and $i_1 + i_2 + \dots + i_m = n$.

2. History

Early Foundations (17th–18th Century)

The foundations of the multinomial distribution can be traced back to early probability theory. Blaise Pascal and Pierre de Fermat's work on binomial outcomes in the 17th century, followed by Jacob Bernoulli's formulation of the Law of Large Numbers, established key principles later extended to multinomial cases.

Formalization and Extension (19th Century)

In the 19th century, mathematicians like Pierre-Simon Laplace and Siméon Denis Poisson explored probability models with multiple outcomes. While not explicitly named 'multinomial' at the time, their work on categorical probabilities set the stage for later formalization.

Rigorous Definition and Statistical Use (Early 20th Century)

The multinomial distribution gained prominence in the early 20th century with Karl Pearson's chi-squared test (1900), which relied on multinomial probabilities for categorical data analysis. Ronald A. Fisher further integrated it into statistical inference, particularly in likelihood-based methods.

3. Proof

There is the way to prove the multinomial distribution. Let's continue using the rolling dice example. Let $\mathbf{A}$ be the event that: among n times of rolling n dice (each has m faces), the 1st face occurs i_1 times, the 2nd face occurs i_2 times, ..., the m th face occurs i_m times, i.e. $\mathbf{A} = \{X_1 = i_1, X_2 = i_2, \dots, X_k = i_k, \dots, X_m = i_m\}$.



There are n trials in total, which means the number of ways to achieve event A is the number of ways to distribute i_1 1st faces, i_2 2nd faces, ..., i_m m th faces in n trials.

To make the distribution, we should firstly compute the number of ways to distribute i_1 items in n trials, i.e. $\binom{n}{i_1}$

As we have already determined the positions of i_1 items (1st face), there are then $(n - i_1)$ positions left, therefore the number of ways to distribute i_2 items in $(n - i_1)$ is $\binom{n-i_1}{i_2}$.

In the same way, the number of ways to distribute i_3 items in the remaining positions are $\binom{n-i_1-i_2}{i_3}$.

We repeat this process, the number of ways to distribute the i_k items in the remaining positions is $\binom{n-i_1-i_2-\dots-i_{k-1}}{i_k}$.

According to **the Product Rule**:

$$\begin{aligned}
 |A| &= \binom{n}{i_1} \binom{n-i_1}{i_2} \binom{n-i_1-i_2}{i_3} \dots \binom{n-i_1-i_2-\dots-i_{k-1}}{i_k} \dots \binom{n-i_1-i_2-\dots-i_{k-1}-\dots-i_{m-1}}{i_m} \\
 &= \frac{n!}{i_1!(n-i_1)!} \times \frac{(n-i_1)!}{i_2!(n-i_1-i_2)!} \times \frac{(n-i_1-i_2)!}{i_3!(n-i_1-i_2-i_3)!} \times \dots \times \frac{(n-i_1-i_2-\dots-i_{k-1})!}{i_k!(n-i_1-i_2-\dots-i_k)!} \times \dots \times \\
 &\quad \frac{(n-i_1-i_2-\dots-i_{k-1}-\dots-i_{m-1})!}{i_m!(n-i_1-i_2-\dots-i_k-\dots-i_m)!} \quad \text{(Expand binomial coefficients)} \\
 &= \frac{n!}{i_1! i_2! i_3! \dots i_k! \dots i_m! (n-i_1-i_2-\dots-i_k-\dots-i_m)!} \quad \text{(Simplify the fractions)} \\
 &= \frac{n!}{i_1! i_2! i_3! \dots i_k! \dots i_m! 0!} \quad \text{(As } i_1 + i_2 + \dots + i_m = n) \\
 &= \frac{n!}{i_1! i_2! i_3! \dots i_m!} \quad \text{(Or } \binom{n}{i_1, i_2, \dots, i_m})
 \end{aligned}$$

Then we need to calculate the size of the sample space Ω . In this example of rolling dice, $\Omega = \{\text{All the possible distributions of every faces of the dice}\}$

The first die has m faces (outcomes), the second die has m faces, ..., the n th die has m faces. By using **the Product Rule**:

$$|\Omega| = m \times m \times m \times \dots \times m = m^n$$

Therefore,

$$P(A) = \frac{|A|}{|\Omega|} = \frac{n!}{i_1! i_2! i_3! \dots i_m!} \times \frac{1}{m^n} = \frac{n!}{i_1! i_2! \dots i_m! m^n}$$

where $0 \leq i_k \leq n$ and $i_1 + i_2 + \dots + i_m = n$.

4. Application: Multinomial Theorem

A brilliant application of multinomial distribution is the powerful Multinomial Theorem. Multinomial Theorem states that for any positive integer n , k and any real numbers $x_1, x_2, \dots, x_k$:

$$(x_1 + x_2 + \dots + x_k)^n = \sum_{i_1+i_2+\dots+i_k=n} \binom{n}{i_1, i_2, \dots, i_k} x_1^{i_1} x_2^{i_2} \dots x_k^{i_k}$$

In the expression $(x_1 + x_2 + \dots + x_k)^n$, there are n terms of $(x_1 + x_2 + \dots + x_k)$. Therefore each term in the product expression is with a total power n . In a term $x_1^{i_1} x_2^{i_2} \dots x_k^{i_k}$, the power of n is distributed into k numbers (i.e. $x_1, x_2, \dots, x_k$). So the coefficient of the term $x_1^{i_1} x_2^{i_2} \dots x_k^{i_k}$ is the number of ways to distribute n powers into i_1 items of x_1 , i_2 items of x_2 , ..., i_k items of x_k . Now apply the multinomial distribution formula, the coefficient of the term $x_1^{i_1} x_2^{i_2} \dots x_k^{i_k}$ is then $\binom{n}{i_1, i_2, \dots, i_k} = \frac{n!}{i_1! i_2! \dots i_k!}$.

For each term in the product expression, they can be written in the general form $x_1^{i_1} x_2^{i_2} \dots x_k^{i_k}$ and apply the conclusion above

For example, let's try to expand $(x_1 + x_2 + x_3)^3$. Let's start with the $x_1^2 x_2$ term, in the product expression of $(x_1 + x_2 + x_3)^3$,

it can be written as $x_1^2 x_2^1 x_3^0$, which means distributing a total power of 3 into (2, 1, 0), therefore the coefficient is $\binom{3}{2, 1, 0}$, which is 3. By the same process, we can work out that:

The coefficient of $x_1^3, x_2^3, x_3^3 = \binom{3}{3, 0, 0} = 1$

The coefficient of $x_1^2 x_2, x_1^2 x_3, x_2^2 x_3, x_2^2 x_1, x_3^2 x_1, x_3^2 x_2 = \binom{3}{2, 1, 0} = 3$

The coefficient of $x_1 x_2 x_3 = \binom{3}{1, 1, 1} = 6$

Therefore, $(x_1 + x_2 + x_3)^3 = x_1^3 + x_2^3 + x_3^3 + 3x_1^2 x_2 + 3x_1^2 x_3 + 3x_2^2 x_3 + 3x_2^2 x_1 + 3x_3^2 x_1 + 3x_3^2 x_2 + 6x_1 x_2 x_3$

5. Conclusion

The multinomial distribution provides a powerful framework for modelling categorical data across numerous applications, from genetics to game theory. By extending the binomial distribution to multiple outcomes, it offers a versatile tool for analysing realworld scenarios where events fall into distinct categories.

6. Reference

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Buffon's Needle: Proof and Other Variations

HEI CHING DAVINA TAM

In 1777, Georges-Louis Le Clerc, better known as Comte de Buffon to history, published “Essai d'arithmétique morale”, a collection of twenty five essays on Mathematics and moral issues (Roger, 1978). In his work, he posed “Buffon's Needle”, a significant problem revolutionising probability by introducing the field of geometric probability (Heyde, 2023). This paper will solve and discuss the Buffon's Needle problem, as well as variations and extensions of Buffon's original problem, including the Buffon-Laplace Needle Problem. In Article 23 of his essay collection, Buffon states, “in a room where the floor is simply divided by parallel joints [a needle's length apart,] one throws a [needle] in the air” (Hey, Neugebauer and Pasca, 2010, pp.245–282) and proceeds to calculate the probability that the needle intersects one of the parallel joints. Experiments show this probability converging to $\frac{2}{\pi}$.

The theoretical probability can, in fact, be proven using trigonometric functions and integral calculus (Klyve, 2019). Let O be the midpoint of the needle; l be the length of the needle; x be the perpendicular distance from O to the closest parallel line; A be the intersection between extension of the needle and its closest parallel joint; θ be the acute angle OA makes with the parallel line. In the case of the needle crossing the joint, the condition $\frac{l}{2} \geq OA$ must be true. Hence, $\frac{x}{OA} \geq \frac{x}{l/2} \Rightarrow \sin \theta \geq \frac{x}{l/2} \Rightarrow x \leq \frac{l}{2} \sin \theta$ as $\sin \theta = \frac{x}{OA}$. Since $0 \leq x \leq \frac{l}{2}$, $0 \leq \theta \leq \frac{\pi}{2}$, the sample size of this event is $\frac{\pi l}{4}$.

It follows that the probability is $\frac{\int_0^{\pi/2} \frac{l}{2} \sin \theta d\theta}{\frac{\pi l}{4}} = \frac{\frac{l}{2} [-\cos \theta]_0^{\pi/2}}{\frac{\pi l}{4}} = \frac{\frac{l}{2} (1)}{\frac{\pi l}{4}} = \frac{2}{\pi}$.

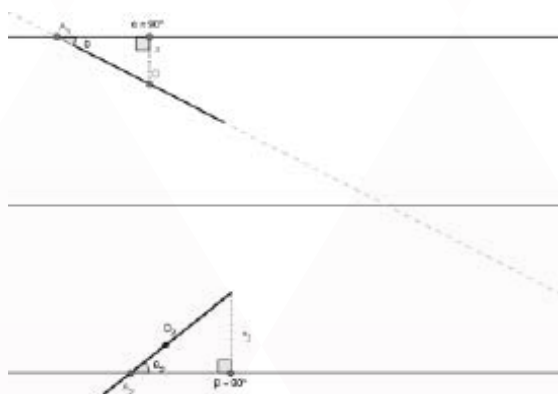


Table 1: The cycle of a five-stroke engine

Note that the method of calculating probability is reminiscent to comparing “areas” of favoured and sample spaces. Such method, known as geometric probability, was introduced by Buffon using this problem (Usu.edu, 2025). This problem was historically significant as an early example of Monte Carlo Methods, which utilises random experiments to approximate mathematical constants (Sharma, 2023). One will find that $\frac{2(\text{total number of trials})}{\text{number of favourable trials}}$ approaches π as the number of trials increase, as seen in figure 2 and 3. Such experiment served as one of the earliest methods of estimating π .

π , approximated by 240 trials of “Buffon's Needle”

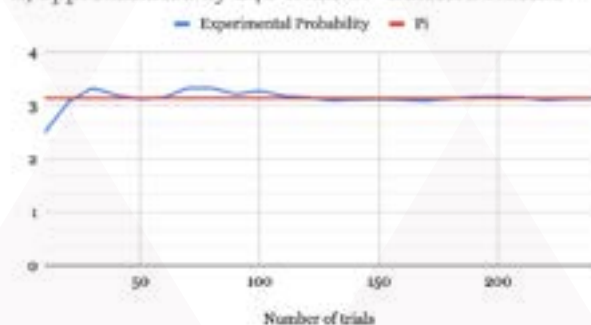


Figure 2: As the number of trials increase, the estimate gets closer to π .

How close would π be when approximated with Buffon's Needle?

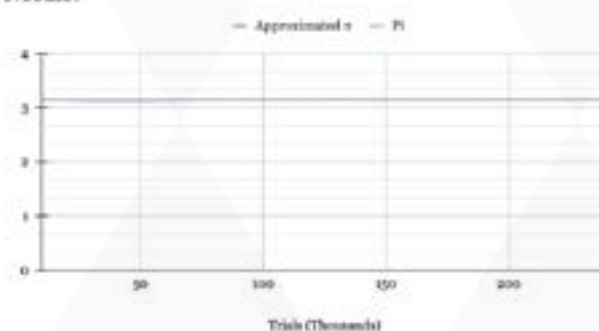


Figure 3: An approximation of π from 240,000 computer generated trials of Buffon's Needle (University of Illinois, 2015)

A logical generalisation that follows is the case where the parallel lines are t apart, where $t \geq 1$. The crossing conditions have not changed, whilst the revised sample space is $\frac{t}{4}$ for $0 \leq x \leq \frac{t}{2}$, as x is the distance between O and the nearest parallel line, which does not exceed $\frac{t}{2}$. Hence the altered probability would be $\frac{2t}{\pi t}$. For $t = 2l$, the probability for a needle crossing a line is $\frac{l}{\pi}$. If the experiment is conducted for a sufficient amount of trials, one finds the probability does indeed converge to the reciprocal of π , supported by the data in figures 3, 4 and 5. A similar formula may be derived for $t \leq 1$, but is more complex and unsuitable for discussion in this paper.

A generalisation of Buffon's Needle

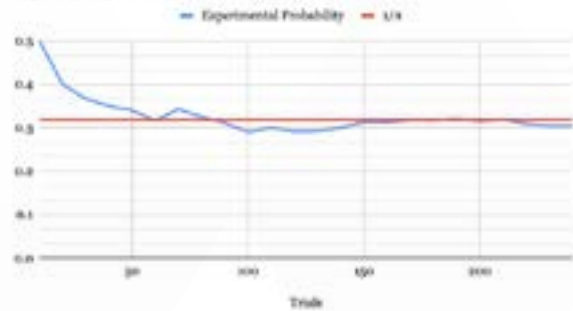


Figure 4: $\frac{1}{\pi}$ approximated by 240 trials of the modified Buffon's Needle

How close would $1/\pi$ be when approximated with Buffon's Needle?

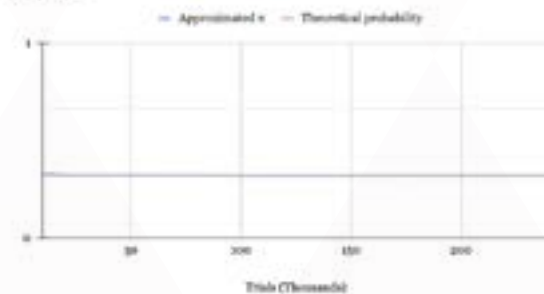


Figure 6: Chart of computer simulated Buffon's Needle variation experiment of 240,000 trials (University of Illinois, 2015)

Trials (Thousands)	Experimental probability of Buffon's Needle with $t = 2l$
10	0.3296
20	0.32385
30	0.3220333333
40	0.3218
50	0.32134
60	0.3209166667
70	0.3204857143
80	0.3206
90	0.3199888889
100	0.31868
110	0.3179181818
120	0.3179083333
130	0.3179769231
140	0.3181142857
150	0.3184066667
160	0.31861875
170	0.3192705882
180	0.3188722222
190	0.3189842105
200	0.31881
210	0.3189333333
220	0.3184363636
230	0.3185565217
240	0.3186708333

Figure 5: Experiment results of computer simulated experiment on Buffon's Needle variation (University of Illinois, 2015)

Buffon's Needle problem can be further extended to grids. This extension, called the Buffon-Laplace needle problem, involves two sets of parallel lines perpendicular to each other, further developing the concept of geometric probability to a higher dimension. Although Buffon initially solved it in 1777, his solution contained an error and was corrected by Pierre-Simon Laplace in 1812 (Weisstein, 2025).

The problem states that on a plane tiled with rectangles of dimensions a and b ($1 \leq a, b$), the probability that a randomly dropped needle crosses one of the lines, or edges of rectangles, is $\frac{2l(a+b)-l^2}{\pi ab}$ (Connelly, 2025).

By symmetry, one can consider one rectangle of length b and width a (Connelly, 2025). Using the defined terms O, A, θ , let the perpendicular distance from O to the closest vertical line be x ; the distance from O to the closest horizontal line be y ; the point where the extension of the needle intersects the closest horizontal and vertical lines be A and B , respectively.

To find the probability, one can use the Inclusion-Exclusion Principle to derive $P_H + P_V - P_{Both}$, where P_H denotes the probability that the needle crosses a horizontal line; P_V denotes the probability that the needle crosses a vertical line; P_{Both} denotes the probability that the needle crosses both the horizontal and vertical line (Cheung et al., 2021).



From the previous generalisation of Buffon's Needle, one can derive $P_H = \frac{2l}{\pi b}$ and $P_V = \frac{2l}{\pi a}$ instantly. For a needle to cross a vertical line, we will have $\cos \theta = \frac{x}{OB} \geq \frac{x}{\frac{l}{2}} \Leftrightarrow x \leq \frac{l}{2} \cos \theta$ as $OB \leq \frac{l}{2}$.

Similarly, a needle crosses a horizontal line if $\sin \theta = \frac{y}{OA} \geq \frac{y}{\frac{l}{2}} \Leftrightarrow y \leq \frac{l}{2} \sin \theta$. Since $0 \leq x \leq \frac{l}{2}$, $0 \leq y \leq \frac{l}{2}$, $0 \leq \theta \leq \frac{\pi}{2}$,

the sample size is $\frac{l^2}{\pi}$. Hence $P_{\text{Both}} = \frac{\int_0^{\frac{\pi}{2}} (\frac{l}{2} \cos \theta)(\frac{l}{2} \sin \theta) d\theta}{\frac{\frac{a}{2} \frac{b}{2} \frac{\pi}{2}}{\frac{1}{2} \frac{1}{2} \frac{\pi}{2}}} = \frac{\frac{1}{2}(\frac{l^2}{4})}{\frac{l^2}{8}} = \frac{l^2}{4\pi}$. By applying the Inclusion-Exclusion Principle,

the final probability that a needle crosses at least one line is $P_H + P_V - P_{\text{Both}} = \frac{2l}{\pi b} + \frac{2l}{\pi a} - \frac{l^2}{4\pi} = \frac{2la + 2lb - l^2}{\pi ab} = \frac{2l(a+b) - l^2}{\pi ab}$.

As Buffon said at the end of his essay, "one could propose several other problems of this type, which do not cease to be interesting and even useful". Buffon's needle is certainly a very insightful problem, providing intuition on geometric probability and approximation of constants. Many extensions of his original problem have since been explored, some of which are discussed in this paper, are perceptive and may still be relevant to modern day.

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04

Middle School – Natural Sciences

Short-Term Hibernation Inspired Space Travel

JIAYI FANG

1. Introduction

Space has always been an advanced field to explore. There have been people being consistently sent to space over the past 60 years. For a long time, some experts in the field have considered space hibernation as a possible implementation one day in the future. However, no successful human space hibernation has occurred. Early studies (Zakharova et al. 2021) have confirmed the possibility of inducing torpor on non-hibernating animals, such as rat, and reviving them a few hours later after experiments. In this essay, several hibernation mechanisms of animals in diverse classes will be analyzed and the reasons why they are impracticable for long-term human application will be discussed. A potential short-term hibernation plan is proposed after excluding the infeasible approaches through careful analysis.

2. Benefits of hibernation inspired space travelling

Microgravity in space has negative effects on muscle strength and endurance, motor control, coordination and balance (Lambrecht et al. 2017). Hibernation could reduce these effects by lowering BMR. People could also gain mental comfort from hibernation, as it reduces the feelings of homesickness when unconscious. Furthermore, space hibernation could reduce the amount of food cargo needed by 75% and the size of spacecraft needed by up to one-third (Fox-Skelly 2023). Wastage is dramatically reduced due to lower food intake, which saves more space. The radiation and tissue-damage could also be minimized. NASA and ESA studies (Bradford et al. 2018) support its feasibility for trauma care and radiation protection.

3. Animal hibernation and why it is feasible/unfeasible on human cases

Here are hibernation mechanisms from animals which are impractical for humans for long-term.

It is necessary for scientists to ensure that human body fluids remain in a liquid state during hibernation. Cryoprotectants are produced by some fish species and marine invertebrates as adaptations to prevent their body fluid from freezing during torpor. However, Human organ systems are more complex, and using cryoprotectants analogs are impracticable owing to their toxicity, which can damage cellular components, affect metabolic functions, and induce epigenetic modification. If circulation becomes impaired, serious problems may occur. Moreover, some animals, such as bears, wake up by the help of their internal clock known as the circannual rhythm, as they have evolved precise yearly biological timers. This mechanism is likely unfeasible for humans, as our ancestors did not develop such a system.

Animals develop a body system with decreased body temperature and HR to reduce energy usage during hibernation. A great example of deep hibernator: edible dormouse, has a significant body temperature decline to 5 Celsius degrees and heart rate reduction from 300 to 5 bpm during hibernation (**“Dormice: Unlocking Secrets of Memory and Cognition – Innovations Report” 2021**). Hypothermia is used in clinical practices for protecting tissue from acute injury such as a stroke. Unlike animals, human cannot enter hibernation naturally, extensive drugs are required. On the one hand, the size of a spacecraft could not support that. On the other hand, human organs are incapable of working in such an environment with extreme low body temperature and insufficient oxygen supply for weeks. Therefore, short-term semi-hibernation with a smaller temperature drop would be a better option if feasible. Scientists could potentially achieve short-term human space hibernation after years of clinical trials and spacecraft reconstruction for pharmacological requirements accordingly.

4. Why short-term hibernation is expected to become possible and how issues above could be solved

'White fat' is the main storage tissue and supplies energy for general body maintenance. Essentially, it keeps the animal alive while it hibernates. Hedgehogs consume white fat for basic survival, and they revive from hibernation by consuming brown fat, which can readily rewarm their body. (Andy, n.d.) Brown fat burns fuel and provides heat energy to help maintaining body temperature during cold situations such as entering a shadow in space. Humans can potentially develop a similar internal energy supply system during short-term hibernation.

Melatonin is explored and used for managing sleep disturbances in astronauts ("**Seven Ways Astronauts Improve Sleep May Help You Snooze Better on Earth - NASA,**" n.d.). With pharmaceutical development, it may be possible to make a controlled, high-quality state of induced hibernation using melatonin analogs—and prevent astronauts from waking up unintentionally at the wrong time.

Researchers have conducted experiments demonstrating the short-term hypothermia of rat for 6 hours, featuring a drop in deep brain temperature to about 22–23 °C and a halving of heart rate. [Fig.1] They appear to resume normal behavior with no long-term adverse effects after revival. (Zakharova et al. 2021) Moreover, Controlled hypothermia and metabolism are already widely used in clinical practice. These experiments strongly demonstrated the potential applications of hour-based hibernation and semi-hibernation with moderate drop in temperature and HR on humans in targeted occasions

Induced torpor reduces metabolism, ensuring survival in crises such as equipment failure, life-support shortages, or medical emergencies. It minimizes resource use and mitigates psychological stress.

5. Risk assessments

First and foremost, the aggressive use of drugs for entering and maintaining a hibernation-state, brings challenges, both financially and technically. Secondly, human ancestors have no adaptations for low-energy supply state unlike other hibernators. The human brain is highly sensitive to a lack of oxygen and so it may be damaged during hibernation regardless of long-term or short-term. Another area scientists need to work on is reducing the risk of memory reorganization after torpor. Studies in animals like bats indicate that most memories remain intact even after extended periods of neural inactivity (Ruczynski and Siemers 2010). [Fig.2] Therefore, if we aim to induce hibernation in humans, it is crucial to further explore how memories are maintained during torpor by learning from other animals such as bats.

6. Conclusions

Although the long-term human hibernation remains uncertain, research looking at potential mechanisms may provide the novel insights needed for short-term hibernation to become a reality. Animal traits like white fat and moderate temperature drop may help, with future advances needing cross-field research.

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8. Supplementation

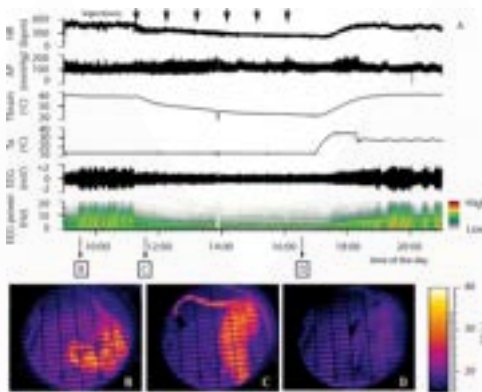


Fig.1. Example of the suspended animation state induced, showing a decline in HR and body temperature and recovers after 6h.

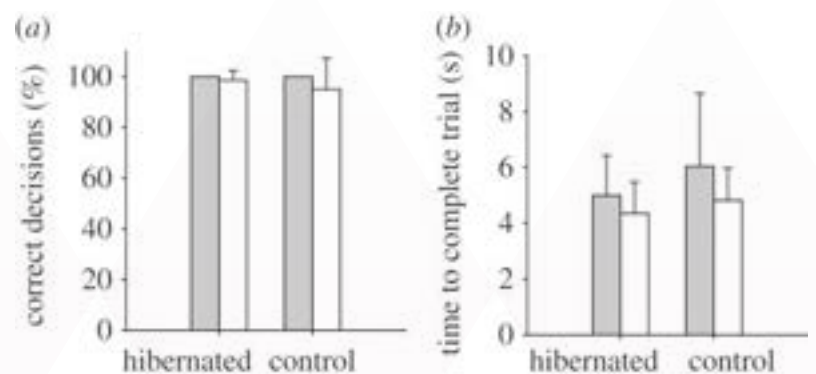


Fig.2 (a) Average proportions of correct decisions and average time to complete a trial (b) for hibernated and control bats before (grey bars) and after (white bars) the hibernation period. Means+1s.d. For statistics, see text.

The Glowing Mystery of Fireflies

LEILANI YAN

1. Introduction

Fireflies are also referred to as glowworms or lightning bugs. They are soft-bodied beetles that have the amazing ability to generate light in their abdomens with minimal emission of heat. It is an incredible biological process fueled by bioluminescence, a chemical reaction discovered by Raphaël Dubois in the late 1800s (Bose, 2024). Fireflies' ability to emit light is crucial to their survival, as it supports the mating process and helps them defend against predators. Furthermore, the mechanism behind firefly bioluminescence has contributed to progressing a number of scientific fields, particularly in LED technology, drug development, and biological imaging.

2. Biochemical mechanism behind the glowing nature of fireflies

The term "bioluminescence" describes the method by which fireflies generate light. The overall process takes place in specialized cells known as photocytes that are found within the light-producing organs of the lower abdomen, and can be divided into 3 major phases: Luciferin activation, oxygenation, and light emission.

Inside photocytes, the molecule ATP (adenosine triphosphate) that gives the reaction energy, combines with Luciferin, which is a substrate that produces light. At the same time, magnesium ions and the enzyme Luciferase are present that catalyze the reaction (Kreier, 2024). This results in Luciferyl-AMP, a high-energy intermediate. Subsequently, when oxygen is introduced into the bioluminescent organ of the firefly, Luciferyl-AMP is oxidized by reacting with oxygen to result in an unstable dioxetanone ring with high energy. Then, the dioxetanone ring breaks, releasing carbon dioxide and oxyluciferin in its electronically excited state. When oxyluciferin returns to the ground state, it releases a photon (particle that is a quantum of the electromagnetic field) of yellow-green light (approximately 560 nm).

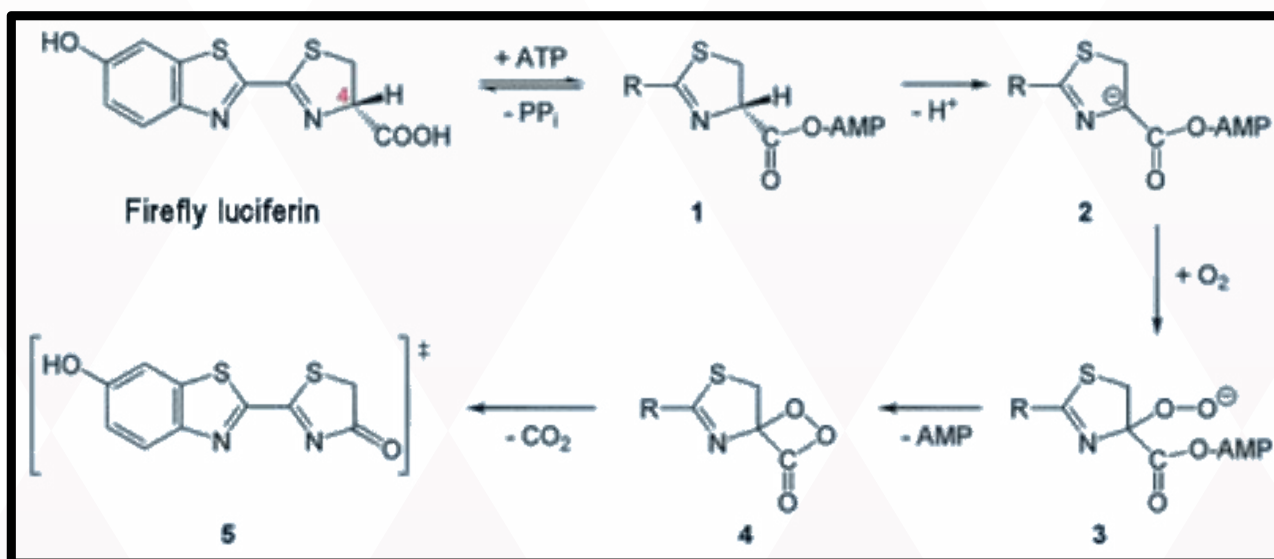


Figure 1. The process of Bioluminescence. Luciferin reacts with ATP to form Luciferyl-AMP (Step 1). Hydrogen ion is lost because oxygenation triggers a series of electron rearrangements; the negative charge on the nitrogen stabilizes the excited-state intermediate (Step 2). Luciferyl-AMP is oxidized, producing the dioxetanone ring (Step 3). An electronically excited state of oxyluciferin molecules and carbon dioxide is produced (Step 4). (Credit: <https://www.chm.bris.ac.uk/motm/luciferin/luciferinh.htm>)

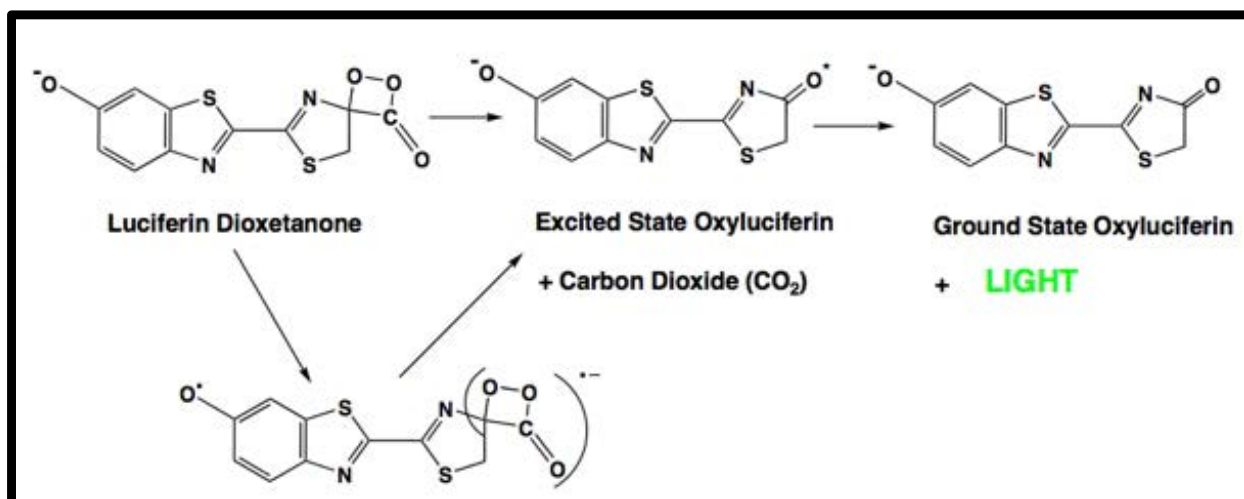


Figure 2. Different states of oxyluciferin. (Credit:<http://photobiology.info/Branchini2.html>)

Fireflies are capable of quickly turning on and off their luminescence through control of the oxygen delivery into the light organ. They have specialized tracheal tubes surrounded by tiny sphincter muscles that can contract or relax to regulate the amount of oxygen that enters their light organs. Moreover, the nervous system of fireflies releases nitric oxide (NO) during light emissions, which inhibits mitochondria in photocytes ("How and why do fireflies light up?", 2005). This indicates that oxygen will go into the light organs instead of being used for energy in the form of ATP by mitochondria during respiration. When fireflies stop producing nitric oxide, their mitochondria utilize oxygen once again, which extinguishes their glow.

3. Reasons for the glowing of fireflies

Fireflies' ability to produce light functions as a way of communication, specifically in finding a mate. Different fireflies have developed their unique flash pattern that enabled males and females to recognize potential mates of the same species in the dark (Genetic Science Learning Center, 2016). Some of the variations in flashing patterns include frequency, duration, and timing of flashes. This sex communication system prevents fireflies from mating with other species by avoiding cross-species mating, thus allowing successful reproduction.

Apart from that, the production of light by fireflies is also crucial for self-defense. The continuous glow serves as a warning signal by displaying their toxicity to predators (a phenomenon known as aposematism) (National Center for Biotechnology Information, n.d.). Additionally, fireflies also use their glow to protect themselves from predators by using lucibufagins. Lucibufagins are defensive steroids that make fireflies unpalatable to predators such as birds and toads (Liesch, 2018). These toxins can result in a bitter taste of fireflies to their predators, as well as causing vomiting and death of their predators.

Furthermore, certain fireflies have evolved to emulate the flashing pattern of harmful species. A harmless species (the mimic) developed to resemble a dangerous or distasteful species (the model), a phenomenon known as Batesian mimicry. For example, female fireflies of the genus *Photuris* imitate the light signals of *Photinus* females to lure and prey on *Photinus* males, stealing *Photinus* male's lucibufagins for their protection (D'Lyn Ford, 2019). The *Photuris* females, which are harmful to *Photinus* males, are the predators in this case, and they are imitating the signals of their prey, the *Photinus* females.

4. Scientific applications inspired by Fireflies

Firefly bioluminescence has led to the development of several tools used for medical imaging, drug screening, and more efficient lighting systems.

Firstly, it has been used to develop bioluminescence imaging (BLI). BLI is a highly sensitive, non-invasive technique that can

detect low light levels. This makes it an excellent tool for monitoring tumor growth and the progress of anti-tumor therapy in animal models (National Center for Biotechnology Information, 2013). Researchers genetically modify cells to express luciferase, which acts as a reporter when exposed to luciferin.

Secondly, firefly's light-emitting reaction is used in luminescence-based tests to screen for potential drugs (Glennade HJ, Dimond A and Fisher AG, 2023). Bioluminescence offers a quick and precise method to screen possible therapeutic candidates, which allows for rapid evaluation of drug efficacy and identification of potential side effects (UT Southwestern Medical Center, 2008). By checking the intensity and location of the light signal, scientists can monitor the extent to which a drug is delivered to the target, knowing how it affects the target cells (e.g., killing cancer cells), and whether it spreads to other parts of the body.

Moreover, bioluminescence is a "cold light", meaning that less than 20% of the light generates thermal radiation, or heat (BiteSized Biology, 2020). It illustrates how fireflies make light through a complicated yet extremely efficient chemical reaction that transforms chemical energy directly into visible light with nearly 100 percent efficiency. By comparison, incandescent light loses 90% of its energy as heat. The high efficiency of firefly bioluminescence has motivated scientists to apply this knowledge to LED designs. This contributed to the development of lighting solutions that last longer compared with incandescent bulbs and keep functioning for longer periods without losing their efficiency. By taking inspiration from fireflies' structure, scientists have been capable to increase LED efficiency by 55% (Yale School of the Environment, 2013). Furthermore, different luciferins in fireflies can emit different wavelengths of light, which allows them to create a variety of glow colors (Lumitex, 2024), leading to the discovery of LEDs with precise color outputs by scientists, where these functions are used in architectural lighting (Firefly Light & Design, 2023).

5. Conclusion

In conclusion, the glowing properties of fireflies are an appealing evolutionary adaptation that involves the interaction of multiple chemical components, among which the results of the luciferin-luciferase reaction play a crucial role in the survival, mating function, and self-defense of fireflies. By further investigating these fireflies, scientists have successfully opened new avenues in the field of biology, creating a better future and a more convenient life for everyone.

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05

Chemistry

Hard and Soft Nucleophiles and Electrophiles: An Inspection into their Reactivities

SKYLER SUTIAN WANG

When learning organic chemistry, students often encounter questions like: Why does this reaction happen at one atom of the molecule but not another? Why do these nucleophiles react differently despite all being nucleophilic? To answer such questions, chemist Ralph Pearson introduced the Hard-Soft Acid-Base (HSAB) Theory (Pearson, 1997, pp. 198). The theory assigns Lewis acids (electrophiles) and Lewis bases (nucleophiles) a hardness characteristic. Hard nucleophiles are characterized by high charges, small size, high basicity, and low-energy HOMO. Soft nucleophiles are characterized by neutral charge, larger size, no basicity, and high-energy HOMO (Clayden, 2001, pp. 357). Experimental data have shown that, generally, “hard nucleophiles tend to react well with hard electrophiles, and soft nucleophiles tend to react well with soft electrophiles” (Clayden, pp. 507). This essay aims to explain the mechanisms that underlie this intriguing property.

The hardness of nucleophiles and electrophiles is important for understanding chemoand regioselectivity, which is “the preferential reaction of a chemical reagent with one of two or more different functional groups” (Muller, 1994, pp. 1097) and “[when] one direction of bond making or breaking occurs preferentially over all other possible directions” (Muller, pp. 1160) respectively. Two main factors influence organic reactions: electrostatic interactions and orbital overlap. Reactions involving hard nucleophiles and electrophiles are mostly governed by electrostatic interactions, while those involving soft nucleophiles and electrophiles are primarily affected by orbital overlap (Clayden, pp. 357). Several simple examples can illustrate this property.

First, the chemoselectivity of substitution vs. addition reactions in certain compounds can be determined by examining the hard-soft properties of the reagents. A simple example of such distinguishment is when a hard and a soft nucleophile reacts with a carbonyl compound like an ester. The typical ester cleavage reaction under basic conditions (saponification) involves a nucleophilic attack by a hard nucleophile, the hydroxide anion, on the carboxyl carbon, a hard electrophile, shown as follows.

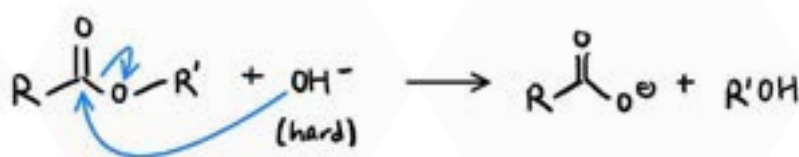


Fig. 1, saponification of esters. The hydroxide ion attacks the carbonyl carbon and an alkoxy group leaves. Later, proton transfer happens, resulting in a carboxylate ion and an alcohol.

However, when the ester reacts with a soft nucleophile, the reaction usually happens on the α -carbons instead. For example, if the R group in the above structure is CH_2Br , a soft nucleophile will attack the saturated α -carbon but not the carbonyl carbon.

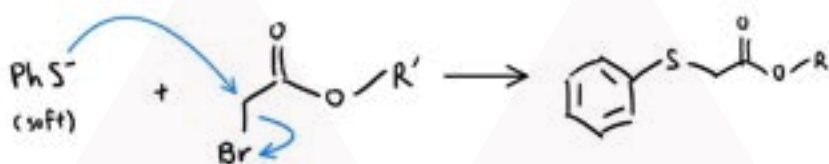


Fig. 2, $\text{S}_{\text{N}}2$ nucleophilic substitution of phenyl thiolate on an ester. The thiolate ion attacks the α -carbon and a leaving group leaves (Clayden, pp. 658).

In soft-soft interactions, where there are almost none electrostatic attractions, the quality of orbital overlap plays a more important role. In a carbonyl group, the LUMO is a π^* antibonding orbital, while in a saturated carbon, the LUMO is a σ^* antibonding orbital. " σ^* antibonding orbitals are, of course, higher in energy than non-bonding lone pairs, but the higher the energy of the nucleophile's lone pair, the better the overlap. The $3sp^3$ lone-pair electrons of sulfur overlap better with the high-energy σ^* orbital of the C-X bond than do the lower energy $2sp^3$ lone-pair electrons on oxygen because the higher energy of the sulfur electrons brings them closer in energy to the C-X σ^* orbital" (Clayden, pp. 356).

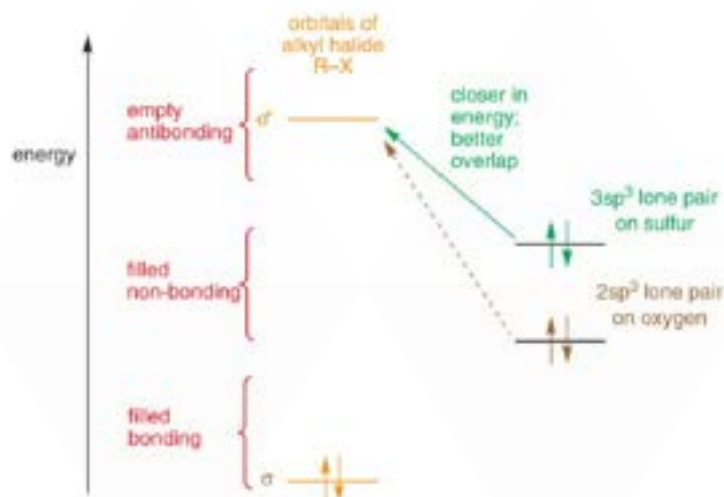


Fig. 3, a simplified molecular orbital diagram of a saturated carbon compared with the energies of the non-bonding lone pairs of sulfur and oxygen (Clayden, pp. 356)

This explains why the hydroxide ion is ineffective at reacting according to the mechanism shown in Figure 2. Therefore, when there are multiple electrophilic sites on a reagent, chemoselective reactions could be predicted by studying the hardness of the reagents.

Second, the regioselectivity of addition reactions can be determined by examining the hard-soft properties of the reagents too. A straightforward example that demonstrates this is the nucleophilic addition on an α,β -unsaturated carbonyl group.

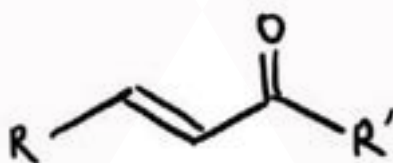


Fig. 4, the structure of an α,β -unsaturated carbonyl group.

In this structure, the carbonyl carbon is a hard electrophile because it's small and partially positively charged. The β -carbon is a soft nucleophile due to conjugation of the π system.



Fig. 5, the resonance structures of the α,β -unsaturated carbonyl group, showing the origin of electrophilicity of the β -carbon.

When this group reacts with a hard nucleophile like hydroxide, the hard nucleophile will react on the carbonyl carbon.

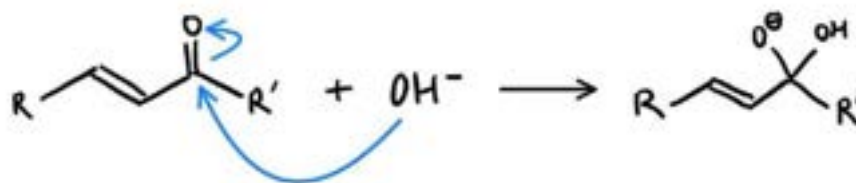


Fig. 6, the mechanism of a hydroxide attacking the carbonyl carbon. This is the general mechanism of hard nucleophilic addition on α,β -unsaturated carbonyl compounds, aka. 1,2-addition.

When this group reacts with a soft nucleophile, like phenyl thiolate, the soft nucleophile will react on the β -carbon.

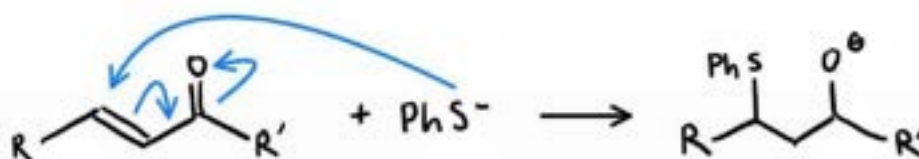


Fig. 7, the mechanism of a phenyl thiolate attacking the β -carbon. This is the general mechanism of soft nucleophilic addition on α,β -unsaturated carbonyl compounds, aka. 1,4-addition, conjugate addition, or Michael Addition.

The two mechanisms differ in the forces that drive the reaction. The 1,2-addition is dominated by electrostatic interactions and kinetic favorability (Clayden, pp. 507). Hard species, as defined, have high charge density, which leads to having stronger Coulombic attraction forces. The attraction forces thus make 1,2-addition more kinetically favorable (Clayden, pp. 581) because the molecules collide easily and thus more frequently. On the other hand, the 1,4-addition is dominated by orbital overlap and thermodynamic stability (Clayden, pp. 507). "As a C=O π bond is stronger than a C=C π bond" (Clayden, pp. 581), losing the alkene makes the product more thermodynamically stable than losing the carbonyl. Thus, the regioselectivity of reactions can also be determined by examining the hardness of the reagents.

Although HSAB theory is useful in certain cases, there are various limitations that could hinder its effectiveness in predicting reactions. These may include kinetic vs. thermodynamic favorability, steric hindrance and electrostatic effects, solvent and reaction conditions, and the specific qualities of reactants. For example, the reaction between tert-butyl bromide and the ethoxide ion is predicted to be a hard-hard S_N1 based on the HSAB theory, but in reality, steric hindrance and ethoxide's high basicity causes it to be an $E2$ reaction that involves the ethoxide ion attacking a hydrogen, producing isobutene and ethanol. As a conclusion, HSAB is a useful tool when working with organic reactions, but the theory is limited to specific cases where there are few other factors influencing the reaction.

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How Do Conjugated Pi Systems and Bleaches Create and Remove Colours?

YIXIN WANG

Introduction

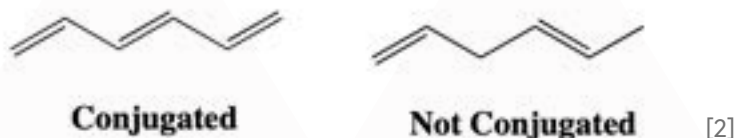
Color is a ubiquitous phenomenon in nature and daily life, this richness of color stems from fundamental chemical principles, specifically the influence of conjugated pi systems within molecules. This essay will explain the molecular mechanism by which conjugated π systems affect the color of substances. This essay will also use examples to illustrate the connection and explore the principle of bleaching in relation to these systems.

1. Color Perception

We perceive color because substances selectively absorb certain wavelengths of visible light (approximately 400–700 nm), while reflecting or transmitting others. The reflected or transmitted wavelengths determine the perceived color. [1] Simple organic molecules which have higher energy gaps primarily absorb high-energy ultraviolet (UV) light, which is invisible to the human eye, appearing colorless or white. However, molecules with extended conjugated π systems have lower energy gaps, enabling them to absorb visible light, thus generating color.

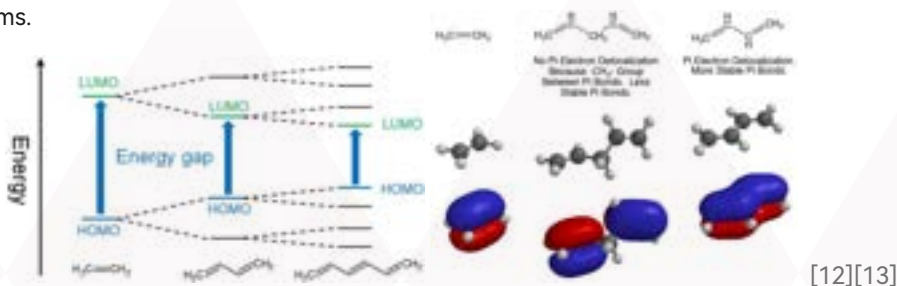
2. Conjugated Pi Systems

A conjugated π system is a system of overlapping p-orbitals where the π electrons are delocalized across the entire system. [9] This delocalization occurs when three or more p-orbitals are arranged in an alternating double-single-double bond pattern. For an isolated double bond (non-conjugated), the π electrons are confined between the two carbon atoms. This positioning results in a higher energy state because the repulsion between the electrons is increased. [3][4]



In contrast, electron delocalization in a conjugated π system spreads electron density over a larger area. This increased distance between electrons reduces electron-electron repulsion. Consequently, the system's overall energy is lowered, therefore increasing stability. In addition, the HOMO (highest occupied molecular orbital) energy level increases because the electrons are less tightly bound. Simultaneously, the LUMO (the lowest unoccupied molecular orbital) energy level decreases because it's less repulsive to an incoming electron. Consequently, there is a reduced energy gap between the HOMO and LUMO in a conjugated pi system in comparison to a non-conjugated pi system. [5][8]

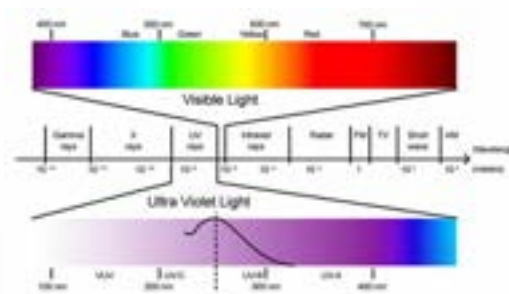
In conjugated systems, the HOMO-LUMO energy gap of longer conjugated systems is smaller because the electrons achieve delocalization over a larger area, thereby further reducing the repulsion between the electrons and lowering the overall energy. Conversely, shorter conjugated systems exhibit a larger HOMO-LUMO energy gap, though this is smaller than that of non-conjugated systems.



3. Mechanism of Color

The absorption of light is directly related to the energy difference (ΔE) between the HOMO and LUMO. When a molecule absorbs light, an electron transitions from the HOMO to the LUMO. The required energy must exactly match the HOMO-LUMO energy gap (ΔE). According to Planck's equation[6]: $\Delta E = hc/\lambda$, where h is Planck's constant, c is the speed of light, and λ is the wavelength of absorbed light. A smaller ΔE corresponds to a longer wavelength of light absorbed.

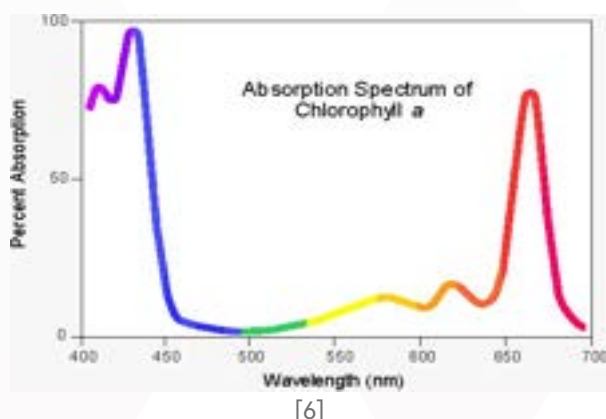
Conjugation within a π -system effectively lowers ΔE , enabling the absorption of longer wavelengths within the visible spectrum. The longer the conjugated system, the lower the energy of absorbed light, shifting the perceived color towards the red end of the spectrum. Conversely, shorter conjugated systems absorb higher-energy light, resulting in colors closer to the blue end of the spectrum. Molecules lacking conjugated π -systems exhibit larger ΔE values, absorbing only high-energy UV light ($\lambda < 400$ nm) therefore appearing colorless. [4]



4. Mechanism of Color

4.1 Leaves in different seasons

The colour change of tree leaves is a typical colour phenomenon associated with the conjugated π -system in nature. In spring, tree leaves appear green, which is mainly due to the large amount of chlorophyll contained in the leaves. The chlorophyll molecule has a large conjugated π -system, and its structure contains a porphyrin ring consisting of four pyrrole rings.[7] The π -electrons in the porphyrin ring can be delocalised in the whole ring system, forming a large conjugated π -system. Chlorophyll's extensive conjugated π system absorbs blue-violet (~430 nm) and red (~660 nm) light, reflecting green light, resulting in the green color of leaves.



As temperatures decrease in autumn, chlorophyll degrades, appearing carotenoids. These molecules have shorter conjugation lengths than chlorophyll. They primarily absorb blue-violet light (around 450-480 nm), resulting in the yellow and orange color of autumn leaves.[2]

4.2 The ink sprayed by the octopus

The dark brown-black color of octopus ink is due to melanin. Melanin contains an extremely large number of conjugated double bonds and aromatic rings. This conjugated structure can widely absorb various wavelengths within the visible light range and reflect very little light, thus appearing black visually.[10]

5. Bleaching

Bleaching agents effectively remove color from a wide range of substances, including grass, ketchup, blood, and many food and vegetable stains by disrupting conjugated pi system. Most bleaching agents achieve this through oxidation, reduction, or chemical cleavage. The principle of these methods increases the HOMO-LUMO energy gap,

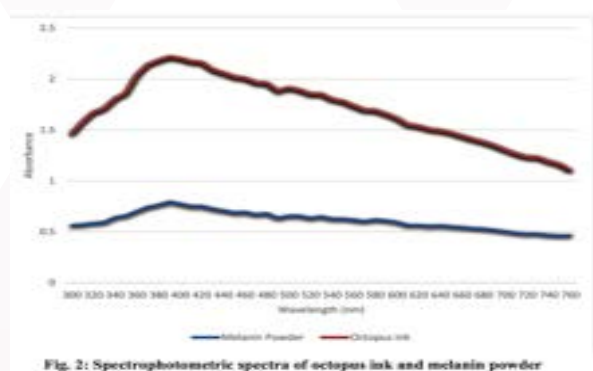


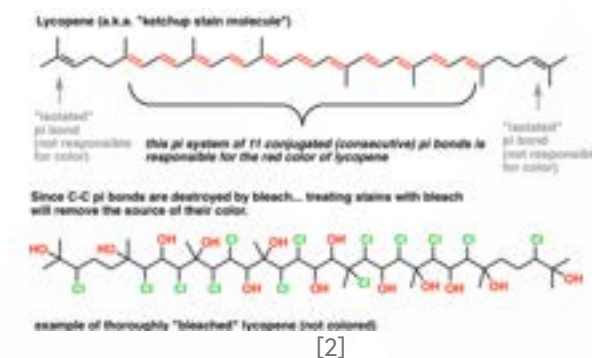
Fig. 2: Spectrophotometric spectra of octopus ink and melanin powder

shifting the absorption spectrum back to the UV range. The molecule no longer absorbs visible light, resulting in a loss of color. [11]

For example, the red color of ketchup originates from lycopene, which contains conjugated π -systems. Sodium hypochlorite (bleach) reacts with the alkenes in lycopene, breaking the C=C π bonds and destroying the conjugated system, rendering the stain colorless.[2]

Conclusion

In summary, Conjugated π systems are fundamental to color. By delocalizing electrons, they adjust the HOMO-LUMO energy gap, dictating the wavelengths of visible light absorbed and, consequently, the perceived color. Bleaching disrupts this system, widening the HOMO-LUMO gap and shifting light absorption to the ultraviolet range, making the substance colorless.



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06

CompSci: Artificial Intelligence

Building a Tic-Tac-Toe Game with Minimax

ROGER YEOH

1. Introduction

Tic-tac-toe is an abstract strategy game that requires both sides to think deeply to win optimally. The game is based on a nine-grid pattern, “O” and “X” act as opponents, they take turns filling the nine gap with their symbol. The first one to get three symbols together in a horizontal, vertical or diagonal line win the game. It’s an excellent and suitable game to start with and delve into artificial intelligence algorithms. Because there exist a range of knowledge and intractable points that need to be focus on. Such as the turn system and the game status detection in the basic structure of game. Furthermore, the minimax algorithm includes the implementation of recursion, the programming of the decision tree, and the whole-process evaluation aimed at finding the best outcomes.

2. Two-player tic-tac-toe

To realize the project, it is necessary to begin with the fundamental two-player tic-tac-toe game. The program uses several fundamental and essential Python programming concepts, including lists, loops, conditionals, and functions. A 2D list was used to store the game state. Each gap is assigned a value of 'X', 'O', or a blank space. The `check_winner()` function checks if any row, any column or any diagonal has the same symbol. The `is_draw()` function checks if all cells are filled without any player winning, by using a one-line expression with a nested loop to check all cells. In general, a large loop is constructed to repeatedly ask for input for each gap in turns, check input validity, place symbols, check the game status and show who’s winning. In order to enhance the playability of the game, the `tkinter` module is imported to display the game in a graphical user interface. Furthermore, three useful buttons— Undo, Redo and Scoreboard are added below the game board.

3. Implementation of minimax

Subsequently, an attempt will be made to implement minimax into the fundamental game. But before proceeding further, it is imperative to review the underlying principle. Minimax is a decision rule that is used in artificial intelligence, decision theory, combinatorial game theory and statistics for minimizing the possible loss in a worst-case scenario. The minimax algorithm works by simulating all possible future outcomes. By using recursion, it builds a decision tree, where each node represents a possible game state. The algorithm assigns a score to each final state (e.g.: +1 for a win, -1 for a loss, and 0 for a draw). Then, it backtracks through the tree and makes the right choice that leads to the best possible outcome for the AI.

In the context of the game tic-tac-toe, the implementation of minimax could be divided into multiple sections. Firstly, the fundamental focus is on the basic minimax algorithms used in turn-based games, between two players — AI and human. Second, AI needs to construct a game tree while evaluating the possible future outcomes. Third, the goals of the algorithms are to choose the move with the minimum score at the min nodes, and choosing the one with the maximum score with at max nodes. Lastly, the perfect strategy is considered throughout the whole module to ensure AI play optimally.

By adding a minimax module to the original code, the game is now available to play against AI in easy mode. It should be mention that easy mode would only adopt one level of evaluating, then the choices are made randomly.



```
def find_best_ai_move(self):
    # Step 1: Check if AI can win immediately
    for i in range(3):
        for j in range(3):
            if self.board[i][j] == "":
                self.board[i][j] = "O"
                if self.check_winner("O"):
                    self.board[i][j] = ""
                    return (i, j) # if the place available for winning is founded, return immediately
                self.board[i][j] = ""

    # Step 2: Block player win
    for i in range(3):
        for j in range(3):
            if self.board[i][j] == "":
                self.board[i][j] = "X"
                if self.check_winner("X"):
                    self.board[i][j] = ""
                    return (i, j) # stop the player if it will win for the best step
                self.board[i][j] = ""

    # Step 3: Choose random move (for easy difficulty)
    available = [(i, j) for i in range(3) for j in range(3) if self.board[i][j] == ""]
    if available:
        return random.choice(available) # choose a blank space randomly
    return None # nowhere to go
```



The module is divided into three steps. Step 1 checks if AI can win immediately in the next move. If the place is found, AI will fill its symbol in and win the game. Instead, if the place is not found, AI will try to block the player in their next move by step 2. In easy mode, AI only detects the current status to find out if the player will win at any places in the next step, and fill the possible gap. Otherwise, if either player or AI can't win in their next step and there are still vacancies, AI will choose a random gap to fill in.

```
def best_move(self):
    best_score = float("-inf") # Start with the lowest possible score
    move = None # To store the best move coordinates
    for i in range(3):
        for j in range(3):
            if self.board[i][j] == "": # Check if the cell is empty
                self.board[i][j] = "O" # Try placing 'O' (AI) in this cell
                score = self.minimax(self.board, 0, False) # Evaluate this move using Minimax
                self.board[i][j] = "" # Undo the move (backtrack)
                if score > best_score: # If this move is better than previous best
                    best_score = score # Update best score
                    move = (i, j) # Save this move as the best one
    return move # Return the best move found

def minimax(self, board, depth, is_maximizing):
    if self.check_winner("O", board): # If AI wins in this state
        return 1 # Return highest score
    elif self.check_winner("X", board): # If player wins
        return -1 # Return lowest score
    elif self.is_draw(board): # If board is full with no winner
        return 0 # Return neutral score

    if is_maximizing: # AI's turn (maximize score)
        best_score = float("-inf")
        for i in range(3):
            for j in range(3):
                if board[i][j] == "":
                    board[i][j] = "O" # Try placing 'O'
                    score = self.minimax(board, depth + 1, False) # Recursively simulate player's response
                    board[i][j] = "" # Undo move (backtrack)
                    best_score = max(score, best_score) # Pick the best (max) outcome
            return best_score
    else: # Player's turn (minimize score)
        best_score = float("inf")
        for i in range(3):
            for j in range(3):
                if board[i][j] == "":
                    board[i][j] = "X" # Try placing 'X'
                    score = self.minimax(board, depth + 1, True) # Recursively simulate AI's response
                    board[i][j] = "" # Undo move (backtrack)
                    best_score = min(score, best_score) # Pick the worst (min) for AI
            return best_score
```

Subsequently, the minimax algorithm would get a boost in difficulty.

The minimax module is divided into two steps. The first step tries all possible moves by using the minimax algorithm to simulate the outcomes. The move that maximizes the chance of winning or at least ends in a draw, so it never plays a suboptimal move. Furthermore, step 2 employs a recursive search algorithm, unsuitable move simulations would be undone through backtracking as a result. By following these methods, AI always chooses the option with the best result. Concurrently, all possible strategies of the player are anticipated and countered in advance.



After conducting a series of trials, the AI was proven to be unbeatable in this game. The reason for this, which differentiates unbeatable mode from easy mode is that the code conducts a range of new logics. It performs a complete game tree search, exploring and evaluating every possible move by both players through recursive decision-making until the end of the game. Score-based strategies are also used in this process. AI always selects the move that maximizes its chances of winning or, at worst, results in a draw, by backtracking. In addition, this AI does not skip any branches (no pruning) and avoids randomness. It guarantees the mathematically best move.

4. Conclusion

Artificial intelligence has successfully solved the game of tic-tac-toe with the minimax algorithm, which makes optimal choices at every turn, ensuring a win against the human player, or at worst, a draw. The AI chooses the most beneficial moves at each decision point through a complete game tree search and a score-based system, ensuring the optimal solution is achieved. On the other hand, although the algorithm is easy to run in the program of tic-tac-toe, the search is relatively inefficient due to the lack of pruning. It is not applicable to more complex and larger games as well as programs. In the process of constructing the game, I gained a deeper understanding of the core concepts of recursion, game trees and decision-making strategies, which provided me with further insight into AI.

Understanding and Addressing Bias in AI Training Data: Sources, Impacts, and Mitigation Strategies

ZIMO HUANG

1. Introduction

Since 2010, AI has undergone rapid and transformative development to make decisions and provide advice in real life, but at the same time AI is subject to several ethical concerns that cause model outputs to be unfair. Bias can be caused by a variety of factors, but it is usually related to training data. [1] Bias of training data refers to the impact that unrepresentative or subjective input data has on the fairness of a machine learning model. Bias in data can have serious consequences not only for individuals and companies, but also for society. The aim of this paper is to analyse bias in training data. The essay will first explain a few different types of training data bias, give examples of how data bias can lead to actual unfair outputs, and finally list the implications of the situation and explore ways to mitigate the effects of the situation.

2. Sources and types of data bias

There are many types of bias in training data, notably selection bias, measurement bias, and historical bias. [2] Selection bias occurs when the collected data is not fully representative of the target population, often due to non-random sampling methods. Measurement bias arises from flaws in the data collection process, such as inconsistent procedures, unreliable instruments, or ignoring demographic differences. Historical bias refers to systematic inequalities embedded in real-world data: even if the data is factually accurate, AI systems will faithfully reproduce these biases. [3] The following section will explore the potential consequences of these biases within AI applications.

Cases of Data Bias

There is ample evidence demonstrating the widespread presence of bias in AI systems across various applications. A notable case of selection bias occurred on Facebook, where the platform's recommendation algorithm inappropriately suggested users "continue watching videos about primates" after viewing a video featuring Black individuals [4]. This incident is closely linked to the limitations of facial recognition technology, which research has shown to be disproportionately inaccurate when identifying people of colour, particularly Black individuals. These inaccuracies often result in discriminatory outcomes due to algorithmic errors and flawed training data [5]

Measurement bias is exemplified by fitness trackers, which are frequently calibrated using data collected mainly from male participants. This calibration method leads to systematically overestimated calorie expenditure for female users during various physical activities [6].

Historical bias is often found in recruitment algorithms. A recent study from the University of Washington revealed substantial racial, gender, and intersectional biases in how three state-of-the-art Large Language Models (LLMs) ranked résumés. The researchers used over 550 real CVs, modifying only the names to represent different races and genders. Results showed a strong preference for names associated with white individuals (85% of the time), a slight preference for women (11%), and no preference for Black men [7]. Another example is Amazon's recruitment algorithm, trained on ten years of internal hiring data, primarily from male applicants. The model ultimately penalised CVs from female candidates, reinforcing gender bias in hiring decisions [8].

3. Impact on Fairness

Bias in AI systems can perpetuate or even intensify existing social inequalities, resulting in discrimination against marginalised communities and restricting their access to essential services [9]. For affected individuals, the widespread

deployment of biased systems can shape public perceptions, reduce employment opportunities, and harm mental well-being by inducing elevated levels of stress and anxiety [10]. In high-stakes domains such as healthcare, algorithmic bias may lead to unequal access to medical treatment and, in some cases, direct harm to patients [11]. For businesses, bias in AI can result in flawed decision-making, such as inaccurate market analyses. Moreover, public exposure of such bias can trigger reputational damage, erode consumer trust, and diminish brand credibility [12]. On a societal level, algorithmic bias further entrenches systemic inequalities, deepens socioeconomic divides, and undermines efforts to foster equity and inclusion [13].

4. Detection and Mitigation Strategies for Data Bias

Detecting and addressing bias in datasets begins with a systematic examination of potential imbalances or skewed patterns [14]. Exploratory Data Analysis (EDA) is often used as an initial step to uncover distributional anomalies, detect outliers, and identify potential sources of error before any modelling assumptions are made [15][16]. Mitigating bias in AI systems remains a complex and multifaceted challenge. Nevertheless, several strategies have been proposed to confront this issue. A widely used approach is data preprocessing, which includes techniques such as oversampling, undersampling, and synthetic data generation to balance representation [17]. For instance, Buolamwini and Gebru demonstrated that increasing the sampling of dark-skinned individuals significantly improved the accuracy of facial recognition models for that group [18]. In addition to data-level interventions, enhancing transparency in AI systems is also critical for building trust and reducing bias [19]. One example is SHAP (SHapley Additive exPlanations), a tool that improves the interpretability of machine learning models while helping to ensure system accountability and security [20].

5. Conclusion

In conclusion, bias in training data constitutes a fundamental source of output bias and discrimination in AI systems. As AI technologies continue to evolve, researchers and practitioners must accord equal emphasis to addressing data bias. However, resolving input bias extends beyond purely technical solutions; it also involves critical questions concerning how data representativeness is defined and by whom. Given that this issue intersects with diverse social and ethical interests, definitive answers remain elusive. Therefore, future AI research should adopt inclusive, multidisciplinary approaches that engage multiple stakeholders to foster the development of fairer and more equitable AI systems.

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07

Engineering: Robotics

Introduction

Natural disasters are unpredictable and unstoppable. We cannot stop the disaster from happening. However, we can contain the damage it does. As the world develops, first responders have worked more closely and employed engineers to utilize robotics in order to create robots that can aid in various tasks during a disaster response as shown in Figure 1. This includes data collection, travelling in tight spaces inaccessible to humans and handling hazardous materials. But most importantly, these robots are able to support rescue operations without risking a single responder's life. Metal parts are expandable, but a life is precious. In this research paper, I will be discussing the development and uses of robotics in past events, and how it may be further advanced.

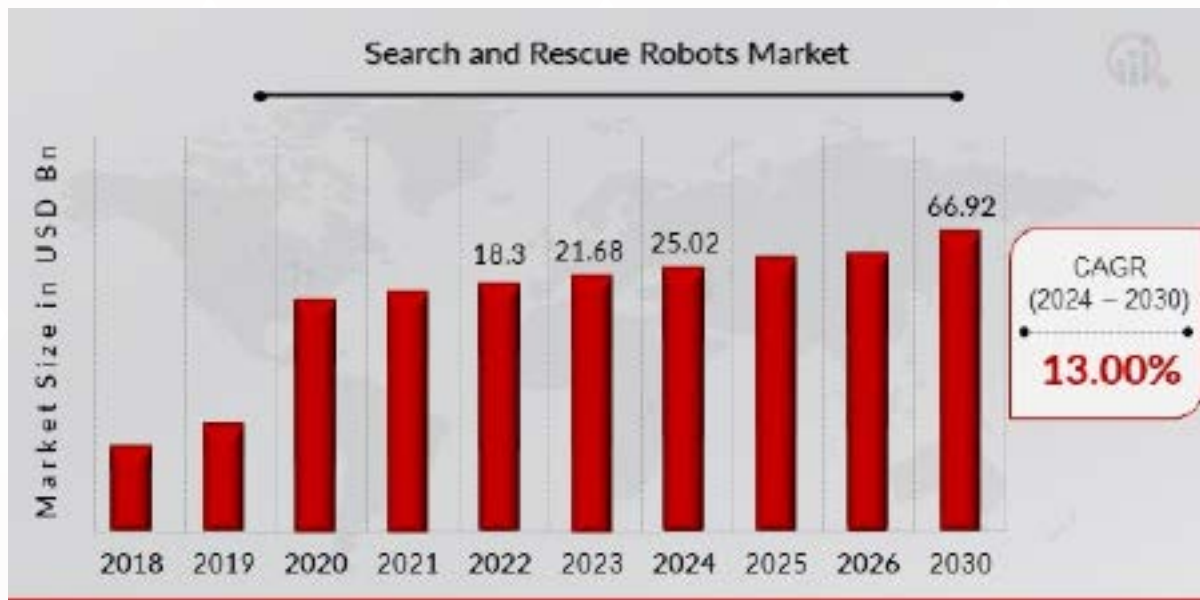


Figure 1: Growth in market size of Disaster Response Robots [1]

General Background

During a response which uses robotics, an ecosystem must be built in order to have the wireless robots run smoothly with their operators in safe locations. This often involves a Robotic Command Vehicle (RobLW), a Network Communication platform and all the drones involved [2]. All these are connected together to allow seamless communication from the operators and back.

Technical dive

The main umbrellas of drones used are Unmanned Aerial Vehicles (UAVs), Unmanned Ground Vehicles (UGVs) and Unmanned Surveillance Vehicles (USVs) [2][3]. UAVs and UGVs come in all shapes and sizes. All UAVs and UGVs have the basic requirements, like 360-degree sensors. However, each and every



Figure 2: An example ecosystem set up by the German Rescue Robotics Center task force [2]

drone is customized for their own job. A smaller UAV, like a First-Person View (FPV) drone may only be equipped with a 360 camera in order to travel through smaller cracks and fallen debris. This drone would then find a route for a larger UAV with a c camera to capture more detailed images for recreation [2]. It then sends the information to the RobLW, giving human responders a map and layout of safe and dangerous areas, as well as any survivors spotted.



Figure 3: a larger UAV and its attachments [2]

A UGV may be similarly equipped with cameras and object detection, but it would be differently built to allow for ground movement. A common UGV is often built with tank treads to allow for easier movement in rubble. It may also be built with 'flippers' to allow for a larger range of movement as well as vertical mobility [3].

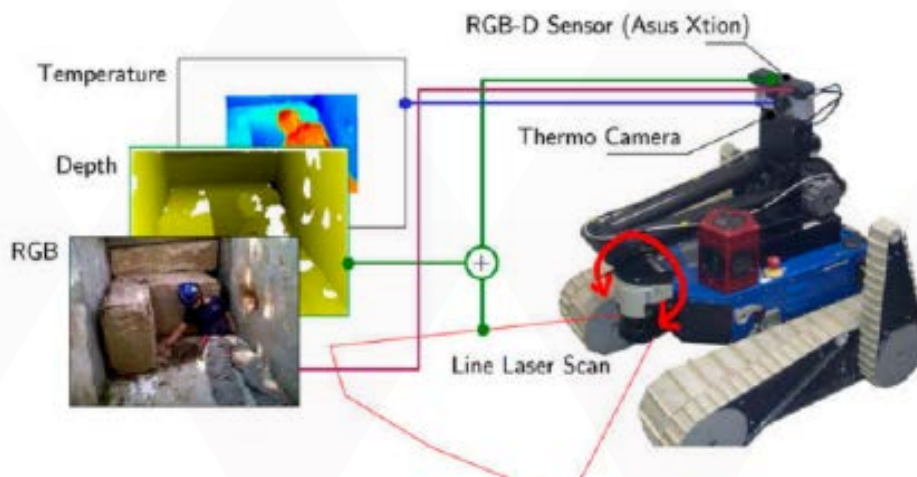


Figure 4: A common UGV with several sensors and cameras attached [3]

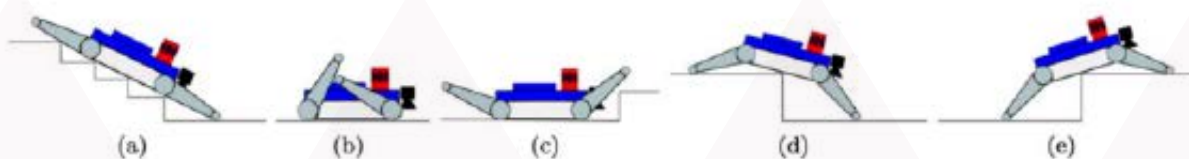


Figure 5: Possible range of motion for a UGV with movable 'flippers'

A more unusual UGV could come in the shape of a snake. This snake-like robot shown in Figure 6 comes from Tokyo, and is designed as a reconnaissance tool to view restricted or dangerous areas. Along with a camera in the 'head', there are

17 attached joints with wheels and sensors on each limb to allow a larger range of motion. This allows it to easily traverse rubble as well as perform simple tasks like turning a gas pipe [7]. Similar snake-like robots have been seen in underwater applications.



Figure 6: The snake-like robot traversing through uneven rubble [7]

Current applications / Case Study

The most notable use of robots as disaster relief was after one of the biggest disasters of them all. The World Trade Center. A platoon of robots sent by the Center for Robot-Assisted Search and Rescue (CRASAR) and US Navy searched the wreckage that day [5]. These were small UGVs, roughly the size of a shoebox. They were attached to a tether and sent nearly 30 meters underground to where no human or dog would ever want to venture. They were specifically equipped with infrared sensors and cameras to allow for detection of heat as well as smooth video footage. Although no survivors were found, they were able to locate 5 bodies and remains in the rubble [5].

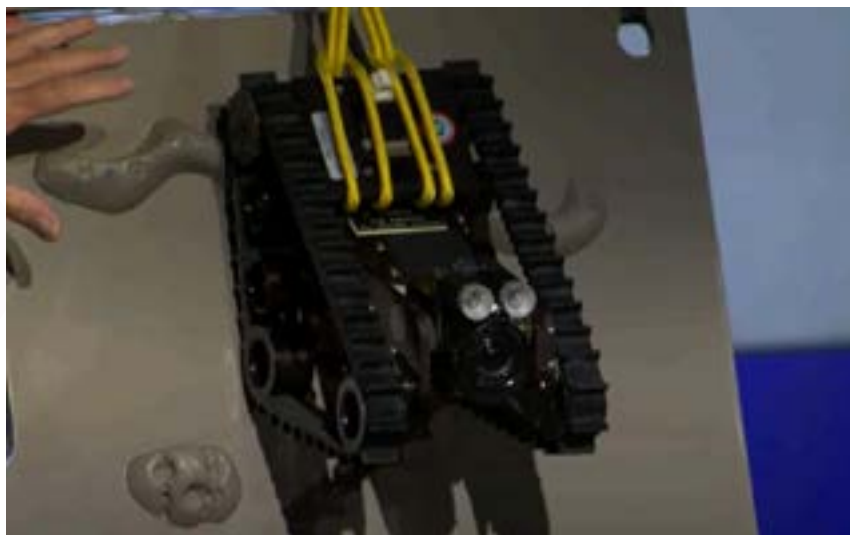


Figure 7: Bujold, a UGV, sent to explore the rubble of the World Trade Center [4]

Challenges and Limitations

Over the years of disaster response, many limitations have been discovered during use. Firstly, it requires up to 3 experts to pilot one drone [6]. This would mean that to have a swarm of drones moving at the same time, a large number of responders must be trained. Professionals must also be called in order to pilot larger drones [2]. When a drone is damaged in the site, it may be unretrievable and problems may arise. Due to unstable and dangerous conditions, it is easy for a drone to be knocked down or hit when moving [2]. Therefore there must be development in ensuring a robot can pick itself back up and continue its mission[7].

Future Developments

Several developments are desired within creations of future drones. Research has been put into developing high level obstacle avoidance algorithms for UAVs to help drones navigate dangerous environments [2]. Improvements in the mobility of UGVs are also being looked into, as they often require more human effort to use than UAVs, as they are easily subjected to destruction in unstable ground [2].

More research is also being committed to allow better collaboration between humans and drones. Treating each drone as a separate team member must be done in order to improve distributed situational awareness as well as allowing smoother communication for each team member's capabilities and needs [3].

Data collected is not only used for recounting and recording the event. Data is the most important thing in these efforts, because they can be used to figure out the cause, and prevent the effect from ever happening.

Conclusion

In summary, the first thing to enter a burning building is no longer a human. It has now become essential for the use of drones and robots during a disaster response. Not only do they provide safety, but they also collect valuable data to prevent a future accident. Each drone is specifically built for their purpose, and they must be used in collaboration with humans and other drones for the best result. In a few years, we can expect autonomous robots coming to our rescue.

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The Ethics of Robotics: How Automation is Changing Jobs and Society

NUO CHEN

Nowadays, as the development of automation industry, the societies system and the modern economy have been reshaped. While it is true that the development of automation increased productivity and efficiency, it also raises the ethical concerns which related to the growth of economic and the social inequality. This essay will talk about how the automation change the jobs and society, and whether this change bring us advantages or threaten.

The most impressive impact of automation is the replacement of jobs. Corresponding to the concentration of benefits is the dispersion of risks to a broader group of workers. Some of the Low-skill and monotone job in industries like logistics manufacturing and simple retail are more likely to been automated. According to the study outcome from McKinsey Global Institute there will be more than 800 million jobs could be lost to automation by 2030. [1] The workers who has been replaced need to turn to different jobs which have high technical requirements. The cost of learning latest and high technical skills are majorly high. Some workers can hardly afford the payoff, so that they might lose their job or been pushed into low-wage job, which is not good for workers life .For example in the 80s of the 20th century, when robotic welding technology became popular, Detroit's three major auto factories fired about 200,000 assembly line workers in only two years, and less than 15% of workers who were over 55 years old and have more than 20 years of service have been reemployed because they could neither program nor repair robots[2]. In this case the unemployment rate of this economic system will undergo a considerable growth, and the maximum productivity may be decline. At the same time , algorithms, chips, and data are in the hands of few people, so that will accumulate more wealth. This unequal distribution of benefits can lead to an increase of gap between the rich and the poor, and a broader social instability. A study by Frey and Osborne found that automation is likely to affect low-income communities disproportionately, as their jobs are more susceptible to being replaced. The rich, often gain from capital investments and new job opportunities created by technological advancement [3].

This trend threatens the ideal of equal opportunity and challenges the ethical foundation of a fair society. If left unaddressed, automation could create a "technological underclass" with limited access to quality jobs, education, and upward mobility.

You might ask: The automation of production lower the cost of produce, so that the price of commodity will decline, which means consumer will benefit from automation. But when a large number of workers lose their jobs, social trust may ruin and consumption may shrink, ultimately leading to a decline in overall social welfare. Here comes the question, how to balance the speed of automation technic and the social welfare of people. Now we already got some policy — tax and Human-machine collaboration. For example the Japanese government has adopted tax incentives to support companies to maintain a human-robot collaboration model when adopting AI technology, ensuring the participation of low-skilled workers. Japan's Ministry of Health, Labor and Welfare requires companies to ensure a labor participation rate of at least 10% to ensure that jobs are not completely replaced by AI. Germany has adopted a progressive tax policy, through the "social innovation fund", which redirects tax revenues back to the public service sector, and provides financial support for retraining low-income groups to help them enter high value-added industries[3]. In short the government can provide financial compensation and low-cost education to those who are directly unemployed due to automation through a "robot tax" or a "data dividend tax". The tax imposed on the manufactory who are high automatic, and pass to the consumer through the higher price. If we have enough data, we can calculate the amount of taxes and subsidies which can leading to a balance of market and society.

Another important phenomenon is about the robots replace humans in dangerous environment, which significantly improves the safety of worker. Jobs that involve working at extreme heights, in underground mines, or in deep-sea environments pose serious risks to human life due to falls, cave-ins, exposure to toxic substances, and extreme pressure or temperature. By deploying robots in these kinds of conditions, we can highly reduce the number of injuries and fatalities. We can not only

save the manpower but also increase the efficiency. In this case, there is another element need to be considered which is called decision-making. Algorithmic bias, for example, can occur when AI systems are trained on inaccurate data. This can lead to wrong decisions. How can we clean up the incorrect data? or can we teach AI system to identify them? We have a long way to go.

For summary, as the automation technology develops, our jobs and society have been reshaped. There will be some problem about the increases unemployment rate and increases inequality. And we already have some policy to deal with them. The automation also ensure the safety of workers in high-risk occupations, but we need some time to ensure that the system can make decision accurate and correct.

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08

Engineering: Sustainable Vehicles

Methods to Optimise Aerodynamic Characteristics and Propulsion System on Electric Aircrafts

YUKAI LIU

Abstract

With the rise of CO₂ emissions from aviation, it is important to consider methods such as electrification of aircrafts to reduce emissions. This work investigates an electric propulsion system based on battery and fuel cells and their impact on performance and emissions. The system is introduced and analysed in both aerodynamics and energy aspects to optimise the efficiency of the system on aircrafts. Both these aspects are explored through improvements to current conventional aircraft designs. The essay demonstrates a modified model of electric aircraft with greater comfort, higher speed, lower noise, longer range and less energy consumption than pure-electric aircrafts. The essay puts forward creative ways to address issues of pure-electric aircrafts to form a second-generation of electric aircrafts and opens a new area for future research based on the combination of batteries and fuel cells.

1. Introduction

Given the fast development of aviation over the world, it is important to consider the electrification of aircraft engines to lower the CO₂ emissions. According to statistics, the emission of CO₂ in aviation has already reached 2.4% of the global emissions and is still increasing [1].

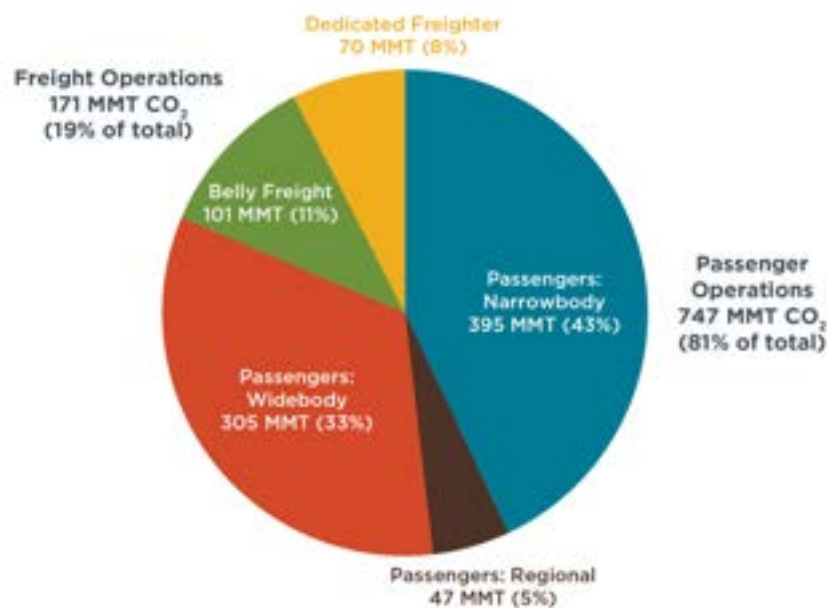


Figure 1: CO₂ Emissions in 2018 by Operations and Aircraft Class [1].

Short passenger flights consume little energy and carry comparatively less weight, requiring less power and energy storage. This fact could facilitate the replacement of fossil fuel aircrafts with fully electric aircrafts for short-distance flights. Different architectures have been proposed, each of them with their advantages and drawbacks [2]. One of the most promising technologies is the use of hydrogen fuel cells to provide electricity along with a battery to drive the aircraft. However, the structure of the aircraft with this energy configuration and the design of the propulsion system are still unclear.

The objective of this work is to investigate electric propulsion systems based on fuel cells and battery, and their impact on performance and emissions.

2. Propulsion System

2.1 Engines

The aircraft uses propellers powered by electric motors as source of thrust. Electric motors provide high power limit so the aircraft can be more flexible with greater power during take-offs and go-arounds. The electricity provided is from fuel cells and batteries. This engine structure is inspired by Airbus Aerostack Project [3], and the possibility and reliability of this energy structure are primarily proved by recent research [4].

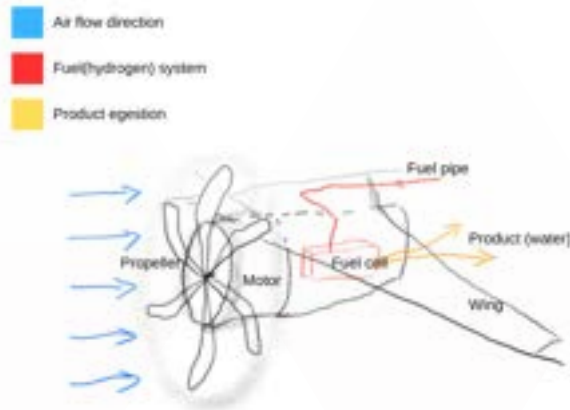


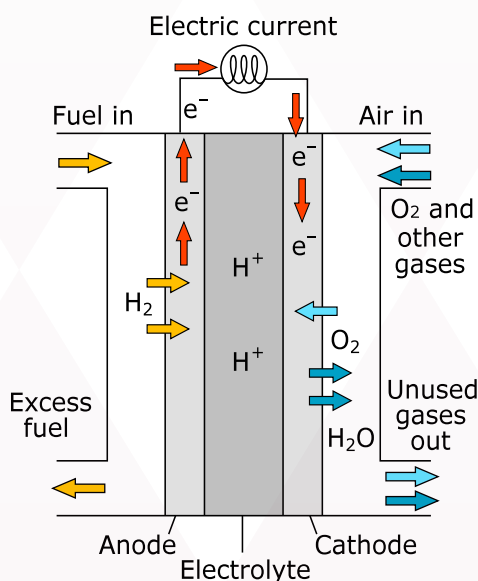
Figure 2: Diagram of Engine Structure.



Figure 3: Power System Diagram.

In a fuel cell, each hydrogen atom is divided into a proton and an electron. The proton goes through the partial permeable membrane (electrolyte) and the electron goes through the load by a separate circuit. At the other end of the structure, these particles combine with each other and oxygen from air to form water, which is finally released into the air. Moving electrons through a separate circuit form voltage.

Fuel cells transfer energy totally through chemical reactions without heat exchange. Less extra heat is produced so the overall efficiency could be far higher than traditional combustion engines. Recent research shows that the overall efficiency of a fuel cell could reach 80% [5], which is significantly higher than conventional combustion engines (around 20%) and advanced combustion engines (around 55–60%) [6]. Moreover, no combustion results in less noise and vibration in the cabin, and no CO₂ or NO_x emissions.



2.2 Hydrogen Tank

The hydrogen to operate the fuel cells is stored in the tail of the aircraft, where the auxiliary power unit (APU) is on fuel-powered aircrafts. Storing gas or liquid in the tail maximises the utilization of irregular tail space. Placing the tank in the tail allows emergency evacuation of high-pressure hydrogen through vent valves to avoid explosions.

Hydrogen shall be stored in a low temperature environment, so a refrigerator is required. Pressure difference would increase as the aircraft climbs, so strong material with high strength at low temperature needs to be used.

Figure 4: Structure of a Fuel Cell (Source: https://en.wikipedia.org/wiki/Fuel_cell, accessed in July 2025).

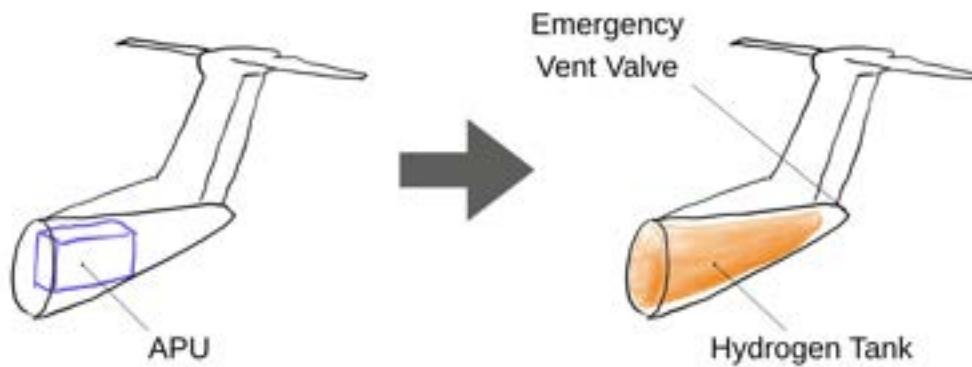


Figure 5: Modification from APU to Tank.

As hydrogen is lighter than fossil fuels, the total variation of weight during flight will be lower. The proportion of energy carried is calculated as:

$$\text{proportion} = \frac{W_{\text{Fuel}}}{W_{\text{Total}}}$$

To meet the same maximum take-off weight (MTOW), a conventional fuel aircraft needs to carry lots of fuel, which lowers the actual commercial payload. Electric aircrafts could reach higher efficiency of carrying passengers as cost for carrying fuel to fly could be neglected.

2.3 Auxiliary Battery

Auxiliary battery creates a buffer between fuel cells and motors to enhance the flexibility during flights. When the aircraft demands sudden power, the battery could intervene before the fuel cells rise their power. In case of emergency when hydrogen is in short supply, battery can still provide basic aircraft controls until landing.

Batteries are placed in the two wings near the motors. Short distance between batteries and motors allows lower cost for electric cables and less electric resistance. Furthermore, the heat produced during operation conducts through the wings, so they could be used as de-ice system to melt ice. This design can improve performance and range of electric aircraft in cold regions comparing to pure-electric aircraft.

3. Aerodynamic Characteristics

3.1 Wings

Wings are above the cabin to enhance the work of the propeller engines by allowing larger diameter of propellers and more distance between the propeller and the ground. This wing configuration is widely applied in propeller aircraft, like DASH-8 series.

Wings contain batteries, so they tend to bend upwards to cancel the weight of the batteries. Each wing is divided into sections of batteries with fireproofing separations and individual fire extinguishers to ensure safety in case of fire. Moreover, weight of battery remains nearly constant, resulting in no change of centre of mass (CG) during flight, which simplifies the payload work on the ground.

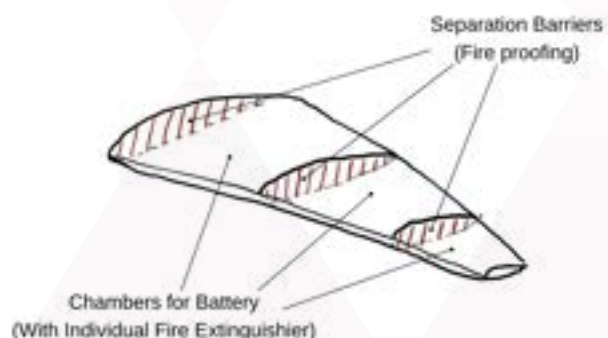


Figure 6: Wing Diagram.

3.2 Fuselage

The fuselage does not require significant changes. The central fuel tank could be changed into an extra auxiliary battery to extend the range or into an extra luggage area to improve payload. Quieter motors powered by electricity cut down the requirement of sound barrier, giving out more weight to focus on commercial payload.

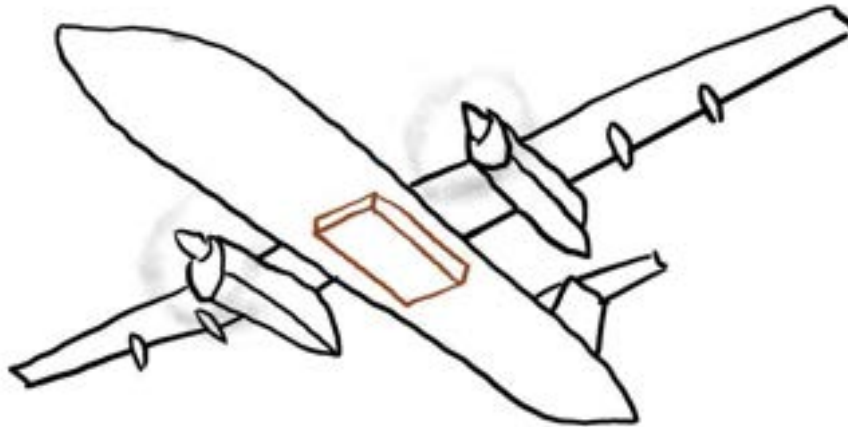


Figure 7: Central Compartment for Different Functions.

The passenger cabin needs to be separated from the hydrogen tank in the tail by a barrier that isolates heat. Furthermore, the size of the hydrogen tank can vary to fit different range and payload requirements for customers.

4. Conclusions

Combination of electric motors and hydrogen-based fuel cells provides broader possibility for electric aircraft than pure-battery aircraft with better performance in range and speed. Some of the proposed methodologies exceed current technology limit, therefore this provides a clear pathway to innovate towards clean aviation. Hydrogen storage and fuel cell efficiency do not have mature technologies so far, which makes the designs proposed here not applicable nowadays. Future innovations are required, and they could focus on building material that can resist high pressure at low temperature, developing efficient fuel cells to meet the high-power requirement of aircrafts and techniques to cool hydrogen to liquid state.

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Is There a Practical Roadmap to Carbon-Neutral Transportation? A System-Based Evaluation of Sustainable Propulsion Systems

ZHUOHENG LI

Abstract

Road transportation plays a crucial role in global warming and must transition toward a sustainable solution. This essay aims to evaluate various vehicle propulsion systems and outline a practical roadmap for sustainable transport using a system-level approach. The evaluation not only focuses on the propulsion technology itself, but the entire lifecycle from energy generation to real-world application is considered.

The essay compares hydrogen, electric, nuclear, natural gas, and biofuel cars. Performance metrics include energy efficiency, environmental impact, technological feasibility, and safety. Hydrogen is shown to be inefficient and carbon-intensive unless generated by electricity from renewable sources. Electric vehicles are already practical, but suffer from limited range and potential pollution of land and water. Nuclear propulsion offers superior range but faces major obstacles, including safety, technological limitations, and client acceptance rate. Biofuels and natural gas provide an overall sustainable solution for short-to-medium term transition options due to lower emissions and compatibility with current infrastructures.

The essay concludes that a dynamic and diversified roadmap is essential. No propulsion system is flawless, but they all provide meaningful contributions depending on the timeframe and context. Moreover, efforts to invest in and research renewable energy resources are always crucial and necessary. No matter which propulsion system is being used, clean energy always contributes to a carbon-neutral roadmap. This essay provides a strategic reference for the direction of sustainable progress toward carbon neutrality in the mobility sector.

1. Introduction

In the new era of sustainable engineering, researchers increasingly assess the road transportation system about environmental responsibility and the efficiency of the system as a whole. However, there is no single propulsion technology that can give a perfect solution. The objectives of this essay are to: (i) evaluate the practicality and sustainability of different propulsion systems and (ii) give a realistic pathway toward carbon-neutral transport. This analysis adopts a system-level approach, from energy generation to energy usage.

2. Evaluating different propulsion systems

Hydrogen

Hydrogen fuel cells offer the attractive promise of zero direct emissions and lower cost for travelling. For example, a petrol-heavy truck requires fuel costs of about 300–500 RMB, while a hydrogen-heavy truck requires only 160–200 RMB per 100 km (Jin, 2025).

However, the problem lies in its extreme inefficiency for hydrogen vehicles. According to Figure 1, only 33% of the energy can be used

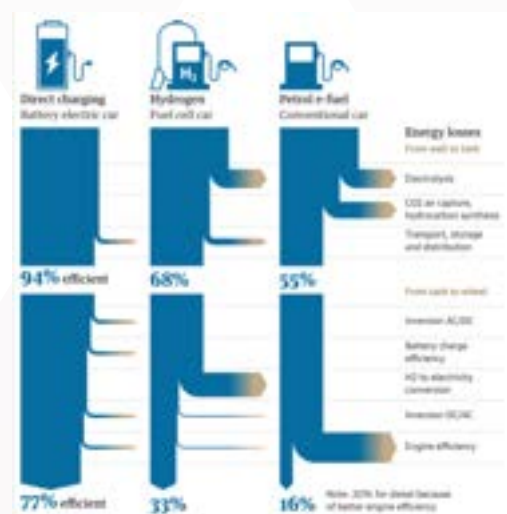


Figure 1: Efficiency of battery electric car, fuel cell car, and conventional car (Source: Guardian Graphic).

from the tank to the wheel in a hydrogen fuel cell vehicle. Furthermore, 99% of global hydrogen comes from fossil fuels, which is grey hydrogen, emitting as much CO₂ as the UK and Indonesia combined. Green hydrogen, which only accounts for 1% of global hydrogen production made via electrolysis, is insufficient as electrolyzers lose 20–40% of the energy input (World Economic Forum, 2021).

Hydrogen, while promising for hard-to-electrify sectors (World Economic Forum, 2021), remains an inefficient and carbon-intensive choice for near-term vehicle propulsion.

Electric Vehicles



Figure 2: Main concerns for electric vehicles; data from a survey distributed to car owners.

Electric vehicles are currently the most widespread clean propulsion technology with a rapidly growing market; there were 17.5 million new electric cars sold in 2024 (Our World in Data, 2025). However, due to limited energy density, a typical electric car can only travel about 600 km with an energy storage capacity of 96 kWh (Mercedes Inc, 2025). In a survey distributed to petrol car owners, 36% of the responses suggested that limited driving range is their reason for not changing to an electric vehicle. Limited driving range remains one of the most frequently cited concerns among clients.

Innovative approaches, such as wireless charging technologies that charge the vehicle while driving, have been explored to address this issue. A car could reach a speed of up to 100 km/h while charging at 20 kW, which means that the vehicle will sustain negligible net energy loss (Vedecom Inc., 2018).

Electric vehicles are only truly sustainable when charged using clean energy and supported by better charging options. Moreover, lithium batteries may contaminate water where high levels of lithium are toxic to organisms (Chow, 2022). EVs require energy for battery production and rare-earth mining (Alonso, 2012), making lifecycle emissions significant unless renewable energy is used throughout.

Nuclear Vehicles

Nuclear propulsion offers unmatched energy density and no direct emissions. Concepts of nuclear-powered vehicles could run for decades without refuelling (Durkin, 2012). It is estimated that 8 g of thorium can allow a Hummer car to drive 1.5 million km, equivalent to over 220,000 litres of petrol being burned in the combustion engine. Range anxiety of electric vehicles can be eliminated as a nuclear reactor that is integrated into the car could produce electricity to drive the motors (Wang, 2017). However, in practice, there are critical drawbacks. With over 400,000 road collisions annually, after collisions of cars, nuclear fuel may leak, which leads to significant radiation that will cause catastrophic consequences for human health and the environment. Furthermore, the presence of fissile materials also raises national security concerns due to the

potential for weaponization in hostile hands. Additionally, public hesitation due to radiation risk also limits its near-term viability. Moreover, it is challenging to miniaturize nuclear reactors with current technology (Wang, 2017).

2. Evaluating different propulsion systems

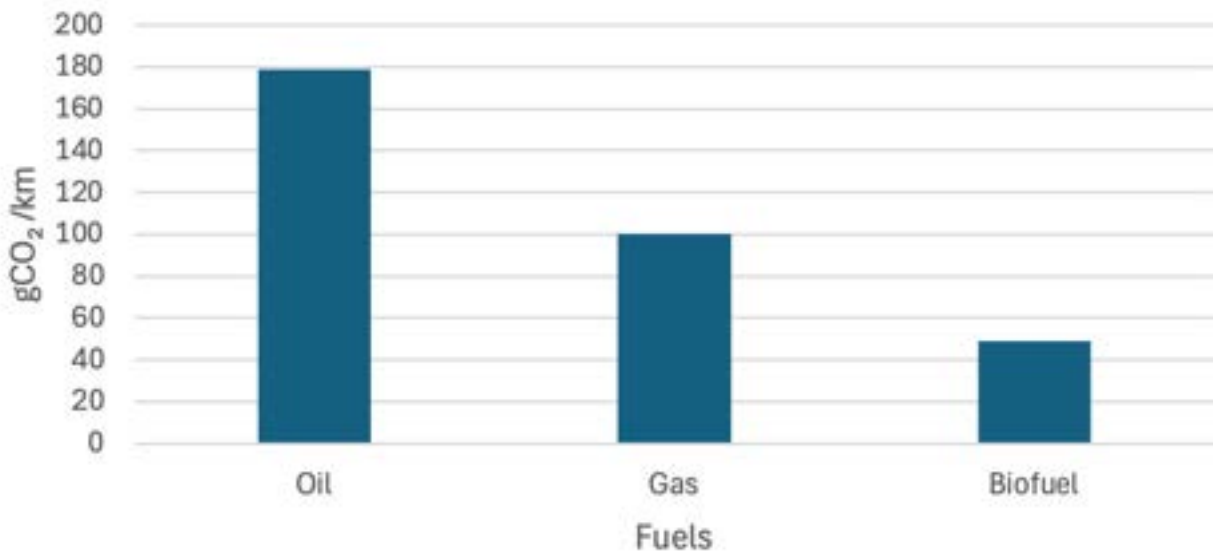


Figure 3: A bar chart illustrating gCO₂ emitted/km for a typical car (Nissan LEAF).

Source: <http://carbonintensity.org.uk/> (accessed in July 2025).

Natural gas vehicles

Natural gas is a relatively green energy source. It contains significantly less air pollutants such as NO_x than diesel (Jelti, F. et al, 2023). Natural gas, having the highest hydrogen-to-carbon ratio among fossil fuels, results in lower CO₂ emissions per unit of energy. It is also practical to implement this fuel economically as the price in caloric equivalents is only 0.4 against diesel fuel (Pascoli et al, 2001). Natural gas also promises safe storage where the possibilities of leakage are negligible since container technology for natural gas has been well developed in recent years (Jelti, F. et al, 2023).

Biofuel vehicles

Methanol can be produced by biomass (Dalena, F. et al, 2017). The use of Biofuels is much more environmentally friendly, emitting only 49 gCO₂/km, while oil emits 179 gCO₂/km. Methanol can allow an M85 vehicle to emit at least 30% less harmful ozone-depleting emissions. Biodiesel is made from vegetable oils or animal waste (Maheshwari, P. et al, 2022). While biodiesel emits more NO_x than conventional diesel, it offers significantly lower CO₂ emissions and safer storage properties since biodiesel is dramatically less reactive than petroleum fuels (Jelti, F. et al, 2023). Though two times more expensive, biofuels work with current engines, avoiding costly redesign (Yuan, 2006). However, deforestation may occur to bring space for planting plants that are suitable for biofuels (Gao, Y. et al, 2011); accordingly, extra CO₂ may be emitted during the transport of wood during deforestation.

4. Conclusion

Balancing the advantages and disadvantages of each propulsion technology is required to engineer a roadmap toward carbon neutrality. Natural gas and biofuels promise a realistic transitional pathway as they have dramatically lower lifecycle emissions and can make use of existing infrastructures. The electric vehicles will remain the most mature short and long-term strategy if battery capacity, renewable grids, and charging systems continue improving. Nuclear propulsion, although its technical and safety aspects remain a crucial challenge, holds promise for the future due to zero direct emissions and far superior range than any other propulsion system. A diversified roadmap is necessary, as no single propulsion system

offers a universally optimal solution. We must prioritise practicality, readiness, and total environmental impact. Achieving sustainability requires not only innovation in propulsion technologies, but also systemic optimisation across energy supply and infrastructure.

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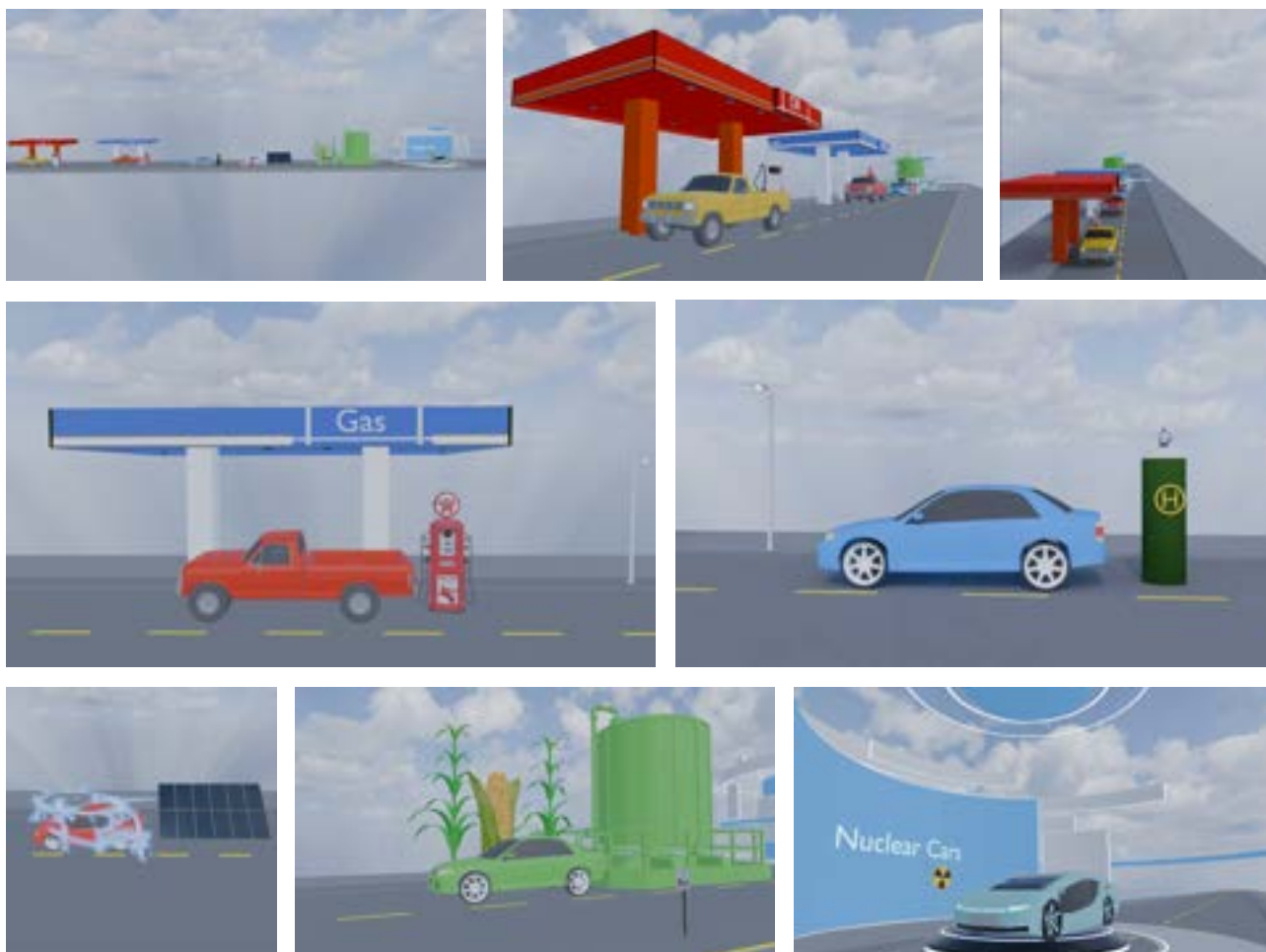
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6. Appendix

Some photos of 3D-modelled concepts of vehicles consisting of different propulsion systems. From left to right of the road, it gives a timeline reference, where oil is the earliest form of cars while nuclear cars are a strong potential for mass application in the distant future.



60

Mathematical Economics

Is It Rational to Trust Economic Theory When Formulating Real-World Policies During a Pandemic?

JAMES ZHANG

In 2020, the COVID-19 pandemic highlighted a gap between theoretical assumptions of rationality and human actions. Economists predicted public submission to lockdowns, consumer rationality and firms accepting profit reductions; expected utility models, costbenefit analysis and rationality theories were all utilised to guide governments to maximise social welfare under uncertainty and global crisis¹. Yet in reality, deviation from these models arose, as people overestimated risks and hoarded scarce products, while the markets oscillated unpredictably. Were these outlying actions irrational behaviour, or was failure rooted within insufficiency of the theories themselves?

This essay critically evaluates whether economic theories can be rationally trusted given pandemic conditions. Using COVID-19 as a case study, it assesses how Neoclassical utility and choice models can both cultivate and distort policymaking. Ultimately, this piece extends by applying ambiguity aversion to reveal how economic assumptions collapse in contact with uncertainty, hence arguing that while theories remain indispensable, to ensure effectiveness, divergences between logic and human behaviour at times of disaster must be acknowledged.

Across COVID-19, governments faced a dilemma of preventing mortalities, containing the disease and steering away from economic mishaps. In doing such, the Expected Utility Theory (EUT) and Rational Choice Theory are integrated when assessing options and tradeoffs during uncertainty, which can be modelled as:

$$EU(L) = \sum_{i=1}^n p_i \times u(x_i) \quad (1)$$

Where L represents a policy with index of possible outcomes i , for a set x e.g. Gross domestic product (GDP) or death tolls, that associates probability p and utility derived from each outcome with $u(x)$ ². The function $EU(L)$ exhibits the expected utility when imposing policy L , this proves pivotal during times of pandemic, as it grants policymakers the ability to weigh trade-offs between choices. Governments face the daunting tasks of setting the rigidity of lockdowns within the country; empirical research has proven that more stringent measures enforce greater public health and lower fatalities³, but concomitantly weaken economic prosperity as restrictions distort labour supplies and consumption patterns, displaying a negative relationship between health and GDP.

Furthermore, the weighted utility outcome of lockdowns can be calculated with the monetary value of utility gained from economic output E , subtracted by the disutility (negative social wellbeing) of death D , and can be modelled through a social-welfare function:

$$U(L_i) = \alpha E_i - \beta D_i \quad (2)$$

The constants, α, β , calibrate the model to satisfy economic reasoning and moral principles that guide decision making. This framework assumes $\alpha > 0$, as greater economic output often translates to positive contributions, like greater employment and overall efficiency. Conversely, $\beta > 0$ reflects the premise that overall welfare is reduced due to decreases in future output and the moral cost of loss of life. This is vital as governments must evaluate policies under scarcity, forcing an optimum solution to balance justifiable outcomes using net-utility contributions, ultimately granting a more transparent and rational analysis⁴.

In application, a utility comparison of two proposed policies in March 2020 within the United Kingdom (UK) can be considered as: strict lockdowns (L_1), and minimal lockdowns (L_2). Empirically, the UK's 'Value of A Statistical Life' is £1.8 million⁵, and with the imposition of measures L_1 , GDP is projected to fall by 9% of its total (£2.219 trillion)⁶, approximating

to £199.7 billion, with an estimated death toll of 100,000, which corresponds to a human cost of £180 billion; hence, it can be derived:

$$U(L_1) = \alpha(-199.7) - \beta(180)$$

In contrast, weaker lockdowns would result in 3% decrease in GDP but lead to almost 500,000 deaths according to Imperial College's model⁷. Thus implying:

$$U(L_2) = \alpha(-66.6) - \beta(900)$$

Following the above, we can derive that while L_1 places a greater economic loss and deaths are reduced significantly in comparison with L_2 . Thus, forcing policymakers to base decisions on different weighted variables α, β . If the social cost of a death is significantly greater to a policymaker, then $U(L_1) > U(L_2)$, displaying that stricter lockdowns should be imposed as they are of greater utility.

However, COVID-19 revealed inconsistency with theoretical models and human behaviour, this is as traditional models assume agents act when facing risk, but pandemics involve ambiguity, in which probability was unknown or ill-contested. This can be viewed with deviating approximations of mortality rates, unknown vaccine development timelines and uncertain societal effects⁸, all contributing to limitations in the EUT.

In addition, Gilboa and Schmeidler's Maxmin Expected Utility (MEU) was developed to replace known probabilities with a distribution and account for the worst-case expected utility. The outcome is achieved by considering the minimal utility of all models, then selecting the greatest valued model. It is depicted as:

$$MEU(a) = \min_{P \in \mathcal{C}} \sum_{i=1}^n p_i u(L_i(a))$$

This depicts that early lockdown measures were not imposed due to known probabilities but due to policies being evaluated under worst-case assumptions, as incomplete data, limited testing, meant that traditional cost and benefit analysis was no longer reliable, concluding that economic theory would be undermined due to ambiguity aversion within times of uncertainty⁹. Alongside such, economic models during the pandemic failed to account for ethical factors and systematic failures. The economic models assume that the government can enact optimal policies that all citizens will guarantee to abide by. However, individuals are affected by behavioural means and attitude towards the leading institutions, meaning that without legal enforcement and social influence, policies may fall short and falter when executed.

In evaluation, economic theories such as the Expected Utility Theory present effectiveness in assessing trade-offs, but are limited during uncertainty within a pandemic, COVID-19 did not just present consumer irrationality, but also displayed that classical models fail to integrate real-world complexities, so by implementing models that account for ambiguity like MEU and considering ethical grounds provides a more rigorous assessment. Lastly, economic theories during a pandemic can be trusted in modelling real-world decisions, with acknowledgements that they oversimplify insights. While they are an indispensable tool, its usage must be contextualised and ethically grounded, to ensure successful policies for the livelihood of those impacted.

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Principle of Budget—How to Manage Household Finances

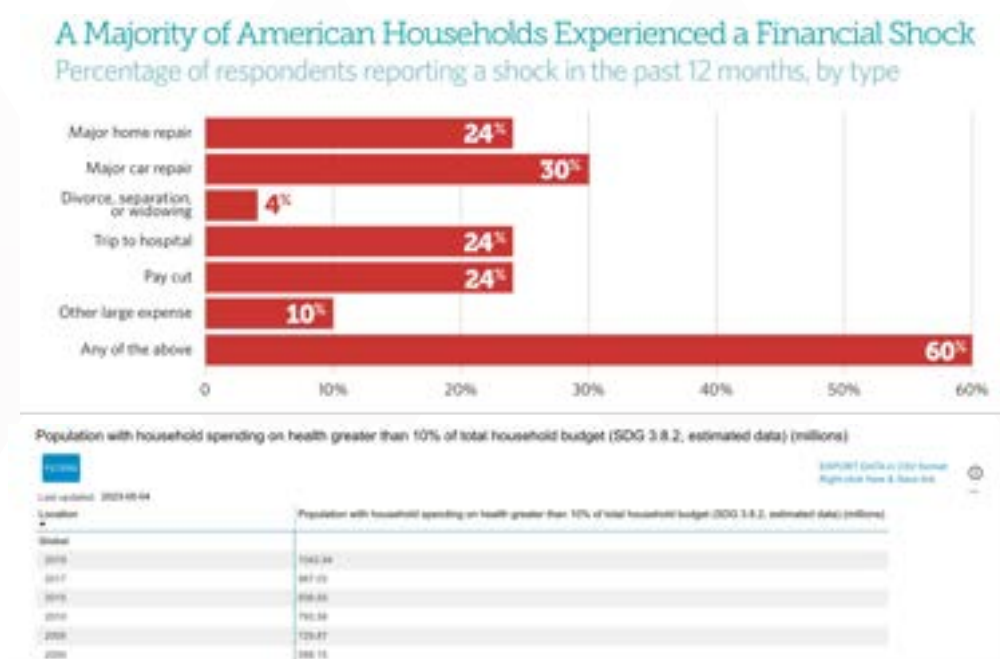
HUIYU DENG

Introduction

Budgeting plays a crucial role in the financial management. It helps individuals, corporations and government to optimize allocation of secure resources. For households, managing family budget appropriately is closely relative to the material well-being of every family members and their resilience when facing potential risks in unknown future. This essay is going to explain three principles of budget about household finances from both economic and mathematical perspectives which can provide you with useful suggestions on management of household budget. First, it explains the importance of saving and how households can increase the quality and completion of saving. Then, it shows how can a family decide to balance current and future consumption. Finally, it explains the essential role of cooperative households financial management on enhancing the quality of decision-making and how a final plan can be found from different decision-makers.

1. Saving-the first defense

Saving plays a important role in preventing financial shock, preparing pensions and accumulating wealth. There are various causes of financial shock that can damage household financial stability. The most common reasons are major car or home repair, pay cut, Catastrophic health expenditure and so forth (The Pew Charitable Trusts, 2015). In 2015, 60% of American households experienced a financial shock (The Pew Charitable Trusts, 2015). As for the population with household spending on health greater than 10% of total household budget, the figure has increased steadily since 2000 and reached about 1042.94 millions in 2019 (WHO 2023).



There are some suggestions for households to save. First, setting long-term goals like pension scheme can help household to improve the self-control ability and to cultivate saving habits.(Ouyang et al, 2025) This can be beneficial to the improvement of household financial safety. Second, separating long-term goals to several more specific, short-term goals make it more likely for people to complete their saving plan.(Hershfield et al, 2019).This is because they find it less psychologically painful to give up money spread out over time in a smaller, more granular format.(Hershfield et al, 2019) Third, making use of functional apps to improve the measurability of your saving, in other words, Clarifying how much planned savings remain uncompleted and their deadlines can increase the completion rate of saving plans.



2. Balance current and future expenditure

People can tend to spend more at present or in the future, in other words, same amount of current and future expenditure can generate different level of utility for them. Also, due to diminishing marginal utility, more consumption made in one time period, lower the marginal utility it can bring to the consumer. Considering these, we can find out the best plan for consumption in different periods of time which can maximize the utility for individuals by adopting the model below:

utility function and its key variables:

C1: the consumption in the first period of time

C2: the consumption in the second period of time

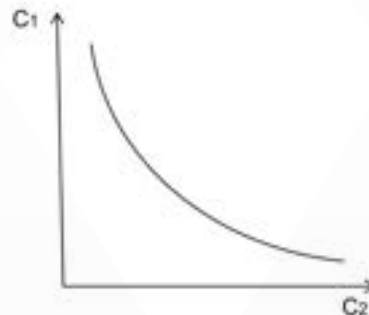
A/(A+B) & B/(A+B) : the utility indexes, they show an individual's different preference to spending in different time periods. A and B can be any extremely positive real number.

$$U(C1, C2) = C1^{a/(a+b)} \cdot C2^{b/(a+b)}$$

construct a graph for the utility function on a rectangular plane coordinate system with vertical C1-axis and horizontal C2-axis

$$dU/dC1 = [a/(a+b)C1^{a/(a+b)-1}] \cdot C2^{b/(a+b)}$$

$$dU/dC2 = [b/(a+b)C2^{b/(a+b)-1}] \cdot C1^{a/(a+b)}$$



the slope of the graph show rate of substituting C1 for C2

Thus:

$$Gu = - dC1/dC2$$

greater the marginal utility that can generated by one more unit of C2 compared to that generated by one more unit of C1, greater units of C1 the individual would like to substitute for. In other words, the rate of substitution is inversely proportional to the ratio of marginal utility.

Thus:

$$Gu = - dC1/dC2 = m / [- (dU/dC1) / (dU/dC2)]$$

m is a constant and m can be any extremely positive real number

set the m to be 1

Then:

$$\begin{aligned} - dC1/dC2 &= - [(dU/dC2) / (dU/dC1)] \\ &= - \{ [b/(a+b)C2^{b/(a+b)-1}] \cdot [C1^{a/(a+b)}] / \{ [a/(a+b)C1^{a/(a+b)-1}] \cdot [C2^{b/(a+b)}] \} \} \\ &= - (b/a) \cdot (C1/C2) \end{aligned}$$

Since households have a limited budget, it always involves an opportunity cost when consuming more in one time period in the term of consuming less in the other time period.

Budget function and its key variables:

P1&P2: the price level in the two different time period

I: the total disposable income the individual can spend

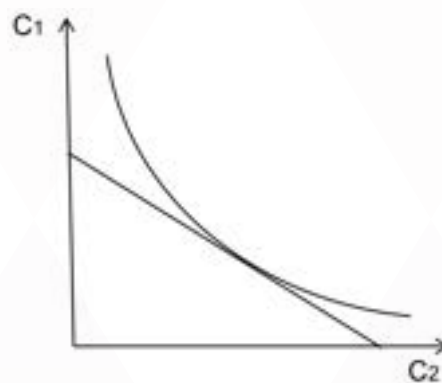
S: the total saving during the periods.

$$P_1C_1 + P_2C_2 = I - S$$

construct a graph for the Budget function on a rectangular plane coordinate system with vertical C_1 -axis and horizontal C_2 -axis together with that of the Utility function.

$$G_b = -P_2/P_1$$

The line of the budget function can only has a single intersection point with the curve of utility function which can provide the greatest level of utility available.



At the intersection point:

$$G_u = G_b$$

$$-P_2/P_1 = -(b/a) \cdot (C_1/C_2)$$

Simply it, we get to know that to maximize utility in this case:

$$(P_2/P_1) \cdot (a/b) C_2 = C_1$$

Assume that there is no inflation during the two time periods and the price level keeps unchanged.

Thus:

$$(a/b) C_2 = C_1$$

However, this model can be subjective to the limitations. First, it takes two time periods only into consideration, which is not enough to response to the more distant future. Also, it is hard for consumers to keep rational all the time. As a result, the utility indexes can fail to truly reflect the level of satisfaction the consumptions in different time periods bring to them. This may be caused by various factors such as promotional activities and blind compliance. Besides, considering utility of consumption only is not comprehensive enough for effective budget as it lacks ability to cope with financial shock. Combing this model with sound saving plans that I have suggested before can strengthen household resilience to potential risks.

3. Cooperative household financial decision-making



Involving every family members in the management of household finance can effectively improve the quality of decision-making. This is because the presence of family members providing humbling and immediate feedback can reduce the knowledge overconfidence of the other decision makers.(Warmath et al, 2019)In addition, the cooperative decision-making enables each member to profit from the knowledge and skills of the others. (Warmath et al, 2019)In this way, individuals' lack of knowledges in a specific domain can be compensated.

Then what should be the final outcome of the consumption pattern when considering individuals with different consumption preferences over time?

First, let's construct the model with three decision-makers with preference indexes $a/(a+b)$ & $b/(a+b)$, $a_1/(a_1+b_1)$ & $b_1/(a_1+b_1)$ and $a_2/(a_2+b_2)$ & $b_2/(a_2+b_2)$.

$$(a/b) > (a_1/b_1) > (a_2/b_2)$$

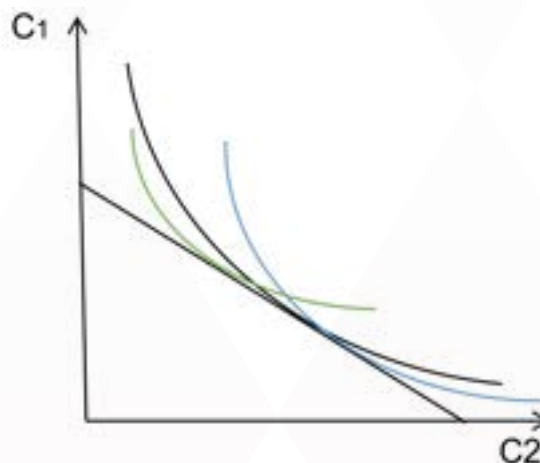
Thus, we get three different outcome of consumption pattern:

$$(a/b) \ C_2 = C_1$$

$$(a_1/b_1) \ C_2 = C_1$$

$$(a_2/b_2) \ C_2 = C_1$$

Then, when the family hold a vote to find out a final outcome and the decision-makers are arranged in the ascending order of (A/B) , the outcome favored by the median decision-maker will be the final outcome.



In this case, it is:

$$(a_1/b_1) \ C_2 = C_1$$

Then, when the number of decision makers is even such as 4. Let's add a new decision-makers with the utility indexes $a_3/(a_3+b_3)$ & $b_3/(a_3+b_3)$.

$$(a/b) > (a_1/b_1) > (a_2/b_2) > (a_3/b_3)$$

Thus, we get four different outcome of consumption parttern:

$$(a/b) \ C_2 = C_1$$

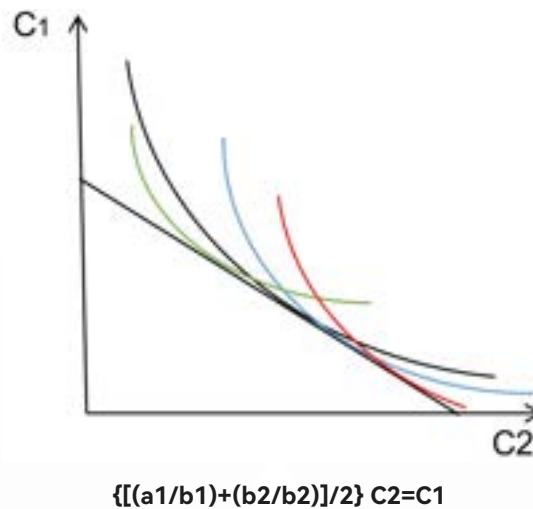
$$(a_1/b_1) \ C_2 = C_1$$

$$(a_2/b_2) \ C_2 = C_1$$

$$(a_3/b_3) \ C_2 = C_1$$

Then, the final outcome should be the mean of two outcomes favored by two median decision-makers.

In this case, it is:



Conclusion

Effective household budget management hinges on three key pillars: saving, balanced present and future consumption, and collaborative family decision-making. Saving is a effective tool of preparing for retirement, accumulating wealth and increasing households' resilience when coping with financial shock. With structured long-term and short-term goals, and digital tools which increase the measurability, household can complete their saving plans more effectively and efficiently. The mode of utility and budget can help household optimizes budget allocation over time, despite some inevitable limitations. The model can be further developed when taking more than one family members' will into consideration, which leads to a outcome favored by the median decision-makers. This can help households achieve better strategies of using the family budget rather than involving only one person in the decision-making.

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Challenges in Measuring and Quantifying Uncertainty

XUANZI HAN

Introduction

Uncertainty exists in every decision we make. To feel more confident in making decisions, we always try to turn uncertainties into countable numbers, such as “a 25% chance of rain tomorrow”, but actually face serious challenges. Using basic math and statistics, this essay explains three key challenges in measuring uncertainty and shows how understanding these limits can help us make better decisions.

Definition of uncertainty

Uncertainty is used to describe the possible range of values reasonably given to an event. It may appear in forms like the width of a confidence interval. The uncertainty can be given by statistical data or personal experiences. Unlike general doubt, it does not question related results but rather increases confidence in their reliability.

Challenge 1: Two Sides of Probability Calculable Objective Probability

Some probabilities are as exact as math formulas. Such as when you roll a standard die, the chance of getting a “5” is 1 over 6 forever. This kind of probability can be determined by multiple experiments and is always objective and accurate, just like physical quantities such as mass and length, which remain constant. Just as such, the probabilities of such events can all be precisely determined.

The Limitation of Subjective Statistical Probability

Most of the time, probability is just an educated guess. For example, when predicting whether the school can hold the sports day next week, the physical education teacher might say there is a 90% chance based on their work KPI and school calendar, but the grade-head teacher might say a 50% chance just in order to prevent students from overexcitement. These various probabilities reflect the influence of different perspectives on reality on the concept of probability. A weather forecast’s “60% chance of rain” sounds scientific. But it usually means 6 out of 10 prediction models say it will rain. The meteorological center derives this number through expert voting rather than following natural laws.

Research in deep learning indicates that such probability derives from two sources: data uncertainty, given that weather is random, and model uncertainty, inasmuch as predictions are subject to measurement limitations.

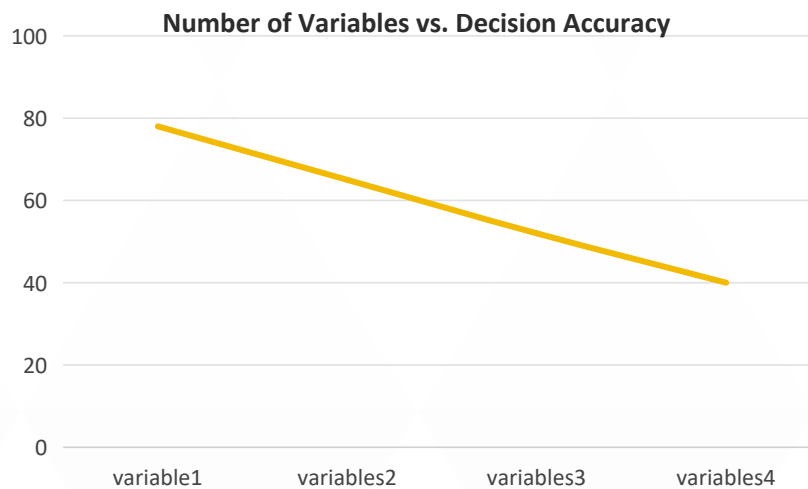
The probability of dice rolls is as stable as physical laws, while subjective probability is like the evaluation of an essay depending on the perspective of the reviewer. This is the first challenge in quantifying uncertainty.

Challenge 2: The Gap between the Probability and the Personal Cognition

In 1738, the experiment devised by mathematician Nicolas Bernoulli shocked the academic community: In the coin-throwing game, the reward is 2 yuan for the first appearance of heads, 4 yuan for the second, 8 yuan for the third, and so on. So, the bonus is and the probability of winning is. Mathematical calculations (the n th positive bonus for the first time $\times$ corresponding probability) $\sum_{n=1}^{\infty} \left(2^n \times \frac{1}{2^n} \right) = 1$ showed that the expected gain is infinite. But in reality, most people are only willing to pay no more than 20 yuan to participate. This is the famous St. Petersburg Paradox.

Daniel Bernoulli proposed a revolutionary theory: what people pursue is not the expected value of money, but psychological

satisfaction. He revealed the law of diminishing marginal utility—when you are extremely thirsty, the first cup of milk tea brings 100 points of happiness, the second cup only adds 30 points, and the third cup leaves only 10 points. This “diminishing return” of satisfaction can be described by the function $U(x) = \ln(x)$, which is called diminishing marginal utility.



Therefore, the gap between precise mathematical calculations and human psychological perception represents the second challenge.

Challenge 3: The Collapse of Multiple Uncertainties

Imagine this: You plan to have a picnic on Friday. The weather forecast for Friday shows a 60% chance of rain and a traffic congestion delay rate of at least 80%. On Saturday, it's sunny, but the temperature reaches 40 degrees Celsius (which can cause food to spoil). Eventually, the chain reaction of these variables leads to a cancellation rate of over 90% for the picnic.

Psychological experiments have shown that when dealing with two uncertain variables simultaneously, the accuracy rate of decision-making can reach 78%; but when there are four variables, the accuracy rate drops sharply to 41%. This cognitive limitation also exists in the field of artificial intelligence—when faced with complex problems, the actual accuracy rate of advanced artificial intelligence is only 40%, but its self-assessed confidence level is as high as 70%.

The third challenge is that the mutual influence between variables and sudden risks makes traditional mathematical models ineffective.

The Three Methods for Solving Uncertainties

When facing uncertainties, there are effective ways to calculate or estimate their probabilities. Here are three helpful steps.

The first step is to distinguish between the two types of probabilities. We need to determine whether the event is “objective” or “subjective and estimated.” For example, in cases like rolling a die, these kinds of events can be repeatedly experimented, and the probabilities are mathematically stable, so the calculated probabilities can be relied upon. However, when predicting the probability of passing the exam, it involves subjective perspective, and the error range tends to be larger.

The second step is to convert possible outcomes into how beneficial it is to oneself. Do not just focus on the cold and impersonal probability figures. Instead, pay attention to the impact on individuals. This way, one's personal emotional experience can also be incorporated into it, which is conducive to making more appropriate decisions.

The third step is to set clear boundaries for uncontrollable risks. For instance, investors will set a limit of a 10% loss and sell out immediately to prevent further losses. When some variables suddenly shift and the situation becomes uncontrollable,



promptly ceasing the bet is often more rational than continuing to gamble.

Conclusion

The certainty is merely a temporary illusion; the uncertainty is the normalcy. And in the fog of probability, humans have gradually learned to seek the light.

As Socrates said, "The only thing I know is that I know nothing." This rational starting point has also been inherited in modern economics. Samuelson, in his article published in the Journal of Economic Literature in 1977, pointed out, "Scientific progress begins when we recognize the limitations of our knowledge."

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10

Mathematics

How to Sum Natural Numbers?

YIFEI GAO

Imagine that there are 100 boxes, the first box has one ball, the second one has two balls, the third one has 3 balls ... which means the n th box has n balls. How can you get the total number of balls quickly?

By writing the series in the opposite order and adding up the original series and the series in the opposite order, like that

$$\begin{array}{r}
 1 \quad 2 \quad 3 \quad 4 \quad 5 \quad \dots \quad 96 \quad 97 \quad 98 \quad 99 \quad 100 \\
 + \\
 100 \quad 99 \quad 98 \quad 97 \quad 96 \quad \dots \quad 5 \quad 4 \quad 3 \quad 2 \quad 1
 \end{array}$$

You will find that the corresponding terms of two series are added up to 101, and there are a hundred 101s in total.

Then we can get equation $(1+100) \times 100$, since half of 101s are repetitive so we need to divide $(1+100) \times 100$ by 2. Therefore, we can get the total number of balls, which is $\frac{(1+100) \times 100}{2}$.

So to get the sum of the first N terms of a series, we can use a formula $\frac{(a+l) \times n}{2}$, where a =first term, l =last term and n =number of terms.

According to sum of the first N terms of a series = $\frac{(a+l) \times n}{2}$, we can get the sum of first N natural numbers easily:

$$\sum_{n=1}^N n = \frac{1}{2}n(1+n)$$

1.1

But this is not all, what to do if we want to get the sum of a series of the form $f(n+1)-f(n)$?

$$\sum_{n=1}^N f(n+1) - f(n)$$

To solve this problem, we can use “difference method” (since there is a “-” here) to get the final answer.

By writing out few terms of $f(n+1)-f(n)$ we can get

$$\begin{array}{l}
 \cancel{f(2)} - f(1) \\
 f(3) - \cancel{f(2)} \\
 \cancel{f(4)} - \cancel{f(3)} \\
 \cancel{f(5)} - \cancel{f(4)} \\
 \dots \\
 f(N+1) - \cancel{f(N)}
 \end{array}$$

By observing these terms, it's not difficult to find that most of these terms are eliminated. (shown above). The remain terms are $f(N+1)-f(1)$. Therefore, the final sum of series of the form $f(N+1)-f(1)$ equals to $f(N+1)-f(1)$.

$$\sum_{n=1}^N f(n+1) - f(n) = f(N+1) - f(1)$$

1.2

After get this formula, we can use it to get the sum of natural numbers squared, cubed and to the fourth power. So how to get these sums? Let's see the sum of natural numbers squared first.

$$\sum_{n=1}^N n^2$$

To get $\sum_{n=1}^N n^2$ we need to use a equation $u_n=(n+1)^3-n^3$, so that n^3 will be canceled when we getting the sum of u_n and

remain $\sum_{n=1}^N n^2$ and other terms which can be expressed by N. Here is the method to find the formula for the sum of natural numbers squared:

By substituting $\sum_{n=1}^N u_n$ in formula 1.2 can get $(N+1)^3-1$.

expanding $(N+1)^3-1$:

$$(N+1)^3-1=N^3+3N^2+3N$$

Now, organizing $\sum_{n=1}^N (n+1)^3 - n^3$:

$$\begin{aligned} \sum_{n=1}^N (n+1)^3 - n^3 &= \sum_{n=1}^N n^3 + 3n^2 + 3n + 1 - n^3 \\ &= \sum_{n=1}^N 3n^2 + 3n + 1 \\ &= \sum_{n=1}^N 3n^2 + \sum_{n=1}^N 3n + \sum_{n=1}^N 1 \\ &= [3\sum_{n=1}^N n^2] + 3 \times \frac{1}{2}N(1+N) + N \\ &= [3\sum_{n=1}^N n^2] + \frac{3}{2}N^2 + \frac{5}{2}N \end{aligned}$$

$$\text{so } 3[\sum_{n=1}^N n^2] + \frac{3}{2}N^2 + \frac{5}{2}N = N^3 + 3N^2 + 3N$$

$$\sum_{n=1}^N n^2 = \frac{1}{3}N^3 + \frac{1}{2}N^2 + \frac{1}{6}N = \frac{1}{6}N(2N+1)(N+1)$$

1.3

Similar to $\sum_{n=1}^N n^2$ we can get the formula for $\sum_{n=1}^N n^3$ and $\sum_{n=1}^N n^4$ with the same method, but at this time, we need use $u_n=(n+1)^4-n^4$ and $u_n=(n+1)^5-n^5$ instead of $u_n=(n+1)^3-n^3$ respectively.

For $\sum_{n=1}^N n^3$:

$$\begin{aligned} \sum_{n=1}^N (n+1)^4 - n^4 &= (N+1)^4 - 1 \\ 4[\sum_{n=1}^N n^3] + 2N^3 + 5N^2 + 4N &= N^4 + 4N^3 + 6N^2 + 2N \end{aligned}$$

$$\sum_{n=1}^N n^3 = \frac{1}{4}(N^4 + 2N^3 + N^2) = \frac{1}{4}N^2(N+1)^2$$

1.4

For $\sum_{n=1}^N n^4$

$$\begin{aligned} \sum_{n=1}^N (n+1)^5 - n^5 &= (N+1)^5 - 1 \\ 5[\sum_{n=1}^N n^4] + \frac{5}{2}N^4 + \frac{25}{3}N^3 + 10N^2 + \frac{31}{6}N &= N^5 + 5N^4 + 10N^3 + 10N^2 + 5N \end{aligned}$$

$$\sum_{n=1}^N n^4 = \frac{1}{5}(N^5 + \frac{5}{2}N^4 + \frac{25}{3}N^3 - \frac{31}{6}N) = \frac{1}{30}N(2N+1)(N+1)(3N^2+3N-1)$$

1.5



By observing the process to get the sum of natural numbers squared, cubed and to the fourth power, we can conclude that can be used to get $\sum_{n=1}^N (n+1)^{a+1} - n^{a+1} = (N+1)^{a+1} - 1$

These formulas for the sums of numbers are really useful in many different areas such as physics, economics and especially in mathematics. For example, it can be used to prove that the area under a curve can be calculated by integration like shown below (if you are interested in it you can take a look). Also, sums of numbers are even essential for us in our daily lives.

One way to calculate the area under a curve is to break it into small pieces, such as rectangles.

By drawing a sketch of $y=x^2$ in the range of $x=0$ to $x=x_1$ and break it into N rectangles. The width of each rectangle will be $\frac{x_1}{N}$. So the height of the n th rectangle will be $n^2 (\frac{x_1}{N})^2$. Since the formula for area of a rectangle is width $\times$ height, so the total area under $y=x^2$ in the range of $x=0$ to $x=x_1$ will be $\sum_{n=1}^N n^2 (\frac{x_1}{N})^3$.

The process to get $\sum_{n=1}^N n^2 (\frac{x_1}{N})^3$:

$$\sum_{n=1}^N n^2 (\frac{x_1}{N})^3 = \frac{x_1^3}{N^3} \sum_{n=1}^N n^2$$

$$\text{since } \sum_{n=1}^N n^2 = \frac{1}{3}N^3 + \frac{1}{2}N^2 + \frac{1}{6}N$$

$$\text{so } \sum_{n=1}^N n^2 (\frac{x_1}{N})^3 = \frac{x_1^3}{N^3} (\frac{1}{3}N^3 + \frac{1}{2}N^2 + \frac{1}{6}N)$$

more rectangles, the area calculated will be more accurate

when $N \rightarrow \infty$

$$\frac{x_1^3}{N^3} (\frac{1}{3}N^3 + \frac{1}{2}N^2 + \frac{1}{6}N) \text{ will become } \frac{1}{3}x^3 \text{ which is equal to } \int x^2 dx.$$

Therefore, the integration of $f(x)$ will equal to the area under the curve of $f(x)$.

GAO YIFEI

Mathematics--Chloe

Taylor's Formula: A Self-Account

YAOZHONG PING

Who discover Taylor series?

Brook Taylor

Who is this?

Brooke Taylor (1685–1731) was an important but little-known English mathematician who discovered Taylor series and finite-difference calculus in his *Methodes Incrementorum, Directa et Inversa*, published in 1715. A graduate of St John's College, Cambridge, Taylor became a Fellow of the Royal Society in its heyday and later served as its secretary, during this time he also formalized the principles of linear perspective in art. Although these ideas had been explored previously by Newton, Bernoulli and others, Taylor's concise and often cryptic presentation meant that his contributions were not fully recognized until decades later (sad but at that time most people think science are the subject only rich people can learn, so the symbol should be as hard as possible so the poor people won't know what it means: a tool of class division), when they were widely acknowledged by mathematicians such as Lagrange. His career was marked by disputes with Johann Bernoulli over priority, and by correspondence with colleagues such as de Moivre. Personally, Taylor suffered huge losses—both of his wives died in childbirth, increasing his mental health issue led to his untimely death at the age of 46. Despite his modest publication record, his legacy lives on because of the important role his work has played in modern analysis.

What is Taylor's theorem?

Taylor's Theorem, provides a way of polynomial approximations which make it easier for computer to understand (e.g. C++ is easier for computer to understand while python is easy for human) like exponentials and trigonometric functions. It is the starting point of the study of analytic functions, and is fundamental in various areas of mathematics, as well as in numerical analysis and mathematical physics. Taylor's theorem also generalizes to multivariate and vector valued functions. It gives the mathematical basis for some landmark early computing machines: Charles Babbage's difference engine calculated sines, cosines, logarithms, and other transcendental functions by numerically integrating the first 7 terms of their Taylor series.

Taylor's theorem provides a way to approximate a k times differentiable function $F(x)$ near a point a using a polynomial of degree k . This polynomial—called the k -th order Taylor polynomial—matched the function's value and its first K derivatives

$$P_k(x) = \sum_{i=0}^k \frac{f^{(i)}(a)}{i!} (x-a)^i$$
$$f(x) = P_k(x) + R_k(x)$$

at a

As $x \rightarrow a$, $R_k \rightarrow 0$ which is the first term.

$$f(x) = f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \dots$$

Taylor series in physics



1: It could be apply in function approximation

e.g. use to apply some complex potentials like harmonic oscillators or some pendulums

2: differential equations & integrals

e.g. represent solutions as power series.

3: classical mechanic & quantum mechanics

e.g. pendulum: for small angle like $\sin\theta \approx \theta$

4: projectile motion

Taylor series can approximate the height for time intervals

5: thermodynamics: used to analyse Helmholtz and Gibbs free energy near equilibrium.

6: gravitational potential

In finance (Black-Scholes model)

Earlier Foundations (Before Taylor)

Taylor Series was prefigured by Scottish mathematician James Gregory, he published an essay on a special case of the expansion for arctangent, arcsine, and arccosine functions in the 1670s, 45year earlier than Taylor, namely

$$\tan^{-1} x = x - \frac{x^3}{3} + \frac{x^5}{5} - \dots$$

Isaac Newton used series expansions to derive solutions in calculus and physics which promote of Taylor's formula (at that time he used a general binomial theorem and a power series).

In the 1700s, mathematicians like Laplace and Lagrange used Taylor expansion to predict the movement of planets, tides and eclipses.

They weren't just mathematical tricks but let us simulate solar system long before computers existed!

We can predict an incoming asteroid that will cause the Earth destruction and apocalypse thanks to the equation they invented.

Promotion of Taylor series:

Lagrange expanded on Taylor's ideas and formalized the remainder term in Taylor's theorem. His update version helped to clarify the theory's convergence and analytical importance.

Where does it apply in the past:

In 18th century it applies on theoretical foundation & early physics

Mathematics: use to approximate functions like sine cosine exponential logarithms... etc.

Physics: newton and other used series expansions (precursors to Taylor series) in some motion equation

Astronomy: helped calculate orbits and corrections to Keplerian motion like perturbation theory

Thermodynamics: use in deriving series approximations for state equation and partition function.

Electromagnetism: expansion of electromagnetic potentials and wave equation.

In 20th century usually applied to technology computation and quantum mechanics

Calculus: Taylor series is essential for numerical analysis (sth. like finite difference approx.)

Quantum mechanics: approx. solutions for the Schrodinger equation, find the probability cloud.

In 21st century it applies on AI & simulation a lot, e.g. in machine learning: optimization algorithm like gradient descent will use Taylor expansions to guide convergence

Computer graphics & simulations: use for smooth curve generation animation interpolation also physics engines.

Engineers use Taylor series to analyze vibrations, stress and curvatures – from electric cables to bridges to buildings, such as a roller coaster's curve that you play in Universal Studios and Disneyland. So we need to thank Taylor for inventing the series for us to have fun riding in the theme parks.

The rails are modelled using polynomials. The electric cables are calculated with $\sinh(x)$ function which can be simulated with a simpler Taylor expansion, preventing over dangling during hot summer and excessive contraction during cold winter.

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Aaron Yaozhong Ping

Wed 23 Jun 23:07



11



Microbiology

Phagetherapy and Effects of Bacteriophages on *Vibrio Cholerae*

XUANLING BAI

Pathogenic strains of Gram-negative bacterium *Vibrio cholerae* cause cholera, a diarrheal disease resulting in extreme dehydration, severe fluid loss (Sack et al., 2004), and mortality rates reaching 25–50% if untreated (Fournier et al., 2007). *V. cholerae* bacteria are categorised into over 200 serogroups based on the lipopolysaccharide (LPS) component of the cell wall, among which only O 1 and O 139 have been found to cause pandemic cholera (Ramamurthy et al., 2022). Though antimicrobial resistance increasingly renders many antibiotics ineffective as cures for cholera, bacteriophages have been found to have significant impacts on *V. cholerae*, and hence phage therapy can potentially be used as an alternative to antibiotics.

Bacteriophages are viruses that infect and replicate within bacteria (Zhang et al., 2022). After attaching to a bacterial host cell, a phage can either enter the lytic or lysogenic replication cycle (Figure 1).

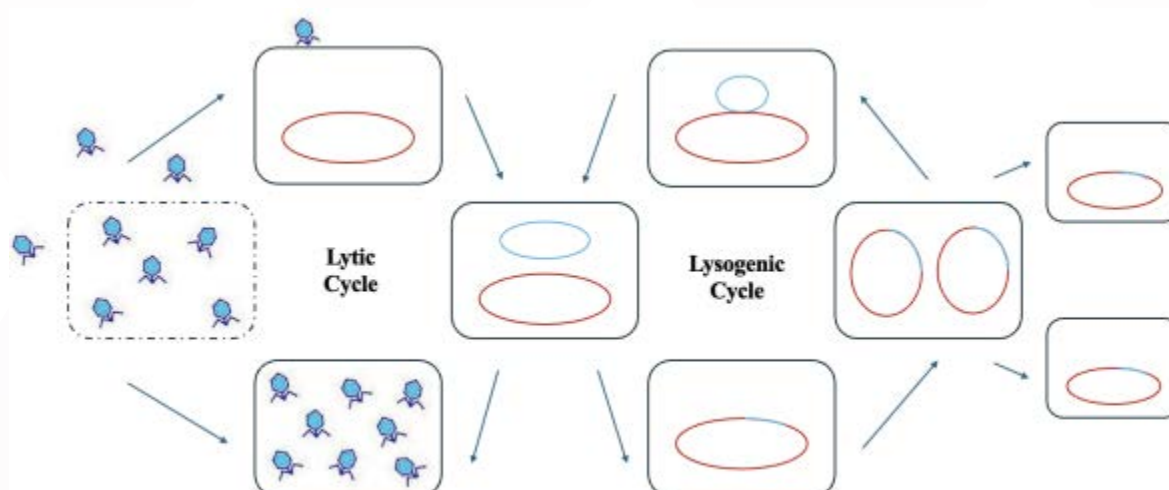


Figure 1: A lytic cycle occurs when the phage transforms the resources of the host cell into viral genomes and capsid proteins, which collect to form replicates of the original phage. The host cell is subsequently lysed and killed, releasing the new phage to infect another host cell. During a lysogenic cycle, the phage genome is incorporated into the cell chromosome and transferred to daughter cells, which does not result in cell death, except when it converts into a lytic cycle in response to environmental stress and hence kills the host cell (Kasman et al., 2022).

Due to the threat of phages on the survival of *V. cholerae* bacteria, natural selection and bacterial evolution have occurred, allowing bacteria with beneficial mutational changes and adaptations against phage predation to survive and reproduce.

Mutations in the receptor genes of *V. cholerae* El Tor strains made these bacteria resistant to temperate phages used in typing, like VP3, VP4, and VP5, and some strains have been found to be resistant to the subtyping phage 919TP (Shen et al., 2016). Additionally, mutations in LPS on the outer cell membrane prevent the absorption of 919TP phage particles into the bacterial cell, and hence protect the cell against infection (Li et al., 2021). The primary function of LPS is to provide structural integrity and a permeability barrier to protect the bacterial cell from the entry of deleterious molecules such as toxins and bile salts during its inhabitation in the gastrointestinal tract (Farhana et al., 2023). Furthermore, Bacteriophage ICP1 has been found to kill *V. cholerae* in many ecosystems, especially aquatic environments with limited nutrition (Silva-Valenzuela and Camilli, 2019). In order to avoid infection by ICP1, phase variation mutations have occurred in the O1–

antigen of *V. cholerae*, including point mutations and substitutions (Seed et al., 2012).

The host range of ICP1 is limited to *V. cholerae* O1, while the host ranges for ICP2 and ICP3 also include non-O1 *V. cholerae* strains. Specifically, ICP2 is capable of forming plaques on *V. cholerae* O 139 strain, MO10, and ICP3 is capable of forming plaques on a non-O1, non-O139 strain, CR034-24 (Seed et al., 2011). The host specificity of phages varies depending on the specific phage; some phages are highly specific, targeting only certain strains of *V. cholerae*, while others have a broader host range, infecting multiple strains. This specificity is influenced by factors like the phage's genetic makeup and the presence of specific receptors on the bacterial surface.

Another defense technique present in *V. cholerae* is present in the Type VI secretion system (T6SS) which uses virulence mechanisms and has a structure similar to that of the T4 bacteriophage's cell puncturing device (Miyata et al., 2011), to deliver toxins into neighboring target cells. It can be hypothesised that the bacterium's T6SS adapted based on phage characteristics, developing this evolutionarily advantageous structure in order to effectively reduce competition by killing nearby cells.

Though bacteria have evolved various resistive mechanisms against phages, phages have responded with equally effective developments, which allows us to use them in phage therapy as an alternative against bacteria, as increasing antimicrobial resistance renders many antibiotics useless.

Phage cocktails, specifically those targeting *V. cholerae*, have shown promise in preventing and treating cholera infections in animal models. These cocktails, composed of multiple bacteriophages, are designed to target different receptors on the bacterial surface, reducing the likelihood of bacteria developing resistance. Unlike antibiotics, phages are replicating viruses capable of auto-dosing, meaning phage replication increases their number and contributes to the dose. An investigated phage cocktail consisting of phages ICP1, ICP2, and ICP3 involves targeting different receptors to reduce the likelihood of multi-phage-resistant *V. cholerae* isolates in the surviving bacterial population. These phages target the LPS O 1 antigen and major outer membrane porin OmpU which is crucial in regulating nutrient entry and exit of waste products. Results of in vitro experiments supported the statement that the ICP cocktail was more effective in killing *V. cholerae* than each phage in isolation. Also, 39% of bacterium isolates were resistant to both ICP1 and ICP3, but none were resistant to all three phages; this further proves the effectiveness of phage cocktails in phage therapy, as it largely increases the chances of effective treatment by lysing a maximum percentage of the bacteria population (Yen et al., 2017).

In conclusion, phage therapy is an innovative approach in treating cholera, caused by bacterium *Vibrio cholerae*. The unique ability of bacteriophages to target and lyse specific bacterial pathogens offers a viable alternative to traditional antibiotics, especially in an era of increasing antibiotic resistance. This strategy not only demonstrates effectiveness in reducing cholera infections, but its research also opens new avenues for understanding host-pathogen interactions and the ecological roles of bacteriophages in microbial communities. Ultimately, the application of bacteriophages against *V. cholerae* paves the way for more effective management of cholera outbreaks, improving global health outcomes and enhancing our response to bacterial diseases as a whole.

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What Evolutionary Pressures – Biological and Anthropogenic – Have Shaped the Way Influenza A Virus Adapts and Persists?

LEE SIK KI

Within the last century, Alphainfluenzavirus Influenzae (Flu A) has claimed over 50 million lives (Kain and Fowler, 2019). In contrast, COVID has cumulatively only killed 7 million (World Health Organization, 2025). Considering influenza's capacity for devastating harm, it is imperative that our global society enhances our understanding of the mechanisms and causes behind influenza's evolution. By doing so, our global healthcare system will be better positioned to mitigate the biological impacts of future pandemics.

Firstly, this paper will explore the biological mechanisms underlying influenza's rapid evolution, highlighting its unique viral structure, high mutation rate, and genetic reassortment. The binomial nomenclature of Flu A is Alphainfluenzavirus Influenzae, a virus belonging to the realm Riboviria (Taubenberger and Kash, 2010). This virus is composed of eight single strands of RNA encased in a lipid envelope, as illustrated in figure 1 by Taubenberger and Kash. Two types of protein spike antigens cover the lipid membrane: haemagglutinin (HA), which facilitates viral attachment to host cells, and neuraminidase (NA), which assists in viral detachment (Long et al., 2018). This process of viral attachment is crucial for initiating the lytic cycle, the mechanism responsible for initiating cell death. Following viral attachment, influenza releases its RNA into the host cell. Host cells synthesize new viral particles based on these RNA instructions, then host cells “explode” to release these new viruses (Long et al., 2018).

However, one variant of HA and NA cannot attach to all cells. Different variants of these antigens are needed to attach to different species (Haran et al., 2014). Therefore, the naming of influenza subtypes are determined by their displayed HA and NA attachments. For example, H1N1 indicates that a cell displays HA1 and NA1 (Shi et al., 2014). A change in HA and NA is formally known as antigenic drift (Shi et al., 2014). This process is driven by two biological mechanisms. Firstly, influenza's genome is very short, consisting of only around 13500 base pairs (Gago et al., 2009). This allows influenza to reproduce very quickly, leading to a high mutation rate and capability of influenza to adapt to new environments (Gago et al., 2009). In addition, influenza exhibits genetic reassortment. As influenza's genome is segmented, meaning it exists in individual, separate parts, two viruses infecting the same cell can randomly mix up their segmented genes during replication, producing a reassorted genome on export (Dugan et al., 2008). The biological mechanisms described have facilitated influenza's rapid evolution and adaptation, allowing influenza to thrive even in rapidly changing external environments.

Next, this paper will explore influenza's transmission and history. Influenza transmits by droplets, through processes including talking, coughing and sneezing (Bean et al., 1982). Influenza can survive for 24–48 hours on hard surfaces, and even longer in cold temperatures due to lower membrane fluidity protecting genetic material (Bisht and Velthuis, 2022). Four major influenza pandemics have occurred in the past: details of these can be found in figure 2 (Chauhan and Gordon, 2020). The first influenza was the deadliest, killing 50 million (Chauhan and Gordon, 2020). A key factor that massively accelerated disease spread was World War I. The cramped conditions of trench warfare created an environment conducive to disease transmission, while global military mobilizations to and from the Western Front accelerated influenza's spread worldwide (Pennington, 2019). Additionally, the absence of influenza vaccines and a limited understanding of transmission methods contributed to widespread devastation (Hagan, 2021).

Next, in the 1950s and 1960s, death rates of influenza pandemics decreased to around 1 million (Chauhan and Gordon, 2020). This drastic reduction was thanks to the invention and distribution of the first influenza vaccine in the 1940s, by Thomas Francis Jr. and Jonas Salk (Barberis et al. 2016). Community vaccinations not only protected the individual but also limited the spread of disease throughout global communities, leading to herd immunity (Kim, 2014). As such, human-to-human transmission decreased massively.

However, animal-to-animal transmission remained prevalent. As the global population skyrocketed (Ritchie et al., 2023), so was the necessity for intense food production. Mass livestock farming of swine (pigs) and avians (chickens) provided ample opportunity for genetic reassortment, as many organisms were packed together in a tiny space (Gao et al. 2024). Eventually, one clone mutated to be zoonotic (see fig. 3), allowing it to transmit from animal to human (Shi et al., 2014). This led to the 2009 Swine flu pandemic, resulting in 300,000 deaths (Smith et al., 2009). Such recent pandemics demonstrate how we still need to be aware of different viruses and their genetic developments, as pathogens still have massive potential to mutate and cause harm.

Currently, studies are ongoing regarding influenza's development. For example, George et al. estimates an average of 40 annual mutations in influenza's genome. In addition, Ganti et al. has found strong evidence of continued antigenic drift within ferrets, pigs, and guinea pigs. Lastly, Rahman et al. has demonstrated that current vaccines are becoming increasingly ineffective against novel influenza variants. This is a major issue because modern vaccines act as selection pressures, leading to the emergence of new seasonal flu variants every winter (Lofgren et al., 2006). Therefore, influenza still cannot be ignored.

In conclusion, influenza A and its subtypes continue to be pathogens of major public health concern. Developments in medicine and genomics have severely limited the spread of influenza throughout human communities, however, massive risk still exists. On an institutional scale, continuous genomic surveillance of pathogen evolution is crucial to mitigate the impact of future pandemics. On an individual scale, practicing proper medical etiquette including staying home when sick, washing hands, and wearing masks, continues to be effective in safeguarding the health of our community and ecosystems.

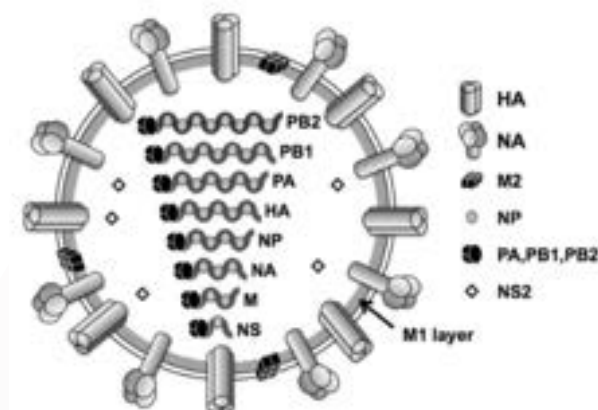


Figure 1: Structure of Alphainfluenzavirus Influenzae (Taubenberger and Kash, 2010). A lipid membrane surrounds 8 single-stranded RNA segments, with HA and NA glycoprotein spikes embedded in the lipid membrane.

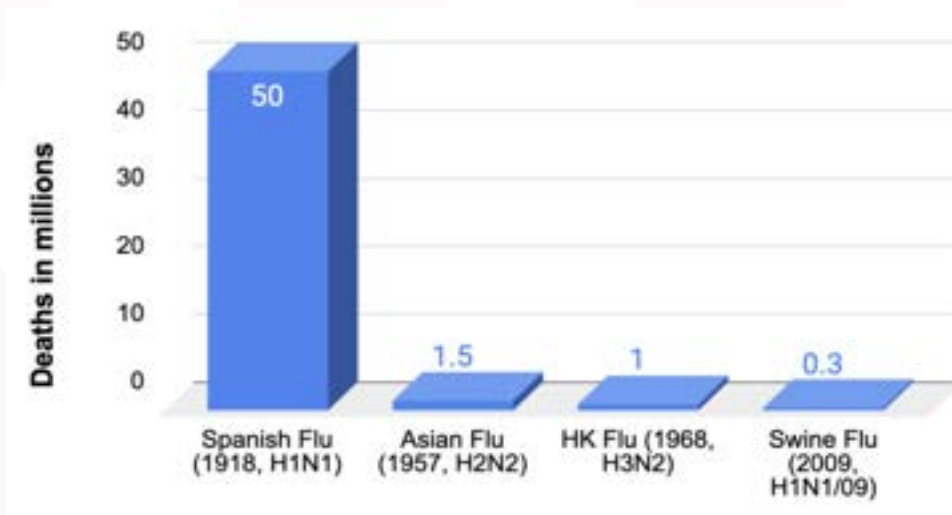


Figure 2: Historical human influenza pandemics, (Chauhan and Gordon, 2020). Figure created by the author.

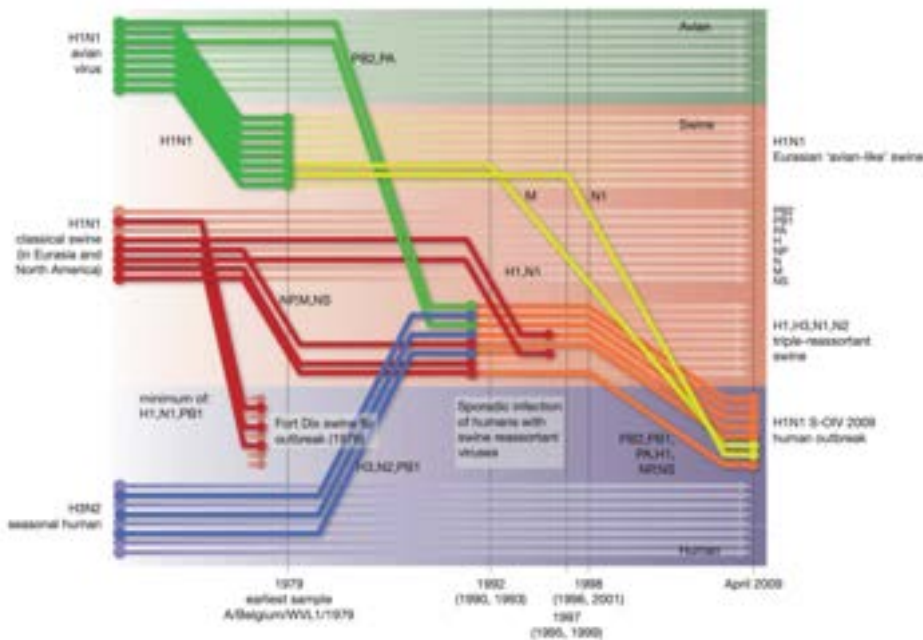


Figure 3: Diagram showing the formation of the 2009 Swine Flu, from the reassortment of swine, avian, and human genes (Smith et al., 2009)

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12

Physics

Is Minkowski Diagram Useful or Solely Confusing?

YUNING FAN

Abstract

Minkowski diagram can be used to derive Lorentz transformation equations. This method seems to consist less abstract calculation but more visual representation. Therefore the Minkowski diagram might be useful for beginners in special relativity to improve their understanding.

Introduction

Minkowski diagram, also called space-time diagram is used to represent the motion of a certain object in terms of its space and time. Some argues this diagram is useless and confusing. This essay will look to explore the applications and advantages of it to show that it is useful.

Construction of a Minkowski Diagram

Have a look at figure 1. Assume the situation under a simple circumstance, where there are two reference frames S and S' and S' is moving away from S with constant relative speed v. Minkowski Diagram contains two axis for each reference frame, a space axis x and time axis ct. The reason to use ct instead of t for the time axis is to keep the units on both axis the same, and c can be simply assumed to hold value 1. Then the axis of reference frame S can be drawn where x and ct are perpendicular to each other, as this is used to represent events with x=0 and ct=0. To draw the S' frame which is for events with x'=0 and ct'=0, it will have its time axis ct' and space axis x' squeezed where the angle θ between these axis and the axis of reference frame S will be equal. To determine the exact value of θ , mark out two points t'A and t'B on ct' axis to represent the duration of an event. To map the event from ct' axis onto ct axis, it is obvious that

$$c\Delta t = \cos\theta c\Delta t'$$

$$\Delta x = \sin\theta \Delta x'$$

$$\tan\theta = \frac{\Delta x}{c\Delta t} = \frac{v}{c}$$

Lastly, the worldline of a light ray in Minkowski diagrams will always be 45° to axis of reference frame S because it has speed c and x=ct. Since the speed of light stays constant regardless of which reference frame it is in, x'=ct'. This

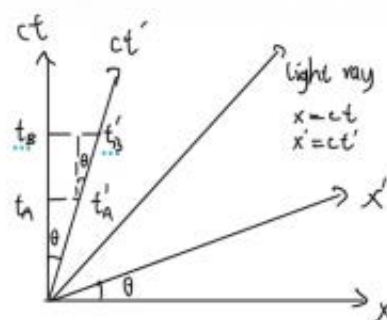


Figure1

means the worldline of light rays will not be affected by Lorentz transformation.

Applications of Minkowski diagrams

After watching a lecture given by Professor Das on YouTube[Das, D. (2021) Time Dilation, Length Contraction & Velocity Addition using SPACETIME DIAGRAM, YouTube. Available at: <https://www.youtube.com/watch?v=1de6GiqcCUC&t=1222s> (Accessed: 24 July 2025).], it seems that the Minkowski diagram is quite useful to help beginners understand Lorentz transformation in special relativity.

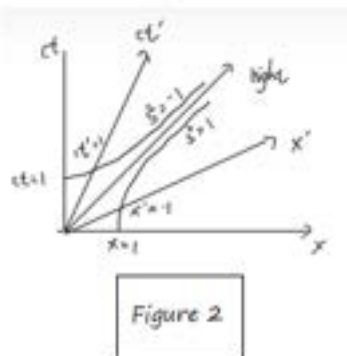
Time Dilation

According to Lorentz transformation, the change in time $\Delta t'$ in reference frame S' moving with constant relative speed v from reference frame S can be calculated using the following equation

$$\gamma^{-1}(t_A - t_B) = (t'_A - t'_B) + \frac{v}{c^2}(x'_A - x'_B)$$

For events that happen at a stationary point in reference frame S' , the value of $x'_A - x'_B = 0$. Therefore the time $\Delta t'$ given by $t'_A - t'_B$ can be written as

$$\Delta t' = \gamma^{-1} \Delta t$$



Where $\gamma^{-1} = \sqrt{1 - \frac{v^2}{c^2}}$ which is always smaller than 1 as v can never exceed speed of light c , $\Delta t'$ will be shorter than

Δt . This means the time taken for the event to happen in reference frame S' will be shorter than observers measured in reference frame S . To use Minkowski diagram to derive the same equation, Look back at figure 1, two points t'_A and t'_B on ct' axis are marked out on ct' axis to represent a time duration that stays stationary in position in reference frame S' .

However, what is special about the Minkowski diagram is that to map coordinates from S' frame onto S frame $\cosh$ and $\sinh$ should be used instead of $\cos$ and $\sin$ which are functions used in a Euclidean space. This is because the magnitude of spacetime in Minkowski diagram works differently to Euclidean space. In special relativity, the magnitude S of a point from the origin on a Minkowski diagram is given by $S^2 = x^2 - (ct)^2 = x'^2 - (ct')^2$ instead of the Pythagoras theorem. Plot the lines on graph as shown in figure 2, and it is obvious that the unit length on axis of S frame and S' frame are different unlike the Euclidean space, thus needing a different method to transfer coordinates from one frame to another. The transformation is given by:

$$c\Delta t = \cosh\theta c\Delta t'$$

$$\Delta x = \sinh\theta \Delta x'$$

$$\tanh\theta = \frac{\Delta x}{c\Delta t} = \frac{v}{c}$$

Then get the value of $\cosh\theta$ to find the relation between $c\Delta t$ and $c\Delta t'$:

$$1 - \tanh^2\theta = 1 - \frac{v^2}{c^2}$$

$$\operatorname{sech}^2\theta = 1 - \frac{v^2}{c^2}$$

$$\operatorname{sech}\theta = \sqrt{1 - \frac{v^2}{c^2}}$$

$$\cosh\theta = \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}} = \gamma$$

Then cancel c on both sides of the equation so we get

$$\Delta t = \gamma\Delta t'$$

Which is the same compared to what is obtained without diagrams.

Length Contraction

Length contraction is very similar to time dilation, but instead of marking 2 points on ct' axis to represent time, 2 points can be marked on x' axis to represent a certain length as shown in figure 3.

Similar to time, the length $\Delta x'$ in reference frame S can be written as

$$\Delta x = \cosh\theta\Delta x'$$

Where we have already obtained

$$\cosh\theta = \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}} = \gamma$$

$$\Delta x = \gamma\Delta x'$$

This is the same to the equation we can get from the equation

$$\gamma^{-1}(x_A - x_B) = (x_A - x_B) + \frac{v}{c^2}(t'_A - t'_B)$$

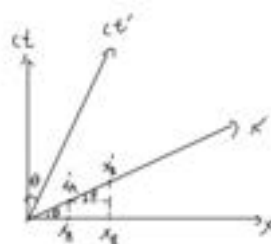


Figure 3

Where $t'_A = t'_B$.

Conclusion and Evaluation

Minkowski diagram provides a different pathway to obtain the Lorentz transformation. It is also more straight forward and easier to calculate compared to the fully algebraic method where 4 equations

$$t' = a_0 + a_{00}t + a_{01}x + a_{02}y + a_{03}z$$

$$x' = a_1 + a_{10}t + a_{11}x + a_{12}y + a_{13}z$$

$$y = a_2 + a_{20}t + a_{21}x + a_{22}y + a_{23}z$$

$$z' = a_3 + a_{30}t + a_{31}x + a_{32}y + a_{33}z$$

are set up to get solved, therefore might be useful for beginners in special relativity to understand the Lorentz transformation.

On the other hand, only Minkowski diagram of motion in 1-dimension has been looked at in this essay. There are circumstances where the motion is in 2 or more dimensions. In this case the Minkowski diagram might not be as helpful.

The reason to use cosh and sinh when changing reference frame is also not explored in this essay. This means with the details of the process of deriving these functions from the diagram the whole process might become complicated again just like the algebraic method. However, due to the limited time I have not looked at these two problems.

How Does the Special Relativity Impact Our Vision

XIAOLE MA

1. Abstract

The impacts of special relativity on how we see the world are always wonderful. Thus, in this article, we will be analyzing how some basic classical mechanics laws such as the reflection laws be adjusted under special relativity conditions. This article will be discussing one special example of optical impacts of special relativity first posted in a science fiction novel 2001: A Space Odyssey and the adjustment of the reflection law under special relativity conditions mathematically.

2. Introduction

2.1 special relativity (speed transformation)

According to the Lorentz transformation:

$$\begin{cases} ct' = \gamma(ct - \beta x) & (Eq1.1) \\ x' = \gamma(x - \beta ct) & (Eq1.2) \\ y' = y & (Eq1.3) \end{cases}$$

There should be a transformation for z' , but that isn't used in the derivation below. We are now transforming the time and position into a frame which is moving at a uniform velocity u along the x -axis, and an object is moving at velocity V_x along the x -axis and V_y along the y -axis. So we can further express x as $V_x t$ and β as $\frac{u}{c}$. Then we can differentiate the three equations on both sides with the respect of t .

$$\begin{cases} dt' = \gamma(1 - \frac{uV_x}{c^2})dt & (Eq2.1) \\ dx' = \gamma(V_x - u)dt & (Eq2.2) \\ dy' = V_y dt & (Eq2.3) \end{cases}$$

Then we can take them into the following equations $v'_x = \frac{dx'}{dt'}$ and $v'_y = \frac{dy'}{dt'}$, and it leads us to the final expression of the velocities as follows.

$$\begin{cases} v'_x = \frac{V_x - u}{1 - \frac{uV_x}{c^2}} & (Eq3.1) \\ v'_y = \frac{V_y \sqrt{1 - \beta^2}}{1 - \frac{uV_x}{c^2}} & (Eq3.2) \end{cases}$$

(Key $\beta = \frac{u}{c}$)

2.2 Transformation of the direction where a beam of light comes from

Assume a beam of light with travelling with a horizontal velocity of $V_x = c \cos \theta$ and vertical velocity of $V_y = c \sin \theta$, (the speed of light has to be c no matter in any frame, so $\sqrt{V_x^2 + V_y^2} = c$) if we were to transform the velocities into a frame with horizontal velocity of u , we can simply use (Eq3.1) and (Eq3.2), we can deduce the following.

$$\begin{cases} v'_x = c \frac{\cos \theta - \beta}{1 - \beta \cos \theta} & (Eq4.1) \\ v'_y = c \frac{\sin \theta \sqrt{1 - \beta^2}}{1 - \beta \cos \theta} & (Eq4.2) \end{cases}$$

With the relationship between $\tan\theta' = \frac{v'_y}{v'_x}$, we only have to take in (Eq4.1) and (Eq4.2), it's simple to deduce the following.

$$\theta = \tan^{-1} \left(\frac{\sin \theta' \sqrt{1 - \beta^2}}{\cos \theta' - \beta} \right) \quad (\text{Eq5})$$

2.3 An example which we are going to analyse

In the 2001: A Space Odyssey, in the latter part of the novel, astronaut David Bowman pilots the spacecraft Discovery One, after experiencing extreme time distortion, he observes the Milky Way in a shape of a bowl. And when their speed becomes faster and faster, in their view the galaxy tends to be wrap up the spaceship up (like shown in the diagram 1.1).



Diagram 1.1

In the following text, we will discuss the boundary functions of the galaxy seen by the astronauts in the spacecraft frame.

2.4 The adjustment of the law of reflection

This example will be about the reflection of lights. In this case we assume that a beam of light is shot on a mirror with a velocity v along the x axis, and the beam of light is thought to be shot with incident angle θ by the people at rest. And the mirror is placed vertically with the x axis. Here we assume that the law of reflection in classical mechanics only work in the proper frame of the mirror.

3. The derivation of the function which describes the boundary

3.1 Some basic settings and the way of thinking

We assume that the spaceship leaves the galaxy with a large velocity of v , and v is parallel to the normal of the surface of the galaxy. The spaceship starts moving away from the galaxy from its centre. At one moment, there is an angle α between the line connecting the spaceship and the outermost star of the galaxy and the line between the spaceship and the centre of the galaxy. As shown in the diagram 1.2.

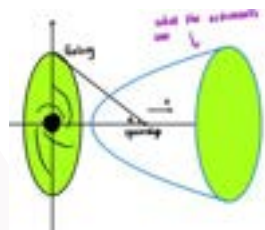


Diagram 1.2

The way of thinking it is to firstly change the objects into the frame of the spaceship, in which the galaxy is moving along the x axis with relative velocity v , and the spaceship is stationary, and we assume the spaceship is at the origin.

3.2 The reason why astronauts sees the galaxy in a strange way

The beams of light from each part of the galaxy is required to reach the observer at the same time to make a full image of the galaxy. And because the distances between each points on the galaxy and the observer are different, it obviously takes different times for the beams of light to reach the observer, what the astronauts see is shown in Diagram 1.1.

3.3 The deduction of the boundary function

First of all, we can set up a 3D coordinate system in the spaceship frame, with the observer at the origin, and the centre of the galaxy at $(x_0, 0)$, where $x_0 = R \cot \theta$, and moving along the x axis with velocity v . Here we can choose one specific diameter (perhaps the one which is parallel to the y axis) of the galaxy and analyse how it appears in the astronauts eyes, after that we only need to spin the curve about the x axis, then we can find out the final expression of the curved surface.

Because the part of the galaxy with coordinate (x_0, y) has a longer distance away from the observer, so it takes more time for light to travel, So we assume that this part of the galaxy releases light Δt ahead of the centre of the galaxy at coordinate $(x_0 - v\Delta t, y)$, Because of the theory mentioned in 3.2, we can simply derive the following equation. (x_0 is a constant)

$$\frac{\sqrt{(x_0 - v\Delta t)^2 + y^2}}{c} - \Delta t = \text{constant} = \frac{x_0}{c} \quad (\text{Eq6})$$

Then we square the two sides up, and leaves us with an expression as follows.

$$(\Delta t)^2 + \frac{2x_0}{c-v}\Delta t - \frac{y^2}{c^2 - v^2} = 0 \quad (\text{Eq7})$$

After solving this quadratic equation, we can find an expression for Δt . (The one with a negative sign is rejected because then it will leave us with a negative Δt).

$$\Delta t = -\frac{x_0}{c-v} \pm \sqrt{\left(\frac{x_0}{c-v}\right)^2 + \frac{y^2}{c^2 - v^2}} \quad (\text{Eq8})$$

$$x = x_0 - v\Delta t = \frac{c+v}{c-v}x_0 - v\sqrt{\left(\frac{x_0}{c-v}\right)^2 + \frac{y^2}{c^2 - v^2}} \quad (\text{Eq9})$$

$$\left(x - \frac{1+\beta}{1-\beta}x_0\right)^2 - \frac{\beta^2}{1+\beta^2}y^2 = \left(\frac{\beta}{1-\beta}x_0\right)^2 \quad (\text{Eq10})$$

Here we have to be careful, because the beam of light that is released from the centre takes some time to travel to the observers, and the galaxy has also traveled for a short distance in that time. So x_0 is divided into two parts, and x'_0 represents the distance that light has traveled, and the value of x'_0 is as follows.

$$x'_0 = \frac{c}{c+v} R \cot \theta \quad (\text{Eq11})$$

Because light requires time to move from the centre of the galaxy to the observers, and we also need to use the Lorentz transformation. When the beam of light reaches the observers, the galaxy has also moved for the same period of time. (Eq9) leaves us with a 2D curve, shown in the diagram1.3, is what the astronauts see the diameter of the galaxy, if we were to find the curve surface, we only need to spin the curve about the x axis, we only need to exchange $y^2 = y'^2 + z^2$ shown in diagram1.4.

$$\begin{cases} \left(x - \frac{1}{1+\beta}x'_0\right)^2 - \frac{\beta^2}{1+\beta^2}(y'^2 + z'^2) = \left(\frac{\beta}{1+\beta}x'_0\right)^2 \\ x'_0 = \frac{1}{1+\beta}R \cot \theta \end{cases} \quad (Eq12)$$

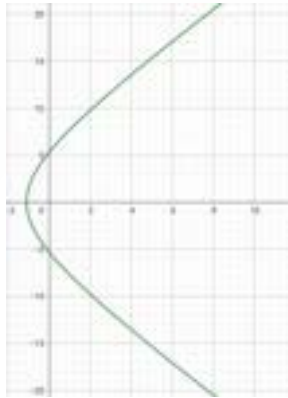


Diagram1.3

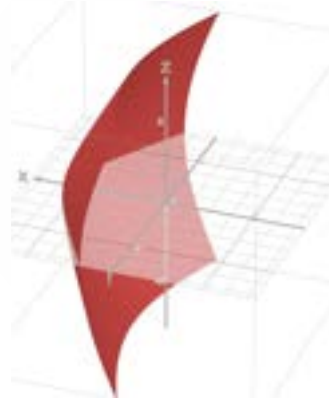


Diagram1.4

4. The derivation of the reflection angle

4.1 Some basic settings and the way of thinking

We set that the mirror is moving along the normal of its surface, and the beam of light is moving in a direction that has incident angle θ . We need to find out the reflection angle in terms of θ . It is obvious that we have to analyse the problem in the proper frame of the mirror, then, we can switch into the observer frame (frame that is not moving).

4.2 Switch into the mirror-proper frame

First of all, we express the velocity of the light in the into two vertical direction $V_x = -c \cos \theta$ and $V_y = c \sin \theta$. Then we can use (Eq4.1) and (Eq4.2), which leaves us with the following.

$$\begin{cases} v'_x = -c \frac{\cos \theta - \beta}{1 - \beta \cos \theta} & (Eq13.1) \\ v'_y = c \frac{\sin \theta \sqrt{1 - \beta^2}}{1 - \beta \cos \theta} & (Eq13.2) \end{cases}$$

After the reflection, because it is in the proper frame of the mirror, it obeys the reflection laws. So $v''_x = -v'_x$. Thus the reflection angle in the mirror frame is $\theta' = \cos^{-1} \left(\frac{\cos \theta - \beta}{1 - \beta \cos \theta} \right) = \sin^{-1} \left(\frac{\sin \theta \sqrt{1 - \beta^2}}{1 - \beta \cos \theta} \right)$

4.3 Switch to the observer frame (the frame that is not moving)

We can directly use (Eq5) to find out the reflection angle seen by the observers.

$$\theta'' = \tan^{-1} \left(\frac{\sin \theta (1 - \beta^2)}{\cos \theta - \beta^2 \cos \theta} \right)$$

Or, we can further use the transformation of the velocity along the x axis

$$v''_x = c \frac{\frac{\cos \theta - \beta}{1 - \beta \cos \theta} + \beta}{1 + \beta \frac{\cos \theta - \beta}{1 - \beta \cos \theta}} = c \frac{\cos \theta - \beta^2 \cos \theta}{1 - 2\beta \cos \theta - \beta^2}$$

$$\theta'' = \sin^{-1} \frac{v''_x}{c} = \sin^{-1} \frac{\cos \theta - \beta^2 \cos \theta}{1 - 2\beta \cos \theta - \beta^2}$$

The relationship between the reflection angle θ'' (y-axis) and θ (x-axis) in diagram1.5.

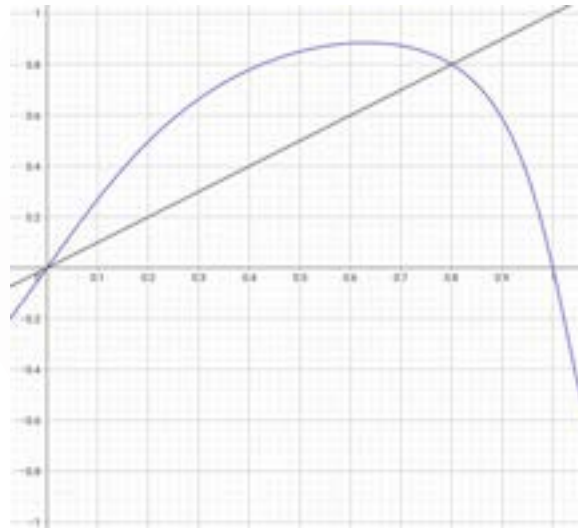


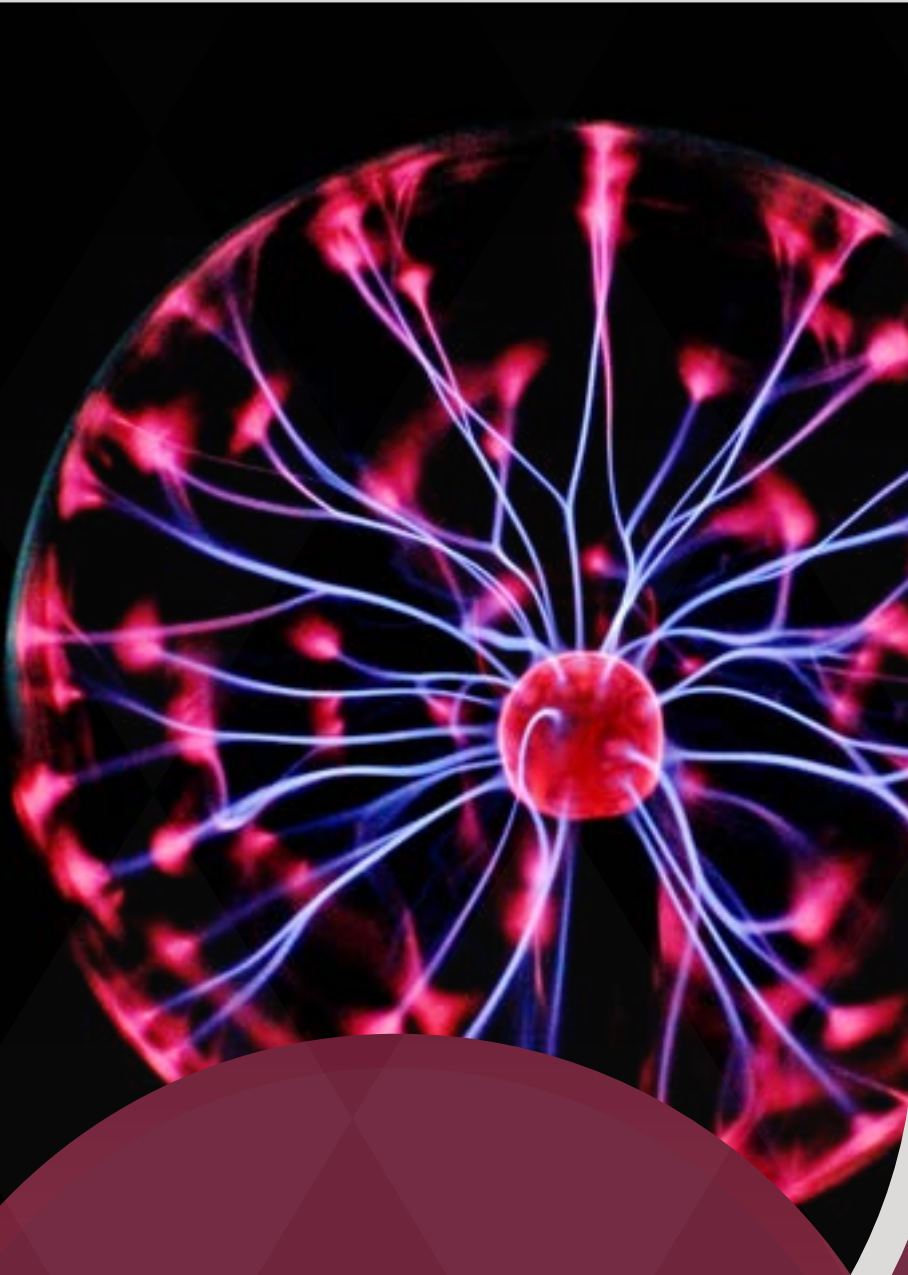
Diagram1.5

5. Conclusion

To conclude, special relativity actually made a great difference to how we see the world. In the two examples above, one explains that some classical mechanics should be adjusted when the speed of the object is moving at a relatively large speed, and there are also amazing affects on what shape will the objects look in our perspective when moving at a very high velocity, just like what the astronaut see the galaxy in the spaceship.



13



Psychology and Neuroscience

How Stress Affects Cognitive Functions Underlying Neural Mechanisms

KEJIA GUO

Introduction

Imagine facing a looming deadline or a sudden crisis: the heart pumps, the adrenaline rushes, and the mind seems to stumble through scenarios hard to control. While in some cases stress enhances concentration and speeds up thinking; in others, it impairs memory, reduces attention and disrupts thoughtful decision-making.

Stress is often defined as a psychological and physical response to perceived threats or challenges in the environment, which is highly prevalent and has been closely linked to reductions in cognitive functions. Specifically, acute or chronic stress tends to disrupt functions through complex neural mechanisms involving the interplay of the hippocampus, amygdala, and prefrontal cortex. These brain regions, along with the hypothalamic-pituitary-adrenal (HPA) axis, the body's primary stress response system, which releases hormones such as cortisol that impacts brain functions, are crucial for regulating response to stress, as well as the subsequent effects on cognition. This essay argues that stress systematically changes cognitive performances by altering the neural systems that support them.

To explore this, the essay will examine four key areas: memory, attention, decision-making, and emotional regulation—each fundamental to daily functioning and particularly vulnerable to stress-related disruption.

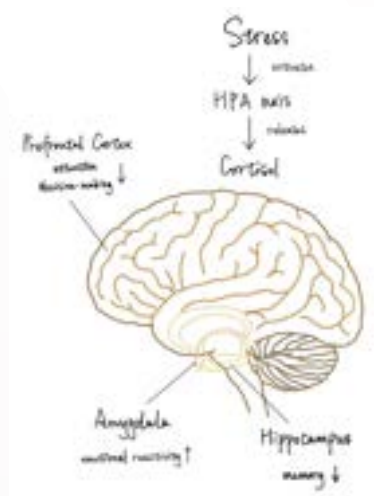


Figure 1. Activation of the HPA axis under stress releases cortisol, which impairs the hippocampus, prefrontal cortex, and amygdala, resulting in memory, attention, and emotional regulation difficulties.

Memory and the hippocampus

Memory is the ability to load, store, and search up information. Stress significantly influences memory by affecting HPA circuit and hippocampus, the brain structure crucial in the formation of long-term declarative memories. When exposed to stress, there is the release of a hormone called cortisol from the hippocampus within the HPA axis. The hippocampus contains a high concentration of receptors for the cortisol such that the region is sensitive and vulnerable (Lupien et al., 1998).

Whereas acute stress can facilitate loading into memory, prolonged or intense stress impairs memory retrieval. Experiments show exposure to high levels of cortisol has been demonstrated to cause hippocampus shrinkage, damaging the structure and connections of the cells in the hippocampus (McEwen, 2007), which directly devastates memory storage. For example, individuals under chronic stress often struggle to recall previously learned knowledge, especially when facing the high-

pressure situations. This demonstrates that the effects of stress on cognition are not only psychological, but also rooted in tangible alterations induced by stress in brain architecture.

While memory illustrates the long-term structural vulnerability of the brain under stress, attention highlights how quickly stress can impair moment-to-moment cognitive control.

Attention and the prefrontal cortex

Attention is the ability to selectively focus on certain stimuli while ignoring others. Selectivity in attention is controlled most by the prefrontal cortex (PFC), the area in the frontal portion of the brain dedicated to executive control, decision making, and behavioral guidance. Stress undermines attention by disrupting the normal functioning of the PFC (Sandi, 2013).

The prefrontal cortex is critical in the regulation of attention maintenance, impulse control, and the performance of complex actions, and stress has negative impacts on these. After exposure to acute or chronic stress, the PFC becomes less efficient due to the flood of chemicals like dopamine and noradrenaline that disrupt its normal functioning (Arnsten, 2009). Such overburden dampens transmission within the PFC and undermines the PFC in communicating with the rest of the brain regions, and the control shifts towards more emotional regions like the amygdala. As a response, attention becomes scattered, cognitive flexibility is lost, and attention on tasks is lost. Such findings are supported in fMRI studies which reveal decreased PFC activation under stress tasks involving executive control (Liston et al., 2009). In real-world scenarios, stressed individuals are distracted more easily, more impulsive, and more unable to concentrate in school, work life, or social environments.

Since disruption in attention affects decision quality, it becomes vital to examine how stress influences the decision-making process.

Decision-making and the Amygdala

Decision-making is the process of selecting between options based on reasoning, preferences, or outcomes. Stress interferes with the process by shifting brain activity from the deliberative system to the habitual system. A main reason behind this is the increased activation within the amygdala, the key region for processing emotional responses, especially fear and threat.

Typically, the PFC helps in the regulation of the amygdala and in making intentional decisions. Stress inhibits the regulating role of the prefrontal cortex, reducing the cognitive control ability of the brain (Schwabe & Wolf, 2011). When stressed, decision-making is more passive: short-term rewards beat long-term rewards and there is a preference for familiar strategies over exploring or adjusting to new information. As a representative example, reward-based task research shows stressed participants are more risky or more impulsive in their choices and show bias towards short-term rewards (Porcelli & Delgado, 2009). This shift illustrates how stress affects decision-making by tilting the balance between cognitive and emotional systems, where heightened emotional reactivity, driven by the amygdala, leads to decisions that are more impulsive, less deliberative, and often riskier.

Given that these decisions are increasingly influenced by emotional responses, it becomes crucial to inspect how stress also disrupts emotional regulation itself.

Emotional Regulation

Emotional regulation is the ability to monitor, evaluate, and regulate emotional reactions in order to maintain well-being. Although it involves a distributed network of brain regions, two of the most critical hubs in this process are the amygdala and the prefrontal cortex. These structures form a core component of the emotion regulatory circuit in the brain.

Under stress states, the equilibrium between the two regions is lost. Generally, the PFC exerts top-down control to suppress the



amygdala's response to emotional stimuli. But during stress, this control weakens and the amygdala becomes overactive. This can lead to stronger emotional reactions and reduced ability to regulate feelings like fear, frustration, or anxiety. As a result, individuals may become more emotional or experience mood swings. These changes are not just behavioral but stem from changes in the functional connectivity between the PFC and amygdala (Urry et al., 2006). This supports the thesis by illustrating how stress disrupts cognitive control, not only excessive attention and decision-making but also overemotional responses, through their impacts on specific neural pathways.

Conclusion

Stress, although in small measures can be handled, it can significantly compromise cognitive functions such as memory, attention, decision-making, and emotional regulation in the long term or when it is at the extreme. The essay illustrates how stress disregulates cognitive functions through the neural systems involving the hippocampus, prefrontal cortex, and amygdala, which are regulated by the HPA axis. Clarification of the mechanism is valuable both in psychology and in areas such as education, mental health, and the workplace where stress is the rule. Present findings reveal the imperative for stress management for the maintenance of cognitive integrity and emotional stability. By deepening our understanding of how stress affects the brain, we can better support interventions that protect learning, judgment, and happiness.

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Why Do We Need to Define and Operationalize Psychological Constructs to Study Them?

YE JIN

Introduction

Psychological constructs, such as happiness, intelligence, or talent, are some abstract concepts that are used to describe mental processes. Unlike traditional physical attributes, psychological constructs are often highly subjective and most of the time multidimensional and cannot be directly measured. Therefore, clearly defining concepts (specifying their precise meaning) and operationalizing them (determining how they will be measured or manipulated) are essential and inseparable components of psychological research. Without clear definitions, concepts remain ambiguous, and without proper operationalization, they cannot be empirically studied. In the study on tacit knowledge, Ambrosini and Bowman (2001) highlight that empirical research on the subject remains limited due to challenges in operationalizing tacit knowledge. They further argue that the absence of clear definitions for cognitive constructs can hinder both theoretical advancement and practical applications in the field. In this paper, we will evaluate the importance of the definition and operationalization of psychological constructs through the example of anxiety. This paper will use anxiety as the psychological concept to be operationalized and show how this allows the possibility of conducting research. For this, we will first explore the importance of a clear definition of the concept to reduce its various dimensions to specific operationalizable variables. To illustrate our point, the essay will explore the design of an experiment that capitalizes on these previous steps.

The importance of definition

A definition refers to the clear explanation of the meaning of a concept. In research fields that rely on psychological constructs, which are inherently abstract and difficult to measure, definitions often serve to delineate boundaries and establish a foundation. Taking anxiety as an example, some scholars define it as tension, while others consider it to mean panic attacks. As a result, although studies may all focus on "anxiety," they are, in fact, examining two entirely different concepts. A precise definition, such as defining anxiety as an emotional state characterized by future-oriented worry, physical tension, and avoidance behaviors (Michelle et al., 2011), ensures that all researchers are studying the same phenomenon. Additionally, definitions are essential when designing measurement scales. For instance, the DSM-5 defines Generalized Anxiety Disorder (GAD) as excessive worry persisting for more than six months, laying the groundwork for subsequent specific diagnostic criteria and enabling consistent diagnosis. In summary, clear definitions remove ambiguity in mental conceptualization and provide a basis for accurate measurements, which are the necessary steps for the operationalization.

The importance of operationalization

The essence of operationalization is the development of a measurable operational program based on the definition of a psychological concept. It reduces a general multidimensional concept into accessible variables that can be measured. You may want to explain what aspects are operationalized in questionnaires. The most common operationalization is the design or use of self-report scales, that is, standardized questionnaires that allow subjects to subjectively report their feelings. For example, the Generalized Anxiety Disorder Scale (GAD-7), a commonly used scale for anxiety, consists of seven questions with a total score of 21, with a score of more than 10 indicating clinical anxiety. However, this type of scale is often considered to be of low validity, which is the extent to which the experiment is measuring what it tends to measure, because of an over-reliance on subjective judgment and social desirability bias (people are trying to appeal to society and project a good image). This is why it is important to measure physiological indicators, such as cortisol levels measured in blood or saliva, in experiments on anxiety, where higher levels indicate more anxiety. Such indicators are more objective and less affected by subjective biases. In addition, behavioral observation is also widely used due to its low cost

and simplicity. For example, in experiments related to anxiety, participants may be asked to give a 5-minute speech while observers assess their overt behaviors such as trembling or sweating. However, such measurements often rely too much on the observer's judgment, and their subjective biases can directly affect the experimental results. In conclusion, the core of operationalization lies in the translation of psychological concepts into measurable procedures, and common methods include self-report scales, physiological measures, and behavioral observations, with experimenters often needing to combine multiple methods to improve validity. Having described the importance of the definition and operationalization of psychological concepts, this essay will then show how this can be applied to research the effect of potential therapeutic procedures for anxiety

Experiment about anxiety

Here is an experiment on the effect of mindfulness on anxiety. Mindfulness is defined as a mental state of focusing on the present moment and being aware of one's feelings, thoughts, and surroundings without judgment. Anxiety refers to excessive worry that interferes with daily functioning. The independent variable is an 8-week standardized MBSR (Mindfulness-Based Stress Reduction) course, and the dependent variables were GAD-7 scores and cortisol levels measured at 10 am. The operational hypothesis of this experiment is that individuals who participate in the 8-week MBSR program will have significantly lower GAD-7 scale scores compared to the control group (which receives no training) and will exhibit a greater reduction in cortisol levels. During the experiment, participants first completed initial assessments of their GAD-7 scale scores and cortisol levels. Then, the experimenter used a matched-pairs design to pair two participants with similar initial GAD-7 scores and cortisol levels, assigning one to the experimental group and the other to the control group. This was done to minimize the influence of initial anxiety states on the experiment. After completing the MBSR course, participants from both groups will be asked to take the GAD-7 scale assessment and undergo cortisol level measurements again. The experimenter will calculate the average values for both groups and compare them with the initial assessment averages. Finally, a conclusion will be drawn to determine whether the hypothesis is supported.

Conclusion

In summary, by defining and operationalizing psychological concepts such as anxiety, researchers can make the leap from theory to empirical research. This greatly ensures the completeness, reliability as well as validity of the experiment. However, it should be noted that operationalization may sometimes use only a single indicator (e.g. cortisol to measure anxiety) while ignoring its complexity, so in the future, researchers should pay attention to multi-method integration when conducting operationalization to ensure the validity of the study. Although operationalization has some limitations, definitions and operationalization are still the cornerstone of empirical research in psychology.

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14

Business Economics

Are Price Caps Good or Bad? The Case of Rent Control

LENA HU

Price caps (price ceilings) are the legal maximum price set by the government for a particular good or service to make goods (e.g. rent, food) more affordable, especially for low-income consumers. Their application in housing markets is as rent control and governments introduce such policy solutions as it will reduce rent prices, making housing that is a basic human necessity and merit good more affordable to benefit everyone on the housing ladder, including individuals who are unable to afford housing and earn the average income in a country. However, the price caps are set below market equilibrium, which leads to excess demand/shortage (Fig.1). The house price to income ratio in London is currently 150x (Statista, 2025), the housing affordability crisis exists due to political failures, market failures (e.g. oligopolistic behaviour of real estate developers) and is so entrenched inherently due to the undersupply of housing. This essay will argue that rent controls are a short-term, unsustainable policy that only benefits incumbent tenants. They are economically inefficient and socially regressive in the long term due to misallocation, suppressed supply, and quality decline that leads to high levels of unaffordability, ultimately failing to address the fundamental issue of an undersupply of housing.

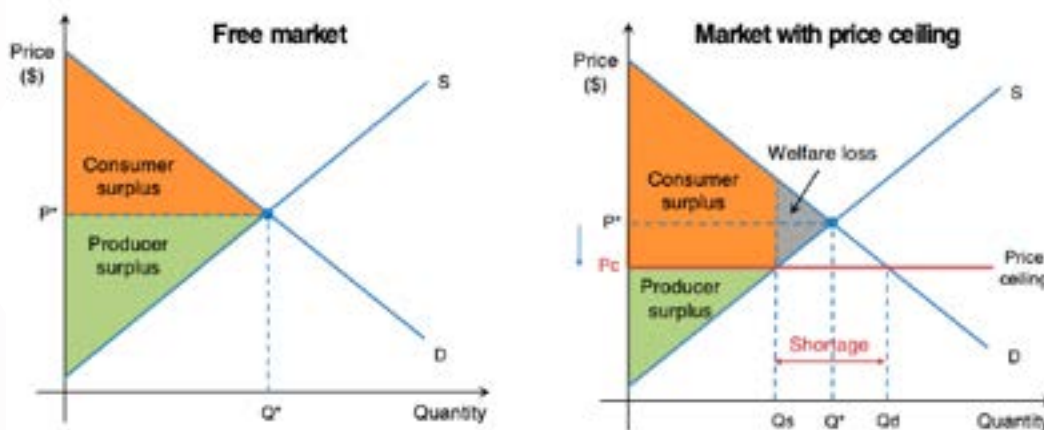


Figure 1: Demand-Supply diagram of Free market and Market with price ceiling/price cap imposed; In the housing market, price caps are as rent controls, where consumers are tenants and producers are landlords (Level7 Education Ltd, 2023)

As economic theory predicts, a rental cap imposed below equilibrium price contracts housing supply while increasing demand, causing a significant shortage, resulting in deadweight loss and unequal welfare distribution between landlords and tenants (Kholodilin, 2024). This theory is supported by the Law of Demand, where a decrease in rent prices will increase demand. Moreover, a mismatch between supply and demand is expected with the reduction in supply of rental accommodation post the implementation of the rent caps (Kholodilin and Kohl, 2023). For instance, Scotland's 2022–2024 rent caps led to a 26% decline in build-to-rent investment in year (Norman, 2025) and the developer's withdrawal of new development projects worth £3 billion (Scottish Parliament, 2025), directly reducing the future housing supply. This clearly shows that a rental cap disincentivises new development and objectively reduces the supply of homes. Therefore, the supply of rental accommodation would be constrained, which would put downward pressure on the current supply of housing that is already low. The policy's temporary nature and weak targeting led to short-term gains but long-term investor flight and market contraction that worsens affordability in the long run (Kholodilin, 2024). Furthermore, investment withdrawn would have had a huge multiplier effect and opportunities for growth in the local economy by generating demand. Moreover, the landlords are also disincentivised with decreased profit incentive as they are unable to charge rents at their own discretion. For example, landlords in San Francisco reduced the rental housing supply by 15% through condo conversions and owner-occupancy shifts in response to rent controls (Diamond, McQuade and Qian, 2019). Similarly, rental units were also converted into residential units in Scotland as it was unprofitable for landlords to rent out their homes. Additionally, gentrification

increased over time as supply tightened, uncontrolled rents rose and wealthier new entrants were priced out, suggesting rent control contributed to a counteracting effect of the inequality it aimed to reduce. Therefore, rent controls deplete both the existing and future housing stock, worsening the initial undersupply. Furthermore, the reduced landlord revenue therefore also reduces incentives for maintenance or new construction, further decreasing quantity and quality of housing supply on the market.

Rent caps disproportionately benefit incumbents at the expense of the new residents. Because benefits will only be reaped by incumbent tenants who do not have to pay extortionately high rents. Whereas new residents are unable to find vacant rental accommodation due to the undersupply of rental accommodation from reduced developers' incentives and landlords' ownership conversions, as stated previously. For example, the case of Catalonia's 2020-2022 high-coverage rent control regulation resulted in rents falling by 4–6% with no immediate supply loss (Jofre-Monseny, Martínez-Mazza and Segú, 2023). But when the regulation was overturned in 2022, rent prices quickly returned to pre-regulation levels (*ibid.*). This reversal illustrates the short-lived nature of rent controls in the absence of structural supply reform with no evidence found for long-term market stability (Diamond, McQuade and Qian, 2019). This indicates how rent caps are not an effectively inclusive policy, as their benefits are primarily captured by only the sitting tenants party, rather than newcomers. New residents are unable to live in rented accommodation due to the still-existing problem of housing undersupply and unresolved affordability issues despite the rent caps.

Furthermore, the decreased incentive oftenants to move due to their benefits gained with the undersupply in the market leads to economic inefficiencies in allocating housing stock (Diamond, McQuade and Qian, 2019; Glaeser and Luttmer, 2003). Firstly, in situations where an incumbent tenant wants to move to another accommodation due to unsuitable size reasons, but in large, rent-controlled units they lack incentive to move with rental cost benefits even when their needs change. Additionally, the lack of housing supply in the market due to the reduced incentives to supply rented accommodations with rent caps means there are fewer available matching options that incumbents are able to rent. Therefore, keeping larger homes and options available off the rental market and away from families who need them. Thus, this results in an allocatively inefficient outcome of misallocation that disproportionately benefits incumbent residents and excludes new residents or those seeking housing with the undersupply of homes and excess demand. Secondly, this 'lock-in' effect disincentivises relocation, rent control reduces geographical labour mobility, preventing workers from moving to areas with better employment opportunities (Kholodilin, 2024). Therefore, this inefficient matching of workers to firms hinders overall national productivity (Hsieh and Moretti, 2019). Meanwhile, rent control can increase unregulated rents via spillover effects and demand displacement, leading to a further decrease in overall national productivity in the long run, with effects not seen in the short term (Autor, Palmer and Pathak, 2014).

Despite the lack of incentive to develop and undersupply being a key issue in successfully implementing rent controls, Germany and San Francisco show us that even when new developments/renovations are exempt from rent caps to alleviate the issue of a fall in supply, the policy remains ineffective with reduced quality houses. For example, in Germany, although the 'Mietpreisbremse' policy temporarily slowed rental growth by 3–5%, it led landlords to falsify renovations simply shifting the purpose to become exempt from the cap (Breidenbach, Eilers and Fries, 2022). This practice reduced the quality-adjusted affordability of housing, as tenants paid more for superficial upgrades in lower-quality homes rather than practical improvements. In the long term, individuals will not purchase homes which maximise their value for money.

In conclusion, rent caps are a 'bad' policy because they disproportionately benefit incumbents only in the short run while exacerbating the housing shortage. The disincentive to supply essentially means the problem of affordability is not solved, as seen in Scotland and San Francisco. It is a politically expedient but economically flawed strategy that treats the symptom (high prices) rather than the root disease (housing undersupply) (Kholodilin, 2024). As governments often aim to implement rent caps as a short-term policy to gain voter support. If the issue of undersupply of housing were solved, then there would be no issue of unaffordability and therefore no requirement for rent caps to exist. Rather than blunt caps, effective policy must target increasing supply through housing vouchers, tax incentives for construction, zoning reform, and

public investment in social housing.

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Can a Business Be Ethical or Profitable?

YUHAN CHEN

In modern economic systems, profit motives are often seen as the core driving force behind market operations and economic growth. In today's society, whether a company can balance its profit objectives with social responsibility has become an important part of how society evaluates a company's capabilities.

Profit-seeking behavior is indispensable in a market economy

In modern economic systems, profit motives are often seen as the core driving force behind market operations and economic growth. In today's society, whether a company can balance its profit objectives with social responsibility has become an important part of how society evaluates a company's capabilities.

Neoclassical economics posits that in market mechanisms, profits serve as “signals” for resource allocation; firms seek to maximize profits by allocating resources to areas that generate the highest value, thereby achieving optimal resource allocation in the market (Varian, 2010; Hayek, 1945). To maximize profits, firms must allocate resources to production areas that generate the highest value, ensuring that marginal costs equal marginal benefits to achieve an optimal economic state.

Under market and policy incentives, firms can pursue profits while aligning with ESG objectives, achieving a win-win outcome of economic and social benefits through innovative management and social responsibility measures. If firms receive rewards in the market or through regulation for sustainable behavior, profit maximization and sustainable development are not mutually exclusive; firms can find a balance between meeting consumer needs and pursuing social responsibility.

Resource flows in the market do not always move toward optimal efficiency but are often constrained by imitation mechanisms and path dependence (Nelson & Winter, 1982). Certain industries may remain in suboptimal states due to historical inertia, leading to resource concentration in areas that appear profitable but are unsustainable, such as fast fashion, spam platforms, or high-energy-consuming industries. This misalignment limits the market's adaptability and suppresses the potential for long-term innovation. Worse still, some “innovations” are not based on genuine needs but driven by profit bubbles. For example, in fields such as NFTs and cryptocurrencies, speculative profits have become the dominant force, distorting the direction of innovation and leading to resource waste and market volatility (Schumpeter, 1942).

In this context, ESG (environmental, social, and governance) practices have become an important case study for measuring institutional maturity. When regulation is lax and institutions are unclear, ESG can easily be distorted into a profit-seeking tool. Companies may publish detailed sustainability reports on the surface, but in reality, they may ignore carbon emissions, exploit labor, or hinder real institutional change. This “greenwashing” phenomenon not only misleads the public and investors but also undermines the trust and justice that ESG was intended to establish (Friede, Busch, & Bassen, 2015). Evidence suggests that when high profits conflict with fair value, some companies are more inclined to sacrifice the latter (Porter & Kramer, 2011; Flammer, 2015). However, in a well-regulated and transparent institutional environment, many companies have already established balancing mechanisms, integrating environmental protection and social responsibility into their business strategies. This suggests that profit motives are not inherently incompatible with a sense of responsibility; the key lies in whether external institutions are sufficiently robust to effectively channel profit-seeking behavior toward long-term public interests (Pedersen, Fitzgibbons, & Pomorski, 2021). Therefore, ESG is not a reflection of a company's “conscience,” but rather an indicator of the robustness of its institutional framework: with proper regulation, ESG can serve as a bridge between capital and society; without it, it may devolve into a mere performance of responsibility. Pursuing ESG is not necessarily hypocritical; profit motives and social responsibility are not mutually exclusive. Some companies

establish clear trade-off mechanisms that allow them to balance social objectives with commercial operations. Whether a company assumes ESG responsibility depends on its stage of development and external market environment. For example, Bosideng, as a leading company in the apparel industry, aims to achieve “net zero emissions” in its operations by 2038, has donated a total of 1.4 billion yuan, has its board of directors directly overseeing ESG, and links executive compensation to carbon emissions reduction. However, in contrast, the fast fashion industry often engages in false environmental claims and exaggerates its emissions reduction commitments. Italy imposed a fine of 1 million euros in August 2025, and France imposed a fine of 40 million euros in July of the same year.

At the same time, the development of a corporate culture in a company becomes an important part of the development of social and ethical responsibility. Training in corporate culture does not directly benefit the company, but employees are motivated to take the responsibility to the whole society.

The sequence of profit and public welfare: not opposites, but interdependent

Profit is the first step, fairness is the second. For profit-oriented organizations to achieve social goals, the prerequisite is “operation + profit”; otherwise, “zero divided by zero is still zero.” It is advocated to view public welfare from a realistic perspective: without an economic foundation, public welfare cannot be sustained.

This perspective aligns closely with Adam Smith's market logic that “self-interest drives public welfare,” and is further supported by Amartya Sen's theory that “development is freedom,” which posits that social fairness must be built on economic development and resource availability. Additionally, from the perspective of mixed value theory (Emerson, 2000), if profit-driven organizations cannot generate basic revenue capacity, their social missions will also be difficult to sustain. This sequential approach of “profit first, then public welfare” reflects the contemporary for-profit sector's high priority on institutional design and resource sustainability.

Adam Smith argued that individuals, in pursuing their own interests, inadvertently promote the interests of society as a whole. In *The Wealth of Nations* (1776), he outlined how market mechanisms coordinate resource allocation and implied that public welfare must rely on a stable market environment. The principle of “profit first, then fairness” actually embodies Smith's logic: profit-making activities in the market are the prerequisite for the creation of social value. Without output, there can be no fair distribution. Amartya Sen also emphasized the core idea that the realization of fairness and welfare must be based on economic development and freedom of choice. “Without an economic foundation, public welfare cannot be sustained,” which aligns with Sen's view that “freedom and fairness are both ends and means of development.”

Overall, the ability of firms to balance profitability and ethical responsibility is related to the environment of the industry in which they operate and their willingness to be socially responsible. This suggests that the profit motive does not naturally exclude a sense of responsibility, but that the key lies in whether external institutions are strong enough to effectively channel profit-seeking behavior toward the long-term public good.

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15



Chemistry

Color in Nature: Made by Conjugation, Removed by Bleach

YICHENG HUANG

1. Introduction

In our lives, color is everywhere: seen in blue skies, green leaves, and vivid flowers. These beautiful natural colors are not just visual pleasures; they arise primarily from conjugated π systems, which absorb specific wavelengths of visible light. The perceived color results from the remaining reflected light. However, these colors are not permanent; chemical bleaching disrupts the conjugated systems, eliminating their ability to absorb light and thus removing color.

2. The physical chemistry of color formation

The vibrant colors we see in many natural compounds arise from electronic transitions within molecules, especially those with conjugated π systems. A conjugated π system is a molecular structure where alternating single and multiple bonds allow p-orbitals on adjacent atoms to overlap continuously, creating a region of delocalized π electrons across several atoms. These electrons can absorb energy from incoming light and jump from a lower-energy orbital to a higher-energy orbital. As the conjugated π system extends, the energy gap between the two types of orbitals becomes smaller, so the energy required for the transition turns lower. Energy and wavelength are inversely related, so lower energy absorption leads to a longer wavelength, according to the formula below:

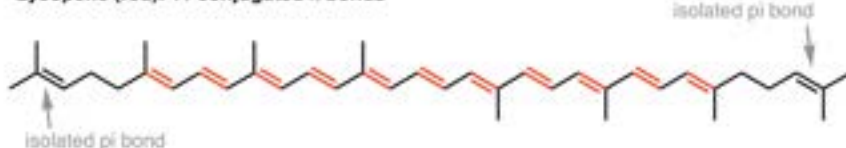
$$E = \frac{hc}{\lambda}$$

As the absorption wavelength increases, the perceived color shifts toward the red end of the spectrum, resulting in a deeper color. That is the color we see. However, the presence of a conjugated system does not necessarily result in visible color. For instance, 1,3,5-heptatriene, despite containing three conjugated double bonds, appears colorless because it absorbs light only in the ultraviolet region, not the visible spectrum. [1]

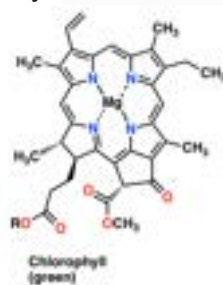
3. Examples of colors in nature

A Lycopene – Linear conjugated system. Lycopene contains an extended linear π -conjugated chain with 11 conjugated double bonds. This long, uninterrupted π system lowers the energy gap between the HOMO and LUMO levels, enabling absorption of blue and green light. The remaining transmitted or reflected light appears red to our eyes.

Lycopene (red): 11 conjugated π bonds



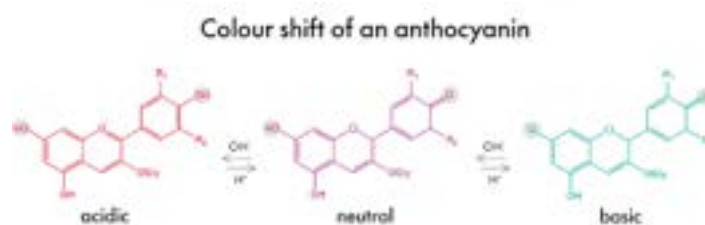
[2]



[3]

B Chlorophyll – Cyclic conjugated system with metal. Chlorophyll's color arises from a large cyclic π -conjugated system called the porphyrin ring, which incorporates a central Mg^{2+} ion. The ring structure allows delocalization of π electrons across the macrocycle, altering the energy levels and resulting in strong absorption of red and blue light. The reflected green light is what we perceive.

C Anthocyanin – Cyclic conjugated system without metal. Anthocyanins contain fused aromatic rings with extensive conjugation. Though they lack a metal center, the delocalized π electrons within the rings still produce strong visible absorption. Moreover, the color is pH-dependent: under acidic conditions, the conjugated form dominates, which is red, while under basic conditions, structural changes alter conjugation, shifting the color toward purple or blue.



[4]

D Human made color: Phthalocyanine – Cyclic conjugated system with metal. Compared with the three examples above, phthalocyanine is artificially synthesized by humans through chemical reactions in the laboratory, which provide a longer conjugated system, and the area of delocalized electrons becomes larger, the average energy in the system turns lower, so it is more stable and that can solve the limitations of natural pigments in terms of easy fading and uneven coloring effect. This also makes artificial dyes widely used in industry.

4. Bleaching agents with color

Bleaching agents are substances that lighten or whiten materials by chemically altering or removing their color-producing components. Most common bleaches, such as hydrogen peroxide, break the π bonds by adding oxygen, or causing cleavage of the carbon-carbon double bonds. Breaking these bonds interrupts the continuous overlap of p-orbitals, then leads to the destruction of the conjugation system. Without the conjugation, electrons become localized, and the energy gap between them increases. This means that if we want to excite electrons now, higher-energy light – ultraviolet light must be needed; therefore, the chromophores gradually lose the ability to absorb visible light, and the color disappears. For the natural examples above, they all used oxidation bleaching agents to bleach.

In addition to oxidation bleach, reducing bleach is also often used in real life. For example, under appropriate conditions, sodium dithionite can reduce anthocyanin dyes. It donates electrons to the colored part of a molecule, and these added electrons can disrupt the chemical bonds within the chromophore: some double bonds may change to single bonds. Then the ability for the molecule to absorb visible light disappears, and finally, the effect of bleaching is achieved.

However, although these bleaches are effective, they are not without limitations. First of all, oxidized bleach, such as hydrogen peroxide, has strict requirements for the bleaching conditions of cotton fabrics; it often requires high temperatures, alkaline conditions, and stabilizers, which increase the energy and chemical demand. Moreover, the process necessitates significant water consumption for post-bleaching washing. [5] From another perspective, the bleaching process is usually not selective – although it removes unwanted colors, it may also lighten the desired tone or damage the surrounding pigment. Lastly, repeated or long-term bleaching may release potentially harmful by-products into the environment, causing hidden dangers to safety.

5. Conclusion

In both natural and human-made materials, color arises fundamentally from conjugated π systems that absorb visible light. While conjugation is central to color, bleaching agents disrupt these systems by breaking key double bonds through oxidation or reduction, leading to colorlessness. The structural stability of man-made dyes often makes them more resistant

to bleaching than natural pigments, reflecting practical differences in daily applications. Finally, recognizing the limitations of current bleaching methods highlights the importance of more selective and sustainable alternatives in color removal.

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The Chemical Reactivity Nature in Hard and Soft Nucleophiles and Electrophiles

YICHENG GUI

Introduction

The principle of chemical reactions is the electron flow from “electron-rich” areas to “electron-poor” areas, and these areas are named as nucleophiles and electrophiles. This reveals that Nucleophiles and electrophiles serve as crucial key in organic reactions. A nucleophile is a reactant that donates a pair of electrons to form a new covalent bond, whereas the electrophile is a reactant that accepts electrons [1]. Since nucleophiles and electrophiles have the same definition as Lewis bases and acids, these reactants are also referred as Lewis bases and acids, respectively.

Generally, it is known that the reactivity of Lewis acids and bases is affected by factors such as their charge, electronegativity, and conjugations. Nevertheless, their hardness and softness will also induce an impact on their reactivity. This essay particularly explores how these contrasting characteristics of softness and hardness of Lewis acids and bases are shaping their chemical reactivity. With the guiding theory of the Hard-Soft Acids-Bases (HSAB), specific examples are presented.

The HSAB Theory

In a chemical reaction, metal ions are generally considered as Lewis acids since they accept electron pairs donated from Lewis bases that acted by ligands. Certain metal ions will have stronger interactions with certain ligands, which gave rise to the Hard-Soft Acids-Bases (HSAB) Theory. The Hard-Soft Acids-Bases (HSAB) Theory is an approach that explains the stability of metal complexes and the mechanisms of their reactions [2]. It is widely used to understand and predict chemical reactivity through classifying acids and bases as hard or soft according to their electron properties.

The softness or hardness of acids and bases is determined by various factors, but most commonly the polarizability of atoms. If the atom is less polarizable, it is ‘harder’; if the atom is more polarizable, then it is ‘softer’;. Specifically, ions or atoms with high charge or small radius, such as Lithium(1+) and Fluoride (F⁻), are considered ‘hard’ because they have high charge density, making them not that polarizable. [3]. Atoms with a smaller degree of charge and larger sizes are more readily polarized, so they are considered as soft acids and bases, such as Copper(1+) and Carbon monoxide (CO).

Chemical Reactivity of Nucleophiles and Electrophiles under different hardness

Chemical reactivity is the ability of a substance to undergo a chemical reaction. The contrasting hardness and softness of Lewis acids and bases will directly impact their reactivity in acid-base interactions.

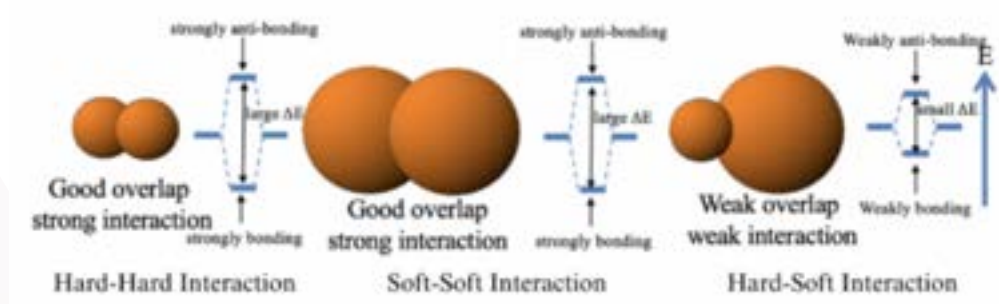


Figure 1. Different types of Orbital overlap (cited from: Landskron, K. 4.2: Hard and Soft Acids and Bases[4])

On one hand, hard acids (hard electrophiles) will tend to interact with hard bases (hard nucleophiles) but not soft bases to form ionic bonds, while soft acids (soft electrophiles) tend to interact with soft bases (soft nucleophiles) to form covalent bonds. These hard-hard interactions and soft-soft interactions react more strongly and quickly than hard-soft interactions. The nature of these interactions can be understood through the orbital overlap between Highest Occupied Molecular Orbitals (HOMOs) of nucleophiles and Lowest Unoccupied Molecular Orbitals (LUMOs) of electrophiles. As HOMOs and LUMOs from different molecules interact, new bonds are formed. As illustrated in Figure 1, the orbitals of two molecules with similar size (for both hard-hard and soft-soft interactions) generally overlap better as well as interact strongly, leading to a stronger bond. Conversely, the two molecules with disparate sizes tend to have weaker orbital overlap because the large orbital can only overlap with a small fraction of the small orbital. Under this circumstance, the orbital overlap weakens, bringing a decline in the reaction's efficiency and speed.

Eventually, the soft-soft and hard-hard Lewis acids and bases react much stronger than hard-soft interactions[4]. In addition to this factor, regarding only hard-hard interactions, they are more driven by electrostatic attraction between the highly electron-rich(δ^-) nucleophiles toward highly electron-poor(δ^+) electrophiles rather than orbital overlaps. This is because hard molecules are relatively small and non-polarizable, leading to higher charge densities [5]. The positive and negative charges of hard acids and bases leads to more intense attraction.

Despite the soft-soft and hard-hard interactions, another shift in the reactivity of nucleophiles and electrophiles is caused by the disparity in hardness of Lewis acid attacking the corresponding Lewis base in a reaction. The typical electrophile "Acid Chloride", one of the most creative carbonyl existed, is utilized as an example.

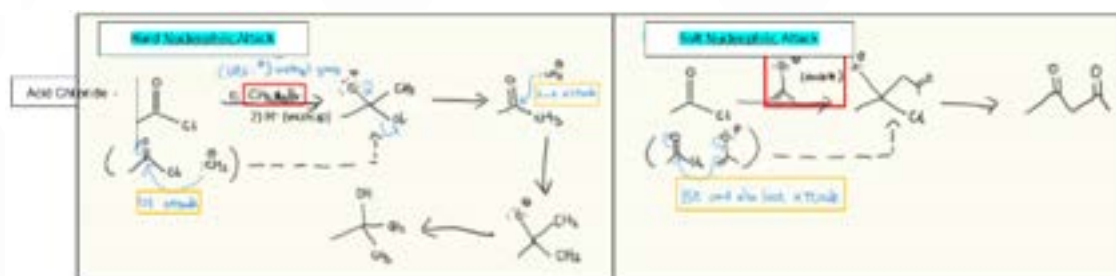


Figure 2. The illustration of Hard(on the left) / Soft(on the right) nucleophilic attack on Acid Chloride

According to Figure 2, an apparent difference between soft and hard nucleophilic attack in reactions is shown. Shown on the left, the CH_3MgBr is a Grignard reagent with a methyl group CH_3 [6]. This methyl group is nucleophilic, and will donate an electron pair to the electrophile 'Acid Chloride', forming a new sigma bond between carbons and breaking a $\text{C}-\text{O}$ π bond. In this reaction, what's abnormal is that the nucleophilic attacks the electrophile twice. During the first attack, the nucleophile adds to a carbonyl group, forming a tetrahedral intermediate, and it is still reactive enough to be attacked by a Grignard agent [7]. Finally, the Oxygen ion is protonated, making the Acid Chloride into a tertiary alcohol. Contrarily, for the soft nucleophilic (enolate) attack reaction shown on the right, it only displays the attack once. Soft nucleophiles are less reactive than hard nucleophiles because their negative charge is more delocalized, making them less attractive to electron-poor electrophiles.

Eventually, the attack process ceased after the first attack.

Conclusion

Understanding nucleophiles and electrophiles is extremely crucial in chemical reactions. Throughout this paper, the Hard-Soft Acids-Bases (HSAB) are presented to strengthen the understanding of nucleophiles and electrophiles. This procedure ensures further development into the understanding of complex interactions and reactivity change between Hard-Hard, Soft-Soft, and Hard or Soft nucleophilic attack. The interactions between HOMOs and LUMOs, electrostatic attraction between atoms, and the varied hardness of nucleophiles attacking electrophiles are all crucial factors that help elucidate the disparity in species' reactivity.

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16



CompSci: Cybersecurity

Meltdown: A Transient Execution Attack

HAORONG XU

Abstract

This essay explores how some performance features of modern processors and operating systems, combined with a cache side-channel attack, can create severe security issues, leaking data from an arbitrary memory location. This compromises the data security on both workstations and cloud servers, as it breaks memory isolation and any protections based on it. I show that current countermeasures, such as KAISER, have significant performance penalties.

1.Introduction

TRUST is a principled concept in cybersecurity. According to Ross J. Anderson [1], a trusted system or component breaks the security policy if it fails, while a trustworthy system or component never fails. In this essay, I present meltdown [2], an attack that exploits various features of modern operating systems and processors and enables a process to access kernel memory space, rendering process isolation completely ineffectual. It turns out that the operating system and hardware trusted by many programs can fail miserably.

The meltdown attack utilizes several design flaws, and the inconsistency in access control in architecture (the design of processors) and microarchitecture (actual implementation) lays the foundation for it. Specifically, the CPU does not check access rights for individual μ ops, letting them access data that is inaccessible on the architectural level. Additionally, the processor does not roll back microarchitectural changes caused by instructions that are terminated, enabling a side-channel attack. Moreover, the layout of the virtual memory address space further extends the attacker's capability, intensifying its impact.

Some defenses against the meltdown attack focus on eliminating the inconsistency and making virtual memory safer, while other solutions enhance the security of the architecture. However, variants of the meltdown attack are constantly proposed, bypassing all countermeasures, and these defenses often contribute to performance penalties.

2.Background

In this section, I explain the mechanisms of virtual memory, out-of-order execution, cache side-channel attack, and transient instructions.

A. Virtual Memory

Virtual memory is a memory management technique that isolates processes from each other, creating the illusion that every process has exclusive access to the memory. [3] The virtual address space is divided into consecutive pages, which are mapped to physical memory using multiple page tables. Meanwhile, there are two parts of the address space, kernel virtual memory and process virtual memory, as shown in Fig.1. The kernel virtual memory is mapped into all processes' address space to speed up system calls, and it is not accessible by a process running in user mode. It is worth noting that the entire physical memory is mapped into the kernel's virtual address space for performance reasons, allowing the kernel to access an arbitrary location in physical memory. In section IV, I highlight that this feature, combined with a meltdown attack, allows attackers to manipulate other processes' memory, invalidating process isolation.

B. Out-of-order Execution

Out-of-order execution is a microarchitecture design that allows later instructions to be executed while a prior instruction is blocked, given that the input data required by the later instruction is ready. [5] Meanwhile, pipelining, an implementation technique in which multiple instructions are overlapped in execution [4], is a standard feature of many modern processors. With pipelining, each instruction is decomposed into micro instructions (a.k.a. μ op) that perform basic operations, and μ ops relying on different hardware resources can run simultaneously. μ ops retire when they have been executed, meaning that they are completed and their results are correct and visible [7]. μ ops retire in order as if they execute in-order, since a μ op will not retire before all preceding μ ops retire, even if it is finished. In addition, any exceptions are handled at this stage to improve performance.

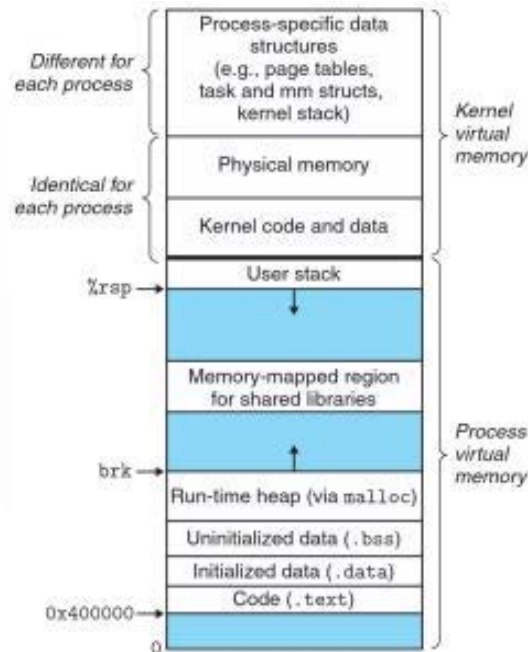


Fig. 1. The virtual memory of a Linux process. [3]

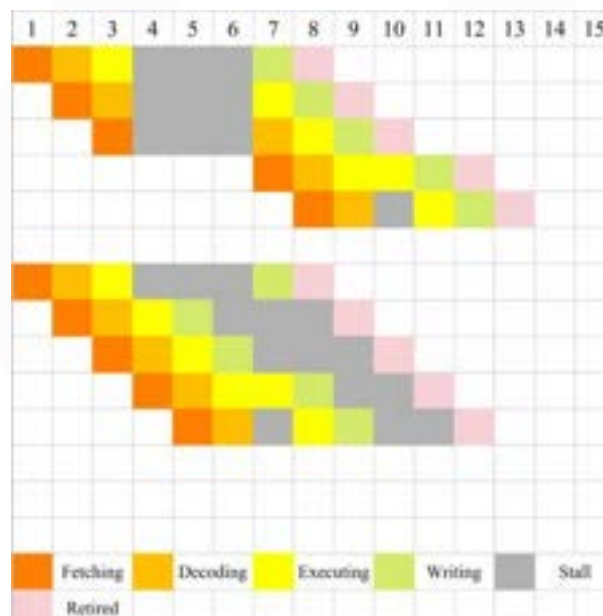


Fig. 2. Each line represents an instruction and each column represents a clock cycle, while each cell stands for a μ op (except for "retired"). The five instructions above illustrate in-order execution, when a prior instruction is stalled, all other instructions have to wait, whereas in out-of-order execution (the instructions below), μ ops can execute as long as hardware resources are ready. However, instructions must retire in order.

C. Cache Side-channel Attack

Side-channel attack generally refers to an attack leading to a leakage of information by measuring physical parameters. Cache side-channel attack involves exploiting the access timing differences introduced by caches [8]. The meltdown attack fundamentally relies on the Flush+Reload attack [9], in which a memory location is constantly flushed(i.e., being removed from cache) and reloaded into cache. By measuring the time taken to reload the data, the attacker can check whether the data has been loaded into the cache by another process or not. Cache side-channel attack is used to create a covert channel, enabling attackers to leak data and bypass any boundaries.

For instance, consider the following program [6]:

```
// FLUSH the array from the CPU cache
for(i=0; i<ARRAY_SIZE; i++) _mm_clflush(&array[i+4096]);

// Access some of the array items
array[3*4096] = 100;
array[70*4096] = 200;
```

After executing the code above, two elements in the array are loaded into the cache, leading to a significantly lower access time, as illustrated in Fig.3.

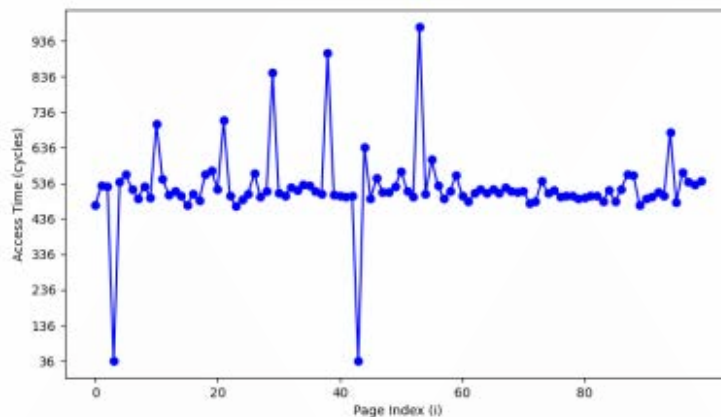


Fig. 3. The access time of each page stored in an array. The access time of page 3 and 70 is significantly lower due to the cache. The result is obtained with Intel Core Ultra 7 155H.

D. Transient Instructions

Out-of-order execution enables the CPU to run instructions in advance. However, the processor can end up running instructions that will never fully execute. For instance, consider the following code:

```
throw runtime_error("exception");

a = mem[1024];
```

The control flow will never reach the second line, because the first line always raises an exception, immediately redirecting the control flow. Nevertheless, the CPU might still execute the second line. Instructions that are always executed out of order and leave side effects (In this case, the second instruction brings `mem[1024]` to cache) are called transient instructions. Similarly, we can turn instructions into transient instructions by exploiting the CPU's branch prediction feature, in which the CPU guesses which one of the two branches will be executed, and runs them in advance. Although the results of out-of-order instructions are discarded if the CPU picks the wrong branch, the execution still leaves side effects. Therefore, if we can ensure the CPU never jumps to one branch but always executes the instructions in the branch speculatively, a sequence of transient instructions is created.

3. Meltdown attack

The meltdown attack involves two parts: executing transient instructions and building a covert channel. In this section, I describe the approach to conducting the attack.

The goal of a meltdown attack is to obtain data stored in kernel space. As described in section II-A, this area is inaccessible to user-mode processes, so an exception will arise after memory access instructions are executed. The attacker can resolve the exception by handling or suppressing it. The attacker can set up a signal handler or put the memory access in a transient instruction sequence, which completely suppresses the exception because the transient instructions are not retired (so no chance to raise an exception).

Afterwards, we need to set up a covert channel to transmit the data obtained by using a cache side-channel attack. Specifically, a probe array is set up and constantly flushed and reloaded. The sending side loads a particular location in the array into cache, based on the value of the data transmitted. By monitoring which part of the array is loaded into the cache, the receiver can decode the data.

In practice, the core mechanism of the meltdown attack can be implemented with the following instructions:

```

; al stands for the least significant byte of rax
; rcx stores the address of data to be obtained from kernel memory space
; rbx stores the base address of the probe array
1 mov al, byte [rcx]
2 shl rax, 0xc ; multiply rax by page size
3 mov rbx, qword [rbx + rax] ; read from the probe array based on the value of rcx
  
```

In line 1, a byte is copied from the kernel virtual memory to a register, and this instruction is broken down to several μ ops to execute. Although the memory access is invalid, the exception is not served until the instruction retires. In the meantime, it is likely that the following instructions have already been decomposed to μ ops and are waiting for the content of rax to be ready. As soon as the content from memory arrives, subsequent μ ops start to execute, and line 3 ensures the memory location [rbx + rax] is cached by copying 8 bytes to another register. Essentially, we utilize the time window between μ ops starting to execute and the exception diverting the control flow and flushing the pipeline (i.e., before the first instruction retires), exploiting the exception handling mechanism in out-of-order execution, illustrated by Fig.4.

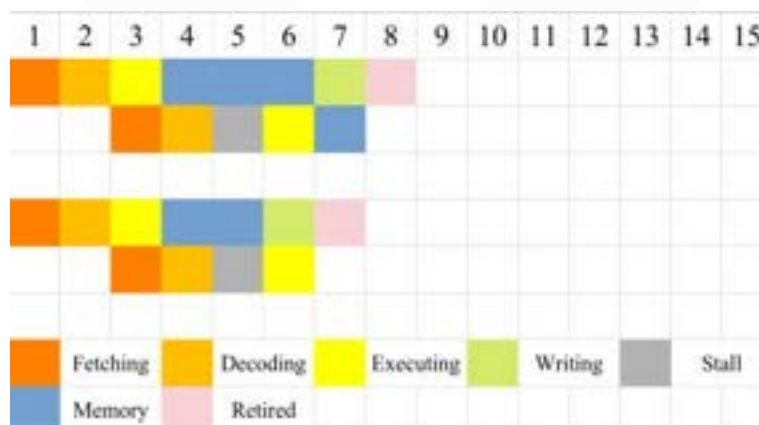


Fig. 4. An ideal model illustrating the μ ops sequences of two MOV instructions. The upper case represents when our attack wins the race. In clock cycle 6, the content of memory is fetched to rax. In clock cycle 7, the last MOV starts to fetch a page

from the probe array, performing side-channel attack. In clock cycle 8, the exception is handled and the pipeline is flushed, but the page has already been loaded into the cache. The two rows below illustrate when we lose the race, because the first MOV retires before the page from the probe array is fetched.

Notice that in line 2, the content of `rax` is multiplied by page size (4KB) to ensure that two adjacent locations that are accessed are far enough apart. This prevents the CPU from prefetching adjacent pages into the cache. Moreover, there is no guarantee that the reading from the probe array instruction wins, so this process might be repeated to improve its success rate and accuracy.

4.Impact

The meltdown attack allows processes in user mode to access an arbitrary location in kernel virtual memory. Thus, the attacker also has access to physical memory, as described in section II-A. With the aid of kernel data structures like page tables, the attacker can access the memory of other processes stored in physical memory. For instance, the attacker can find a process's virtual address space layout in `mm_struct`, and look up the corresponding physical address in page tables.

The Institute of Information Security (ISEC) demonstrated that the meltdown attack can be used to spy on password input, leak physical memory content, and reconstruct a photo from memory [10]. This significantly threatens the confidentiality and secrecy of data. In practice, the meltdown attack can dump kernel or physical memory with 3.2 KB/s to 503 KB/s, depending on specific parameters, and this means critical data like RSA private key can be leaked within 0.1 seconds. On cloud servers, this can lead to hazardous consequences as many cloud service providers rely on containers to provide separation of resources, while these containers may be running on the same physical server [11]. As the attacker has access to physical memory and the kernel is shared among all containers, isolation techniques, including namespace, will not offer protection. Therefore, the attacker can read other clients' secret data and runtime status, violating data privacy.

5.Countermeasures

In previous sections, I have demonstrated that the meltdown attack depends on the layout of virtual memory, transient instruction execution, timing differences due to cache, and a race condition to work. Therefore, the attack can be defended by breaking any of these conditions.

Proposed by Gruss et al. [12], KAISER was originally used to mitigate side-channel attacks by not mapping kernel memory into user space, and it was implemented in Linux as kernel page-table isolation. As shown in Fig.5, KAISER creates two sets of page tables, the one for user-mode contains minimal information about the entry and exit points of, while the one for kernel consists of the complete mapping table. When a process enters kernel mode via system calls, the page tables are switched to the full ones. [13]

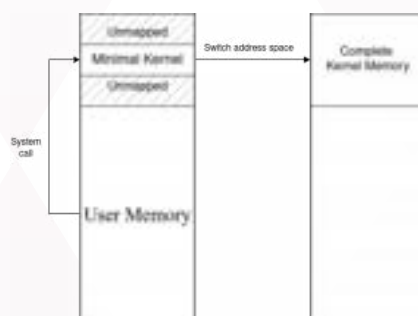


Fig. 5. The KAISER mechanism, preventing accessing kernel memory in user mode.

Despite blocking the meltdown attack to a large extent, it has some limitations because of the implementation of the x86 architecture and performance overhead. Meanwhile, major CPU vendors introduced updates to microcodes to mitigate a variant of the meltdown that leaks the contents of inaccessible system registers. Most defenses attempt to either enforce

access control on the microarchitectural level or suppress the exception (i.e., the illegal instruction returns valid dummy data instead of raising an exception). Nevertheless, the defenses mentioned above cannot completely block all variants of the meltdown attack, and most of them introduce a significant performance penalty, according to Canella et al. [14]

6. Conclusion

In this essay, I explained the meltdown attack, which enables unprivileged processes to leak data stored in kernel memory. I also discussed its hazardous impact on personal users and cloud service providers. Finally, I examined some defenses against the meltdown and addressed their flaws. Therefore, I stress that future defenses should be tested thoroughly on all variants of the meltdown and focus on minimizing performance cost.

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Human Factor in Aerospace Cybersecurity: A Critical Risk and Its Mitigation

ZHENG TANG LI

Abstract

The aerospace industry is characterized by its sensitivity and complexity, with the rising reliance on digital and network connections making cybersecurity essential for routine operations and maintenance. However, nearly 75% of incidents or accidents in the global aviation industry are caused by human error. This paper introduces two human factor frameworks: The HFACS model and SHELL model to analyze a real-world case-- The Qantas Data Breaching. And suggests several specific mitigation and solution strategies with expectations for future developments in this industry.

Introduction

Cybersecurity is increasingly critical in modern aerospace industries where navigation, flight control and communication systems rely heavily on digital and network technologies. Failures in these domains can lead to severe consequences ranging from data breaches to disruption of airport and or even safety-critical incidents. Among the various vulnerabilities, human factors—such as credential misuse, lack of cybersecurity awareness, and procedural lapses—contribute to 70%-80% of occupational accidents in both military and civil aviation [1]. This paper investigates how human errors contribute to cybersecurity vulnerabilities in the aerospace industry and explores mitigation strategies by identifying security requirements, threat actors, and security policies.

Human Factor in Cybersecurity

In 2024, Continental Aerospace Technologies was attacked by the PLAY ransomware, which led to the encryption of critical assets, including manufacturing schematics and employee information. Human factors error frameworks have been developed and refined to identify and mitigate risks. As digitalization rises across the industry, these models expand more application in cybersecurity context.

Human Factors Analysis and Classification System (HFACS) was initially developed from Reason's (1990) "Swiss-cheese" model of latent and active failures. The HFACS model classifies human error into four hierarchical levels:

- (1). Organizational influences,
- (2). Unsafe supervision,
- (3). Preconditions for unsafe acts,
- (4). Unsafe acts of operators

These layers form a pyramid (see Fig 1) which shows that failures in the upper layer propagate downward to erode safety barriers. HFACS-Cyber is extended from the original model to identify cybersecurity vulnerabilities within a system. For instance, insufficient security training in phishing email recognition can be classified in the Organizational influences level. Unsafe supervision includes failures in supervising third-party access permissions. Job stress of employees and misleading user interface design fall under Preconditions for unsafe acts. Unsafe acts of operators such as misclick of phishing email and misconfiguration.

Research indicates that the dominant causal factor in HFACS framework are physical environment and skill-based errors



Fig 1. Pyramid structure of HFACS framework

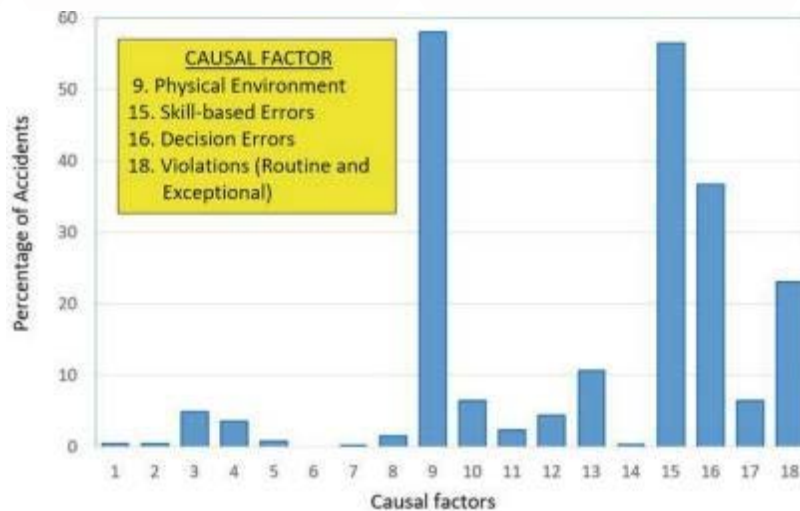


Fig 2. Example of HFACS analysis applied to 1020 commercial aviation accidents[7].

with similar percentages near 56% – 58% [7]. On the other hand, difficulty in identifying complex interaction between layers and potential for oversimplification cannot be ignored in practical research.

The SHELL model is another framework that helps to study interactions between human and other systematic components and how mismatches between these elements can lead to human error. It comprises four components:

- Software (e.g. flight control program),
- Hardware (e.g. aircraft, machines, digital devices),
- Environment: both internal and external environment such as airport,
- Liveware: Human, including individuals and group (e.g. the pilots)

The SHELL model emphasizes that human error typically does not result from humans alone but from interaction between human and other components of the system (e.g. Liveware-Software). This general view is fundamental in analyzing how incompetence of human-system interfaces (e.g. poorly designed procedures) can undermine cybersecurity performance in aerospace settings [3]. For example, the absence of measures preventing unauthorized devices from accessing physical ports (e.g. USB) is the failure in Liveware-Hardware component which can lead to malware penetration. Research in Indonesia shows that mismatches between liveware and liveware are a prevalent cause of aviation incidents and accidents.

[4]. Limitation of the SHELL model lacks considerations beyond the scope of liveware-centered interaction.

Both frameworks are designed to achieve effective human factor analysis. The HFACS focus on identifying the paths of systematic failures. The SHELL model emphasizes match of cognition and interface and is suitable for research on



Fig 3 SHELL model and the interfaces

combination of mistake in human decision and defect in technical interfaces. The following content of this paper will use these two models to analyze specific real-world aerospace cybersecurity cases and plan a feasible mitigation. Nevertheless, both models have disadvantages including lack of quantitative analysis and broad categories and it is worth noting that two models are designed for reaction rather than proaction of incidents, they focus on problem recognition rather than resolution.

Case Study: Qantas Social Engineering Attack

On June 30, 2025, Australian Qantas Airlines experienced a cyberattack on a third-party call center, resulting in a data breach affecting approximately 6 million customers. Personal information, including names, email addresses, phone numbers, birthdates and frequent flyer numbers is stolen [5].

Attackers used vishing — voice phishing — to impersonate an employee and gain access to a customer support platform [6]. In this attack, the human vulnerability exploited was the trust and urgency bias inherent in the call-center workflow. The security requirement in this case should include standardized identity verification process, Multi-Factor Authentication (MFA), detection of abnormal behaviors and layered security models.

The threat actor is suspected to be Scattered Spider ransomware group who employs social engineering techniques such as vishing (voice phishing) [6].

By introducing two previously mentioned human factor models, the human errors in this incident can be identified clearly.

Table 1 HFACS Analysis of Qantas Social Engineering Attack

Layer	Reason
Organizational Influences	Lack of social engineering response and defense method and insufficient helpdesk training.
Unsafe Supervision	Absence of systematic re-check and report mechanisms.
Precondition for Unsafe Acts	Strain and stress of helpdesk agents, speed-oriented procedure omission.
Unsafe Acts	Helpdesk agent grantee access request from unknown phone call before identity verification.

Table 2 SHELL Analysis of Qantas Social Engineering Attack

SHELL Interfaces	Reason
Liveware-Software	Lack of security warning in platform UI (User Interface).
Liveware-Environment	High-Stress environment in call-center increases possibility of occurrences of human errors
Liveware-Liveware	Ineffective team communication and insufficient training.

Discussion and Reflection

Mitigation strategies and solutions for the Qantas case including implementing strengthened third-party governance and rearrange call center to trusted local locations so it can be integrated into regulatory supervision and supply chain control. To address unsafe supervision in the liveware-environment dimension, introduce multi-level approval for access changes and conduct cybersecurity simulation drills. In preconditions and liveware-software part, enhancing UI with embedded security notification and stress resistance training are feasible solution. In the unsafe acts level, detection and prevention of anomalous access patterns and dual-authorization for high-risk requests are required. Improved communication procedures between staff and third parties are solutions at the liveware-liveware level.

The case of Qantas reveals that human factor vulnerabilities still exist in well-established aviation system. In current developments, introducing automation and AI-based control seems to be a trend. Although this implementation can reduce the number of human decision points, the complete removal of human participation is neither practical nor desirable. Human oversight and training are essential in responding to unpredictable AI or automatic system failure or manipulation. Hybrid approaches such as strengthened Human-AI collaboration and dynamic switch of control permission should be the future of aerospace industrial upgrade. Hybrid systems have demonstrated a 54% reduction in errors compared to traditional methods [2]. Fig 4 also shows that the proportion of human-factor-related incidents keeps declining from 1970 to 2020 [2].

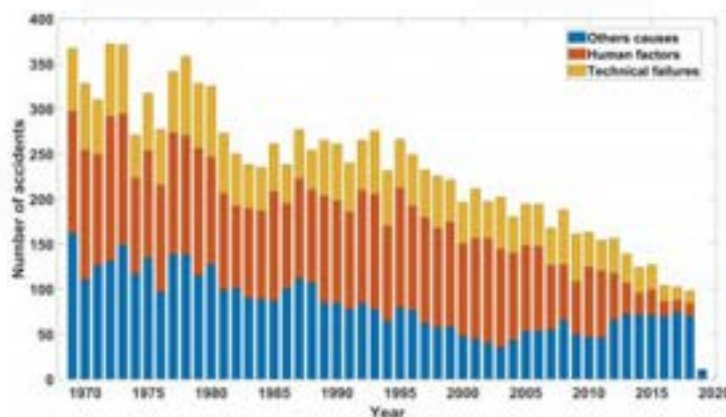


Fig 4. Aeronautical accidents from 1969 to 2019 with human factors, technical failure and others causes as contributory factors. These data were retrieved from Bureau of Aircraft Accidents Archives (www.baaa-acro.com)[8].

Conclusion

Through application of frameworks such as HFACS-Cyber and SHELL, human factor risks can be identified and mitigated by organization, while these frameworks are effective, their limitations regarding quantitative analysis must also be acknowledged. The Qantas incident demonstrates the severe consequences of failures across multiple layers. Effective cybersecurity requires not only technological solutions but also human-centered interventions. Although AI and automation

is developing, human factor will remain significant in future aerospace industry especially in cybersecurity issues.

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17

Mathematical Economics

Optimal Category of Energy under Uncertainty of Technological Choice

FANGYE JIANG

1.Introduction

Each era of humanity has its own representative creation. When 1840 came, the improvement of steam engine represents a long time period named as The First Industrial Revolution, the time that machines started to replace the traditional labor and capitals like livestock. As long as The Second Industrial Revolution started, the creation of light bulbs and the application of electrical energy changed people's life. Nowadays with the climate change and the scarcity of fossil fuel, the new types of energy are talked frequently. Policies about energy are published, such as the goal 7 in the 17 Sustainable Development Goals (SDGs) by the United Nations Department of Economic and Social Affairs, which aims to make sure it is accessible for all people to have affordable, reliable, sustainable and modern energy (United Nations Department of Economic and Social Affairs, 2015).

To firms, the climate change is not only a crisis, but also an opportunity. Because the trend of using new types of energy can inspire companies about new products and motivate them to grab the burgeoning market. However, sometimes the chance means challenges, not all the opportunities are suitable and low-risk to a certain firm to develop. The estimation of the risk under uncertainty is crucial for entrepreneurs to trade of and make decisions.

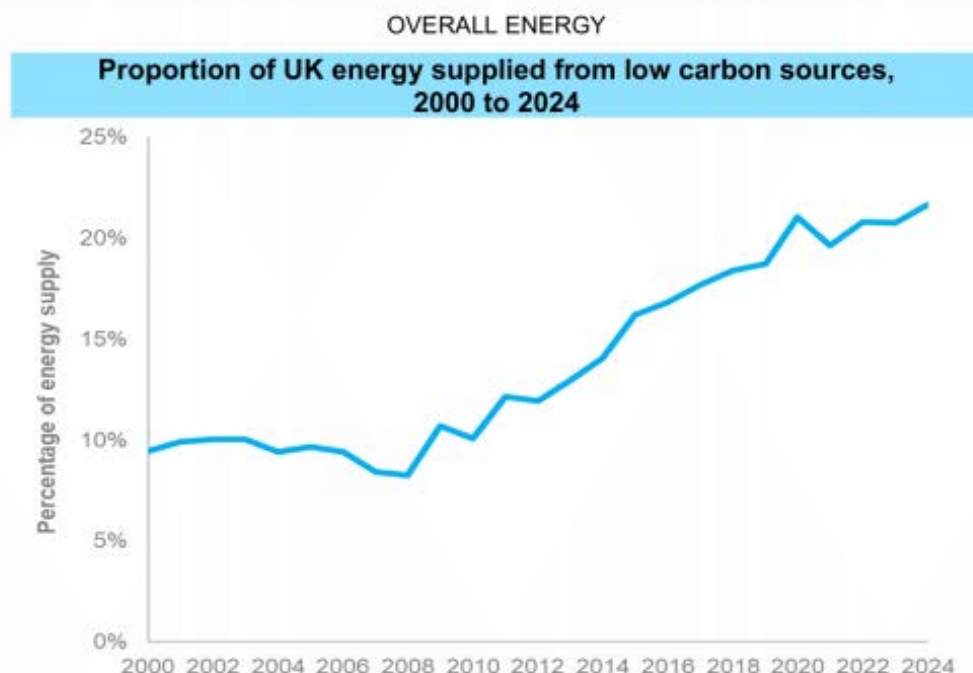
This research lists several variables and does an analysis to estimate if it is worth to choose a certain technique.

2.Literature Review

To do estimation, reliable data are necessary. Data will offer a sense of the situation of energy in the country and will be of help to do analysis.

2.1 New Energy Resource

Today, numerous new types of energy have been developed. To ease the analysis, we consider the market is in the United Kingdom (UK) and analyze the risk under the uncertainty of future energy trend for the firm.



	Percentage					
	2000	2010	2020	2022	2023	2024
Nuclear	8.4%	6.3%	6.5%	6.1%	5.4%	5.4%
Wind	0.0%	0.4%	3.9%	4.1%	4.3%	4.4%
Solar	0.0%	0.0%	0.7%	0.7%	0.8%	0.8%
Hydro	0.2%	0.1%	0.4%	0.3%	0.3%	0.3%
Bioenergy	0.9%	2.3%	7.4%	7.1%	7.2%	8.0%
Transport fuels	0.0%	0.6%	1.1%	1.4%	1.6%	1.7%
Heat pumps	0.0%	0.4%	0.7%	0.7%	0.8%	0.8%
Other	0.0%	0.0%	0.3%	0.3%	0.4%	0.4%
Total	9.4%	10.1%	21.0%	20.8%	20.8%	21.7%

Figure 1: Department for Energy Security and Net Zero. (2025). Proportion of UK energy supplied from low carbon sources, 2000 to 2024[Figure]. GOV.UK. <https://www.gov.uk/government/collections/uk-energy-in-brief>

According to the Department for Energy Security & Net Zero (DESNZ) (2025), the percentage of energy supplied which is from low carbon source in period between 2000 and 2024 did not change in a large scale from 2000 to 2008. From 2008 to 2024, this percentage increased. In detail, except the nuclear energy, which had a decreasing proportion of energy supply, other kinds of low carbon energy source including wind energy, Solar energy, hydrogen energy, bioenergy, transport fuels and other energies had a rise in the percentage. The total proportion of the low carbon energy source also increased from only 9.4% in 2000 to 21.7% in 2024, which is beyond one fifth of all energy categories. This result clearly shows the progress in the new types of energy besides traditional fossil fuel, meanwhile, it reflects the rising demand of new energy resources and a potential market for companies indirectly.

3.Data Analysis

Nowadays, there are numerous tools for the firm to adjust risk and make decision. Some of these tools will be used to estimate the risk and potential in energy categories.

3.1 Energy Categories and Standard

To do further analysis, energy categories and different aspects of energy to analyze need to be confirmed. The categories will be Offshore wind energy, Solar energy, and Hydrogen energy. In this research, aspects been considered will be Cost, Technology Readiness Level (TRL), and Proportion in Renewable Energy Source.

3.2 Cost

For the Levelized Cost of Energy (LCOE) of offshore wind energy in 2016 in the United Kingdom, it was £97/MWh (Duan, 2017). The cost for solar energy for a large scale photovoltaic system in 2021 is £51/MWh (Mandys, Chitnis, & Silva, 2023). The estimated LCOE of hydrogen energy in H-Class Combined Cycle Gas Turbine (CCGT) in 2025 is £114/MWh (DESNZ, 2023).

Use the equation:

$$\text{Score} = \frac{\text{Maximum Cost} - \text{Cost}}{\text{Maximum Cost} - \text{Minimum Cost}} \times 100$$

to calculate the score in the aspect of cost, the higher score is better.

(1)

3.3 Technology Readiness Level (TRL)

According to Big Bang Partnership (n.d.), the fixed-bottom turbine for offshore wind energy which is commercial technology is TRL 9. The TRL for Photovoltaics is 9 and the deployment of green hydrogen equipment for site services is 5-7 (consider it as TRL 7) (North Sea Transition Authority, 2024). The equation:

$$\text{Score} = \text{TRL Level} \times 10 \quad (2)$$

is used to calculate the score for TRL.

3.4 Proportion in Renewable Energy Source

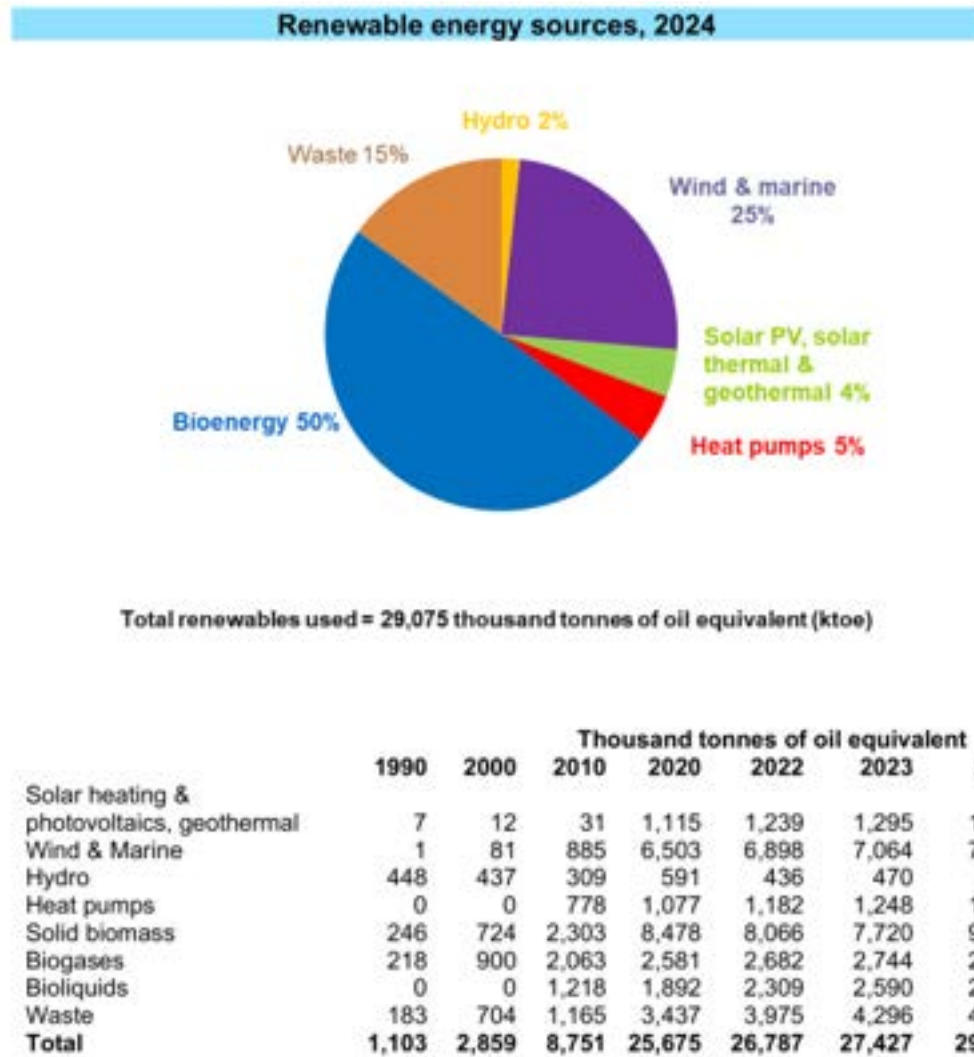


Figure 2: Department for Energy Security and Net Zero. (2025). Renewable energy sources, 2024[Figure]. GOV.UK. <https://www.gov.uk/government/collections/uk-energy-in-brief>

According to the data in Figure 2, it is able to know different proportion of energy in renewable energy source. The equation:

$$\text{Score} = (1 - \text{Proportion of Energy in Renewable Energy Source}) \times 100 \quad (3)$$

to calculate the score of each category of energy to make decision. Because the higher the proportion, the harder it is to enter the market, using 1 to minus the proportion then multiplying 100 make that the higher the score, the easier it is to enter the market.

3.5 Coefficient of Variation

Using data from subsection 3.1 to 3.4, a table can be shown as:

	Offshore Wind Energy	Solar Energy	Hydrogen energy
Cost Score	26.98	100.00	0.00
TRL Score	90.00	90.00	70.00
Proportion Score	75.00	96.00	98.00

Figure 3

The coefficient of variation is calculated as the weight for each category using equations:

$$\bar{x}_j = \frac{1}{m} \sum_{i=1}^m x_{ij} \quad (4)$$

$$\sigma_j = \sqrt{\frac{1}{m} \sum_{i=1}^m (x_{ij} - \bar{x}_j)^2} \quad (5)$$

$$v_j = \frac{\sigma_j}{|\bar{x}_j|} \quad (v_j = 0 \text{ when } \bar{x}_j = 0) \quad (6)$$

$$w_j = \frac{v_j}{\sum_{k=1}^n v_k} \quad (7)$$

where x_{ij} is the original value of score No.j of energy category No.i, $\bar{x}_j$ is the average value of score No.j, σ_j is the standard deviation of score No.j, v_j is the coefficient of variation of score No.j, and w_j is the weight of score No.j.

3.6 Result

Use the coefficient of variation as weight, a table can be shown as:

	Weight	Offshore Wind Energy	Solar Energy	Hydrogen energy
Weighted Cost Score	0.8134	21.95	81.34	0.00
Weighted TRL Score	0.09226	8.303	8.303	6.458
Weighted Proportion Score	0.09434	7.076	9.057	9.245
Total Score		37.33	98.70	15.70

Figure 4

4. Conclusion

Compare the total score, $98.70 > 37.33 > 15.70$. So the Solar Energy will be the best choice.

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Optimal Balance Between Consumption and Investment

ABDULLAH AL MUHID

Introduction

What is consumption and investment? Consumption is defined as “The action of using up a resource.” But in economics it is “The use of goods and services by a household”. Investment is defined as “an asset or property acquired to generate income or gain appreciation”. What we will learn is how to balance these two, so that we can still have fun and live buying things we want to buy but also at the same time building wealth for the future and compounding your investments in index funds and other financial assets that give steady returns. Additionally we will learn how to budget money for comfort now and financial freedom later.

Body

Balancing consumption and investments is one of the most important financial habits one can develop. If you spend your whole paycheck now, you will enjoy the money now and short term pleasures but you are risking having no savings for emergencies, no passive income, and no security later in your life. On the flip side, if you invest every single penny, you will suffer and burn out.

Having the right balance helps you live well now without sacrificing your future later. It allows you to build wealth slowly and steadily, while you are enjoying your present, watching your investments compound.

Budgeting: The Foundation Of Financial Balance

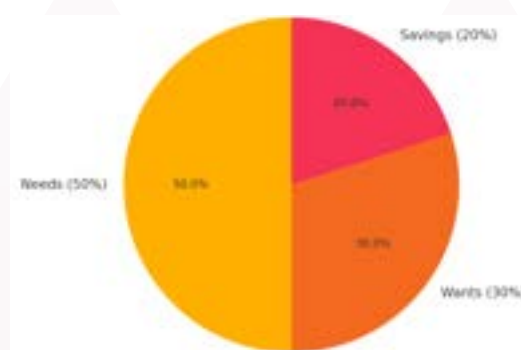
The first step to balancing your finances is learning how to budget. A popularized method is the 50/30/20 rule. It breaks down your income as follows:

- 50% for needs like (rent, food, bills, etc...)
- 30% for wants like (shopping, entertainment, hobbies, etc...)
- 20% for savings and investments

This rule is simple and flexible to fit to different people's needs for example if you live in a high cost of living area you could raise it to 60/30/10 or something else that fits your lifestyle. What matters mostly is tracking your expenditure and prioritize saving or investing 15-20% of your income for the future.

There are many apps that help with this like Mint, YNAB, or even a simple spreadsheet can help you monitor your expenditure and stick to your plan.

This graph helps you visualize it:





Understanding Your Investment: Making Money Work For You

Once your budgeting and saving consistently, (you have at least 3 months of living expenses in a savings account that gives you returns), it comes the time when you have to start investing, investing means putting your money into assets that will grow over time. These assets include,

- index funds (which tracks the stock market) like the S&P 500
- ETF's (exchange traded funds)
- stocks and bonds
- real estate
- retirement accounts (like IRA's or pensions)

The greatest advantage to investing like this is compound interest. This means your money gets interest, and then that interest earns more interest and so on. For example, if you invest 1,000\$ in a index fund that averages 7% interest per year, it doubles roughly every 10 year without you doing anything to it.

This means the earlier you start the more time your money has to grow. That's why starting small and now is better than waiting until you have more money later.

The One Third Rule: A Strategy To Follow

One powerful strategy/rule backed by research is the one third rule. It says to divide your income into thirds: One third for essentials, one third for debt payments or future investments, one third for enjoyment and wants.

This strategy works well for lots of people because it is flexible, and easy to remember and encourages balance between fun, responsibility, and long term financial growth. It also helps avoid financial stress.

Behavioral Tricks To Help You Stay Consistent

Managing money isn't about math and numbers only but also about behavior. Even if you know how to budget and invest, emotions can get in the way of them. States below are a few techniques that help:

- Mental Accounting: Create different accounts for spending, saving, and investing. This makes it easier to resist overspending.
- Automation: Set up automatic transfers into savings and investment accounts right after you get your paycheck.
- Save more: Commit now to increase your savings rate with every future raise. This lets your savings add up.

These strategies help in building discipline without needing great willpower.

A Real Life Example: Two Different Paths

Let's imagine 2 people:

The first person, Person A spends most of their money on clothes, food, tech, and rarely saves. They say they will invest "later on".

The second person, Person B also enjoys life but follows one of the rules stated before. They invest monthly into an index fund like the S&P 500 and build an emergency fund.

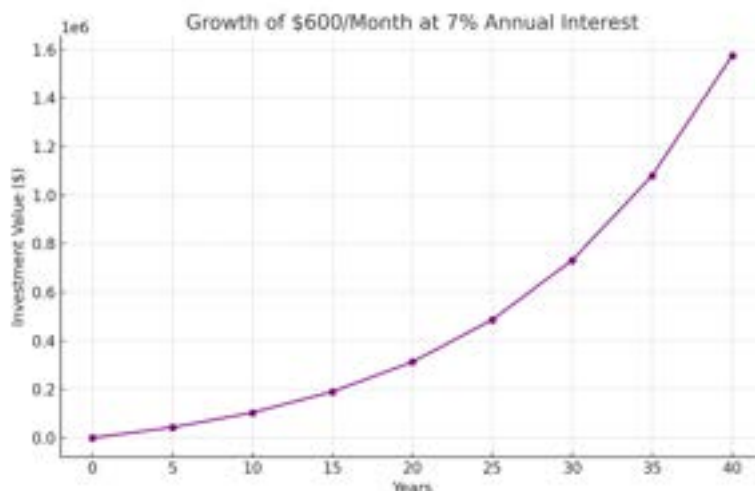
After 10 years, Person B would have a portfolio growing in value. Person A would have nothing saved. The difference between them? Person B found a balance and stuck with it.

Adjust As Life Changes

Balance in finance is not a one time decision but a habit. As you grow your income or your expenses change, your budget should adapt too. You have to check in with your goals each month and hold yourself accountable. Are you saving enough? Are you enjoying yourself? Are you investing enough? If not, adjust everything, the best plans to make are ones you stick to long term not one that feels like a punishment.

Theoretical Example

According to Josephson, Amelia, 24 Feb. 2025, the average salary for a young adult in the United States of America, is 3,397\$ per month. Theoretically, if a young adult were to invest a little less than 20% of their salary which comes to around 600\$ per month into an index fund with an annual return of 7% this is how it would look like.



The graph shows a young adult investing 600\$ per month into an index fund that returns 7% annually.

In forty years of investing less than 20% of their paycheck, they would have almost 1.6 million dollars. That's if they stayed at the same job, with the same exact salary, for all that time which is unrealistic, since they would get raises and invest even more.

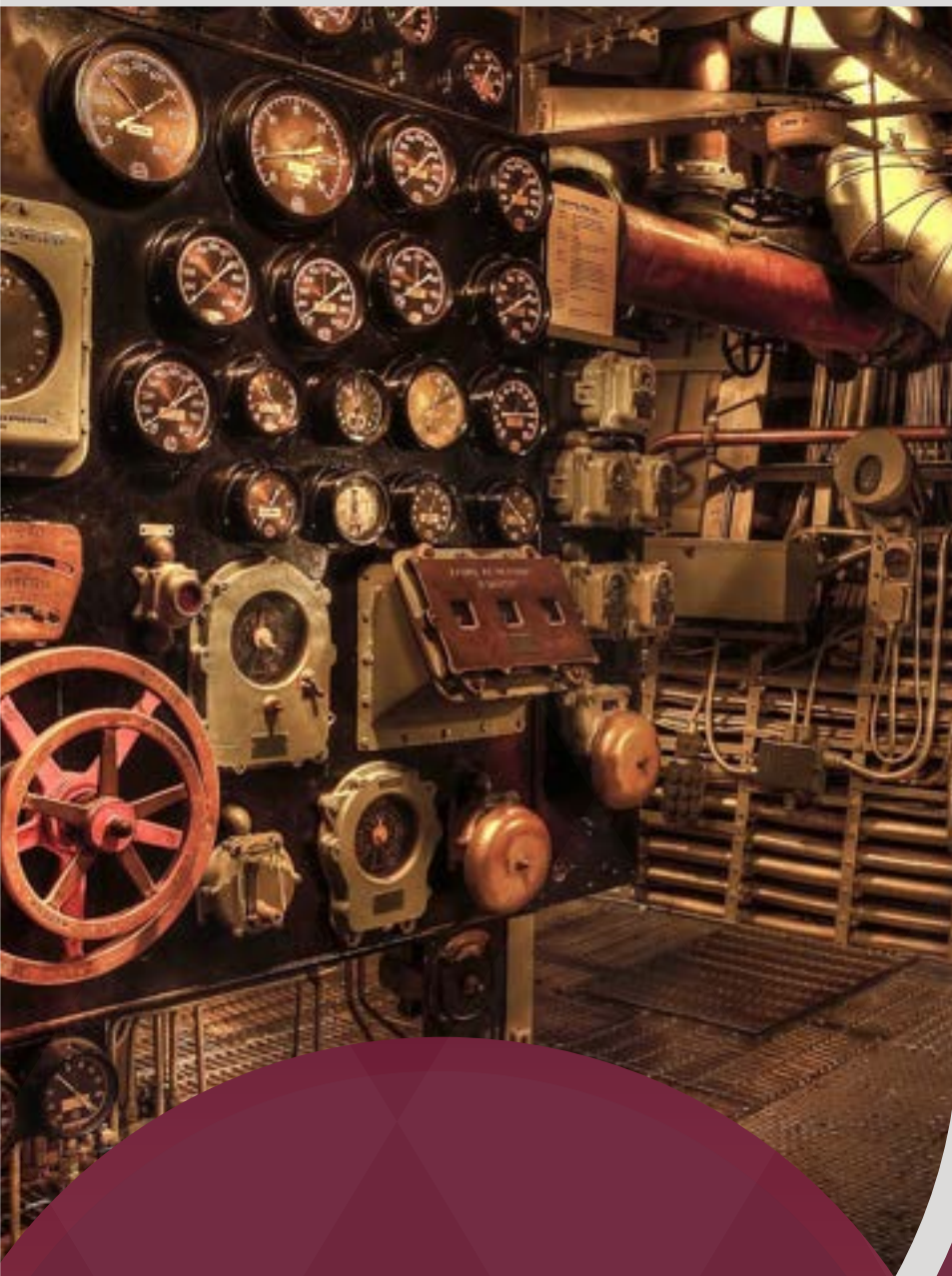
Conclusion

Finding the right balance between consumption and investment is about living intentionally. It means enjoying your money now and still preparing for the future. With a smart budget and consistent investing, even if your investing in small amounts, you can build wealth, reduce stress about the future and overall stress, and reach financial independence.

Whether you're a student or just starting your first job, it is never too early to learn these things, since the earlier you start the more time your money gets to compound. You can start with a basic budget open a simple account at a bank, and automate everything. Overtime it will all compound and grow.

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18

Engineering

Challenges of Supersonic Air Travel for Passenger Aircraft

ZHIYING LU

Abstract

The retirement of the Concorde in 2003 marked the end of an era for commercial supersonic passenger travel, leaving unresolved technical and economic challenges that continue to constrain its revival. This paper systematically examines the key barriers facing next-generation supersonic passenger aircraft, such as the sonic boom, supersonic engine efficiency, sustainability, and aircraft materials. The feasibility of potential solutions, including advanced composites, optimised aerodynamic designs, and new propulsion systems are also discussed. It is found that using variable sweep wings can help to mitigate sonic boom, whilst applying programmed lapse rate and optimised flaps and slats use can increase engine efficiency and sustainability. Sustainable aviation fuels will make supersonic travel more sustainable, with advanced composites seen as the appropriate materials for engine components and the fuselage, helping improve efficiency, though such advances still need to be optimised prior to application.

1. Introduction

Supersonic air travel has long been regarded as a revolutionary advancement in global transportation, as it greatly reduces flight time. Concorde once achieved this goal, flying at twice the speed of sound and reducing transatlantic journeys to just 3.5 hours. However, its retirement in 2003, caused by restrictions on sonic booms and limitations on accessibility, exposed crucial flaws in modern supersonic air travel. Inspired by the Concorde programme, this paper aims to identify the challenges faced by passenger aircraft travelling faster than the speed of sound, in other words, at supersonic speeds. By reviewing existing literature, this paper summarises the obstacles to supersonic transport (SST). From the destructive noise of sonic booms to economic viability and environmental concerns, it aims to explore a new path towards future SSTs that aligns supersonic speed with today's demands for accessibility, sustainability, and efficiency.

2. Results

2.1 Sonic Boom Noise

In supersonic aviation, the speed of aircraft exceeds the speed of sound, faster than 1236km/h at sea level, with a Mach number greater than 1. As the sound barrier is broken, a pressure wave is produced. Each successive pressure wave produced by the aircraft interferes with a previous one and merge into a single shock wave, creating a "Mach cone" that eventually causes a sonic boom.

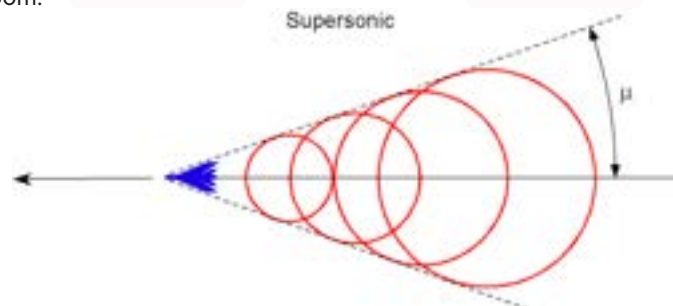


Figure 1: Wavefronts when Source moves at Supersonic Speed. Source: Embry-Riddle Aeronautical University

Based on Henne's analysis [1], the sonic boom was catastrophic for the Concorde's commercial viability. Producing pronounced N-wave boom signature with about 90 PLdB (Perceived Level in decibels) intensity at a speed of Mach 2. The Anglo-French Concorde SST caused widespread disturbances, breaking windows, affecting populated areas and damaging

the ecology.

Consequently, as Candle's analysis on Concorde and the future of supersonic transport [2] pointed out, both international and regional laws began to restrict SST's legality. The International Civil Aviation Organization (ICAO), the Federal Aviation Administration (FAA), and the European Union Aviation Safety Agency (EASA) all set up legislation that banned SSTs travelling over land. Henne stated that 75% of Concorde's typical small-civil-jet operations occur over land, so such bans considerably limit its market and economic viability, leading to its retirement in 2003. Henne identified several solutions to limit the intensity and character of sonic booms in small supersonic civil aircraft. Principally, by applying a variable wing sweep geometry. The core concept of this is to shape a more favorable sonic boom signature while improving the ability to optimise aerodynamic performance across diverse flight regions, balance high-speed fuel efficiency and low-speed operational needs.



Figure 2: Variable Sweep Wing, Gulfstream Aerospace

During take-off and low-speed flight, the wings are set to a minimum sweep angle, which increases their aspect ratio and enhances lift generation to meet take-off field length requirements and improve subsonic efficiency. Meanwhile, the broader wing surface reduces induced drag, enabling quieter and more efficient airport operations.

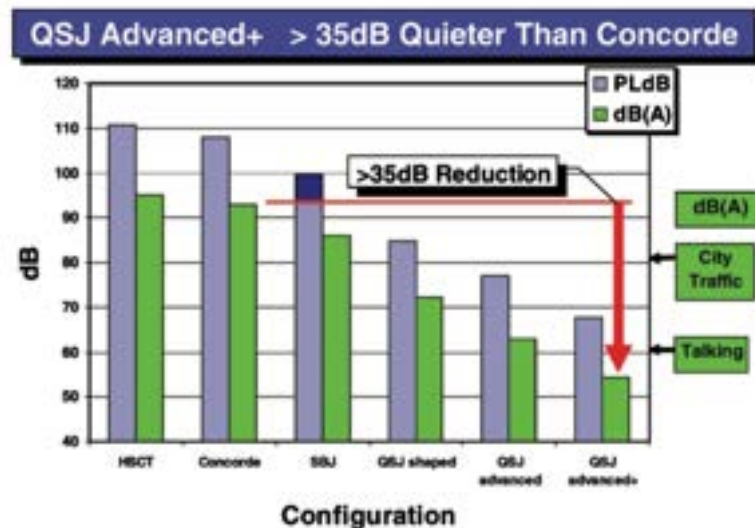


Figure 3: Progress in Lowering Sonic Boom Strength

In contrast, during supersonic cruise, the wings are swept to a steeper angle, minimising wave drag by aligning the wing with supersonic airflow to reduce the strength of shock waves generated at the wingtips and leading edges. The optimised sweep angle, combined with specially designed fuselage details, produces a ground signature with a uniform shape and lower initial and final pressure increases, thereby mitigating the noise of the sonic boom due to its closer sinusoidal waveform and lower frequency.

This adaptive design addresses both performance and sonic boom suppression. Figure 3 shows that the Quiet Supersonic Jet (QJ) manufactured can be at least 35dB as quiet as the Concorde.

2.2 Supersonic Engine: Noise and Sustainability

2.2.1 Supersonic Engine Noise

As technology improves, new materials, advanced engines and innovation in aerodynamics can all render civil supersonic flight economically viable. However, as *The Bridge: Linking Engineering and Society* [3] stated, the environmental concern is of equal importance. Being at a speed faster than sound, the shock wave (sonic boom) produces wave drag in return, resulting in a less fuel-efficient engine that requires more fuel to generate thrust. In order to mitigate wave drag, one may use lower-bypass engines to improve fuel efficiency when cruising as well as carefully optimising the cross-sectional area of supersonic engines. However, this produces more noise pollution near to the ground, since engine efficiency depends on the jet exhaust. On the other hand, the paper stated that the computer-managed device, programmed lapse rate (PLR), is able to solve this problem.

Being designed to dynamically manage thrust and reduce engine noise, PLR is normally switched on during critical flight phases, particularly take-off and initial climb, where noise pollution is most regulated. This noise reduction occurs because a lower jet exhaust velocity weakens turbulent shear noise, the main contributor to engine noise. Though climbing time increases slightly, the aircraft is still able to accelerate during the climbing phase and maintain performance at cruise conditions due to the excess thrust released previously.

Furthermore, the paper provides an alternative approach, to vary the wing's flaps and slats with computer-assistance, maximising aircraft performance at its best lift-to-drag ratio. The positions and functions of flaps and slats are shown in Figure 4.

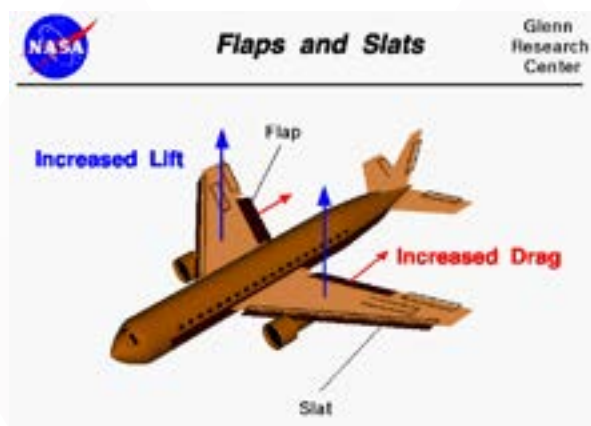


Figure 4: Flaps and Slats on an Aircraft. Source: NASA

2.2.2 Sustainable Aviation Fuel (SAF)

Compared to supersonic fuel, the subsonic fuels are 1.5% more renewable [3]. Research has found that instead of relying largely on liquid hydrocarbons, sustainable aviation fuels (SAF) are the key solution for jet fuels. SAF can reduce emissions from production, transportation by employing renewable sources. Most optimistically, they can achieve a reduction in lifecycle CO₂ up to 80%.

Several companies source SAFs from innovative sources:

- Fulcrum Bio Energy converts municipal waste.

- Lanza Tech uses bacteria to synthesize fuel from industrial waste gases.
- Carbon Engineering captures atmospheric CO₂.

Complementing traditional biofuels, which are produced from crops, algae and forest residues, this new technology ensures that the production of SAF avoids generating fuels from food crops or harmful land use. For example, turning virgin forests into crop land, or growing more crops intentionally.

According to Forbes [4] and Lufthansa [5], SAF is roughly three to five times more expensive than fossil jet fuel, resulting in a low demand and high cost of production, making it currently hard to be profitable. The airline responded that one possible solution is to focus on long-term potential and increase the demand so they can reduce long-term average cost (LRAC) by reaching economies of scale. In general, modern manufacturing techniques must achieve a balance between economic viability and technical feasibility.

2.3 Materials: Engine and Fuselage

The Bridge [3] also discussed the challenges and solutions of materials. Supersonic transport requires structural materials with better temperature capability and chemical resistance in order to enhance thermal efficiency in both the compressor and turbines in the engine.

2.3.1 Materials for Supersonic Engines:

For high temperature components such as turbine blades, combustion chambers and nozzles, Ceramic Matrix Composites (CMCs) are now targeted to reach 1500° C without degradation. Such CMCs have already entered into service at 1300° C. Being a heterogeneous material that consists of a ceramic matrix reinforced with carbon or ceramic fibers, CMC is a perfect solution because of its physical properties. It is particularly stable at high temperatures and in aggressive environments with high strength and low density.

There are still challenges present without solutions. Titanium-Aluminide are applied to manufacture low- and medium-temperature components like compressor blades. Being significantly lighter than traditional nickel-based alloys, it should be used in weight-sensitive components with lower temperatures. Despite the advantages, it is susceptible to corrosion from atmospheric pollution; therefore, constant maintenance is currently inevitable.

2.3.2 Materials for Fuselage Design

Although Concorde's aerodynamic engineers had already understood relative concepts like area ruling, which reduces wave drag and sonic boom by smooth, continuous transitions in cross-sectional areas from nose to tail, they did not manage to take full advantage as aluminium is hard to configure (forming a curved, area-ruled shape) as designed. As a result, a new material is needed to achieve the original objectives. It is found that carbon composites, especially carbon-carbon composites behave to exhibit appropriate physical properties. Being mechanically strong, high-temperature resistant, it can be used for manufacturing more area-ruled fuselage in the following generation.

3. Conclusion

In conclusion, this research analysed major difficulties encountered in recent supersonic transportation for passenger aircraft that have been revealed since the Concorde programme ceased. The challenges include

- Destructive sonic booms that result in legislation bans of supersonic overland air travel.
- Noise of supersonic engines, closely related to its fuel inefficiency and environmental problems

- Material choices for engine components and fuselage.

Each is analysed with possible solutions according to recent technological development using modern SST design concepts. The paper then discusses

- Using variable sweep wings to mitigate sonic boom
- Applying PLR as well as operate flaps and slats to increase engine efficiency and sustainability.
- SAF as renewable fuels despite some possible concerns in the future.
- CMC and carbon-carbon composites as appropriate materials for engine components and fuselage.

While some of them show significant improvement, others, on the contrary, still need to be optimised due to potential drawbacks.

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Are Vacuum Tube Amplifiers Better than Solid-State Amplifiers?

WENAO XIONG

Abstract

Amplifiers are used in a wide array of electronic devices and systems to increase the power, voltage, or current of a signal. They are essential components in audio equipment such as stereos and musical instrument amplifiers, as well as in radio and television transmitters and receivers. There are two major types of amplifier: the vacuum tube amplifier (valve) and the solid-state amplifier (transistor), which are used in different areas due to their differing characteristics. This paper finds that solid-state amplifiers are preferable if accurate, realistic sound is desired, while valve amplifiers are better suited for those seeking a more colourful musical tone. This paper discusses the processes by which valve amplifiers and solid-state amplifiers respectively amplify signals. It then introduces the concept of harmonic distortion present in valve amplifiers, which makes their sound more favourable in musical applications. Issues arising with valve amplifiers are discussed, such as cost and maintenance. This paper shows that hybrid amplifiers, which aim to utilise the advantages of valves while mitigating their drawbacks, appear to be the better option, accommodating the needs of all consumers using a combination of the two technologies.

1. Introduction

Components for audio amplification have continued to develop in recent decades. However, many amplifiers use components that are far from novel. Vacuum tubes (valves) have been used since the first computers, and although their use in other systems has been replaced by transistors, vacuum tubes remain a popular choice in audio amplification systems. For individuals such as guitarists in particular, the right amplifier can make all the difference to one's sound, and the choice between solid-state and vacuum tube amps is not one to be taken lightly. Therefore, in this paper, we shall compare solid-state and valve amplifiers in order to explore the considerations involved in choosing between them.

2. The Principle of amplifiers

2.1 Vacuum tube amplifier

The structure of a vacuum tube is shown in Fig.1. The heater warms the cathode, generating a cloud of negatively charged electrons in a vacuum. A high positive charge at the anode attracts these electrons, creating a current from the cathode to the anode. A negative voltage applied to the control grid repels electrons back towards the cathode. If 100 volts is applied to the top resistor and the control voltage is grounded (0 volts), all electrons flow to the anode. Applying a negative control voltage to the grid reduces electron flow, increasing resistance in the voltage divider. For example, with -1.5 volts on the grid, the tube behaves like a 100,000-ohm resistor, leaving 50 volts at the output. Amplification occurs when raising the grid voltage from -1.5 to 0 volts reduces the tube's resistance to a minimum (e.g., 10,000 ohms), yielding about 9 volts at the output. Lowering the control voltage to -3 volts increases resistance (e.g., 900,000 ohms), raising the output to over 90 volts. Thus, a 3-volt input change causes an 81-volt output change. This is amplification: the vacuum tube transforms a voltage divider into a voltage multiplier.

Fig.1 Components of a vacuum tube

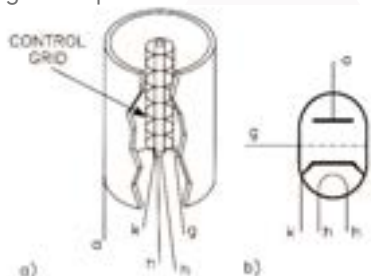
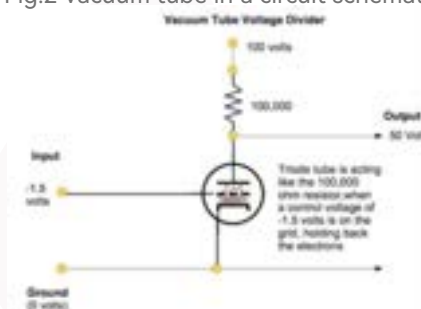


Fig.2 Vacuum tube in a circuit schematic



2.2 Solid-state amplifier

A solid-state amplifier employs transistor circuits to transform an electrical signal into an audio wave. Before any signal is applied, a DC bias voltage is set on the base, forward-biasing the base-emitter junction. This creates a small base current I_B typically in the μA to low mA range. The transistor operates in its active region, where: $I_C = \beta \cdot I_B$. In this equation, β is the current gain. This setup ensures the transistor is ready to amplify small signals linearly.

If one applies a small AC voltage on top of the base bias- this is the audio or signal voltage we want to amplify. The input is:

$$v_{in}(t) = V_0 \cdot \sin(\omega t)$$

Solid-state tubes have an exponential base-emitter characteristic. a small increase in base voltage causes a proportional change in base current ΔI_B :

$$\Delta I_C \approx \beta \cdot \Delta I_B$$

So even a tiny change in I_B creates a much larger change in collector current I_C . The collector current flows through a collector resistor R_C , creating an output voltage:

$$v_{out} = V_{CC} - I_C R_C$$

As I_C increases due to input, the voltage drop across R_C increases, so the output voltage decreases. This means the output signal is inverted (180° out of phase) and larger in amplitude than the input. The transistor modulates the collector current based on the input signal at the base, and this variation creates a larger voltage swing at the output due to Ohm's law across R_C . Thus, a small AC input voltage causes a large variation in collector current, which causes a large AC output voltage across the load.

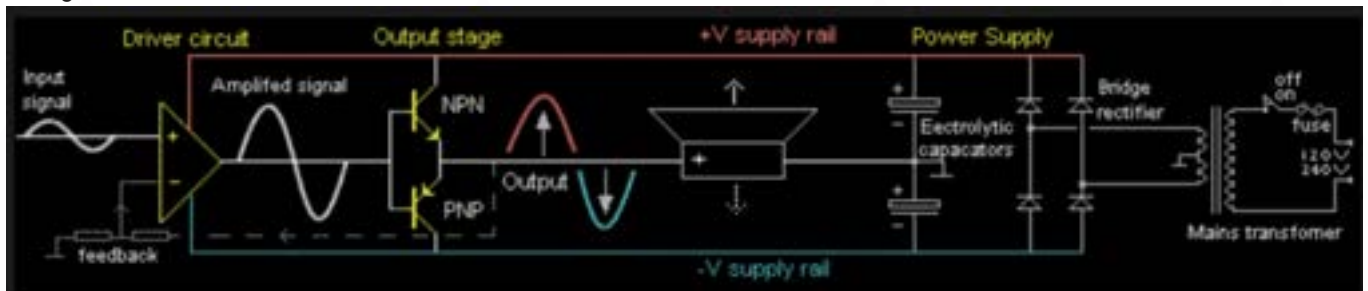


Fig.3 Solid-state tube amplifying circuit schematic, including the steps introduced above and a power supply

3. Harmonic Distortion: the major factor of differences

When evaluating an amplifier, the primary consideration should be how the sound is altered after amplification — a concept known as harmonic distortion. Within a valve, electrons flow from the cathode to the anode, controlled by the voltage applied to the grid. As the input voltage increases, the relationship between the input voltage (at the grid) and the output current (flowing to the anode) is not perfectly linear.

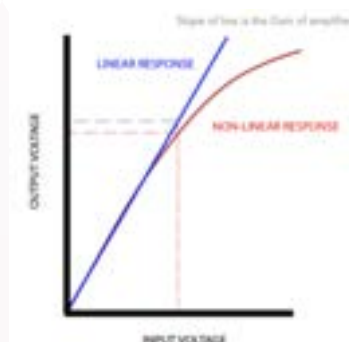


Fig.4 output voltage- input voltage relationship curve of a vacuum tube

The output current and the input voltage satisfies:

$$I_p = k \cdot \left(V_g + \frac{V_p}{\mu} \right)^2$$

Using Taylor's expansion, suppose the output current with respect to input voltage can be modeled as:

$$I_a = a_0 + a_1 V_g + a_2 V_g^2 + a_3 V_g^3 + \dots$$

If the relationship is linear, we will consider only $I_a = a_0 + a_1 V_g$. When a signal $V_g = A \sin(\omega t)$ is input, the period of I_a would be the same with it of V_g . However, due to the existence of nonlinear terms, this will bring:

$$a_2 V_g^2 = a_2 A^2 \sin^2(\omega t) = \frac{a_2 A^2}{2} [1 - \cos(2\omega t)]$$

And the third harmonic;

$$a_3 V_g^3 = a_3 A^3 \sin^3(\omega t) = \frac{a_3 A^3}{4} (3 \sin(\omega t) - \sin(3\omega t))$$

Similarly, a term in the form $a_n V_g^n$ will cause an output function containing $n\omega t$, thus, a harmonic with frequency $n f$. This is the harmonic distortion: Frequencies with integer multiples of the input signal are introduced and altering the waveform, thus coloring the sound.

Here lies the difference: tube amplifiers respond to playing dynamics — striking the strings more forcefully pushes the signal through more rapidly, creating distortion in the valve sound, while playing softly keeps the sound clean. In contrast, solid-state amps simply produce a louder output without distorting. As for which is better, it depends on the intended application. If amplifiers are to be used with instruments in a musical band, a vacuum tube amplifier is preferable for its colourful sound. However, if the focus is on car radios or microphones for speech, where a clean sound is desired, solid-state amplifiers are the better choice.

4. Other considerations: why transistors cannot be replaced

Now focusing on musical usage: although the harmonic distortion produced by tubes offers favourable tones and dynamic response, there are factors beyond sound quality that prevent valve amplifiers from becoming more widely adopted.

4.1 Efficiency and heat

Tubes tend to run hot. They require time to “warm up” and can be less energy-efficient. They also have a limited lifespan and will eventually need replacing. Transistors run cooler and are generally more energy-efficient. They are ready to deliver peak performance at a moment's notice.

4.2 Durability and maintenance

Tubes have a limited lifespan. They degrade over time and eventually burn out, requiring replacements. Moreover, compared to their solid-state counterparts, tube amplifiers often require periodic bias adjustments and maintenance to ensure optimal performance.

4.3 Cost and portability

Vacuum tubes, generally speaking, cost more than solid-state amps. Tubes make amplifiers bulkier. They are generally heavier and take up more space, making them less ideal for compact setups or frequent relocation.

5. Hybrid amplifiers as an alternative

Valve amplifiers are valued for their warm, rich sound, but still present challenges in terms of cost and maintenance. To better harness these benefits, hybrid amplifiers have been introduced.

A hybrid amplifier typically features a valve stage for preamplification and a solid-state stage for power amplification. The valves in the preamp stage introduce the desired warmth and coloration, while the subsequent power amplification, handled by solid-state components, drives speakers or headphones with high efficiency and low heat output. Hybrid designs can offer the characteristic valve warmth without the inefficiencies of an all-valve power section. They are also generally more affordable than high-end valve amplifiers and require less maintenance.

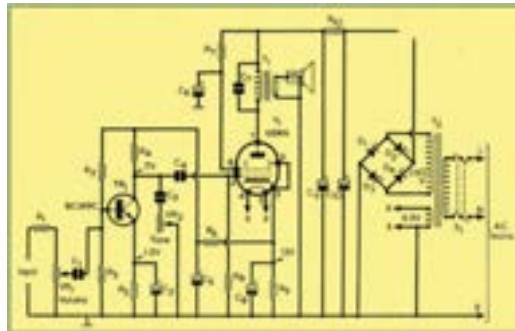


Fig.5 Circuit of a hybrid amplifier

6.Conclusion

This paper compares vacuum tube amplifiers and solid-state amplifiers in terms of their timbre after amplification, as well as other factors beyond sound quality. It demonstrates that whether vacuum tubes are superior depends on the context of use: in scientific and everyday applications, solid-state amplifiers are preferable. In musical contexts, vacuum tubes appear to be superior only in terms of tonal quality — but since tone is the crucial factor in such cases, they can be considered better than solid-state amplifiers. In the latter scenario, to address the drawbacks associated with vacuum tubes, this paper introduces the hybrid amplifier as a balanced solution that maximises the tonal advantages of valves while minimising their disadvantages.

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Mathematics

Fitzwilliam College Winter and Summer School Journal · 2025

Introduction to Laplace Transforms

JINGWEN LEI

Table of Contents

- 1 What is Laplace Transform?
 - Definition
 - Properties
- 2 Use Laplace Transforms to solve Differential equations
 - SHM
 - Damped Pendulum
- 3 Conclusion

From Fourier to Laplace

Pre-requisites for this presentation: differential equation, integration by parts, exponential integration, complex numbers, Fourier transform.

Start with Fourier transforms!

Definition

See the definition of Fourier transform

$$F(\omega) = \mathcal{F}(f(t)) = \int_{-\infty}^{+\infty} f(t)e^{-i\omega t} dt$$

In which, $f(t)$ is called **time domain** function and $F(\omega)$ is called **frequency domain** function.

Restrictions of Fourier transform

Restrictions

- 1 Generally, function $f(t)$ should follow

$$\int_{-\infty}^{+\infty} |f(t)| dt < \infty$$

- 2 For periodic functions, in one period: limited breaking points limited maximum/minimum, and also

$$\int_T |f(t)| dt < \infty$$

, where T is one period of t .

In a word, **converge**. But what if it does not?

Broaden the conditions

Definition

See the definition of Fourier transform

$$F(\omega) = \mathcal{F}(f(t)) = \int_{-\infty}^{+\infty} f(t)e^{-i\omega t} dt$$

When the condition fails, we add a **negative exponential factor** e^{-at} to create a new function, where $a > 0$,

$$g(t) = \begin{cases} e^{-at}f(t) & t > 0 \\ 0 & t < 0 \end{cases}$$

Illustration of f to g

See a visualization:



Figure: The green line shows $f(t) = t^2$ Figure: The blue line shows $g(t) = e^{-t}t^2$

Process page

This is the process of applying Fourier transform to $g(t)$ and leads to the definition of Laplace transform.

$$F(\omega) = \mathcal{F}(f(t)) = \int_{-\infty}^{+\infty} f(t)e^{-i\omega t} dt$$

$$g(t) = \begin{cases} e^{-at}f(t) & t > 0 \\ 0 & t < 0 \end{cases}$$

So for $t > 0$, we have

$$\begin{aligned} \mathcal{F}(g(t)) &= \int_0^{\infty} (f(t)e^{-at})e^{-i\omega t} dt \\ &= \int_0^{\infty} f(t)e^{-(a+i\omega)t} dt \end{aligned}$$

Definition of Laplace transform

Definition

The Laplace transform can be defined by the integral

$$\mathcal{L}\{f(t)\} = F(s) = \int_0^{\infty} f(t)e^{-st} dt$$

s is a complex number which can be written as

$$s = \alpha + i\omega$$

where $\alpha, \omega \in \mathbb{R}$

New restrictions

Definition

The Laplace transform can be defined by the integral

$$F(s) = \mathcal{L}\{f(t)\} = \int_0^{\infty} f(t)e^{-st} dt$$

Although $f(t)$ can be a divergent function now, to hold the transform, the new function integrated should satisfy

$$\int_0^{\infty} |f(t)e^{-st}| dt < \infty$$

Therefore, in Laplace transform, $|f(t)| \leq Ce^{s_0 t}$, where $0 < s_0 < s$.

i.e. The increase of $|f(t)|$ should not overcome the attenuation of e^{-st} for $t > 0$.

Inverse Laplace transform

Definition

To go back from $F(s)$ to $f(t)$, we use the inverse Laplace transform

$$f(t) = \mathcal{L}^{-1}\{F(s)\} = \frac{1}{2\pi i} \lim_{T \rightarrow \infty} \int_{\alpha - iT}^{\alpha + iT} F(s)e^{st} ds$$

Briefly,

$$f(t) = \mathcal{L}^{-1}\{F(s)\}$$

See deduction next slide.

This means that if we can find the frequency domain function $F(s)$, by applying the inverse Laplace transform, we can get $f(t)$.

Deduce the expression for inverse Laplace transform

Begin with $g(t)$

$$g(t) = \begin{cases} f(t)e^{-st} & t > 0 \\ 0 & t < 0 \end{cases}$$

1. Apply Fourier transform to $g(t)$, with frequency $\sigma \in \mathbb{R}$.

$$\begin{aligned} G(\sigma) &= \mathcal{F}\{g(t)\} = \int_{-\infty}^{\infty} g(t)e^{-i\sigma t} dt \\ &= \int_{-\infty}^0 0 dt + \int_0^{\infty} f(t)e^{-st}e^{-i\sigma t} dt \\ &= \int_0^{\infty} f(t)e^{-(s+i\sigma)t} dt \\ &= \mathcal{L}\{f(t)\} \\ &= F(s), s = \alpha + i\sigma \end{aligned}$$

Deduce the expression for inverse Laplace transform

Therefore, $G(\sigma) = F(s)$

2. Inverse $G(\sigma)$ back to $g(t)$ using **inverse Fourier transform**

$$g(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} G(\sigma)e^{i\sigma t} d\sigma$$

To relate this equation with $f(t)$, multiply e^{st} on both sides

$$g(t)e^{st} = \frac{1}{2\pi} \int_{-\infty}^{\infty} G(\sigma)e^{st}e^{i\sigma t} d\sigma$$

Since $g(t) = f(t)e^{-st}$, $f(t) = g(t)e^{st}$, and also $G(\sigma) = F(s)$,

$$f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(s)e^{st}e^{i\sigma t} d\sigma$$

Deduce the expression for inverse Laplace transform

3. Write the integral on RHS with respect to $s (= \alpha + i\sigma)$, while $\frac{d\sigma}{ds} = i$

$$\begin{aligned} f(t) &= \frac{1}{2\pi} \int_{-\infty}^{\infty} F(s)e^{st}e^{i\sigma t} d\sigma \\ &= \frac{1}{2\pi} \int_{\alpha-i\infty}^{\alpha+i\infty} F(s)e^{st} \frac{ds}{i} \\ &= \frac{1}{2\pi i} \int_{\alpha-i\infty}^{\alpha+i\infty} F(s)e^{st} ds \\ &= \frac{1}{2\pi i} \lim_{T \rightarrow \infty} \int_{\alpha-iT}^{\alpha+iT} F(s)e^{st} ds \end{aligned}$$

Which is exactly the expression of $\mathcal{L}^{-1}\{F(s)\}$ shown in the definition of inverse Laplace transform.

Have a rest

There are too many equations above, so here is a rest page.

Next section: Properties



Properties-Bijection

Most time domain functions and frequency domain functions are one to one.

$f(t) = \mathcal{L}^{-1}\{F(s)\}$	$F(s) = \mathcal{L}\{f(t)\}$
1. 1	$\frac{1}{s}$
2. e^{at}	$\frac{1}{s-a}$
3. $t^n, n = 1, 2, 3, \dots$	$\frac{n!}{s^{n+1}}$
4. $t^p, p = -1$	$\frac{\Gamma(p+1)}{s^{p+1}}$
5. $\sqrt{t}$	$\frac{\sqrt{\pi}}{s^{3/2}}$

This property allows us to apply the Laplace transform to both sides of the equation but the equality still holds.

Properties-Derivative theorem

Theorem

For the derivative and the second derivative of the time domain function $f(t)$

$$f'(t) = sF(s) - f(0^-)$$

$$f''(t) = s^2 F(s) - sf(0^-) - f'(0^-)$$

And in general,

$$f^{(n)}(t) = s^n F(s) - \sum_{k=1}^n s^{n-k} f^{(k-1)}(0^-)$$

This is the key property that allows us to turn an ordinary differential equation into an algebraic equation.

These figures are illustrative. Mathematical content is not to scale. Page 100 11/18

Example 1-SHM

Example

Q: Simple Harmonic Motion of a block attached to a spring on a smooth desk, given that $x = x_0$ and $dx/dt = 0$ for $t = 0$, find $x(t)$.

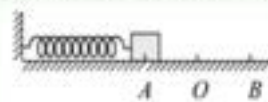


Figure: A block attached to a spring is pressed from O to A and then released.

These figures are illustrative. Mathematical content is not to scale. Page 100 12/18

Example 1-SHM

Example

Simple Harmonic Motion of a spring on a block attached to a spring on a smooth desk, given that $x = x_0$ and $dx/dt = 0$ for $t = 0$, find $x(t)$.



Figure: A block attached to a spring is pressed from O to A and then released.

Now apply Laplace transform to both sides and use the properties of second derivative:

$$\frac{d^2 x(t)}{dt^2} = -\omega^2 x(t)$$

$$s^2 X(s) - sx(0) - x'(0) = -\omega^2 X(s)$$

Rearrange:

$$X(s) = \frac{sx_0}{s^2 + \omega^2}$$

These figures are illustrative. Mathematical content is not to scale. Page 100 13/18

Example 2-Damped Pendulum

Here is another example from isaacphysics.org.

Example

Damped Pendulum (2nd Order) The equation describing the displacement x of the bob of a damped pendulum from its equilibrium position is given by

$$\frac{d^2 x}{dt^2} = -\omega_0^2 x - 2\gamma \frac{dx}{dt}$$

where ω_0 is the angular frequency of undamped oscillations of the pendulum and γ is related to the damping.

Assuming $\omega_0 > \gamma$ find $x(t)$ given that $x = x_0$ and $\frac{dx}{dt} = 0$ at $t = 0$.

These figures are illustrative. Mathematical content is not to scale. Page 100 14/18

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These figures are illustrative. Mathematical content is not to scale. Page 100 15/18

Example 1-SHM

Example

Simple Harmonic Motion of a block attached to a spring on a smooth desk, given that $x = x_0$ and $dx/dt = 0$ for $t = 0$, find $x(t)$.



Figure: A block attached to a spring is pressed from O to A and then released.

Start by writing down the ordinary differential equation:

$$m\ddot{x} = F$$

$$m \frac{d^2 x}{dt^2} = -kx$$

$$\frac{d^2 x}{dt^2} = -\frac{k}{m}x$$

$$\frac{d^2 x}{dt^2} = -\omega^2 x, \omega^2 = \frac{k}{m}$$

These figures are illustrative. Mathematical content is not to scale. Page 100 16/18

Example 1-SHM

Example

Simple Harmonic Motion of a block attached to a spring on a smooth desk, given that $x = x_0$ and $dx/dt = 0$ for $t = 0$, find $x(t)$.



Figure: A block attached to a spring is pressed from O to A and then released.

Inverse back:

$$X(s) = \frac{sx_0}{s^2 + \omega^2}$$

Look up in the Laplace transform table:

$$\frac{s}{s^2 + \omega^2} \rightarrow \cos \omega t \quad \text{Real } t > 0$$

$$\frac{s}{s^2 + \omega^2} \rightarrow \cos \omega t \quad \text{Real } t > 0$$

$$x(t) = \mathcal{L}^{-1} X(s) = x_0 \cos \omega t$$

These figures are illustrative. Mathematical content is not to scale. Page 100 17/18

Example 2-Damped Pendulum

Example

The equation describing the displacement x of the bob of a damped pendulum from its equilibrium position is given by

$$\frac{d^2 x}{dt^2} = -\omega_0^2 x - 2\gamma \frac{dx}{dt}$$

Assuming $\omega_0 > \gamma$, find $x(t)$ given that $x = x_0$ and $\frac{dx}{dt} = 0$ at $t = 0$.

Step 1: Move the derivatives to one side

$$\frac{d^2 x}{dt^2} + 2\gamma \frac{dx}{dt} = -\omega_0^2 x$$

Step 2: Apply derivative theorem

$$\frac{d^2 x}{dt^2} = s^2 X(s) - sx(0) - x'(0)$$

$$\frac{dx}{dt} = sX(s) - x(0)$$

These figures are illustrative. Mathematical content is not to scale. Page 100 18/18

Example 2-Damped Pendulum

Example

The equation describing the displacement x of the bob of a damped pendulum from its equilibrium position is given by

$$\frac{d^2x}{dt^2} = -\omega_0^2 x - 2\gamma \frac{dx}{dt}$$

Assuming $\omega_0 > \gamma$, find $x(t)$ given that $x = x_0$ and $\frac{dx}{dt} = 0$ at $t = 0$.

Step 3: Substitute back with initial displacement $x(0) = x_0$ and initial velocity $x'(0) = 0$.

$$s^2 X(s) - sx_0 - 0 = -\omega_0^2 X(s) + 2\gamma(sX(s) - x_0)$$

Rearrange the equation!

$$X(s) = \frac{(s + 2\gamma)x_0}{s^2 + 2\gamma s + \omega_0^2}$$

Example 2-Damped Pendulum

Example

The equation describing the displacement x of the bob of a damped pendulum from its equilibrium position is given by

$$\frac{d^2x}{dt^2} = -\omega_0^2 x - 2\gamma \frac{dx}{dt}$$

Assuming $\omega_0 > \gamma$, find $x(t)$ given that $x = x_0$ and $\frac{dx}{dt} = 0$ at $t = 0$.

Step 4: Inverse!

Let's try to find out something on the table first,

$$\frac{1}{s^2 + 2\gamma s + \omega_0^2} = \frac{1}{(s + \gamma)^2 + \omega_1^2}$$

Figure: $f(t)$ on left and $F(s)$ on right

The denominator of these two equations above seems quite relevant to the denominator of $X(s)$

$$X(s) = \frac{(s + 2\gamma)x_0}{s^2 + 2\gamma s + \omega_0^2}$$

Example 2-Damped Pendulum

Example

The equation describing the displacement x of the bob of a damped pendulum from its equilibrium position is given by

$$\frac{d^2x}{dt^2} = -\omega_0^2 x - 2\gamma \frac{dx}{dt}$$

Assuming $\omega_0 > \gamma$, find $x(t)$ given that $x = x_0$ and $\frac{dx}{dt} = 0$ at $t = 0$.

Step 5: Try to fit in the table by rearranging $X(s)$.

$$X(s) = \frac{(s + 2\gamma)x_0}{s^2 + 2\gamma s + \omega_0^2} = \frac{(s + 2\gamma)x_0}{(s + \gamma)^2 + \omega_1^2} = \frac{(s + \gamma)x_0}{(s + \gamma)^2 + \omega_1^2} + \frac{\gamma x_0}{(s + \gamma)^2 + \omega_1^2}$$

Let $\omega_0^2 - \gamma^2 = \omega_1^2$.

$$X(s) = \frac{(s + \gamma)x_0}{(s + \gamma)^2 + \omega_1^2} + \frac{\gamma x_0}{(s + \gamma)^2 + \omega_1^2}$$

Example 2-Damped Pendulum

Example

The equation describing the displacement x of the bob of a damped pendulum from its equilibrium position is given by

$$\frac{d^2x}{dt^2} = -\omega_0^2 x - 2\gamma \frac{dx}{dt}$$

Assuming $\omega_0 > \gamma$, find $x(t)$ given that $x = x_0$ and $\frac{dx}{dt} = 0$ at $t = 0$.

Which can then be split into two parts:

$$X(s) = \frac{(s + 2\gamma)x_0}{(s + \gamma)^2 + \omega_1^2} = \frac{(s + \gamma)x_0}{(s + \gamma)^2 + \omega_1^2} + \frac{\gamma x_0}{(s + \gamma)^2 + \omega_1^2}$$

So that we can find corresponding time domain function in the table:

$$\frac{(s + \gamma)x_0}{(s + \gamma)^2 + \omega_1^2} \rightarrow e^{-\gamma t} \cos(\omega_1 t) x_0$$

$$\frac{\gamma x_0}{(s + \gamma)^2 + \omega_1^2} \rightarrow \gamma x_0 e^{-\gamma t} \frac{1}{\omega_1} \sin(\omega_1 t)$$

Example 2-Damped Pendulum

Example

The equation describing the displacement x of the bob of a damped pendulum from its equilibrium position is given by

$$\frac{d^2x}{dt^2} = -\omega_0^2 x - 2\gamma \frac{dx}{dt}$$

Assuming $\omega_0 > \gamma$, find $x(t)$ given that $x = x_0$ and $\frac{dx}{dt} = 0$ at $t = 0$.

Finally, due to the property of linearity combination:

$$x(t) = x_0 e^{-\gamma t} \cos(\omega_1 t) + e^{-\gamma t} \frac{\gamma x_0}{\omega_1} \sin(\omega_1 t)$$

which will be the solution.

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- 2 Laplace Transform of some differential equations
 - 2.1 Spring
 - 2.2 Damped Pendulum
- 3 Conclusion

Conclusion

Conclusion

- 1 What is Laplace transform?
- 2 Use Laplace transform to solve differential equations
- 3 More application

This is only a small part of the properties and applications of Laplace transform, and the process behind the Laplace transform table can be rather complex, while using Laplace transforms could save us lots of efforts especially in some complex cases. For example, solving partial differential equations.

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The Blackbody Radiation

ERI YOSHITAKE

The Blackbody Radiation

What did Planck do and what maths did he need?

Eri Yoshitake

What is the Blackbody Radiation?

The electromagnetic radiation emitted by a blackbody ¹.

¹an idealised object that absorbs all incident electromagnetic radiation regardless of its frequency or angle.

Image source: <https://commons.wikimedia.org>

The Blackbody Radiation

The Ultraviolet Catastrophe: A Classical Physics Conundrum

Rayleigh-Jeans Law Prediction:
Classical physics predicted that as the frequency of radiation increased, the emitted energy would also increase without limit.

The Ultraviolet Catastrophe:
While the Rayleigh-Jeans law matched experimental results at low frequencies, it sharply diverged at high frequencies, particularly in the ultraviolet region of the electromagnetic spectrum. This profound discrepancy between classical theory and experimental observation was called the ultraviolet catastrophe.

Image source: <https://commons.wikimedia.org>

Who is Max Planck?

I am Max Planck

Image source: <https://en.wikipedia.org>

Max Planck and the Birth of Quantum Mechanics

- Quantized Energy: Energy is absorbed and emitted in discrete packets called quanta, not continuously.
- Specific Energy Levels: Atoms can only absorb or emit energy at certain fixed levels, leading to specific frequencies of light.
- Planck's Constant (h): Energy of a quantum is given by

$$E = nhf$$
 where n is an integer and f is frequency.
- Planck's Radiation Law:

$$E = \frac{8\pi hf^3}{c^3} \cdot \frac{1}{e^{\frac{hf}{k_B T}} - 1}$$

Image source: <https://commons.wikimedia.org>

Rayleigh-Jeans and the UV catastrophe

- Count standing EM modes in a cavity: number of modes per unit frequency (per volume) $\propto f^2$.
- Classical equipartition $\Rightarrow$ each mode has average energy $k_B T$, so energy density grows like $f^2 k_B T$.

Rayleigh-Jeans result (classical):

$$u_{RJ}(f) = \frac{8\pi k_B T}{c^3} f^2$$

This predicts ever more energy at high frequency — the "ultraviolet catastrophe" (clearly at odds with experiment).

Image source: <https://commons.wikimedia.org>

Planck's Key Idea: Energy is Quantised

- Planck assumed each resonant oscillator can only have energies that are integer multiples of a quantum.

$$E_n = nhf \quad (n = 0, 1, 2, \dots)$$
- This single assumption changes the statistical average energy per mode (no longer $k_B T$ for all modes).

Image source: <https://commons.wikimedia.org>

Average Energy Of A Quantum Oscillator

We compute the thermal average using Boltzmann weights:

$$\langle E \rangle = \frac{\sum_{n=0}^{\infty} E_n e^{-\frac{E_n}{k_B T}}}{\sum_{n=0}^{\infty} e^{-\frac{E_n}{k_B T}}} = \frac{\sum_{n=0}^{\infty} (nhf) e^{-\frac{nhf}{k_B T}}}{\sum_{n=0}^{\infty} e^{-\frac{nhf}{k_B T}}}$$

Using the geometric-series identities yields the compact result:

$$\langle E \rangle = \frac{hf}{e^{\frac{hf}{k_B T}} - 1}$$

(For low frequencies $hf \ll k_B T$, this reduces approximately to $k_B T$, recovering the classical limit.)

Image source: <https://commons.wikimedia.org>

Planck's Radiation Law

Combine the oscillator average energy with the density of modes.

$$u(f) = \left(\frac{8\pi f^2}{c^3} \right) \langle E \rangle = \frac{8\pi h f^3}{c^3} \cdot \frac{1}{e^{\frac{hf}{kT}} - 1}$$

This is Planck's law (spectral energy density per unit frequency).

It matches experiments and falls off exponentially at high f , thus removing the ultraviolet divergence predicted by Rayleigh-Jeans.

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Alternate Forms and Remarks

- Often expressed per unit wavelength (use change of variables $\lambda = c/f$) for plotting vs. wavelength.
- Planck introduced the constant h (Planck's constant) — the fundamental scale of quanta.
- Historical note: This single assumption launched quantum theory and explains the blackbody curve.

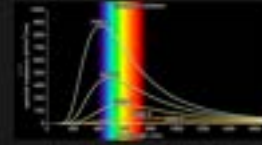


Image source: https://en.wikipedia.org/wiki/File:Blackbody_radiation_spectrum.png

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Planck's Key Ideas

- Quantized Energy:** Energy comes in fixed packets — quanta.
- Discrete Levels:** Atoms can only absorb/emmit specific amounts.
- Planck's Constant h :** $E = hf$ links energy to frequency.
- Radiation Law:** Explains blackbody spectrum, fixes ultraviolet catastrophe.

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What the Discovery of Quantum Physics did to Planck...

Planck before the discovery of Quantum Physics



Image source: https://en.wikipedia.org/wiki/File:Max_Planck_1900.jpg

Planck during the discovery of Quantum Physics



Image source: https://en.wikipedia.org/wiki/File:Max_Planck_1900.jpg

Planck after the discovery of Quantum Physics



Image source: https://en.wikipedia.org/wiki/File:Max_Planck_1900.jpg

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References and further reading

Chapter notes (derivation sketches) (handbook PDF)

Helpful explanation: Physics StackExchange discussion ¹, HyperPhysics ², An article from Britannica ³, An article from Medium ⁴

Photo references: Wikipedia ⁵, Medium ⁶, Study.com ⁷

¹ To derive Planck's law from the quantization of energy with the Ultraviolet Catastrophe (handbook)

² HyperPhysics Blackbody Radiation

³ Planck's radiation law — Definition, Formula, Table — Britannica

⁴ The history of Planck's Blackbody Radiation by Thomas R. Shull — by Ben (David) Heger — (2018), SCIENCE 10100 — Medium

⁵ Max Planck — Wikipedia

⁶ The history of Planck's Blackbody Radiation by Thomas R. Shull — by Ben (David) Heger — (2018), SCIENCE 10100 — Medium

⁷ Planck's radiation law — Britannica

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Appendix

- Planck says that energy levels are quantized with the equation $E_n = (M)n$, i.e. the average energy of a system is the energy of each level multiplied by the number of oscillators with that energy divided by the total number of oscillators.

- The number of oscillators with a specific energy level can be represented as $e^{-\frac{E_n}{kT}}$ where E_n is the energy of that state and kT represents how hot the surroundings are.

- The total number of oscillators is therefore $\sum_{n=0}^{\infty} e^{-\frac{E_n}{kT}}$

- Substituting these values into the equation in step 1, you would get $\langle E \rangle = \frac{\sum_{n=0}^{\infty} E_n e^{-\frac{E_n}{kT}}}{\sum_{n=0}^{\infty} e^{-\frac{E_n}{kT}}}$

$$\langle E \rangle = \frac{\sum_{n=0}^{\infty} E_n e^{-\frac{E_n}{kT}}}{\sum_{n=0}^{\infty} e^{-\frac{E_n}{kT}}}$$

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Appendix

- Factor out constants by replacing E_n with $(M)n$, so it would be:

$$\text{Numerator: } M \sum_{n=0}^{\infty} n e^{-\frac{Mn}{kT}} \quad \text{Denominator: } \sum_{n=0}^{\infty} e^{-\frac{Mn}{kT}}$$

Let $r = e^{-\frac{M}{kT}}$ ($0 < r < 1$)

- Denominator as geometric series would be:

$$\sum_{n=0}^{\infty} r^n = \frac{1}{1-r}$$

- Numerator via derivative trick would form:

$$\sum_{n=0}^{\infty} n r^{n-1} = \frac{1}{(1-r)^2} \Rightarrow \sum_{n=0}^{\infty} n r^n = \frac{r}{(1-r)^2}$$

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Appendix

- Putting together, it would form:

$$\langle E \rangle = M \cdot \frac{\frac{r}{(1-r)^2}}{\frac{1}{1-r}} = M \cdot \frac{r}{1-r}$$

- Replacing r and simplifying using $e^{\frac{1}{x}} = e^{\frac{1}{x}}$ would produce:

$$\langle E \rangle = \frac{M}{e^{\frac{M}{kT}} - 1}$$

- With the formulas above you obtain Planck's law.

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Medicine

Where Do Acoustic Neuromas Occur, What Do Patients Notice, and What Complications Might They Have After Surgery?

ANNA CHIA

Introduction

Acoustic Neuromas (also known as vestibular schwannoma) are non-cancerous brain tumours that develop from malfunctioning Schwann cells covering the vestibular nerve. The vestibular nerve (cranial nerve VIII), sends signals from the inner ear to the brain, and transmits information from the superior olivary complex in the brainstem to the cochlea,¹ playing an important role in sound modulation and body balance. While Acoustic Neuromas are non-cancerous and rare (with only 1 in 100,000 people developing one each year)², the growth of this tumour can impact quality of life, and even become life threatening depending on their development. In this essay, I will be discussing the locations of which Acoustic Neuromas commonly develop, symptoms that patients commonly notice, and potential complications that may occur following surgical intervention.

Location of acoustic neuromas

Acoustic neuromas often originate near the porus acusticus, where the Obersteiner-Redlich zone, or the transition zone of a nerve where glial cells are replaced by Schwann cells, is found.³ While acoustic neuromas develop from the internal auditory canal, they can expand into the cerebellopontine angle, and are usually found near the cochlear or vestibular nerves, particularly the inferior division of the vestibular nerve.⁴

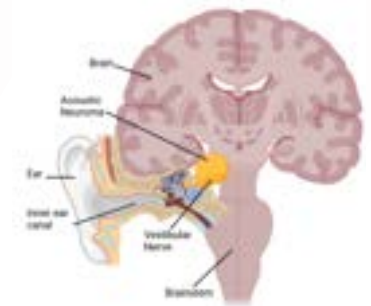


Figure 1: A diagram created by Biorender showing the location of an acoustic neuroma (author owned)

As the tumour grows, it compresses critical structures, including cranial nerve VIII (responsible for auditory functions), cranial nerve VII (a facial nerve), cranial nerve V (involved in sensory functions), cranial nerve IX and vagus nerves (which control muscles for jaw movement), the cerebellum, and the brainstem. The growth of the acoustic neuroma also contributes to increased intracranial pressure. All of these factors contribute to the signs and symptoms that patients often observe.

What do patients notice?

Symptoms of acoustic neuromas are easy to miss, and may even take years to develop, depending on the tumour's growth. However, as the tumour progresses, the symptoms get increasingly more noticeable and severe.

1. Hearing Loss and/or Tinnitus

Progressive loss in hearing in one ear is one of the most common and earliest symptoms of acoustic neuroma, where over 90% of individuals presenting acoustic neuroma observe one-sided hearing loss.⁵ Tinnitus is another common symptom of acoustic neuromas, where affected people perceive constant or intermittent ringing sounds in the ear. This is due to the acoustic neuroma compressing important auditory nerves such as cranial nerve VIII.

Moreover, small acoustic neuromas that do not compress nearby structures can also negatively impact hearing by secreting toxic molecules that damage the inner ear.⁶

2. Facial weakness or numbness

Facial symptoms, such as facial weakness or numbness, often present late, except for large acoustic neuromas. Such symptoms are caused by the compression of cranial nerve VII, which is involved in controlling facial expressions and

movement of the jaw.

3. Vertigo

As acoustic neuromas arise from vestibular nerves, which are involved in maintaining the body's balance, vestibular symptoms such as instability, especially when moving the head, may show.

4. Problems with limb co-ordination (ataxia)

Large acoustic neuromas can compress the cerebellum, which can result in problems with limb co-ordination.

Surgical treatments

Depending on the size and growth rate of the acoustic neuroma, and the patient's overall health, treatment for the acoustic neuroma would vary. Small acoustic neuromas are usually monitored by healthcare professionals. However, if the tumour becomes very large or causes symptoms, surgery may become a prominent treatment option.

There are three main surgery approaches:

- Middle Fossa craniotomy, which involves accessing the Acoustic Neurosa via the middle fossa

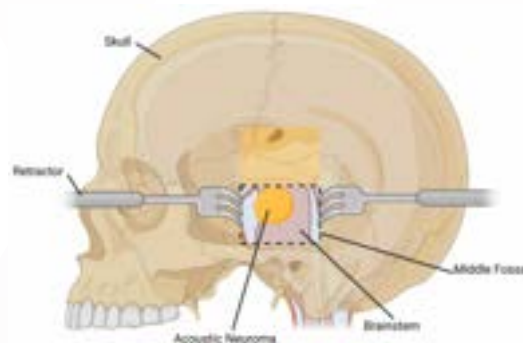


Figure 2: A diagram created by Biorender showing how a Middle Fossa craniotomy is approached (author owned)

- Translabrynthine craniotomy, where an incision is made behind the ear and portions of the mastoid bone and the bone over the dura of the brain is removed to access the tumour

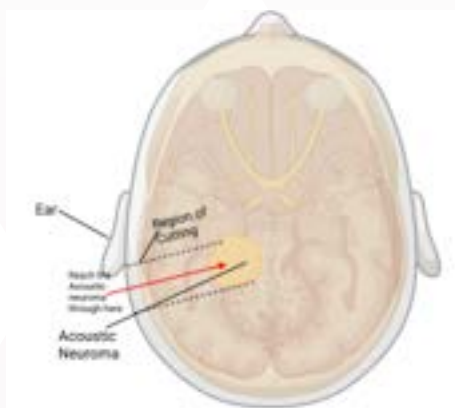


Figure 3: A diagram created by Biorender showing how a Translabrynthine craniotomy is approached (author owned)

- Suboccipital craniotomy, which allows a surgeon to access the Acoustic Neuroma by removing a small piece of bone at the base of the skull (occipital bone)

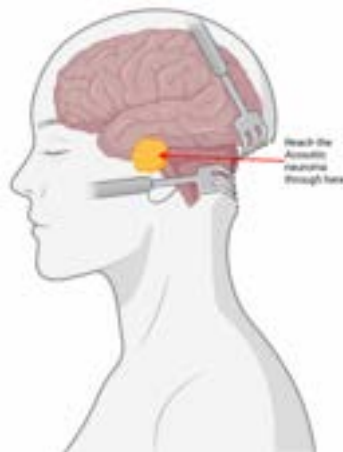


Figure 4: A diagram created by Biorender showing how a Suboccipital craniotomy is approached (author owned)

Although each approach ultimately allows for the removal of the tumour, they also come with their own complications and limitations.

Middle Fossa craniotomy

A common complication of middle fossa craniotomy is cerebrospinal fluid (CSF) leakage (In a study of middle fossa craniotomy complications observed CSF leakage in 13% of participants).⁷ This can be caused by the accidental damaging of the dura mater during surgery.

The retraction of the temporal lobe during the procedure also risks damaging the brain, resulting in complications such as brain swelling and temporary paralysis.

Translabyrinthine craniotomy

As translabyrinthine craniotomy involves the removal of internal ear structures (the mastoid bone), hearing loss is an inherent complication of this type of surgery.

Translabyrinthine craniotomy also occurs close to cranial nerve VII (the facial nerve).

Consequently, complications during this procedure include the potential damage of this nerve, which can result in facial weakness or numbness.

Suboccipital craniotomy

Suboccipital craniotomy occurs at the occipital bone close to the brainstem, where many nerves are found. Potential damage to the brainstem during surgery can impair motor function, and induce vertigo, amongst other symptoms.

Shifting the cerebellum allows for access to the acoustic neuroma in suboccipital craniotomy. Cerebellum retraction risks brain retraction injuries such as cerebellum edema (swelling) or cerebellum ischemia (lack of blood to the cerebellum).⁸

Furthermore, suboccipital craniotomy occurs close to sigmoid sinus, and damage to the sigmoid sinus can lead to either venous sigmoid thrombosis (formation of a blood clot in the sigmoid sinus) or stenosis (the narrowing of the sigmoid sinus).⁹

Conclusion

In conclusion, acoustic neuromas are, while benign, can result in severe symptoms due to their proximity to critical brain

structures. While there are varying approaches to remove this tumour, each procedure has its own distinct advantages and limitations. Overall, the optimal approach depends on the size and location of the acoustic neuroma. However, in scenarios where hearing is not needed, Translabrynthine craniotomy may be the safest option, while Middle Fossa and Suboccipital craniotomy are preferred to preserve hearing.

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How Does the Body Respond to Hemorrhage? Focusing on the Short-Term Responses From Cardiovascular Aspects

YUECHEN WANG

1.Introduction

Hemorrhage is the leakage of blood from blood vessels. When the blood loss reaches 15-20% of the total blood volume, the risk of hemorrhagic shock increases significantly. General symptoms of massive hemorrhage include pale and cool skin, dizziness, which can progress to hemorrhagic shock in severe cases.

According to Darcy's law, $Q=(P1-P2)/R$ (Q: flow; P1: pressure in; P2: pressure out; R: resistance inside the tube), researchers can calculate blood flow, blood pressure, and resistance of fluid passing through a uniform tube (blood vessel) adapted from the original formula, the formula of $CO=(MAP-RAP)/TPR$ (CO: Cardiac Output; MAP: Mean Arterial Pressure; RAP: Right Atrial Pressure; TPR: Total Peripheral Resistance). Resulting in an RAP at approximately zero, the formula simplifies to $CO=MAP/TPR$. This formula provides a fundamental framework for understanding the body's compensatory mechanisms in response to blood loss. This study aims to explain the short-term responses to hemorrhage in the human body in cardiovascular system.

2.Process of hemorrhage responses

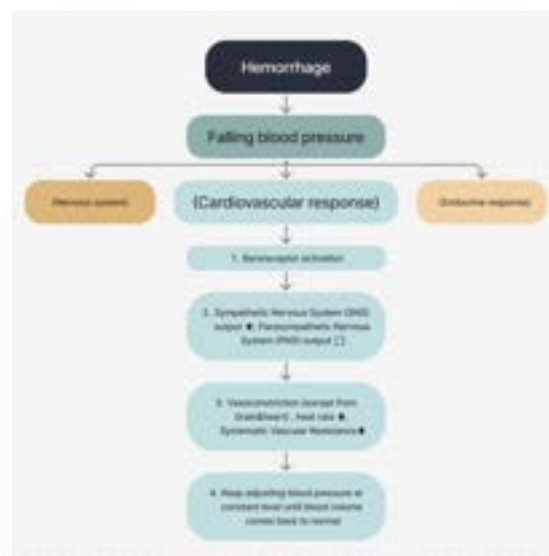


Fig2.1 A general flow chart showing the cardiovascular responding process in massive hemorrhage.

The overall process of the human body for maintaining blood pressure is illustrated in Fig2.1. When low blood pressure (resulting from massive hemorrhage) is sensed by baroreceptors, Sympathetic Nervous System (SNS) activity is amplified to increase heart rate and myocardial contractility. Simultaneously, Parasympathetic Nervous System (PNS) activity is inhibited which reduces the vagal tone sent from the Vagus nerve and allowing the heart rate to increase. In this situation, CO increases due to higher heart rate and stroke volume (Fernandes et al., 2010). However, to provide a more sustained compensatory mechanism, peripheral vasoconstriction (body apart from the brain and the heart) is also essential to increase TPR, thereby helping to maintain MAP.

2.1 Baroreceptor responses

Baroreceptors are pressure receptors located in the carotid sinus and aortic arch, (see on Fig 2.2).

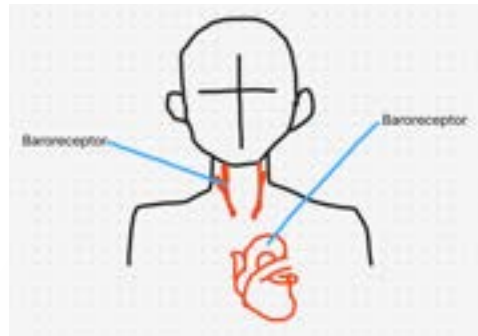


Fig 2.2 A simple graph showing approximately where the baroreceptors lie on

The receptors function primarily to transmit signals via sympathetic and parasympathetic nervous system by sending action potentials at a certain rate to the brain that indicates real-time blood pressure level. Therefore they play a crucial role in maintaining MAP homeostasis in response to fluctuations (Thrasher, 2004).

2.2 SNS & PNS responses

2.2.1 SNS response

2.2.1.1 Cardiac effects:

SNS activation increases heart rate by releasing Norepinephrine (NE) from sympathetic nerve endings. NE is then received by β_1 -adrenergic receptors on the Sinoatrial (SA) node, which stimulates it to generate action potentials in a faster rate. Studies show that the increase in magnitude of heart rate is directly proportional to the concentration of NE (Shimizu et al., 2010).

2.2.1.2 Vascular effects:

SNS activation causes vasoconstriction in arterioles throughout the body (excluding those in the heart and brain) occurs when NE binds to α_1 -adrenergic receptors on vascular smooth muscle, increasing TPR consequently MAP.

Additionally, renin is released from the kidneys to promote sodium and water retention, to help restore blood volume

Fig 2.3 shows the structure of SNS responding logic.



Fig 2.3 Simplified mind map showing the logic of SNS response (Jean-Luc Elghozi et al., 2007)

2.2.2 PNS response

Under normal conditions, PNS releases acetylcholine to inhibit SA node activity which prevents heart rate from getting excessively high. When the blood pressure falls below normal, Vagus nerve activity is inhibited, allowing heart rate to

increase.

2.2 Summary table see on Fig 2.4

Changes	SNS	PNS
Heart	• Heart rate (β receptors) • Cardiac muscle constriction	• Vagus nerve activity inhibited, allowing heart rate to increase.
Blood vessels	• Vasoconstriction - α receptors (except in heart and brain)	No direct effect
Kidneys	• Renin secreted (β receptors)	No direct effect
Overall effects	1. Heart Output (CO) 2. TPR 3. MAP	• Assisting Heart rate

Fig 2.4 Summarizing the key information in section 2.2 mentioned above

2.3 Vasoconstriction

Vasoconstriction in hemorrhagic response is a method applied on the arterioles within the body excluding blood vessels in the heart and brain. NE is released and bind to α_1 -adrenergic receptors on smooth muscles in “non-essential” tissues such as skin, gut, kidneys, etc. Fig 2.5 is showing the regions vasoconstriction are applied in. Venoconstriction also contributes to increase TPR by increasing venous return to the heart, therefore helps to maintain CO. Nevertheless, vasoconstriction has its limitations. For instance, prolonged vasoconstriction may lead to hypoxia in the constricted tissues and organs, eventually leading to tissue ischemia. This is particularly harmful to the kidneys, where ischemia endangers the patient due to impaired renal filtration (Hideaki Watanabe et al., 2023). In conclusion, vasoconstriction is a vital short-term mechanism to prevent circulatory collapse in massive blood loss but could not be a sustainable way for long-term blood pressure maintenance.

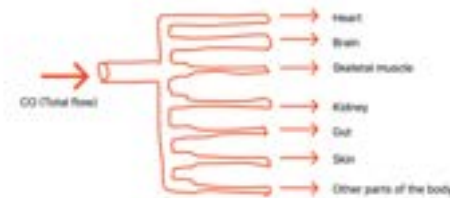


Fig 2.5 showing the area of vasoconstriction in hemorrhage (Prof. Mason, M.J., 2025)

2.4 Long-term responses to hemorrhage

Hematopoietic stem cells stored in bone marrow are activated to produce new blood cells and gradually restore the volume of blood loss. However, the resulting anemia may take weeks to recover. In this period, cardiovascular strain may remain elevated, necessitating close medical care. Following high severity hemorrhage or trauma, heart failure, kidney failure or the development of chronic kidney, heart diseases could occur due to myocardial ischemia and prolonged hypoxia if the consequences are not adequately managed (Seyed Ali Gaskari et al., 2015).

3. Conclusion

In summary, the human body's short-term cardiovascular responses to massive hemorrhage are primarily centered on the equation derived from Darcy's law, $MAP = CO \times TPR$. Building on this foundation, baroreceptors, SNS, PNS, smooth muscles in the arterioles, as well as the endocrine system, work synergistically to increase CO, TPR, or both, helping the body to maintain its blood pressure during hemorrhage. These mechanisms all buy valuable time for survival, also provide clinicians with insights into treating hemorrhage and other conditions involving blood pressure dysregulation.

4. Acknowledgements

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Microbiology

Microbiome Dysbiosis and *Clostridium difficile*: From Pathogenesis to Microbiome Restoration Therapies

TROOPER LIU

Clostridium difficile (*C. diff*) is a prime cause of healthcare-associated infections,¹ responsible for nearly 500,000 U.S. cases annually.² This bacterium exploits a major flaw of modern medicine — antibiotic overuse. When protective gut microbes are wiped out, *C. diff* invades a weakened gut microbiome, causing symptoms from diarrhea to life-threatening colitis, with 20% of patients suffering recurrent infections.³

This essay explores the pathogenicity and clinical relevance of *C. diff*, focusing on the pivotal role of gut microbiota and its disruption. Furthermore, exploring current and emerging microbiome-based therapies, which are redefining treatment approaches and clinical outcomes.

What is *C. diff*?

C. diff is a Gram-positive, spore-forming, obligate anaerobic bacillus adapted to survive in oxygen-limited environments, like the human colon. Its resilient endospores persist in harsh conditions, enabling long-term survival on surfaces and transmission in healthcare settings.⁴

Pathogenicity and symptoms are driven by toxin A (TcdA) and toxin B (TcdB),⁵ some hypervirulent strains also produce a binary toxin (CDT), whose role remains under investigation.⁶ However, these virulence factors are only expressed after successful colonization — an event linked to the disruption of the gut microbiota.

Pathogenic Mechanism and Risk factors

C. diff infection (CDI) begins when ingested spores survive barriers like gastric acid and germinate in the colon, triggered by primary bile acids.⁷ Healthy gut microbiota resists this by competing with *C. diff* and suppressing spore growth. However, broad-spectrum antibiotics eliminate microbial competitors. Antibiotic resistant *C. diff* exploits this dysbiosis (imbalance in microbiota) to proliferate, releasing principal exotoxins TcdA and TcdB which enzymatically glucosylate Rho-family GTPases, disrupting the host cytoskeleton by causing actin depolymerization, triggering cytoskeletal breakdown, cell rounding, apoptosis and a strong inflammatory response.⁵ Further, CDI has a high recurrence rate, driven by lingering spores and incomplete microbiome recovery.⁸

Microbiome and Antibiotic Resistance

At the core of *C. diff* pathogenesis, lies the collapse of gut microbial ecology and antibiotic resistance. The gut microbiome is a dense, diverse microbial ecosystem of commensal obligate anaerobes which are crucial for nutrient metabolism, immune modulation and colonization resistance — preventing pathogenic overgrowth.⁹

In healthy individuals, colonization resistance operates mainly through: 1. *Clostridium scindens* and other Firmicutes which convert primary bile acids (which trigger *C. diff* spore germination) into secondary bile acids (germination inhibitors); and 2. Bacteroid species outcompete *C. diff* for critical nutrients, starving the pathogen of resources.¹⁰ Antibiotics disrupt this balance by reducing microbiome diversity.¹¹ This creates an ecological void where *C. diff* spores, thanks to antibiotic resistant genes, are allowed to germinate unchecked.

Microbiome recovery post-antibiotic therapy is slow (Figure 1), key microbes such as *Clostridium* cluster XIVa and *Bifidobacterium* often don't repopulate quickly, sustaining vulnerability to reinfection.¹² CDI further exacerbates

dysbiosis through toxin-induced inflammation, perpetuating microbial imbalance and tissue damage.¹³

Emerging evidence also suggests that microbial metabolites function as signaling molecules, helping suppress *C. diff* virulence gene expression.¹⁴ Additionally, the microbiome shapes host immunity, promoting protective IgA and Th17 responses critical for mucosal defense, functions often impaired in dysbiosis.¹⁵

Therefore, *C. diff* exploits a collapsed microbial ecosystem, highlighting the need for therapeutic strategies that help restore microbial diversity. Animal models have shown that restoring microbiome composition is essential to prevent recurrence.¹⁶

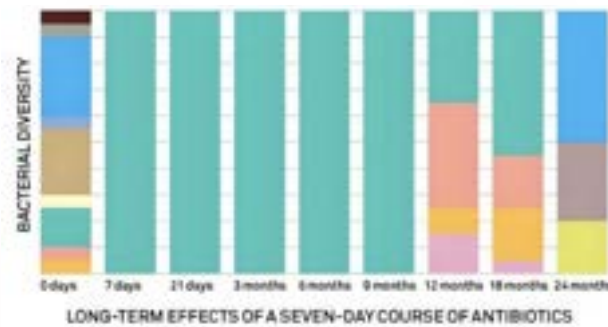


Figure 1

Study of human microbiome diversity after a 7-day course of clindamycin. Bacterial diversity in the microbiome decreases drastically after treatment, showing reduced microbiome to single antibiotic-resistant strain of *Bacteroides*. Original microbiome diversity not recovered in 2 years following treatment

Source: <https://lynceans.org/talk-107-11232016/>

Transmissions, risk factors and prevention

C. diff transmits mainly via the fecal-oral route, with spores persisting on hospital surfaces, clothing, and medical equipment for months.¹⁷ Hospitals pose high-risk due to frequent antibiotic use and vulnerable patients, making nosocomial transmission most common.¹⁸

Spores' germination in the colon when colonization resistance is compromised, often due to recent antibiotic use. Other risk factors include advanced age, hospitalization, chemotherapy, and immunosuppression, all compromising microbiome integrity.¹⁹

Effective prevention relies on strict infection control: hand hygiene with soap, use of sporicidal disinfectants, avoiding unnecessary antibiotic use, and patient isolation.²⁰

Diagnoses and Clinical detection

CDI diagnosis differentiates active infections from asymptomatic colonization, especially in hospitals, where spore carriage is common. Diagnosis typically begins with clinical suspicions — typically ≥ 3 unformed stools in 24 hours, coupled with recent antibiotic use or hospitalisation.²¹

Laboratory confirmation involves stool testing. A common two-step method detects glutamate dehydrogenase antigens, a sensitive marker for *C. diff* presence, followed by toxin enzyme immunoassay to confirm and detect toxin production.²¹

In more severe or uncertain cases, colonoscopy can reveal characteristic pseudomembranous colitis, while CT imaging aids in diagnosing complications, helping ensure appropriate treatment and prevent further transmission.²²

Treatment and Microbiome Rehabilitation

CDI treatment aim to eliminate the pathogen while restoring disrupted gut microbiomes. Therapies can be broadly classified into four strategies: 1. Direct killing, 2. Microbiome restoration, 3. Toxin neutralization, and 4. Immunomodulation (Figure 2).²³

First-line therapies for mild to moderate CDI involve oral antibiotics like vancomycin or fidaxomicin, which target *C. diff* while preserving microbiota. However, high recurrence rates have shifted clinical attention towards microbiome-target therapies.²⁴

Fecal microbiota transplantation (FMT) is a promising therapy, involving transplanting stool from a healthy donor via colonoscopy, nasogastric tube, or oral capsules, restoring microbial diversity and colonization resistance. Clinical trials show >85% success in recurrent CDI cases, though challenges in donor screening and standardization limit widespread use.²⁵

Newer microbiome therapies include SER-109, containing purified firmicutes spore, were recently FDA-approved for recurrent CDI. Other developments, like RBX2660 and CP101, offer standardized, encapsulated microbial consortia to ensure reproducible delivery and avoid the variability of raw donor stool.²⁶



Figure 2

Schematic representation of four different groups of strategies for the treatment of *C. diff* infections.

Source: <https://pubmed.ncbi.nlm.nih.gov/38591915/>

Furthermore, toxin-neutralizing agents like bezlotoxumab, which binds to *C. diff* toxin B, is approved to prevent recurrence in high-risk patients.²⁷ Vaccine research targeting *C. diff* toxins are promising but are still in clinical trial stages.²⁸

Overall, the treatment for CDI is shifting from short-term eradication of the pathogen to an approach centered around long-term restoration of the gut microbiome, preventing recurrence and improving patient outcomes.

Conclusion

Clostridium difficile is both a major Healthcare-associated infection, and example of antibiotic overuse plaguing our healthcare system. Its virulence is not solely due to toxins and spore resilience, but its ability to exploit the ecological void left by antibiotic-induced dysbiosis. CDI is a disease of disrupted colonization resistance — where disrupted microbiomes allow opportunistic pathogens to thrive.

Traditional antibiotics treatments, while important, often perpetuate dysbiosis, enabling recurrence. Therefore, the therapeutic landscape has shifted from suppression to microbiome restoration. Emerging interventions like fecal microbiota transplantation, live biotherapeutics like SER-109, and precision antimicrobials aim to re-establish microbial resilience, with cure rates exceeding 85% in recurrent cases.

Ultimately, CDI highlights the need to reconceptualize infectious disease treatment through a microbiome-centered lens. Future success depends on integrating microbiome rehabilitation, targeted antimicrobial stewardship, and host

immunomodulation. Showing how sustaining microbial homeostasis is just as critical as eliminating pathogens themselves.

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Treatments of *Helicobacter pylori* Infections

JINGTONG HE

Abstract

Helicobacter pylori (*H. pylori*), a bacterium that affects over half of global population, lives in gastric regions of human body. They cause disease by releasing toxins: the oncoprotein CagA and vacuolating cytotoxin VacA. Generally, the treatment of *H. pylori* includes antibiotic therapy and dietary control. Antibiotic therapy includes three consecutive lines (stages) and use of triple therapy. Meanwhile, dietary control accounts for lower level of salt and higher level of microelements such as iron and antioxidants. This essay concludes that it is best for recovery when both treatments are in sync.

Key Words: *H. pylori*, infection, clinical treatment, nutrient

1.Introduction

H.pylori, most found in the stomach's pyloric region, in the glands beneath the mucosal surface, is a type of bacteria that causes chronic gastric infection (National Center for Infectious Diseases (U.S.), Division of Bacterial and Mycotic Diseases, Health Communications Activity, 1998; Talley & Vakil, 2005). It is generally 0.5–1.0 μm wide, with length between 2.0 and 4.0 μm , having a short helical, S-shape and is Gram- negative (Öztekın et al., 2021). It was estimated that *H. pylori* impacts over half of the world population (Hooi et al., 2017; Stefano et al., 2018). *H. pylori* was categorized into group 1 carcinogens by the International Agency for Research on Cancer (IARC) (Shatila & Thomas, 2022). Besides its infamous name as a carcinogen, it is also responsible for high percentages of ulcer cases in the stomach and duodenum (Elbehiry et al., 2023).

The diseases of *H. pylori* are caused by the released toxins. A toxin from a bacterium is defined as an antigenic poison that causes disease (Cambridge Dictionary, 2019). Currently, there are two types of toxins that *H.pylori* releases, including (i) the bacterial oncoprotein produced by the cytotoxin-associated gene A, CagA, and (ii) the protein of the vacuolating cytotoxin A, VacA. The chronic infections caused by the toxins could lead to cancer occurrence (IARC, 2000). There are currently three main ways to treat the infections caused by *H. pylori*, including antibiotic therapy, antisecretory agents, and dietary control. This essay focuses on the treatment of *H. pylori* infections.

2.Treatments of *H. pylori* Diseases

Treatment of *H. pylori* diseases includes antibiotics, antisecretory agents with proton pump inhibitors (PPI), as well as dietary adjustments (Fig. 1) (Öztekın et al., 2021; Elbehiry et al., 2023; Peterson, 1997).

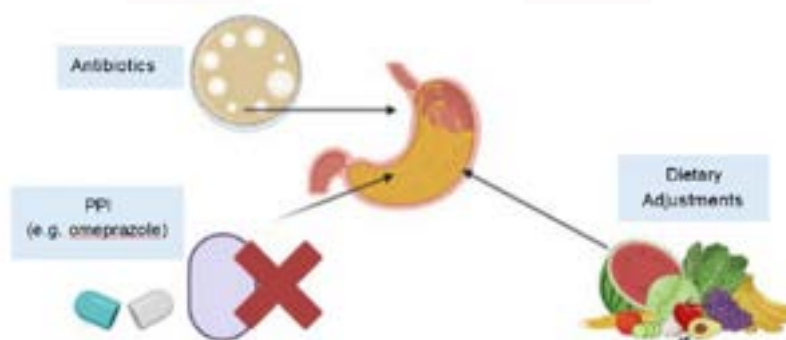


Fig. 1: Summary of available treatments of *H. pylori* diseases, including antibiotics, proton pump inhibitors (PPI) such as Omeprazole representing antisecretory agents, as well as dietary adjustments. Created in BioRender.

2.1 Antibiotic Therapy

Antibiotic therapy is the use of antibiotics in treating infections, initiating with parental antibiotics, and is adjusted according to the result of following symptom improvement and reduction in inflammatory marker (Science Direct, 2020).

As antibiotic resistance can be easily developed while treating *H. pylori* infection, it was suggested to use a combination of antibiotics (Pohl et al., 2019). Substances such as clarithromycin, amoxicillin, levofloxacin, metronidazole, tetracycline, rifabutin, and bismuth-containing compounds have all been suggested for treating *H. pylori* diseases (Fig.2) (Yang et al., 2014; Nguyen et al., 2019). Öztekin et al. summarised the first-line, second-line, and the third-line treatment options of *H.pylori* (2021). For the first line therapy, this involves a triple therapy of clarithromycin, any one from amoxicillin and metronidazole, and a PPI or ranitidine bismuth citrate as an alternative (Suzuki et al., 2010; Fischbach et al., 2009). When the first-line therapy fails, second-line therapy according to individual antibiotic sensitivity and resistance is carried out, generally involving tetracycline, metronidazole, a bismuth salt, or PPI (Öztekin et al., 2021). The third-line follows with testing the antimicrobial susceptibility and local antibiotic resistancy of *H. pylori* culture from the taken gastric biopsy, and continuing the treatment (Malfertheiner et al., 2017). However, long-term PPI usage can have adverse impact on human health as it may inhibit the absorption of micronutrients, increasing the risk of having hypomagnesemia and fractures (FDA, 2019).

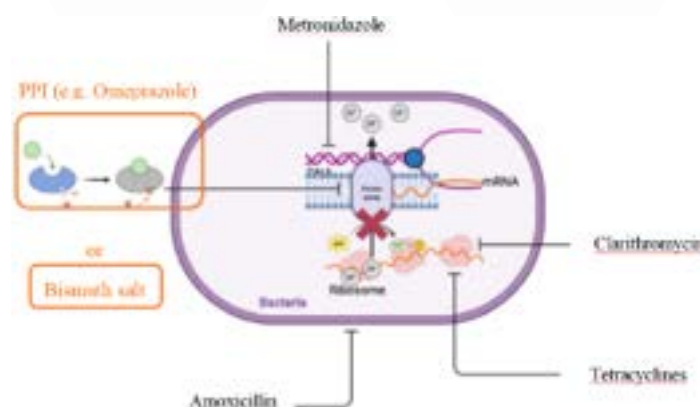


Fig. 2: Summary of most common agents used for treatments of *H. pylori* diseases, including multiple classes of antibiotics (targeting DNA synthesis such as Metronidazole, targeting protein translation at the 50s ribosomal subunit such as Clarithromycin, targeting protein translation at the 30s ribosomal subunit such as Tetracyclines, and targeting cell wall synthesis such as Amoxicillin), as well as proton pump inhibitors (PPI) such as Omeprazole or Bismuth salt as an alternative. Created in BioRender.

PPI and metronidazole can be used in both first and second line. The function of PPI is to prevent the pumping out of protons from the H⁺/K⁺ ATPase, thus reducing stomach acid secretion, so auxiliates the increase of pH needed for recovery (Ierardi et al., 2019). Meanwhile, metronidazole can diffuse into cells, bind to the DNA and disrupt protein synthesis (Weir & Le, 2023). To be therapy-specific, the role of the involved antibiotics should be considered. In the first-line, clarithromycin binds to *H. pylori* 50s ribosomal subunit to inhibit protein synthesis (National Center for Biotechnology Information, 2012) while amoxicillin binds to the penicillin-binding proteins to inhibit transpeptidation, which is crucial in cell-wall synthesis (Akhavan et al., 2023).

In second-line therapy, tetracycline inhibits cell replication by binding to the 30s ribosome subunit, hindering binding from aminoacyl-tRNA to the acceptor of mRNA- ribosome complex (Shutter & Akhondi, 2023). Bismuth salt affects *H. pylori* in four ways: disrupts the cell wall, disrupts the periplasmic space, inhibits enzyme activities and ATP synthesis, and adhere the bacteria to gastric mucosa (Alkim et al., 2017). Since third-line therapy is the continuation of treatment, it will not be further discussed.

2.2 Nutrient Adjustment

H. pylori infection causes digestive, growth and immunological problems, as well as changes in lifestyle (Budzyński & Kłopotka, 2014; Aimasso et al., 2019; Franceschi et al., 2014). Adjustment on the amount of salt, iron, antioxidants, nitrite, and nitrosamines are used to auxiliate patient recovery (Fig. 3) (Öztekin et al., 2021).



Fig. 3: Summary of nutrient factors incorporated in treatment of *H. pylori* diseases. High salt intake can increase the risk of permanent infection. *H. pylori* can cause iron deficiency and decrease the bioavailability of vitamin C. Some nutrients (allicin, lycopene, capsaicin, etc.) have positive effects on *H. pylori* (Öztekin et al., 2021).

Low-salt intake is suggested to prevent the infection developing into permanent because high-salt level is synergetic with *H.pylori* (Loh et al., 2012; Gancz et al., 2008). In comparison, iron, which plays important roles in various chemical reactions, is required for recovery (Flores et al., 2017; Ge & Sun, 2011). Iron deficiency is a consequence of *H.pylori* infection (Merrell et al., 2003; Boyanova, 2011). Antioxidants can repair DNA damage to prevent cancer (Poplawski et al., 2013). For antioxidant such as vitamin C (ascorbic acid), high intake of it could help preventing gastric cancer (Waring et al., 1996). Although vitamin C can protect cancer cells from oxidative stress, its pro-oxidant activity increases the risk of cancer (Zhang & Farthing, 2005). Nitrite and nitrosamines are carcinogens that can be inhibited by antioxidants (Toh & Wilson, 2020; R. A., 1983).

3. Conclusion

H. pylori is a dangerous bacterium that generally lives in human gastric regions (Mayo Clinic, 2022). Its outer membrane structure and relatively large number of flagella makes it difficult to combat. Overall, antibiotic treatment targets the bacteria, whereas food adjustments focus on the interaction between gastric environment and bacteria. Therefore, it should be noted that the major approach to *H. pylori* eradication should be using antibiotic therapy and auxiliate with dietary control. Three lines of treatment should be consistant and in concordance with diet adjustment to prevent cancer occurrence.

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2

Physics

The Twin Paradox: Explanations and the Doppler Effect Approach

NANCY ZHONG

"The Twin Paradox" is one of the most famous thought experiments in special relativity, which illustrates a phenomenon of time dilation. This experiment involves two twins: one A stays on Earth, while the other B travels at high speed near light speed to a distant planet and returns. When they meet again, B is younger than A, despite the apparent symmetry of their relative motion. This paradox challenges classical intuitions of time and motion, but it is fully explained within the framework of special relativity. This paradox recognizes the asymmetry of the twins' experiences, particularly the role of acceleration and the Doppler effect in explaining the observed time differences.

So twin A stays on Earth, which is in an inertial reference frame, while B travels at a constant speed near light speed v to a distant planet and then returns, still at a speed of v . According to special relativity, the faster an object moves through space, the slower it moves through time compared to an unmoving observer. This relationship of time dilation can be quantified with something called the Lorentz factor. The length of a moving object also contracts as measured by an observer at rest.

Key Implications:

1. **Time Dilation:** Moving clocks run slower by a factor of γ .

$$\Delta t' = \gamma \Delta t$$

2. **Length Contraction:** Objects contract along the direction of motion.

r

Formula:

The Lorentz factor is given by:

$$\gamma = \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}}$$

where:

- v = velocity of the object,
- c = speed of light in a vacuum ($\sim 3 \times 10^8$ m/s).

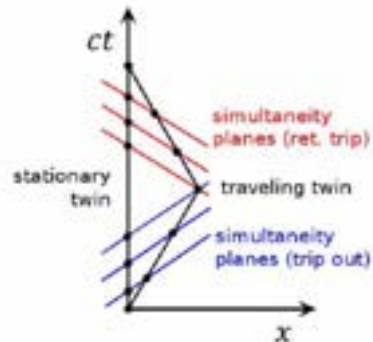
Therefore, when B returns, B's age will be younger than A's. However, from B's perspective, it seems that A is the object that moves away and then returns, so B expects A to be younger. This seemingly contradictory situation is the core of this paradox.

This contradictory phenomenon arises because, if motion is relative, both twin brothers should observe each other's aging rate slowing down. The key to solving this problem lies in the fact that the situations of the two twin brothers are not completely symmetrical, because B experiences acceleration when turning around, while A does not. This difference breaks the symmetry and explains why only B experiences time dilation.

The explanations of the paradox is due to acceleration and the asymmetry of reference frames. Special relativity strictly applies to inertial reference frames, which means they always have to move in constant velocity. B's journey is divided into three stages: the departure stage at speed v to the destination, the turning stage where B accelerates and no longer stays in

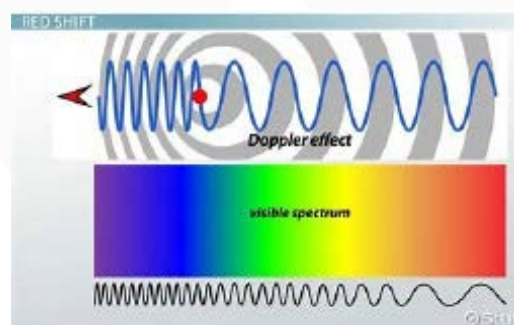
a constant v to change direction, and the return stage at speed v . While A is throughout in a single inertial reference frame. B's acceleration means that B switches between two inertial reference frames, fundamentally changing B's experience.

The time experienced by each twin is derived from the integration of their initial time. Since B's speed is non zero for most of the journey, B's initial time is shorter. In the spacetime diagram, A's world line is a straight line as inertial motion, while B's world line consists of two straight segments and has a "bend" at the turning point. B's path length in spacetime is shorter, meaning time passes more slowly for B.¹



Another way to understand this paradox is by analyzing the Doppler shift of the light signals exchanged between the twins. The Doppler effect is the change in frequency or wavelength of a wave such as sound or light, in relation to an observer who is moving relative to the wave source. Suppose both twins are in their respective stationary reference frames and send light pulses at a frequency $f = 1$ pulse/second. During B's outward journey, the frequency of B's signal received by A will decrease, which is a redshift phenomenon, that its light shifts to lower frequencies. For $v = 0.8c$, this will be $1/3$ pulse/second. During the return journey, the frequency of the received signal by A will increase, which is a blueshift phenomenon, that its light shifts to higher frequencies, reaching 3 pulse/second.

If B's journey consists of a 3-year outward trip plus a 3-year return trip, totaling in 6 years, then A will receive 1 pulse during the outbound phase and 9 pulses during the return phase, totaling 10 pulses, meaning A's time will pass 10 years. This Doppler effect method quantitatively confirms the existence of the time difference without directly mentioning the acceleration phenomenon. The reason why the Doppler effect is important is that it provides direct experimental evidence for the time dilation phenomenon, which is different from the abstract spacetime diagrams. The different frequencies during the departure and return processes reflect the asymmetry of B's motion. Although acceleration causes the turn, the Doppler analysis focuses on the observed signal, thereby explaining why A's clock runs faster overall.²



One common misunderstanding is that since motion is relative, both twin brothers should observe each other's age slowing down. The key insight is that special relativity only guarantees symmetry between inertial observers. B's acceleration breaks this symmetry, meaning B cannot always consider its reference frame as an inertial frame.

The "Twin Paradox" is not a true contradiction; rather, it is a demonstration of the time dilation phenomenon in relativity. This asymmetry occurs because B, due to acceleration, changes the inertial reference frame, while A does not. In addition, the Doppler effect provides a measurable way to observe the passage of time.

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Revisiting the DoubleSlit Experiment: Observation, Complementarity, and the Ontological Implications of Delayed Choice Quantum Phenomena

YIZHEN HE

Abstract

The double-slit experiment remains central to the conceptual foundations of quantum mechanics, challenging classical notions of causality, locality, and objectivity. This paper revisits the experiment in light of delayed-choice configurations and the two-state vector formalism (TSVF), revealing the critical role of measurement—not only in determining observable outcomes but in potentially redefining temporal boundaries. Drawing on foundational literature and experimental realizations, we examine how observer-dependent phenomena, time-symmetric interpretations, and quantum retro causality inform our understanding of reality. We argue that the conjunction of delayed-choice effects and TSVF suggests a shift from ontological realism to a relational, context-dependent quantum framework. These developments not only enrich the philosophical discourse surrounding quantum mechanics but also demand renewed attention to the interplay between epistemology, causality, and physical law.

Introduction

The doubleslit experiment continues to stand as one of the most illuminating and philosophically provocative experiments in modern physics. Originating with Young's early 19thcentury demonstration of wave interference, it evolved in the 20th century into a central probe into the quantum behavior of light and matter (Young, 1802). The essential paradox persists: even individual particles such as photons or electrons, when sent one at a time through a pair of slits, generate interference patterns indicative of wavelike behavior, yet the pattern collapses when any measurement attempts to determine “which path” the particle traversed. This dual behavior has continually upset classical notions of determinism, locality, and objectivity.

In this essay, we revisit the doubleslit paradigm through the lens of foundational quantum experiments and interpretations. Our core questions are: How does measurement define quantum reality? To what extent can future measurement choices retroactively influence past events? Does quantum mechanics demand a revision of causal and ontological assumptions? By synthesizing classical analogies, Copenhagen interpretation, delayedchoice experiments, and the twostate vector formalism, we aim to present a coherent and rigorous account suitable for publication-quality discourse.

Conceptual Foundations: Classical Analogy and Quantum Duality

Thomas Young's doubleslit experiment first established the wave nature of light via the interference fringes produced by coherent light passing through two slits (Young, 1802). Later quantum mechanics recast this behavior: electrons and photons, when individual and unobserved, continue to exhibit interference, indicating intrinsic superposition of paths.

Feynman famously described the doubleslit setup as “the only mystery” of quantum mechanics (Feynman et al., 1965). To bridge intuition, he introduced analogies: golf balls aimed at two apertures and water waves passing through two slits. In the golfball scenario, opening slit L yields a probability distribution $PL(x)$ of impact positions on a distant wall; similarly, opening slit R yields $PR(x)$. Classical reasoning leads to $P(x)=PL(x)+PR(x)$. In contrast, water waves passing through both slits interfere, producing intensity proportional to the square of the resulting amplitude. The resulting interference pattern has no classical particle analog.

Quantum systems behave like the water waves when unobserved—generating interference—yet revert to the classical sum $PL+PR$ when “which-path” information is obtained. The measurement outcome thus determines the phenomenological behavior. This stark dependence of behavior on measurement underscores the measurement problem in quantum

mechanics.

Copenhagen Interpretation: Measurement and Complementarity

The Copenhagen interpretation, formalized by Niels Bohr and Werner Heisenberg in the mid-1920s, remains the mainstream theoretical framework for understanding quantum measurement (Bohr, 1928; Heisenberg, 1927). Here, a quantum system is fully described by its wave function ψ evolving unitarily via the Schrödinger equation. Upon measurement, ψ undergoes a nondeterministic “collapse,” with outcome probabilities given by the Born rule: $P = |\psi|^2$.

Bohr’s principle of complementarity asserts that mutually exclusive properties—such as wave-like interference and particle-like localization—cannot be observed simultaneously. The observable nature of a system depends fundamentally on the experimental context. For instance, if detectors identify which slit a photon passes through, interference disappears. Thus, one cannot meaningfully attribute both path and wave-function behavior simultaneously.

According to this interpretation, quantum systems lack definite properties prior to measurement. The act of measurement does not simply reveal preexisting values, but actively brings certain aspects of reality into being. While Einstein and others criticized this view as subjective or incomplete, Bohr emphasized that quantum theory ensures objective reproducibility within defined measurement contexts.

Delayed-Choice Experiments and Time-Symmetric Interpretation

Wheeler’s Delayed-Choice Thought Experiment

John Wheeler extended the paradox by proposing that the experimenter could choose whether to observe particlelike or wavelike behavior after the photon has entered the interferometer (Wheeler, 1978). In a Mach–Zehnder configuration, a single photon is emitted and reaches beam splitter BS1, becoming a superposition:

$$|\psi\rangle = \frac{1}{\sqrt{2}}(|\text{path 1}\rangle + |\text{path 2}\rangle).$$

Critically, the choice to insert or remove the second beam splitter BS2 is delayed until the photon is already “in flight.”

If BS2 is present, the two paths recombine and produce interference—wave behavior.

If BS2 is absent, detectors reveal which path the photon took—particle behavior.

This setup, particularly implemented with quantum random number generation to ensure space-like separation between the choice and photon entry, eliminates any possible “retroactive adaptation.” It forces the conclusion that the photon’s behavior remains undefined until the final measurement decision, even if made after the photon passes the slits (Jacques et al., 2007) (PubMed, ResearchGate).

Experimental Realizations

The 2007 experiment by Jacques et al. provided a near-ideal realization of Wheeler’s delayed-choice proposal. A single-photon source, a fast random number generator, and high-speed modulators ensured that measurement configurations were determined only after the photon had entered the interferometer. The results aligned perfectly with quantum predictions: the system behaved as a wave or a particle depending on the delayed choice, with no classical causal explanation (Jacques et al., 2007) (PubMed, ifujrj.br).

Subsequent implementations extended this paradigm to atoms, ions, and entangled-photon systems, consistently demonstrating that measurement choice—even post facto—defines the observed quantum reality, in direct conflict with

classical intuition (e.g., macroscopic atomic implementations in Nature Physics, 2015).

Two-State Vector Formalism (TSVF) and Retro causality

To make sense of these results, the twostate vector formalism (Aharonov, Bergmann & Lebowitz; Lev Vaid man) provides a time-symmetric description: a quantum system at intermediate time ‘t’ is fully determined by a pair of states, one evolving forward from the initial preparation $|\Psi\rangle$, and one evolving backward from the final measurement outcome $\langle\Phi|$ (arXiv, MDPI). Together, they form a two-state vector $\langle\Phi| \quad |\Psi\rangle$, encoding information from both past and future boundary conditions.

Under TSVF, weak measurements performed in between pre- and post-selection obey a modified probability rule (ABL formula), leading to weak values that can lie outside eigenvalue spectra. These features provide coherent and predictive descriptions of delayed-choice and quantum-eraser experiments without invoking retro causal paradox or signaling faster than light (tau.ac.il, arXiv, MDPI).

Recent reviews emphasize that retro causal descriptions arising from TSVF avoid contradictions with relativity or causality, as no information can be transmitted superluminally; instead, the formalism maintains internal consistency in which future boundary conditions play a vital explanatory role without violating operational constraints (SSRN).

Philosophical and Ontological Implications

Observer-Dependence and the Nature of Reality

These quantum phenomena force us to confront whether reality is observer-independent. Wheeler’s “participatory universe” metaphor suggests that the observer—and even future measurement choices—actively shape the physical phenomena observed. If a photon does not “choose” to behave as a wave or a particle until its measurement context is decided—even after it has traversed various paths—then classical realism is untenable.

Bohr’s principle of complementarity extends into a broader epistemological claim: reality is not predetermined but is brought about through context-specific observation. The TSVF refines this by indicating that both past and future measurements jointly determine outcomes, challenging classical linear causality.

Causality and Time Symmetry

Classical physics presumes a forward arrow of cause and effect. Yet delayedchoice and TSVF suggest time-symmetric influences: postselections are as essential as preselections in determining intermediate events. While information cannot be sent back in time to influence past events in a manipulative sense, quantum correlations respect a deeper symmetry: the future can influence the meaning—not the manipulable facts—of intermediate measurements. This aligns with interpretive frameworks like the two-state vector and transactional interpretations and undermines naive causal realism.

Complementarity Revisited

The traditional dichotomy between wave and particle behavior relies on complementary experimental setups. However, delayed-choice realizes complementarity in time rather than space. The entanglement swapping and quantum-eraser variants elevate complementarity to multi-particle entangled systems, showing that even entanglement structure can be “decided” after particles have separated.

Impacts Beyond Physics

Philosophically, these findings intersect with debates in metaphysics, epistemology, and even philosophy of mind. If observables depend on future contexts, notions of objective reality, free will, and causal agency come under scrutiny. While

mainstream physics remains cautious, earnest treatments such as Cramer’s transactional interpretation and TSVF offer coherent models that maintain experimental consistency while providing ontological transparency.

Conclusion

The doubleslit experiment, once a striking demonstration of wave behavior, has become a profound window into the epistemology and ontology of quantum mechanics. Through classical analogies, Copenhagen interpretation, delayedchoice realizations, and time-symmetric formalisms, we see that measurement context—and even future measurement decisions—cannot be ignored in understanding quantum events.

These results collectively challenge classical assumptions: determinism dissolves, locality intertwines with complementarity, and time’s arrow softens. The quantum world revealed through these experiments is relational, context-dependent, and subject to observer-defined reality.

Continued experimental advances—such as weak measurement tomography, entanglement-based delayed-choice, and macroscopic implementations—promise further illumination. Philosophers and physicists alike must confront the implications: quantum mechanics does more than predict outcomes—it compels us to rethink reality itself.

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23



Psychology and Neuroscience

About the Associations of Psychiatric Disorders and the Frontal Lobe

KUNLUN LI

From countless and repeated studies, we have discovered that the frontal lobe is responsible for language understanding and use, motor function, emotions, executive functions, consciousness, and essentially everything that forms one's personality (Hadjikhani, 2021). Therefore, malfunctioning of the frontal lobe can lead to a variety of psychiatric disorders such as bipolar disorder (Moebus, Quirin, and Ehrlenspiel, 2023), dissociative disorders (Modesti et al., 2022), attention deficit hyperactivity disorder (ADHD) (Leisman and Melillo, 2022), autism spectrum disorder (ASD) (Hadjikhani, 2021), narcissistic personality disorder (Nenadic et al., 2015), and others. This essay will discuss the prominence of frontal lobe dysfunction in psychiatric disorders and how the frontal lobe is associated with these disorders, which might inspire future studies.

First, the frontal lobe, as mentioned before, is responsible for executive functions, consciousness, language, emotions, motor control, and personality. It takes up 25% of the brain's area and is a lobe that is unique to humans compared to other animals. The prefrontal cortex is a prominent area of the anterior frontal lobe where most higher cognitive functions, such as executive functions, are carried out. In different disorders, this area will display different problems, as discussed in the following paragraphs.

In bipolar disorder, studies have found that the frontal lobe of the left hemisphere is more related to elevated moods and high motivation, while the right hemisphere is associated with depression and low motivation. This frontal cortex asymmetry correlates with the manic and depressive states and the severity of bipolar disorder (Moebus, Quirin, and Ehrlenspiel, 2023). Although the results are inconclusive, mania is correlated with increased activity in the left frontal lobe as a whole, particularly in the prefrontal cortex, while depression is correlated with increased activity in the right frontal and temporal lobes. These associations show that frontal lobe activity correlates with bipolar disorder states, which may aid in diagnosis.

In dissociative disorders such as dissociative identity disorder (DID), depersonalisation disorder (DPD), and dissociative amnesia, changes in frontoparietal lobe activation, especially in the prefrontal cortex, are prominent. DID patients in the traumatic identity state display reduced activation of the ventromedial (middle and lower) prefrontal cortex, which is mostly in charge of self-awareness, social awareness, emotion regulation, and consciousness. This may result in symptoms such as flashbacks, anxiety, re-experiencing traumatic memories, and others, but also shows hyperactivation in the prefrontal cortex during the resting state, which might be triggered by trauma re-evocation. Both DID and DPD patients show hyperactivation in the prefrontal and parietal lobes during verbal working memory tasks. For DPD patients, hyperactivation in the bilateral medial (middle) prefrontal cortex is significant during face processing of both stranger and self faces, which may indicate deficits in implicit self-processing, self-recognition, and self-awareness. As for dissociative amnesia, it usually co-occurs with the dissociative disorders above, so the results can also apply to these patients. Dysfunction of the frontal lobe, particularly the prefrontal cortex, is correlated with traumatic dissociative disorders, especially with the traumatic identity state of DID and the depersonalisation seen in DPD (Modesti et al., 2022).

Aside from emotions and consciousness, the frontal lobe is responsible for motor function and the inhibition of reactions to external stimuli. It is connected to the diencephalic region, which regulates arousal. Dysfunction of the frontal lobe can partially disrupt this circuit. Attention deficit hyperactivity disorder (ADHD) during childhood can result from delayed maturation of this region. ADHD patients of all ages display decreased activity in the frontal lobe, especially in the prefrontal cortex. Lesions in this area can cause an inability to inhibit reactions to task-unrelated stimuli, hyperactivity, and hyperreactivity (Leisman and Melillo, 2022).

For autism spectrum disorder (ASD), it has been found that autistic children have a greater frontal lobe volume than non-

ASD participants at an early age (two to four years old), but no difference is observed in other age groups, suggesting that this early development difference dissipates with age (Crucitti et al., 2022). Another hypothesis presents the possibility that excessive connections within the frontal lobe, caused by malformation during early development, lead autistic individuals to “talk to themselves,” resulting in fewer long-range connections with other parts of the brain that are responsible for responding to external stimuli (Courchesne and Pierce, 2005).

Narcissistic personality disorder (NaPD) is associated with grey matter loss in the frontal lobe, which means there are fewer neurons in this area for higher cognitive functions, as well as alterations in the right frontal white matter. Other evidence shows that antisocial personality disorder (ASPD) also correlates with white matter alterations (Sundram et al., 2012). Borderline personality disorder (BPD) patients exhibit structural and functional abnormalities in both the frontal and limbic areas (Krause-Utz et al., 2014). This suggests that frontal lobe abnormalities are associated with these personality disorders (PDs).

Conclusively, the frontal lobe is the part of our brain that distinguishes humans from animals and one person from another. Many disorders such as bipolar disorder, dissociative disorders, ADHD, ASD, and PDs are prominently associated with malfunction, malformation, alterations, delayed maturation, and abnormal activity in the frontal lobe. These associations might help further investigations into the functions of our brain and lead to advancements in methods of diagnosing these disorders.

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How Can Reinforcement Learning Be Used to Investigate Cognitive Function in Psychiatric Disorders?

ANGELINA TSZ YING WONG

Introduction

Reinforcement learning (RL) is the process of altering one's behaviour by providing positive or negative stimuli. Recent studies suggest that reinforcement learning can be applied in psychiatry to investigate cognitive functions in people with mental disorders (Maia & Frank, 2011). Schizophrenia, depression, and other psychiatric disorders present impairments in reward processing and decision-making. It is thought that by studying reinforcement learning, we can eventually discover effective and personalised treatment methods for patients with these mental disorders (Reddish & Gordon, 2016).

Model-free vs model-based reinforcement learning

The two types of reinforcement learning, model-free and model-based, are relevant to understanding cognitive functions. Model-free learning is when people learn from natural past experiences, while model-based learning involves building or using a model of the environment to predict future outcomes (Dayan & Berridge, 2014). The two systems correspond to different neurological mechanisms. Model-free learning is associated with the ventral striatum, which is involved in reward processing and motivation. On the other hand, the model-based system correlates with the prefrontal cortex in terms of planning and adapting behaviour based on the environment (Daw, Niv, & Dayan, 2005). Notably, studies have found a clear connection between dopaminergic signalling and the reward prediction error—the difference between the expected reward and the actual reward received. As a result, RL can aid in investigating cognitive functions in mental illnesses, supported by the link between neural pathways and reinforcement learning models.

The cognitive dysfunction of various psychiatric disorders has been extensively studied using reinforcement learning models, which examine how people with these conditions differ in their processing of rewards, punishments, and decision-making strategies.

Schizophrenia

Schizophrenia is a chronic psychiatric disorder characterised by symptoms such as hallucinations, avolition, and deficits in attention (WHO, 2022). This is associated with irregularities within the dopaminergic pathways, particularly involving the prefrontal cortex and striatum.

Individuals with schizophrenia tend to exhibit impaired model-based learning, resulting in difficulties interpreting models of the environment to guide decisions. In a study conducted by Culbreth et al. (2016), participants with schizophrenia struggled to complete the two-step decision task. While healthy individuals can adjust their behaviour to maximise rewarding outcomes based on understanding the structure of the task, individuals with schizophrenia often rely on model-free strategies, choosing based on whether the last action was rewarded instead of planning ahead.

Waltz et al. (2007) investigated the responsivity of patients with schizophrenia to positive and negative feedback using a probabilistic selection task. People with schizophrenia performed significantly worse than the control group in distinguishing the most rewarding stimulus, with about 50% of patients failing to reach the standard. Interestingly, their performance in identifying negative stimuli was unimpaired.

In conclusion, these studies demonstrate how RL models can predict symptoms of schizophrenia, such as impaired model-based learning and disrupted reward processing.

Depression

Major depressive disorder (MDD) has typical symptoms such as anhedonia, exhaustion, and cognitive deficits (NHS, 2023). It involves abnormalities in the prefrontal cortex, anterior cingulate cortex, and ventral striatum, all of which are essential for reward processing and decision-making (Huys et al., 2013).

A study by Pizzagalli et al. (2008) used a probabilistic selection task to measure how well participants respond to rewards. It has been demonstrated that depressed individuals frequently struggle to choose the course of action that will offer the greatest reward. This illustrates how the reward response is disrupted in those with depressive illnesses. The inability to identify positive stimuli is linked to anhedonia, a common symptom of depression. This study improves our understanding of the specific areas of brain function that are impaired in depression.

Drug Addiction

Substance use disorders (SUDs) involve compulsive drug-seeking behaviour despite harmful outcomes. Addiction is associated with the ventral tegmental area, a vital part of the reward system (News, 2022).

Studies have investigated drug addiction in rats by assessing their flexibility in adapting to environmental changes. Learning abilities were tested using reinforcement learning models such as reversal learning (Groman, Thompson, Lee, & Taylor, 2022). This model can identify problems in decision-making, as subjects with poor decision-making are more likely to develop drug addiction. The findings also apply to humans, as studies show that low sensitivity to expected reward in early adolescence may predict future drug abuse.

In animal experiments, rats were injected with drugs and tested again to determine any disruption in learning abilities. The results showed that mice exposed to drugs had an insensitive response to negative outcomes. This relates to characteristics of substance use disorders, such as antisocial behaviours towards others.

RL Treatment

Overall, by targeting reward-based learning mechanisms, reinforcement learning provides a method for developing targeted treatments for psychiatric disorders. Through adaptive feedback, RL-based therapies can reduce reliance on negative habitual behaviours, increase sensitivity to positive outcomes, and adjust dysfunctional decision-making processes. These therapies can address cognitive impairments and alleviate symptoms, for example, increasing cognitive flexibility in schizophrenia, resetting reward sensitivity in depression, or shifting drug-related reward biases in addiction towards healthier options (Maia & Frank, 2011).

There are still drawbacks to this approach. For example, its effectiveness may be limited for patients with severely impaired cognitive capacity. Additionally, the transfer of learned behaviours from training tasks to real-life situations can be slow. Nevertheless, reinforcement learning remains a crucial method for understanding cognitive functions in mental disorders.

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